

# IAMPerformance

## PHYSICS OF METHYLATION: LANDAUER METROLOGY

GAPE Issue 003 · measuring how far above the thermal noise quantum each cell class writes and holds its state, against a fixed physical zero

---

**Issue 003 RC1 · September 2026** · The Healthy Range of the Cellular Write Process — Eight Architecture Classes, Forty H<sub>min</sub> Values, One Atlas of 115 Cell Types, and the First Written Specification of the Running Engine

Heath W. Mahaffey · IAMPerformance · Entiat, Washington

Repository: [github.com/hmahaffey/IAM-Validation](https://github.com/hmahaffey/IAM-Validation) · engine as of the commit this PDF was built from (see git log; the chain has been rebuilt through 2026-09-21)

Prepared with Claude Science, 2026-09-19. **Release candidate 1.** Every number is loaded from the runtime files or from a dated run named in the text. Rows marked OPEN are unresolved and are printed as such.

### WHAT'S NEW IN ISSUE 003

**The claim, restated.** This is not a detection tool and not a treatment. Cells compute and write to a two-dimensional surface as a semiconductor or a quantum processor does; the physics of that write process has a healthy operating range per architecture class, calculated once and frozen (2026-04-06); departures from that range correlate with the hypo- and hyper-methylation patterns seen in disease *and* with normal variation in healthy individuals. Disease is corroboration, not product.

**The IAM Atlas.** IAMAtlasREBUILD: 483,092 CpGs, 115 cell types mapped to the eight classes, built 2026-05-28. Every class card now lists its cell types. The atlas did not exist when Issue 002 was written.

**Two instruments, stated.** The GAUGE (entropy of one mean  $\beta$  over per-class identity loci) and the SEPARATION statistic (mean per-CpG entropy over per-cell discriminative markers) are different instruments answering different questions. Issue 002 had one A-score.

**Per-patient cellular age is closed as not reportable (PROC-AGE-01); the aging TRAJECTORY stands and is reproduced.** The healthy immune curve rises 0.47 mA/yr, monotone by decade, on 1,379 donors from four laboratories - CPG-VAL-015's slope on Hannum, reproduced on labs it never saw. That drift is 0.047 across a lifetime against a within-lab spread of 0.0235, so one array resolves a person's age to about 50 years (16 of 100 healthy within 10 yr); the report prints the resolution, never an age. The trajectory is a population measurement and now IS the reference age curve. Sign differs by surface (RECON D2).

**Stage 5 is commissioned on the identity gauge (PROC-MAHA-02): one honest number, with the laboratory's own false-alarm rate printed beside it; the chip term is row 5b.** Karolinska: 10 of 100 healthy beyond p95 - chip-driven (per-chip median SD 0.020 vs 0.012 elsewhere); chip-centring brings every lab to 2-4%. A per-lab constant cannot reach the chip; a reference on the chip can.

**The gauge switch is done (PROC-SWITCH-02): the reported A is the identity-loci gauge with the three-layer reference; the marker-union statistic is diagnostic only.** Row B commissioned 2026-09-21. 40 held-out synthetic healthy patients read 1.001 (100% in band) where the old statistic read 1.125 (0%). Finding on the way: the atlas posterior is itself a fifth laboratory with zero -0.0146.

**Healthy is a band, not a line - and the reference has three layers (s3.5).** A = 1.0 is the architectural commitment line. A patient is read against FLOOR (H<sub>min</sub>, physics, universal) + PIPELINE MAP (one affine per pipeline onto the Roadmap scale; Stage 1s COMMISSIONED, PHASE 1c) + LAB ZERO (40 healthy arrays per lab, read against the reference age curve, set once - PROC-PANEL-03, COMMISSIONED; the control-probe route was tested on four labs and closed, LAB-ZERO-01/02), then placed in a pooled p10-p90 band of lab-zeroed healthy A (identity\_band\_v3, four laboratories, COMMISSIONED PROC-SWITCH-02). The shipped age\_reference\_matrix.json was compiled on the marker-union statistic and is NOT the reference for the identity-loci gauge (PROC-N7-01). The per-class input offset explored in July is retired.

**Presence before score.** A class's A-score is a reading only if the deconvolver finds the class in the sample (DETECT\_FLOOR). Scoring absent classes produced spurious BREACH calls in 4/4 healthy plasma samples on 2026-09-19; the rule is now printed.

**Where the tools come from (§5A).** The Mahaffey number (20.94 = dG<sub>ATP</sub>/RT, the cell's ATP budget over k<sub>B</sub> T; the name n<sub>bio</sub> retired); the forty MCMC-confirmed floors and their forty ceilings; the atlas as a posterior; the Planck toolkit borrowed for the methylome - brightness sky, masking and matched filter (built in spring), and the four disciplines that did the work of 2026-09-19/20: end-to-end simulation with known truth (PROC-N7-01), cross-method disagreement as diagnostic (PROC-NILC-01 - the second deconvolver was cut in July, rerun in September, and vindicated; RUNBOOK s11 now forbids disabling either relay), absolute calibration against an unseen lab (PHASE 1c), and instrument housekeeping from the array's own control probes (LAB-ZERO-01); and why this is not a cohort method (s0, s1.6).

**Reconciliation table (§1).** Every constant and rule that changed between April and repo HEAD, with the commit that changed it. Nothing silently updated.

**Sections 7–8 replaced.** The detection-trajectory and deployment-readiness sections are replaced by Substrate Characterization (§7) and Procedures (§8): what each substrate carries, what is verified, what is open, and the exact steps that reproduce each verified number.

**Stage 1 executed from raw IDATs - the chain is closed.** All eleven test IDAT pairs through the repo's own calibrator reproduce the cached betas bit-for-bit (11/11, max diff 0.000000, both array types). Raw IDAT -> beta -> class fractions -> sealed anchor: every link now reproduced from the repository on a stranger's machine (PROC-CAL-01).

**The end-to-end simulation caught the production gauge reading the wrong CpG set.** A synthetic healthy patient - a pure mixture of healthy Atlas posteriors - reads BREACH through the conductor, because Stage B computes  $H(\beta_{\text{mean}})$  over the bimodal class marker union, not the identity loci its docstring and the SOP name; the age band was compiled the same way, so real blood never showed it. The identity-loci gauge reads the synthetic at 0.99 and real adenoma at 1.10 but has no band yet. Phase 1 was run as the prerequisite and sealed FAIL on its own terms - which is how the pipeline-scale offset (LESSON-SCALE-01) was found (PROC-N7-01, RECON A4, PHASE 1).

**The sealed breast anchor reproduced from raw GEO.** GSE51032, 460 samples, 115 cell types: mean-of-H over the v0\_2 discriminative markers returns the sealed CSV at  $r = 1.00000$ , max diff 0.00004, on a machine that had never seen the project.  $H(\beta_{\text{mean}})$  on the same markers does not ( $r = 0.45$ ). The seal rests on the repo-HEAD markers, not the chrX-removed copy (PROC-ANCHOR-01).

**The aggregation conflict between the current SOP and the running code, measured.** SOP v1.4.0 says never  $H(\beta_{\text{mean}})$ ; the chain at HEAD computes  $H(\beta_{\text{mean}})$ ; the age band is compiled as  $H(\beta_{\text{mean}})$ . On whole blood with the class present the two aggregations are rank-identical with a +0.029 offset; on mixed or absent-class panels  $H(\beta_{\text{mean}})$  inflates as §105 says. Ruled in §1.5: one aggregation per surface, fixed by the reference each was built with (RECON A3) - and the analyst's own first reading of the tissue values is retracted in §10.

**First real-data ground truth for the deconvolver.** Nine known genomic-DNA mixes (Moss 2018): the terminal class recovers a neuronal spike ( $r = 0.945$ ); secretory and cycling do not recover hepatocyte or colon spikes. Recorded as PASS/FAIL in §8 and §10, not smoothed over.

**Operating rules and falsification record (§9–10).** The lessons that govern how the instrument may be read, dated and sourced; and every outcome that went against the framework, kept on the record.

---

#### COVERAGE OF THE CHAIN, BY STAGE (as of this RC)

Issue 003 is the one document for this work; there is no "next issue". Every stage of the chain at HEAD is described in the switching order (Part II) and carries a status in CHAIN\_COMMISSIONING.md. As of this RC: **commissioned** - Stage 1 calibration (PROC-CAL-01), Stage 1s scale map (PHASE 1c), Stage 2 deconvolution (PROC-DECON-01, PROC-ANCHOR-01, N7). **Run and recorded with open defects** - Stage 0 intake (fail-open closed; intensity QC hand-off unwired), Stage 2b lineage splitter (PROC-SEP-03), Stage B class gauge (identity-loci statistic emitted alongside the wired marker-union gauge; band gated on the lab zero), Stage 4.6 patient CMB (four-skies plate; assessability gate to fix), Stage 5 Mahalanobis (driven by the stem\_adult false alarm; key names reconciled), Stage 6 cellular age (pinned at the curve floor; not reportable), Stage 7 tiers, Stage 8 disease matching (separation surface reproduces the sealed anchor), Stage 9 report. **Built, not re-run this cycle** - Stage 4.5 bidirectional. **Not wired by decision** - Stage 3 foreground (SOP s104). Row N (nulls and simulation) is verified and runs on every release (RUNBOOK s11).

**Described only by reference in this RC:** the disease-signature matrix v1.13 (read in s3 for the myeloid rows; the full matrix is a repo file), the crown-jewel disease wall, the directional AD detector (Stage 4.5, VAL-050/051 - reproduce from the kit is an open commissioning row), and the report builders (Stage 9 - the report is commissioned only when Stages 5 and 6 are). These are open rows, not deferred chapters.

#### WHAT THIS PAPER IS NOT

Not clinical validation. Not a diagnostic. Nothing here should inform patient care. The instrument is at a research stage; any clinical use requires prospective validation, regulatory review and qualified clinical oversight. Public retrospective cohorts only; sample sizes in the new runs are small and are printed beside every number.

# CONTENTS

Every chapter by its exact name, with the page it starts on. Generated against the rendered document, so a page number here is the page the section is on.

## FRONT MATTER

- What's new in this issue . . 1
- What this paper is not . . 2
- What this document claims, and what it does not . . 5
- Prior art - the door into the conversation . . 7

## PART I - THE INSTRUMENT

- s1 Reconciliation: Issue 002 to repository HEAD . . 8
- s2 The IAM Atlas: 115 cell types, eight classes . . 19
- s3 Two instruments and the presence rule . . 20
- s4 Framework from Issue 002: ranking, substrates, saturation . . 24
- s5 The physics: Landauer, the Mahaffey number, the reference, the gauge . . 116
- s5A Where the tools come from . . 108

## THE EIGHT ARCHITECTURE-CLASS CARDS

- #1 Immune & Hematopoietic . . 33
- #2 Cycling Epithelial . . 41
- #3 Secretory Glandular . . 49
- #4 Stromal & Connective Tissue . . 56
- #5 Adult Tissue Stem . . 63
- #6 Progenitor / Transit-Amplifying . . 71
- #7 Terminal / Post-Mitotic . . 78
- #8 Pluripotent Stem . . 85

## PART III - THE CHAIN IN DEPTH

- III.1 The chain, stage by stage . . 93
- III.2 The atlas, and the cells it can speak about . . 96
- III.3 The cosmology toolkit, tool by tool . . 97
- III.4 What the chain refuses, and its cosmology twin . . 99
- III.5 The engine, exactly - every formula the chain computes . . 100
- III.6 The screens, in the order the report prints them . . 102
- III.7 The healthy reference - who it is, and what is not in it . . 103
- III.8 Coverage - what is lit, and what lighting one cell requires . . 104
- III.9 The guards - what each refuses, and its last result . . 105
- III.10 The files of the chain, by role . . 106
- III.11 Running it on your own sample . . 107

## RECORD AND APPENDICES

- s7 Substrate characterisation: the substrate x class grid . . 119
- s8 Procedures . . 122
- s9 Operating rules . . 137
- s10 Falsification record . . 139
- s11 Engine map: every stage, and whether this issue covers it . . 142
- s12 For the clinician: the instrument in plain terms . . 144
- Appendix V - the validation index . . 145

|                                                        |            |
|--------------------------------------------------------|------------|
| Appendix VI - the CMB to methylome translation map . . | <b>157</b> |
| Appendix VII - the completion sprint, scored . .       | <b>167</b> |
| Future goals - what is worth the effort, in order . .  | <b>168</b> |
| Glossary . .                                           | <b>176</b> |

# WHAT THIS DOCUMENT CLAIMS, AND WHAT IT DOES NOT

## WHEN WE SEAL

SEALING RULE (author, 2026-09-21). We seal when we have a working tool. A pre-registration is for a BUILT tool whose question is whether it performs against a bar; building it - finding the zero, the scale, the floors, the surface - is exploration and gets a dated working note, not a seal and not a falsification row. Retrospective marking: PROC-CMB-01 to -04 and PROC-MATCH-01 in this document were exploration; the sky was sealed once as PROC-CMB-05. The falsification register records claims that failed, not construction that iterated.

## WHAT THE REPORT SAYS

WHAT THE REPORT SAYS (author, 2026-09-21). Cells detected and their percentages; the A-score of each cell the chain could calculate and of each class on the three-layer reference, with band placement and tier; the Stage 5 departure with the laboratory's false-alarm rate; the patient's sky; every flag, logged. It names no disease and gives no age in years. The disease matrix and its cards, compiled from the preliminary record, are record-side: cohort cellular behaviour is logged as trusted-chain cohorts accumulate. Our job is not to say 'the pattern in this patient's immune cells looks like colon cancer'.

## WHAT THIS DOCUMENT MAY SAY ABOUT DETECTION

RULE (author, 2026-09-21). Issue 003 describes what the chain IS and what has been MEASURED through it. A statement about what the commissioned chain can or cannot detect is made only from a sealed procedure that ran the chain on that question. Until then the language is 'not yet tested', never 'cannot'. Recording a measured DEFECT (e.g. the retired sky formula reading 61% of a healthy genome as anomalous, PROC-CMB-01 C1) is a measurement and stays. Retired the same day by this rule: 'only breast has a pre-diagnostic window' (the record holds VAL-046 2-5 yr multi-cancer, VAL-032 2 yr fragment size, VAL-083 CLL queued); 'healthy blood carries only immune' (Stage 2 reports terminal, stem\_pluri, secretory or cycling above 2% in some healthy arrays, and the record has scored them); 'no organ-level claim is supported by either channel' (cohort-level attribution is in the record; the absolute reading is not yet tested).

## WHAT THE INSTRUMENT REPORTS

In the author's words (2026-09-21): CPG reports whether a person's cellular write process is operating within the healthy range for their age, by architecture class, against a fixed physical zero. It is not a clock: a clock asks how old a person looks and attaches no action to the answer; this asks whether the healthy range is held, and - when it is not - which class of cells slipped. Each clause is a commissioned layer: the range is the floor + identity\_band\_v3 (four laboratories, 1,379 healthy); 'for their age' is reference\_age\_curve\_v1 (measured; the slope CPG-VAL-015 found); 'sitting in it' is the Stage 5 departure with the laboratory's false-alarm rate beside it; 'which cells' is Walther's eight class fractions and the 115-cell atlas beneath them. The last clause is DEMONSTRATED AT THE COHORT LEVEL and not yet absolutely: the atlas and deconvolver have named the moved cells and their organ against a control centroid - breast pre-dx (GSE51032/51057, a coordinated 115-cell shift 10+ yr before diagnosis, anchors reproduced  $r = 1.00000$ ), AD/FTD/PSP-CBD by direction on GIFT, and pre-Atlas VAL-007 (9/9 tissue-of-origin in plasma), VAL-008 (19/19 cancer types), VAL-003 (field effect, 28 tumour types). What remains is the same reading with no control group in the room: one patient against the class band and the laboratory zero. Today one axis (immune, whole blood) carries a band; per-class bands and the absolute breast and CRC readings (paper two) are the gates, in order.

## WHAT THIS FIELD IS CALLED

This work is LANDAUER METROLOGY: the measurement of how far above the thermal noise quantum  $k_B T$  an information-writing process operates, against a fixed physical zero, in any substrate. Applied to the methylome it is the PHYSICS OF METHYLATION. Thermal noise is not the nuisance here; it is the unit:  $M = E_{\text{drive}} / k_B T$ . The Landauer bound is the peer-reviewed anchor (s0b); the metrology - calibration, transfer, absolute reading, reference interval - is what this document adds.

### Not claimed.

- the world's largest atlas, or a new MCMC atlas built to sell
- that one atlas is the right structure - a future builder may well use one per cell class, or a classification not yet conceived; that is fine
- detection of all diseases, or of any disease ten years early
- detection of the myeloid cancers (MDS, CML, CHIP) - those matrix rows are unvalidated and say so
- clinical readiness of any kind - nothing here is validated for patient care

**Claimed.** A method biology did not know it needed: the physics of methylation - a per-class thermodynamic reference, an MCMC atlas as the posterior, and the CMB toolkit (end-to-end simulation, component separation, cross-method comparison, split-half replication) as the validation discipline. On a very limited public dataset with no funding or institutional support, this method already resolves things that cohort comparison

structurally cannot (s1.6). For the immune class in whole blood there are reproducible signals worth investigating with proper support. That is the claim, and all of it.

**Invitation.** The record is written so that a reader can break it: every constant loaded from a file, every procedure runnable from the kit, every reversal kept. The intended reader is the one who says 'cohorts, circular' first - and then pulls the repository.

# PRIOR ART - THE DOOR INTO THE CONVERSATION

Established physics we stand on, what it left open, and why this is cited but not built upon

**Cited.** Sanchez R, Mackenzie SA (2016). Information Thermodynamics of Cytosine DNA Methylation. PLoS ONE 11(3): e0150427. Sanchez R, Yang X, Maher T, Mackenzie SA (2019). Discrimination of DNA Methylation Signal from Background Variation for Clinical Diagnostics. Int J Mol Sci 20:5343. Software: MethylIT (R).

**Independent arrival.** This work did not build on, borrow from, or arrive via Sanchez & Mackenzie. The author reached the Landauer floor from the other end: cosmology (an entropy functional on the Friedmann background, 18 converged MGCAMB/CAMB chains, 2025-26), then particle physics, quantum computing (QAPE) and semiconductors (SCAPE), and only then the cell (GAPE, 2026). Their 2016 paper was first read on 2026-09-20, after every  $H_{\min}$  value, the Atlas, the deconvolver and the validation record existed. It is cited here because it is peer-reviewed physics that the methylome obeys Landauer's bound, and because a reviewer who knows MethylIT would otherwise assume dependence.

## WHAT WE AGREE ON

- Landauer's principle -  $k_B T \ln 2$  per irreversible logic operation - is the physical anchor for cytosine methylation as an information-writing process. (Sanchez & Mackenzie 2016 tested adherence to it on Arabidopsis ecotypes and 93 human tissues; IAM's  $H_{\min}$  is the same bound expressed as a per-class entropy floor.)
- Methylation carries thermal background: fluctuations at operating temperature produce a distribution of beta that is not regulatory signal. (Their Weibull/generalized-gamma model of that background; IAM's  $k_B T$  denominator of the Mahaffey number.)
- Information-theoretic measures on beta (Shannon entropy; Hellinger / total-variation divergence) are the right vocabulary, not raw differences.
- Cohort-comparison methylome statistics have not translated to the clinic (their 2019 paper says so in its first paragraph; this document says so in s0 and s1.6).

## WHAT WAS MISSING - AND WHAT THIS DOCUMENT ADDS

- A FIXED ZERO. MethylIT's reference is the centroid of the control group in the study at hand - a crowd-relative zero re-etched per cohort. IAM's reference is  $H_{\min}$ : one per class per substrate, set by physics and frozen (G-002 MCMC, 2026-04), never re-fit on a cohort. Their instrument has no absolute zero; this one does.
- A SINGLE-SAMPLE ABSOLUTE READING. MethylIT asks 'which sites moved, relative to controls, beyond thermal background' - a group question. IAM asks 'how far is THIS sample from the healthy floor of its cell class' - a per-patient question, answerable with no other patient in the room (s3, the potassium analogy).
- COMPOSITION FIRST. Whole blood and tissue are mixtures; MethylIT scores the mixture. IAM deconvolves against the MCMC Atlas (115 cell types, 8 classes) and reports a class gauge only where the class is present and determined (s108).
- THE CMB TOOLKIT for navigating thermal fluctuations - end-to-end simulation with known truth (PROC-N7-01 found the wired gauge reading the wrong CpG set), cross-method disagreement as diagnostic not defect (PROC-NILC-01), absolute calibration against an unseen lab (PHASE 1c), instrument housekeeping from the array's own control probes (LAB-ZERO-01). None of these exist in a cohort-relative pipeline because a cohort-relative pipeline has nothing to calibrate against.

## IN ONE LINE

Both use  $k_B T \ln 2$ . Sanchez & Mackenzie use it to model the thermal background and FILTER it away to find regulatory sites. IAM uses it as the UNIT of measure and asks how far above it a healthy cell holds its pattern. One filters, one calibrates. The filter is a legitimate and complementary tool - a principled way to ask which identity loci carry regulatory rather than thermal signal - and is noted as a Future Goal.

## THERMAL NOISE IS THE RULER

Thermal noise is not a nuisance in IAM; it is the ruler.  $M = E_{\text{drive}} / k_B T$ : the M1 transistor pays ~117 thermal quanta per switch at 348 K, the cell nucleus 20.94 per ATP at 310 K, the aluminium transmon exactly 1 at its gap temperature - the saturated case qubit engineers reach by cooling until  $k_B T$  is the only scale left. The cell cannot cool;  $H_{\min}$  is where a healthy cell holds its pattern against noise it cannot escape.

SECTION 1

RECONCILIATION — ISSUE 002 → REPO HEAD

Issue 002 was written in April 2026, before the atlas and before the engine was rebuilt (commits 2026-06-25 to 2026-07-03). This table states, for every constant and rule that differs, what 002 said, what the running engine says, why the engine is current, and where to check. Rows marked **OPEN** are real inconsistencies at HEAD that this document does not resolve. The chain-of-custody SOP v1.3.x predates the rebuild; commit d7b0e1f explicitly supersedes its §41. Until this section, the commits were the only specification of the running engine.

| #                    | Item                                                  | Issue 002                                                                      | Repo HEAD (66f37fe)                                                                                                                                    | Why HEAD is current                                                                                 | Evidence                                                                       |
|----------------------|-------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| H1                   | 40-cell H_min table (8 classes x 5 substrates)        | as printed 002 pp.7,105                                                        | byte-identical in cpg_gauge_engine.py<br>H_MIN_TABLE                                                                                                   | unchanged                                                                                           | G-002/G-003b freeze<br>2026-04-06; diffed<br>2026-09-19: 40/40 equal           |
| A1                   | Where healthy sits                                    | A ~ 1.0 with an age curve (AGE_REF, 7 points/class)                            | A = 1.0 is the COMMITMENT LINE, not healthy. Healthy = age_reference_matrix.json p10-p90 band (10 decades/class, n per decade, source per decade)      | engine docstring; Issue 002 \$'NOT forced to 1.0'                                                   | cpg_gauge_engine.py lines 26-29; commit 8cdf352<br>2026-07-01                  |
| A2                   | Age reference values                                  | AGE_REF immune<br>40y=0.946, 60y=0.966, 80y=0.992                              | age_reference_matrix immune<br>44y=0.9477 [p10 .9021, p90 .9934],<br>64y=0.9652, 84y=0.9873                                                            | HEAD carries percentiles + n + citation per decade; 002 carried means only                          | age_reference_matrix.json; Hannum 2013, Horvath 2013, Alisch 2012              |
| O1                   | Per-class input-scale offset                          | not present                                                                    | explored (immune/methyl = 0.055, commit 143704d) then RETIRED                                                                                          | it was reinventing the age band and forcing healthy onto 1.0                                        | cpg_gauge_engine.py lines 46-53                                                |
| T1                   | Tier ladder (gauge)                                   | NORMAL<1.01<br>MARGINAL<1.05<br>DETECTABLE<1.07<br>URGENT<1.10<br>BREACH>=1.10 | identical                                                                                                                                              | unchanged                                                                                           | cpg_gauge_engine.py lines 38-44; 002 tier_label()                              |
| T2                   | Tier ladder (Stage 7 customer, tier_breakpoints.json) | not in 002                                                                     | SUPPRESSED<0.95 NORMAL<1.01<br>ELEVATED<1.07 [Warburg line 1.07]<br>SIG_ELEVATED<1.10<br>BREACH>=1.10                                                  | NORMAL->ELEVATED edge moved 1.04->1.01 to match gauge MARGINAL onset (last commit)                  | commit 66f37fe<br>2026-07-03; tier_breakpoint.s.json _meta.marginal_onset_note |
| T3                   | Two tier vocabularies coexist                         | one                                                                            | gauge says MARGINAL/DETECTABLE/URGENT; Stage 7 says ELEVATED/SIGNIFICANTLY_ELEVATED                                                                    | OPEN - same breakpoints except the gauge's 1.05 split; one vocabulary must be chosen for the report | both files at HEAD                                                             |
| S1 (LESSON-SCALE-01) | Substrate floors for cfDNA                            | cfDNA listed as a substrate with % contributions (CFDNA_PCT)                   | 'cfDNA is the frag/wps/nucl substrates - score it on those floors, NEVER the methyl floor'                                                             | engine docstring                                                                                    | cpg_gauge_engine.py lines 56-58                                                |
| I1                   | Two instruments                                       | one A-score                                                                    | GAUGE = H(beta_mean)/H_min over per-class IDENTITY loci;<br>SEPARATION = mean_i(H(beta_i))/H_min over per-cell DISCRIMINATIVE markers                  | feeding the gauge discriminative markers produced the all-BREACH bug of 2026-06-11                  | cpg_gauge_engine.py lines 16-25; iamatlas_a_scoring.py                         |
| D1                   | Deconvolver role                                      | not in 002                                                                     | Walther NNLS on IAMAtlasREBUILD -> class + cell-type fractions;<br>PRESENCE gate: a cell/class below DETECT_FLOOR is absent and its A is not a reading | cpg_conductor.py 2026-07 '<br>Replaces the confusing walther_clinical.py'                           | cpg_conductor.py<br>DETECT_FLOOR = 0.01                                        |

| #   | Item                                                      | Issue 002  | Repo HEAD (66f37fe)                                                                                                                                                                                                                                                                                                                                                                                              | Why HEAD is current                                                                                                                                                                                                                                        | Evidence                                                                  |
|-----|-----------------------------------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| D2  | Presence floor value                                      | -          | 1% (cpg_conductor.py) vs 3% (README_FOR_FUTURE_AI Mahalanobis adjudicator gate)                                                                                                                                                                                                                                                                                                                                  | OPEN - two floors give different answers on plasma terminal class                                                                                                                                                                                          | both files at HEAD; run 2026-09-19                                        |
| D3  | NILC second deconvolver                                   | -          | CUT from Stage 2 on 2026-07-02; REINSTATED 2026-09-22 as the class-level second opinion (stage_2b_second_opinion in the conductor: compared with Walther by class, agreement bar $L1 \leq 0.10$ , reported as a flag and never as the composition) MEASURED (PROC-NILC-01): $\kappa(\text{blood sub-problem}) = 30.6$ , $r(\text{progenitor, stem\_adult}) = +0.99$ ; NILC L1 vs Walther 0.55-0.65 on all blood. | the divergence was the diagnostic NILC was designed to give: stem_adult/progenitor are not separately determined by the Atlas in blood. Report them jointly for whole blood until they are; a second deconvolver's disagreement is to be read, not failed. | commit c1be0c3 2026-07-02; PROC-NILC-01 2026-09-19; reinstated 2026-09-22 |
| D4  | Deconvolver validation scope                              | -          | N7: MAE 0.0076-0.0093 on SYNTHETIC Dirichlet mixtures; conformance MAE 0.0004 vs TEST_DATA_MANIFEST on 3 real EPIC tissue samples                                                                                                                                                                                                                                                                                | never validated against real tissue of KNOWN composition; GSE122126 in-vitro mixes are the available ground truth                                                                                                                                          | N7_OUTCOME.md; run 2026-09-19                                             |
| K1  | Atlas                                                     | not in 002 | IAMAtlasREBUILD: 483,092 CpGs, 115 cell types -> 8 classes, built 2026-05-28                                                                                                                                                                                                                                                                                                                                     | predecessor IAMAtlas.csv.xz retired (collapsed; flatness bug)                                                                                                                                                                                              | IAMAtlasREBUILD_provenance.json                                           |
| K2  | Atlas CpG count stated                                    | -          | provenance + file: 483,092; SOP v1.3.3 §28: 481,966                                                                                                                                                                                                                                                                                                                                                              | OPEN - SOP figure is stale or counts a filtered set                                                                                                                                                                                                        | diffed 2026-09-19                                                         |
| M1  | iamatlas_celltype_markers_v0_2.json                       | -          | trial zip 200,311 B vs repo HEAD 237,545 B - DIFFERENT files, same name                                                                                                                                                                                                                                                                                                                                          | OPEN - which is current must be stated; version the filename                                                                                                                                                                                               | sha256 diff 2026-09-19                                                    |
| SOP | Chain-of-custody SOP                                      | -          | SOP v1.3/1.3.3 is PRE-atlas; commit d7b0e1f says 'supersede SOP §41'; no post-atlas SOP exists                                                                                                                                                                                                                                                                                                                   | the commits 2026-06-25..07-03 ARE the spec; this document is the first written one                                                                                                                                                                         | git log; README_FOR_FUTURE_AI.md                                          |
| SEC | Sections 7-8 (detection trajectory, deployment readiness) | present    | REPLACED by Substrate Characterization + Procedures                                                                                                                                                                                                                                                                                                                                                              | user decision 2026-09-19: not a detection tool; the claim is the healthy range of the write process                                                                                                                                                        | this issue                                                                |
| F1  | Stage 3 foreground subtraction                            | not in 002 | NOT WIRED by decision (SOP s104 production foreground firewall): the production chain subtracts no age-, sex- or smoking-driven change. Sex: 131 chrX markers removed 2026-06-11. Age: the reference band. Smoking: Stage 7 bin floor shifts (v1.2 interim). Subtraction modules exist as test-only tooling.                                                                                                     | a stated design decision, not a gap; the flowchart's 'NOT BUILT' should read 'NOT WIRED - by s104'                                                                                                                                                         | SOP v1.4.0 s104; flowchart_vKISS                                          |
| F2  | Stage 6 cellular age module                               | not in 002 | iam_cellular_age_scoring.py present in CPG_TRIAL_CODE.zip, ABSENT from repo engine directory                                                                                                                                                                                                                                                                                                                     | OPEN — the canonical repo does not carry a file the trial bundle does                                                                                                                                                                                      | sha256 listing 2026-09-19                                                 |
| F3  | walther_clinical.py origin gate                           | not in 002 | disease_origin_cells.json loaded inside a bare except; on failure the cell-of-origin specificity rule is silently disabled                                                                                                                                                                                                                                                                                       | OPEN — patient-safety gate fails open; three-line fix                                                                                                                                                                                                      | walther_clinical.py ~lines 972-996 (CPG_first_read.md finding 3)          |

| #   | Item                                                         | Issue 002                                                                                                                   | Repo HEAD (66f37fe)                                                                                                                                                                                                                                                          | Why HEAD is current                                                                                                                                                                                                                                                                                        | Evidence                                                                                                      |
|-----|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| F4  | Stale strings describing the retired mean-of-entropies gauge | —                                                                                                                           | cpg_conductor.py Stage A comment ' (mean-of-per-CpG H/H_min)'; cpg_gauge.py line 158 axis label prints on the patient report figure                                                                                                                                          | OPEN — neither changes a number; both are how the 2026-06-11 bug gets reintroduced                                                                                                                                                                                                                         | CPG_first_read.md finding 2                                                                                   |
| F5  | Gauge carries no sign                                        | —                                                                                                                           | A(beta)=A(1-beta); INVERSION below band is degenerate between hyper- and hypo-methylation. beta_mean is computed one line before A and not reported                                                                                                                          | OPEN — report beta_mean or signed departure beside every A; no revalidation needed                                                                                                                                                                                                                         | iamatlas_a_scoring.py; verified A(0.61)=A(0.39)=1.1501 on immune                                              |
| F6  | Two A-score definitions in the corpus                        | 002 §2.5 C1/C2/C3                                                                                                           | A = H/H_min (all numerical results) vs A = 1 + C3/H_actual (three-component decomposition) — equal only at A=1                                                                                                                                                               | OPEN — one must be retired or the relation stated                                                                                                                                                                                                                                                          | 002 §2.5; IAM_for_physicists §4                                                                               |
| F7  | Immune H_min revision history                                | 0.838889                                                                                                                    | calibration value 0.795000 -> 0.8389 +/- 0.0012 after MCMC reanalysis of six immune cell types; immune A moved down 0.055                                                                                                                                                    | state the freeze date and provenance so 'frozen before disease testing' is checkable                                                                                                                                                                                                                       | EDEAR s4.6 (0.795 -> 0.838889, 6.44 sigma); Issue 002 immune card ('revised downward by approximately 0.055') |
| A3  | A-score aggregation: gauge vs separation                     | A = H(beta)/H_min, beta = the cell's mean methylation; H_min_global = H(0.790)                                              | SOP v1.4.0 s105: mean-of-H, never H(beta_mean). Code at HEAD: H(beta_mean) over identity loci; age band compiled the same way.                                                                                                                                               | RULED (see RULING A3): gauge on identity loci = H(beta_mean)/H_min, matching H_min's own definition and the band; separation on discriminative markers = mean-of-H, matching s105 and the sealed anchors; s105 amended to scope its 'never' to marker panels; runtime bimodality guard added to the gauge. | PROC-FORMULA-01; PROC-ANCHOR-01; RULING A3                                                                    |
| M1b | iamatlas_celltype_markers_v0_2.json - which is current       | -                                                                                                                           | TRIAL file carries the 2026-06-11 chrX removal; REPO HEAD does not; the 2026-05-29 seal rests on the pre-removal file.                                                                                                                                                       | RULED (see RULING M1b): adopt the chrX-removed file, commit it, re-seal both anchors (done 2026-09-19: 32/115 cells shift, max 0.059, r = 0.9996; v1 seal retained as SUPERSEDED).                                                                                                                         | PROC-ANCHOR-01; kit/anchors_v2/RESEAL_REPO.json                                                               |
| A3b | SOP v1.4.0 s105 open decisions vs the code                   | -                                                                                                                           | s105 leaves three decisions open (Recipe s6.3 source; marker convention discriminative-vs-most-methylated; s46.5 re-derivation). The 07-01 code answered none of them and introduced a fourth convention (identity loci + H(beta_mean)) with the age band compiled to match. | CLOSED by RULING A3 for decisions 2 (marker convention: identity loci for the gauge, discriminative for separation) and 3 (s46.5 stands, gauge keeps H(beta_mean)). Decision 1 (Recipe s6.3) remains the author's to record in the vault.                                                                  | RULING A3                                                                                                     |
| M2  | 'Mahaffey number' - definition and value                     | glossary: 'n_bio = dG_ATP/(RT) = 20.94'                                                                                     | author 2026-09-19: M = 20.94, one fixed number, no ln2; n_bio is an early QAPE-era form, no longer used; H_min is a separate per-class quantity                                                                                                                              | RULED BY AUTHOR. Open: Hubble2GAPE l.2192 ln2 form (~30) to be reconciled.                                                                                                                                                                                                                                 | author statement; 002 glossary p251; H2G l.2192, l.3504                                                       |
| B1  | class bands are cohort statistics of unequal provenance      | age_reference_matrix.json: immune from hundreds of samples across decades; stem_adult n=28 single source; others in between | the FLOOR (H_min, 40 values) is physics and is one number per class-substrate; the BAND (age_reference_matrix) is a cohort statistic and is only as good as its n and pipeline. Every 'healthy reads wrong' case of 2026-09-19 traces to a band, never to a floor.           | Phase 1: rebuild all eight class bands from ONE healthy cohort (GSE87571, n~730, ages 14-94) through ONE pipeline (Stage 1); record n/source/pipeline per band; classes without a defensible band from blood -> NOT_ASSESSABLE                                                                             | PROC-CHAIN-01                                                                                                 |

| #  | Item                                             | Issue 002                                                                     | Repo HEAD (66f37fe)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Why HEAD is current                                                                                                                                                                                                                                        | Evidence                                                |
|----|--------------------------------------------------|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| V1 | what 'A' is in VAL-001/003/008 (the 27/28 claim) | report: 'A = H(beta)/H_min(class)'                                            | numerator = COHORT-MEAN GENOME-WIDE beta (TCGA tumor / solid-tissue-normal means, Xu 2019; healthy from one Roadmap cell per class); divisor = H_min of the cancer's tissue-of-origin class (LGG->terminal, BRCA->secretory, COAD->cycling ...). Class-correct divisor, global-mean numerator, cohort-level - a different instrument from the identity-loci gauge, sharing only the divisor. The script itself notes a '~10% global entropy offset expected'.                                                                                                                                                                                                                                                                                                                                                                          | state this wherever 27/28 is quoted; it explains why direction holds while absolute A is offset                                                                                                                                                            | GAPE_Evidence_Report_UPDATED VAL-003 script I.5208-5544 |
| A4 | which CpG set the production gauge reads         | docstring / SOP s41 / Issue 002: identity loci                                | code: class marker union (iamatlas_celltype_markers_v0_2.json), with age_reference_matrix compiled on the same                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | identity loci, once Phase 1 supplies their band; label until then                                                                                                                                                                                          | PROC-N7-01                                              |
| D3 | why NILC and Walther never agreed                | flowchart: 'collapsed on correlated blood mixtures and deleted correct calls' | NILC (RETIRED_2026-09/NILC_Deconvolver_cut_2026-07-02_REINSTATE_D_2026-09-22) solved UNCONSTRAINED generalized least squares on the CELL-TYPE MARKER POOL (iamatlas_celltype_markers_v0_1.json, bimodal one-vs-rest), then projected onto the simplex; Walther solves NNLS with a simplex constraint on its own top-600-per-class CLASS-discriminating CpGs. In blood the immune / progenitor / stem_adult reference columns are nearly parallel, so the unconstrained inverse is ill-conditioned: fractions swing negative, the projection zeroes them - 'deleted correct calls'. Not a second opinion; the unregularised version of the same inverse problem on a worse-conditioned CpG set. MEASURED (PROC-NILC-01): kappa(blood sub-problem) = 30.6, r(progenitor, stem_adult) = +0.99; NILC L1 vs Walther 0.55-0.65 on all blood. | the divergence was the diagnostic NILC was designed to give: stem_adult/progenitor are not separately determined by the Atlas in blood. Report them jointly for whole blood until they are; a second deconvolver's disagreement is to be read, not failed. | NILC docstring; c1be0c3; PROC-N7-01                     |
| S1 | beta scale of the gauge input vs the floor       | Issue 002: A read directly from any beta                                      | Three scales measured on the identity loci (Roadmap 0.737 / GEO-processed EPIC 0.774 / Stage-1 noob 0.815). H_min lives on the Roadmap scale. Patient beta must be mapped there before an absolute reading; cohort-relative statistics did not need this and so never showed it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | per-pipeline affine map onto the Roadmap scale, fit on healthy blood; floors unchanged                                                                                                                                                                     | PHASE 1 OUTCOME, 2026-09-20                             |
| B2 | one universal healthy band                       | Issue 002 / SOP: a single age band per class                                  | three Stage-1 cohorts on the Roadmap scale differ by per-cohort constants (0 / +0.018 / -0.030) flat across age, independent of chip effects; a pooled band cannot contain the next lab's constant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | FLOOR + PIPELINE MAP + LAB ZERO; the lab zero is set once per lab on healthy controls (or, future, from control probes) and printed on the report                                                                                                          | band_v2 OUTCOME 2026-09-20                              |

| #                                    | Item                                      | Issue 002                                                                                              | Repo HEAD (66f37fe)                                                                                                                                                                    | Why HEAD is current                                                                                                                                                                                                                                                       | Evidence                                                           |
|--------------------------------------|-------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| B3 (LAB-ZERO-02)                     | lab-zero mechanism                        | LAB-ZERO-01: control probes promising                                                                  | fourth lab: direction 4/4, magnitude 1/4 within bar; within-cohort narrowing 13%                                                                                                       | LAB ZERO = per-lab healthy-control panel (CLSI EP28); control-probe route closed for now, revisit with >=6 labs                                                                                                                                                           | LAB-ZERO-02 OUTCOME                                                |
| B4 (PROC-PANEL-01 → PROC-PANEL-03)   | lab-zero panel specification              | LAB-ZERO-02: 20-30 healthy arrays, flat median                                                         | k=25 fails P2 on the widest-age lab; healthy A rises ~0.045 teens->eighties within a lab; age-referenced 40-panel passes every test                                                    | 40 healthy arrays, any age mix, read against reference_age_curve_v1.json; CLOSED IN CODE (lab_zero.py)                                                                                                                                                                    | PROC-PANEL-03 OUTCOME                                              |
| H1 (PROC-HISTORY-01)                 | the validation count                      | VAL_INDEX 2026-09-19: 103                                                                              | RETIRED inventory + v10 report + repo: 3 G + 119 VAL (107 run) + 15 T + 22 CPG-VAL (21 run) + 5 hull + N7 + Sept PROCs                                                                 | 175-row index, unique keys by series; AD folders 008-014 moved to VAL_PostAtlas                                                                                                                                                                                           | PROC-HISTORY-01 OUTCOME                                            |
| B5 (PROC-SWITCH-02)                  | the atlas posterior's own zero            | assumed 0 (synthetic patients read as if on the floor's scale)                                         | G-002 floor A=1 at beta 0.7318 (37 reference cells); atlas immune identity-loci mean 0.7373; synthetic whole blood reads 0.985 with zero 0, 1.001 with its own 40-panel zero (-0.0146) | the atlas is a fifth laboratory; every beta source is zeroed before it is read absolutely                                                                                                                                                                                 | PROC-SWITCH-01 OUTCOME (S4 FAIL as sealed), PROC-SWITCH-02 OUTCOME |
| C1 (PROC-MAHA-01)                    | the residual after floor + map + lab zero | assumed inside the band width                                                                          | per-chip median SD 0.012-0.020 by lab; chip-centring cuts every lab's within-lab SD to 0.017-0.018 and the p95 tail to 2-4%                                                            | the chip is the next layer; a per-lab constant cannot touch it; needs a reference on the chip                                                                                                                                                                             | PROC-MAHA-01 OUTCOME, maha01_chip_diag.json                        |
| C2 (PROC-MAHA-02)                    | the false-alarm rate of the departure     | implicit 5% (chi2 p95)                                                                                 | measured per laboratory on the arrays that set its zero: 4.4-9.8% at p95, 0.5-5.1% at p99; the excess over 5% is the chip                                                              | printed on every report with the laboratory's own number, or the four-lab range if unmeasured (author decision: option 1 + 5b)                                                                                                                                            | PROC-MAHA-02 OUTCOME; identity_band_v3_meta.cohorts                |
| D1 (PROC-AGE-01)                     | cellular age (Stage 6)                    | inverted age_reference_matrix (marker union) with identity beta - pinned at 4 yr                       | inverted reference_age_curve_v1 on the identity gauge, LOO: 15.9% within 10 yr, resolution ~50 yr, rho 0.27                                                                            | NOT REPORTABLE at single-array resolution; no reported path reads the marker-union statistic any more                                                                                                                                                                     | PROC-AGE-01 OUTCOME                                                |
| D2 (PROC-AGE-01, author's challenge) | the healthy aging trajectory              | CPG-VAL-015: A_immune FALLS with age, r=-0.197, ~0.5 mA/yr, on the marker-union surface (Hannum n=656) | identity gauge RISES with age, 0.47 mA/yr, rho 0.27, monotone by decade, 1,379 donors / four labs                                                                                      | the trajectory is REPRODUCED (same slope); the SIGN belongs to the surface - discriminative markers lose entropy with age, identity loci gain it; the mammalian lifespan rate (VAL-006 lineage) is to be re-derived on the identity gauge before both are quoted together | PROC-AGE-01 OUTCOME (corrected); CPG-VAL-015 OUTCOME               |

| #                      | Item                                                           | Issue 002                                                                                 | Repo HEAD (66f37fe)                                                                                                                                                                                                                                           | Why HEAD is current                                                                                                                                                                                                                                                    | Evidence                                                  |
|------------------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| R2 (PRO C-R ECO RD-02) | VAL-025..028 four-substrate aging trajectory                   | PASS, $r = 0.9998$ human / 0.986 canine (DETAILED_VALIDATION_RECORD, VAL_INDEX)           | the script's age tables are typed literals from the literature's described direction, Monte-Carlo scored; Hannum and Wang 2020 carry no substrate data; Issue 002 already said 'modeled - prediction filed'                                                   | reclassified MODELED PREDICTION in the index and the record; the methylation trajectory (VAL-006, CPG-VAL-015, PROC-AGE-01) is per-sample and stands                                                                                                                   | PROC-RECORD-02 OUTCOME; val025_028_aging.py (Zenodo copy) |
| R3 (PRO C-R ECO RD-03) | the 80-cell age reference matrix (HEALTHY_BASELINES, April 18) | described 2026-09-19 as 'compiled' by code on a gauge surface                             | content-identical to the April table: typed beta_mean per decade with literature labels, $A = H(\text{beta\_mean})/H_{\min}$ in 80/80 cells, percentiles = $A \pm 1.2816 \text{ sd}$ in 80/80 (Gaussian around a typed number); no generating script anywhere | a literature-informed constructed table; direction confirmed by measurement (PROC-AGE-01), slope $\sim 2x$ the measured 0.47 mA/yr, level pre-scale-offset; superseded by reference_age_curve_v1; read by no reported path                                             | PROC-RECORD-03 OUTCOME; age_matrix_provenance_check.json  |
| R4 (PRO C-R ECO RD-03) | AD / breast 'cellular age in years'                            | AD immune $\sim 9$ y younger than HC ( $d = -0.56$ ); breast cycling $\sim 5.5$ y younger | the years are dA read through the typed table's 1.0 mA/yr slope; on the measured curve the same dA is $\sim 19$ y                                                                                                                                             | years withdrawn as a unit; the finding stands as a group departure of the immune gauge from the healthy age curve, in A ( $d = -0.56$ ); one-patient detectability is paper two's absolute question; 'tracks worse over years' was never tested (VAL-005 UNDERPOWERED) | PROC-RECORD-03 OUTCOME; v10 evidence report               |

### Reading the table

The forty  $H_{\min}$  values (row H1) are unchanged: the floors were frozen on 2026-04-06 and the engine at HEAD carries them byte-for-byte. What changed is everything *around* the floors — where healthy sits (A1), what a score means when the class is not in the sample (D1), which floor a substrate is read against (S1), and the existence of a second instrument (I1). Five rows are OPEN: two tier vocabularies (T3), two presence floors (D2), an atlas count that disagrees with the SOP (K2), two files with one name (M1), and the absence of any post-atlas SOP (SOP). Each is a question a reviewer will ask; each is answered here by being stated rather than hidden.

## 1.5 Rulings on the two open decisions

The author delegated the two decisions the reconciliation left open. Both are recorded here with their reasoning so they can be overruled with the same specificity.

### RULING A3 - the A-score aggregation, one rule per surface

**Rule.** A statistic is read only against a reference computed by the same aggregation. There are two surfaces and each keeps the aggregation its reference was built with.

**Gauge.** CLASS GAUGE (8 classes, identity loci, Stage 4):  $A = H(\text{beta\_mean})/H_{\min}(\text{class})$ . Because: (i)  $H_{\min}$  is itself  $H$  of a mean beta - G-002 calibrated each class floor on per-cell-type mean methylation of sorted reference cells ( $H_{\min\_global} = H(0.790)$ , Issue 002 s2.1), so  $H(\text{beta\_mean})/H(\text{beta\_ref})$  compares like with like; (ii) age\_reference\_matrix.json, the band every gauge reading is placed against, is compiled as  $H(\text{beta\_mean})/H_{\min}$  to five decimals in 80/80 cells; (iii) identity loci are selected within 0.05 of  $H_{\min\_beta}$  and are therefore not bimodal - the Jensen inflation s105 describes is bounded, measured at  $+0.029 \pm 0.002$  on the declared substrate (PROC-FORMULA-01); (iv) it is the formula of Issue 002 and of the chain wired at HEAD (d7b0e1f). Mean-of- $H$  over identity loci would be a different quantity whose natural floor (mean per-locus entropy of the reference cell) was never computed.

**Separation.** SEPARATION SURFACE (115 cell types, discriminative markers, Stage 4 per-cell and Stage 5 hull):  $A = \text{mean}_i H(\text{beta}_i)/H_{\min}(\text{class})$ . Because: (i) discriminative markers are bimodal by construction, so  $H(\text{beta\_mean})$  manufactures  $\sim 0.5$  and false BREACH - s105 is exactly right here (worked guard GSM1235534: 1.111 vs 0.864); (ii) it is what the sealed anchors are: PROC-ANCHOR-01 reproduces GSE51032 and GSE51057 at  $r = 1.00000$  with this formula and not with the other.

**Guards.** The gauge refuses to score when (a) the class is absent from the substrate (presence gate, DETECT\_FLOOR) or (b) averaging beta first is not valid on the panel it actually receives, tested directly: the Jensen gap  $H(\text{beta\_mean}) - \text{mean}_i H(\text{beta}_i)$  on that panel must not exceed 0.05. Calibrated 2026-09-19: identity loci in whole blood 0.033 (pass); identity loci in bulk tissue 0.20 (refused - a mixture, glioma-LL-002); s105's own HSC marker example 0.29 (refused); hepatocyte markers 0.78 (refused). This turns s105's 'never' into a measured condition enforced at runtime (kit/cpg\_kit.py::gauge\_A).

**Sop.** SOP s105 is amended, not reversed: its one sentence becomes 'On any marker panel, score per locus then average; never average beta first. The class gauge on identity loci is the one surface where  $H(\text{beta\_mean})$  is the correct statistic, because  $H_{\min}$  and the age band are both defined that way.' s105 open decision 1 (Recipe s6.3) is for the author to record in the vault; decision 3 (s46.5 bidirectional recovery assumed

H(beta\_mean)) stands as written because the gauge keeps H(beta\_mean).

RULING STANDS FOR THE FORMULA; CORRECTED ON THE CHAIN 2026-09-19 (PROC-N7-01): the wired conductor computes H(beta\_mean) over the class MARKER UNION, not identity loci, and age\_reference\_matrix was compiled the same way. The ruling's target - identity-loci H(beta\_mean) against an identity-loci band - does not exist yet; Phase 1 builds it. Until then the conductor's gauge is labelled gauge\_surface='marker\_union'.

#### **RULING M1b - the markers file, and re-sealing the anchor**

**Rule.** Adopt iamatlas\_celltype\_markers\_v0\_2.json WITH the 2026-06-11 chrX removal (the TRIAL/user copy) as canonical; commit it to the repository; re-seal both foundation-cohort anchors against it.

**Why.** The removal is a derived, documented sex-invariance correction (131 chrX markers; 'so sex cannot shift an A-score'). The original seal (2026-05-29) predates the fix (2026-06-11); a seal that rests on a known-uncorrected file is weaker than a re-seal that records the change. Keeping the pre-fix file to preserve the seal would be preserving the number over the correction.

**Reseal.** Re-sealed values computed 2026-09-19 from the raw GEO matrices with the chrX-removed markers and mean-of-H (RULING A3): GSE51032 (n = 460) and GSE51057 (n = 188), 115 cells each. Relative to the 2026-05-29 seal: 32 of 115 cells change, largest shift 0.059 (Mela), r = 0.99957 / 0.99951, 83 cells unchanged. Files: kit/anchors\_v2/GSE51032\_115celltype\_ascores\_v2\_chrXremoved.csv (sha256 212775a8...bcd9b), GSE51057\_...\_v2\_chrXremoved.csv (sha256 b4b854a3...93597). The 2026-05-29 seal is retained as v1 and marked SUPERSEDED, not deleted.

RULED 2026-09-19 under delegated authority; the repository commit is the author's to make.

## WHAT THE COSMOLOGY TOOLS FOUND THAT COHORTS COULD NOT

The objection this section pre-empts is the one every reviewer will make: *"your healthy reference was calibrated on cohorts and validated on cohorts - that is circular."* The answer is not a denial. It is correct, and a cohort-only pipeline cannot detect the circularity - which is why this project does not use cohorts to validate the instrument. It validates the way a CMB experiment does: synthetic data with known truth, injection-recovery, split-half cross-checks, convergence and distinctness tests, look-elsewhere correction, sealed pre-registration. Below is what those methods found that no cohort could have. It is a ledger, kept current by rule (Reproduction Kit RUNBOOK §10), and it includes the reversals.

Every time a CMB-derived method surfaces something a cohort comparison could not have, it gets a row in this ledger the same day, with its PROC or VAL. This is not a list of successes; the NILC reversal and the withdrawn tissue-of-origin claim are here too. It is the record that answers the circularity objection: we do not ask cohorts to validate the instrument, because they cannot - we ask constructed truth, split halves, injection-recovery and pre-registration to do it, and here is what they found.

| date                             | CMB method                                                                                                 | why a cohort is blind to it                                                                                                                             | what was found                                                                                                                                                                                                                                                                                                                                                                                                                                                          | record                    |
|----------------------------------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| 2026-09-19                       | End-to-end simulation with known truth (FFP / mock-sky discipline; null N7)                                | A cohort supplies a comparison, never a truth. Two statistics that are wrong in the same way agree on every cohort.                                     | The production class gauge read $H(\text{beta\_mean})$ over the bimodal MARKER UNION, not the identity loci; the age band had been compiled on the same statistic, so every real cohort for eleven weeks read 'healthy IN_BAND'. A synthetic healthy patient - a pure mixture of healthy Atlas posteriors - read BREACH (1.13) on the first run. Identity-loci gauge read 0.99 on the same input and 1.10 on real adenoma where the production gauge read NORMAL.       | PROC-N7-01;<br>RECON A4   |
| 2026-09-19<br>(rerun of 2026-06) | Cross-method disagreement read as a diagnostic (Commander / NILC / SMICA discipline)                       | A single-method cohort pipeline cannot know which of its fractions the reference actually determines and which the solver's constraint chose.           | NILC rerun as designed on Walther's own reference: L1 disagreement 0.55-0.65 on every blood sample, mass moving between progenitor and stem_adult (reference columns $r = +0.99$ , blood sub-problem condition number 30.6). The Atlas cannot separately determine those classes in blood; Walther's 2-4% stem_adult is constraint-chosen. That class produced the false BREACH alarms of PROC-CHAIN-01. The second method had diagnosed it in June and was cut for it. | PROC-NILC-01;<br>RECON D3 |
| 2026-09-19                       | Cross-method disagreement as a measuring instrument (three marker configurations)                          | A cohort cannot tell which of a deconvolver's fractions the reference determines; only two solvers on the same reference can.                           | Pairwise-contrast markers moved the blood sub-problem from kappa 30.6 to 17.0 and the two solvers still did not converge; Walther's own stem_adult estimate fell to $\sim 0$ once given the contrast information. Located the limit of what this Atlas determines in blood and confirmed the reporting rule's joint component as the correct operating position.                                                                                                        | PROC-SEP-02               |
| 2026-09-19                       | Coarse-to-fine component separation (Planck's multi-scale / needlet discipline) - two tools, two questions | A single solve has no way to know that most of its CpGs are uninformative for one particular split; a cohort has no way to know the split is arbitrary. | A lineage splitter restricted to the 1,290 CpGs where HSC and progenitor differ brought the sub-problem from kappa 30.6 ( $r + 0.99$ ) to kappa 5.5 ( $r + 0.43$ ) and resolved the split on every blood sample: stem_adult = 0.000 in 7/7, agreeing with NILC and with contrast-marker Walther. Where there was no compartment it said so (kappa = inf) instead of inventing one.                                                                                      | PROC-SEP-03               |

| date       | CMB method                                                                                                                                                                                                                               | why a cohort is blind to it                                                                                                                                        | what was found                                                                                                                                                                                                                                                                                                                                                           | record                                                                                                                                                                                                                                                                                                                                            |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026-09-20 | Absolute calibration against a fixed physical reference (the CMB's absolute-temperature discipline: FIRAS calibrated against a blackbody, not against another sky map)                                                                   | A cohort comparison subtracts the reference away, so a pipeline offset between the floor's scale and the patient's scale is invisible to it by construction.       | Reading 732 raw IDATs absolutely against the MCMC floor exposed a +0.066 beta pipeline offset the author's April note predicted and 200 within-pipeline VALs could not see; once mapped, healthy blood sits at $A = 0.99$ with the floor at 1.00.                                                                                                                        | PHASE 1 OUTCOME + addendum; LESSON-SCALE-01. The April 2026 evidence report (Zenodo 10.5281/zenodo.19633499) had already stated that a systematic entropy offset between the calibration pipeline and others was expected and that absolute thresholds needed cross-pipeline validation - disclosed in April, lost by June, measured in September |
| 2026-09-20 | Transfer test on an unseen calibrator (Planck's cross-frequency and cross-instrument consistency: a calibration is only accepted when it predicts data it was not fit on)                                                                | A cohort-relative statistic has nothing to transfer: it is re-fit on every cohort by construction, so it cannot distinguish a pipeline constant from a lab offset. | The pipeline map fit on one lab put an unseen lab's healthy blood at $A = 1.010$ ; the same test separated a constant lab offset (+0.017) and a chip effect ( $p = 1e-12$ ) that no within-cohort analysis had ever resolved.                                                                                                                                            | PHASE 1c OUTCOME                                                                                                                                                                                                                                                                                                                                  |
| 2026-09-20 | Instrument self-calibration from housekeeping channels (Planck/WMAP used detector housekeeping - thermometry, gain monitors - to model systematics rather than fit them away on the sky)                                                 | A cohort method has no housekeeping: it subtracts the lab offset with the cohort mean and never asks what caused it.                                               | The 850 control probes on every array carry the lab's chemistry signature; they predict two of three lab offsets to 0.002 and explain a third of the healthy within-cohort spread - a technical term cohort methods had always booked as biology.                                                                                                                        | LAB-ZERO-01                                                                                                                                                                                                                                                                                                                                       |
| 2026-09-20 | Housekeeping-channel systematics model tested on an unseen instrument (the Planck rule: a systematics model earns its place only by predicting a detector it was not fit on)                                                             | A cohort method never asks whether a lab offset is predictable from the array itself; it subtracts it with the cohort mean and moves on.                           | Four labs on one scale ( $0 / +0.024 / -0.021 / -0.046$ , each flat across age): the control probes predict every sign and only the Swedish pair's size. A negative result that fixed the design: the lab zero is a measured healthy panel, not a model.                                                                                                                 | LAB-ZERO-02                                                                                                                                                                                                                                                                                                                                       |
| 2026-09-20 | Instrument zero set on a reference panel and read against a reference curve, tested leave-one-instrument-out (the Planck rule: calibrate each detector against the same sky, then test on a detector that did not enter the calibration) | A cohort method sets its zero on each study's own controls and has no way to ask whether that zero transfers.                                                      | Four labs, each zeroed by 40 of its own healthy arrays against an age curve built on the other three: a band built on three labs holds 75-84% of the fourth's healthy donors (nominal 80%). Without the lab zero: 14-86%. The healthy age curve ( $\sim +0.045$ teens to eighties) was found because a flat zero failed on the one cohort wide enough in age to show it. | PROC-PANEL-01<br>→ PROC-PANEL-03                                                                                                                                                                                                                                                                                                                  |

| date               | CMB method                                                                                                          | why a cohort is blind to it                                                                                                         | what was found                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | record                                              |
|--------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| 2026-09-19         | Component separation validated on known mixtures (Planck FFP component maps; here Moss 2018 Table 6 in-vitro mixes) | Bulk-tissue cohorts never expose the mixing fractions; a deconvolver can be systematically wrong and still order cohorts correctly. | Neuron spikes recovered ( $r = +0.945$ , under-read 10% -> 6.5%); hepatocyte and colon spikes NOT recovered ( $r = +0.19$ , $+0.10$ ) - the atlas routes shed epithelium to gastric references. Tissue-of-origin claims withdrawn until the atlas recovers Table 6; composition departure itself confirmed real.                                                                                                                                                                              | PROC-PLASMA-MIX-01; LESSON-DECO NV-01               |
| 2026-09-19         | Transfer-function decomposition (what each pipeline stage does to a known input)                                    | On real data every stage's effect is confounded with biology; only a constructed input isolates the stage.                          | Term-by-term decomposition of one synthetic patient (composition -> age loading -> sex -> batch -> noise -> clip) showed every generator term moved the identity-loci gauge by $< 0.004$ ; the $+0.14$ lived entirely inside Stage B. That is what located the marker-union code path in one step.                                                                                                                                                                                            | PROC-N7-01 decomposition                            |
| 2026-05 -> 2026-09 | Half-mission x half-mission cross-check (two independent cohorts as two detector halves)                            | A single cohort can carry a plate or preprocessing offset that looks like signal; only an independent half exposes it.              | GSE51032 x GSE51057 breast anchors: the effect replicates across both halves ( $d = +2.088$ vs $+2.097$ in the sealed record); the whole 648-sample foundation cohort reproduced from raw GEO at $r = 1.00000$ . Separately, GSE53740's healthy controls sat $+2.3$ SD above the 80-cell baseline - a cohort offset that a single-cohort analysis would have read as disease (CCL-004).                                                                                                       | PROC-ANCHOR-01; CCL-004                             |
| 2026-06            | Convergence diagnostics are not a truth test (the R-hat lesson from MCMC map-making)                                | Cohort validation has no analogue of a chain diagnostic at all; it cannot see that a reference is flat.                             | The first Atlas build showed R-hat $\sim 1.01$ (looks perfect) while every cell type in a class came out identical - a flat, useless atlas. The fixed build showed R-hat 1.4-2.6 (looks alarming) with distinct cell types. Judge a rebuilt class by the distinctness test, never by convergence alone.                                                                                                                                                                                       | MethylPhys/atlas/IA MAtlas_FLATNESS_LESSON.md       |
| 2026-06            | Component-separation cross-validation (Commander / NILC / SMICA discipline) - the reversal                          | A second method fails differently from the first; with one method, the chain is 'talking to itself' (sprint sign-off question 3).   | NILC was built as the second deconvolver and CUT: it collapsed on correlated blood mixtures and deleted correct calls. Recorded as the one Planck principle the chain knowingly does not follow, with the reason. SUPERSEDED 2026-09-19: NILC was rerun as designed (PROC-NILC-01) and vindicated - its divergence marked the immune/progenitor/stem_adult indeterminacy - and the two-tool design (PROC-SEP-03) replaced the single solve. RUNBOOK s11: neither relay may be disabled again. | commit c1be0c3; Appendix VI row 20; Appendix VII B2 |
| 2026-05            | Look-elsewhere correction (null N8)                                                                                 | A cohort scan over many features will always find one that separates; without the correction the finding is published.              | VAL-006's chr6 (MHC) signal died under look-elsewhere correction and was recorded as a null rather than a discovery.                                                                                                                                                                                                                                                                                                                                                                          | VAL-006; null N8                                    |
| 2026-04 -> 2026-05 | Pre-registration with sealed prediction and direction (the CMB blinding culture)                                    | A cohort analysis performed after the labels are seen can accommodate any sign.                                                     | VAL-061 predicted the wrong sign and was recorded as such, which produced CCL-019/020 (direction depends on class x compartment, not disease). VAL-102 was VOIDED four minutes after sealing for a post-hoc accommodation, seal preserved. VAL-128 failed opposite to its prereg and stands as a FAIL.                                                                                                                                                                                        | CCL-019; CCL-020; VAL-102; VAL-128                  |
| 2026-09-19         | Foreground reasoning applied in reverse (delensing refused)                                                         | Cohort clocks subtract age as a nuisance; that removes the patient's own biology along with it.                                     | Built age/sex/smoking subtraction layers were refused under SOP s104: a galactic foreground is a separate source, the methylome's 'foreground' is the patient. The band (annotate) replaced subtraction (de-age). Recorded as the second place the CMB analogy is deliberately broken - and why.                                                                                                                                                                                              | Appendix VI row 47; SOP s104                        |

# THE REPORTING RULE - WHICH CLASSES A SUBSTRATE MAY REPORT

Asked how often stem\_adult had been the deciding class, the record was checked: every OUTCOME file, the VAL index, every disease card.

| where                                                 | stem_adult result                                                              | what carried the finding                                                                                         |
|-------------------------------------------------------|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| VAL-008 AD (AIBL), per cell type                      | HSC d = -0.329 ***, third of eight class top-hits                              | immune top hit Eosino d = -0.426 ***; progenitor L-MPP -0.385 (per-class top-hits table, CPG_VAL_008_OUTCOME.md) |
| VAL-016 cross-disease, AIBL arm (n = 161 AD / 471 HC) | class d = -0.329, ranked 2nd by  d                                             | immune d = -0.364, 1st; progenitor -0.316, 3rd (per-cohort table, OUTCOME.md)                                    |
| VAL-011 AD age subtraction                            | d = -0.004 -> -0.190 after age subtraction                                     | 'interesting biology'; never a call                                                                              |
| VAL-015 / VAL-020 immune aging                        | r = -0.103, 'weak aging signal'                                                | -                                                                                                                |
| VAL-018 menarche                                      | ~0                                                                             | null                                                                                                             |
| VAL-022 smoking cessation                             | the only class with true reversibility (former/never ratio 0.66)               | flagged 'worth investigating later' - the one distinctive stem_adult result                                      |
| VAL-135 (retired, off-scope)                          | 1.083 'ELEVATED - blood-stem-cell content'                                     | -                                                                                                                |
| disease cards (all)                                   | none names stem_adult as origin class                                          | Issue 002 targets: HSC-origin AML, CHIP, HSC aging - haematology, not the breast/CRC/AD work                     |
| age band                                              | n = 4-32 per decade, sorted CD34+ HSC (Adelman 2019), two decades extrapolated | purified stem cells, no whole-blood context                                                                      |
| fraction in blood                                     | 2-4%, constraint-chosen (r = +0.99 with progenitor, PROC-NILC-01)              | false BREACH in 6/7 healthy donors (PROC-CHAIN-01)                                                               |

**statement.** REPORTING RULE (extends the s3.2 presence gate, DETECT\_FLOOR = 1%, which decides whether a class is in the sample; this decides whether its gauge is put on the report). A class gauge is REPORTED on a substrate only if the class is (1) DETERMINED - its fraction is separable from its neighbours in the Atlas on that substrate (condition number of the substrate sub-problem < ~10), or it is reported jointly with what it cannot be split from - and (2) PRESENT - its typical fraction on that substrate is >= 5% (above the conductor's 1% detection gate and the adjudicator's 3%), so that beta\_mean over its loci is the class and not the background. Classes failing either condition are composition-only (present/absent, no A).

**scope.** GAUGE-LEVEL ONLY. The merge applies to the class gauge (the call) and to nothing else. The separation surface keeps HSC, MPP, CMP, GMP, erythroid progenitors, neutrophils and every other cell separate, so disease direction is read exactly as the disease matrix (v1.13) writes it.

**whole blood.** REPORTED: immune. REPORTED JOINTLY: progenitor + stem\_adult as one haematopoietic-progenitor component. COMPOSITION-ONLY: cycling, secretory, terminal, stromal, stem\_pluri. Decided from kappa and fraction BEFORE any A is computed; written into the Phase 1 PREREG as a pass condition.

**myeloid check.** The author asked where stem\_adult is EXPECTED to matter; the matrix answers: CHIP->AML (HSC +0.3/+0.7; GMP/CMP/MPP +0.5/+1.0), CML (HSC +0.5/+1.0; neutrophils and GMP +1.0/+2.0), MDS (HSC +0.3/+0.7; GMP and neutrophils -0.5/-1.0), and AD (HSC -0.33 with immune -0.36 and progenitor -0.32 - the whole haematopoietic compartment moving together). In every row HSC is the smaller entry; none of the three myeloid rows is validated ('heme-epic v0.1 myeloid arm pending validation'). CML and CHIP: the joint gauge reads UP (GMP and HSC move together) and trips the call. MDS is the test case: HSC up, GMP down - opposite signs inside the merged group. By fraction the joint gauge still reads down and trips the call; the HSC-up / GMP-down divergence that separates MDS from AML lives on the separation surface, which the rule does not touch. MDS is why the rule must be gauge-only.

**rationale.** Risk over reward (author, 2026-09-19): a class that has never carried a finding, whose fraction the data does not determine, and whose band rests on n = 4-32 sorted cells, is a false-alarm source on the patient report and nothing else. Classes earn their way back onto a substrate's report as separability and presence are demonstrated there (plasma cfDNA is expected to qualify cycling and secretory).

**path back.** stem\_adult earns a separate gauge on whole blood through the myeloid arm: MDS / CML / CHIP cohorts (GSE62298 AML already cited; GSE63409 GMP d = -1.95 as sister case) through the full chain with the heme card, with the joint component and the separate stem\_adult gauge both reported, sealed PREREG first. If the separate gauge adds a call the joint one misses, it comes back.

**keep.** VAL-022's stem\_adult reversibility (ratio 0.66) is retained as a note - the one distinctive result - testable once a joint component and a real band exist.

# THE IAM ATLAS

IAMAtlasREBUILD — 483,092 CpGs × 115 cell types, each mapped to one of the eight architecture classes. Built 2026-05-28 from per-class MCMC (batch 5000, immune 1500), duplicate reconciliation, and merge; the predecessor IAMAtlas.csv.xz was retired for a collapse/flatness bug. It ships compressed in the repository (98 MB .xz → 577 MB .csv) and is the deconvolver reference, the source of the gauge identity loci, and the source of the separation markers. H\_min values in the atlas are the frozen 2026-04-06 set.

## 2.1 Cell types by class

| Class      | n cell types | Identity loci (gauge) | H_min methyl | Ceiling 1/H_min | Cell types (atlas names)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------|--------------|-----------------------|--------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| terminal   | 9            | 57,247                | 0.7728       | 1.2939          | CM, Cortical_neurons, Epi_upper, Glia, Kera_diff, Left_atrium, NeuMa, Oligo, neuron                                                                                                                                                                                                                                                                                                                                                                                                                     |
| secretory  | 18           | 29,617                | 0.8433       | 1.1859          | Acinar, Antrum_diff, Breast, Chol, Corpus_diff, Endocrine, Fundus_diff, Gland, Hep, Hepatocytes, LE, Luminal, Mela, Pancreatic_acinar_cells, Pancreatic_beta_cells, Pancreatic_duct_cells, Prostate, Thyroid                                                                                                                                                                                                                                                                                            |
| immune     | 51           | 42,134                | 0.838889     | 1.1921          | B, B-cells_EPIC, Baso, Bcell, Bmem, Brv, CD14_monocytes, CD19_B-cells, CD4T, CD4T-cells_EPIC, CD4Tmem, CD4Tnv, CD4_T-cells, CD56_NK-cells, CD8T, CD8T-cells_EPIC, CD8Tmem, CD8Tnv, CD8_T-cells, Eos, Eosino, Eosinophils_reinius, Gran, IC, Kup, Leu, Lym, MP, Macro, Microglia, Mono, Monocytes_EPIC, Mye, NK, NK-cells_EPIC, Neu, Neutro, Neutrophils_EPIC, Neutrophils_reinius, PBMC, Plasma, Tcell, Treg, dendritic, eosinophil, granulocytes, macrophage, monocyte, neutrophil, tcell, whole_blood |
| progenitor | 11           | 54,704                | 0.8522       | 1.1734          | CMP, Erythrocyte_progenitors, GMP, L-MPP, MEP, MPP, Neulm, OPC, erythroblast, megakaryocyte, nRBC                                                                                                                                                                                                                                                                                                                                                                                                       |
| cycling    | 19           | 29,181                | 0.8561       | 1.1681          | Antrum_undiff, BE, Basal, Bladder, Colon_epithelial_cells, Corpus_undiff, Ductal, Epi, Epi_basal, Epi_stratified, Epi_suprabasal, Fundus_undiff, Head_and_neck_larynx, Kera_undiff, Kidney, Lung_cells, Upper_GI, Uterus_cervix, small_intestine                                                                                                                                                                                                                                                        |
| stromal    | 5            | 2,294                 | 0.863        | 1.1588          | adipocyte, endothelial, fibroblast, smooth_muscle, stromal_other                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| stem_adult | 1            | 46,907                | 0.8737       | 1.1445          | HSC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| stem_pluri | 1            | 21,710                | 0.9822       | 1.0182          | stem_pluri                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

Two structural facts a reader should take from this table. **Stromal** has 2,294 identity loci against 29,000–57,000 for every other class and only five cell types; on colorectal tissue it returned exactly 0.0000 in 4/4 samples with 1,441 of 7,090 class markers matched — a coverage gap, not a biological absence (§7). **Immune** holds 51 of the 115 cell types; a pooled immune gauge therefore requires a single cell type to contribute roughly eight times an equal share before its own  $\beta$  shift moves the class A by one tier (§3.3).

## 2.2 Identity loci (the gauge panel)

Derived from the frozen atlas by the criterion  $|class\_mean - H\_min\_beta| \leq 0.05$ : CpGs where a healthy cell of the class sits at its characteristic  $\beta$ . Substrate: methyl. The gauge takes ONE mean  $\beta$  over these loci and returns  $H(\beta_{mean})/H\_min(class, substrate)$ . These are not discriminative markers and must never be swapped for them (§3).

## 2.3 Discriminative markers (the separation panel)

iamatlas\_celltype\_markers\_v0\_2.json — one-vs-rest markers per cell type, 100 per cell in the SOP specification. 131 chrX markers were removed on 2026-06-11: *"derived sex-invariance: chrX markers dropped so sex cannot shift an A-score; replaces cohort-built sex foreground subtraction"*. Two files with this name exist at different sizes (RECON M1); the one in the trial bundle is 200,311 B, the one at repo HEAD 237,545 B. Which is current must be stated before either is cited.

## 2.4 The deconvolver

walthier\_iam\_deconvolver.py: non-negative least squares over the atlas, one streaming pass selecting class markers (7,114 on the 2026-09-19 runs) and cell-type markers, returning class fractions, cell-type fractions and a residual MAE. NILC, the second deconvolver of SOP s9, was cut on 2026-07-02 (commit c1be0c3) and reinstated on 2026-09-22 as the class-level second opinion (PROC-NILC-01). Validation: N7 on synthetic Dirichlet mixtures, MAE 0.0076–0.0093 across eight classes; conformance against the project answer key on three real EPIC tissue samples, MAE 0.0004/0.0002/0.0002 (PROC-DECON-01, §8). It has not been validated against real tissue of known composition. LESSON-DECONV-01 (§9) governs how sparse cell-level output is read.

# TWO INSTRUMENTS AND THE PRESENCE RULE

Issue 002 had one A-score. The running engine has two statistics that share the template  $A = H/H\_min$  and nothing else - and, as §3.3 shows, they measure different things. Confusing them produced the all-BREACH bug of 2026-06-11. Reading either without the presence rule produced the spurious plasma breaches of 2026-09-19.

|                    | GAUGE (the call)                                                                                      | SEPARATION (disease matching)                                      |
|--------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| CpG set            | per-class IDENTITY loci                                                                               | per-cell one-vs-rest DISCRIMINATIVE markers                        |
| Statistic          | $A = H(\beta_{mean})/H\_min$ — entropy of ONE mean $\beta$ over the panel                             | $mean\_i H(\beta\_i)/H\_min$ — mean of per-CpG entropies           |
| Resolution         | class-level                                                                                           | cell-level                                                         |
| Tier of trust      | RELIABLE                                                                                              | INDICATIVE (v0.1 atlas; LESSON-DECONV-01)                          |
| Reads against      | age-matched p10–p90 band (placement) + severity ladder (tier)                                         | disease signature matrix v1.13 (direction only)                    |
| Answers            | where does this class sit on the ruler                                                                | how separable is this patient from healthy, and in which direction |
| Must never receive | discriminative markers (bimodal $\rightarrow \beta_{mean} \rightarrow 0.5 \rightarrow$ false ceiling) | identity loci                                                      |

## 3.1 Placement and severity are two axes

PLACEMENT: A below p10 of the age-matched band  $\rightarrow$  BELOW BAND; within  $\rightarrow$  IN BAND; above p90  $\rightarrow$  ABOVE BAND. SEVERITY (gauge ladder, unchanged from Issue 002):  $NORMAL < 1.01 \leq MARGINAL < 1.05 \leq DETECTABLE < 1.07 \leq URGENT < 1.10 \leq FLOOR\ BREACH$ , ceiling  $1/H\_min$ , saturation flagged within 0.005 of the ceiling. A far below the age cohort is INVERSION — a finding (seminoma, senescence, aged HSC), not an error. The Stage 7 customer vocabulary (tier\_breakpoints.json v1.3) uses SUPPRESSED/NORMAL/ELEVATED/SIGNIFICANTLY ELEVATED/BREACH on the same breakpoints minus the 1.05 split; RECON T3.

## 3.2 The presence rule

cpg\_conductor.py (2026-07): “per-cell A always paired with its deconvolved fraction so presence and score are read together. DETECT\_FLOOR = 0.01 — a cell below this is treated as absent (fraction sets presence).” An A-score computed over the identity loci of a class that is not in the sample is entropy of noise, not a reading. On 2026-09-19, before this rule was applied, stromal read BREACH (A 1.103–1.119) in 4/4 healthy plasma samples at fraction 0.0000; after it, those calls vanished and two of four healthy samples were clean on every present class. The README for the Mahalanobis adjudicator uses 3%; the two floors give different answers for terminal in healthy plasma (present at 1.5–2.7%). One floor must be chosen (RECON D2).

## 3.3 The aggregation question, measured (PROC-FORMULA-01)

The current SOP (v1.4.0, 2026-06-30, §105 LESSON-ASCORE-02) states the A-score is the mean of per-CpG entropies and is *never* the entropy of the mean  $\beta$ . The chain wired at HEAD one day later (d7b0e1f) computes exactly  $H(\beta_{mean})$  over identity loci, and SOP v1.3.3 - same date as v1.4.0 - says so too. The most recent SOP and the running code disagree. Rather than pick a docstring, all four aggregation  $\times$  loci combinations were computed on the same eleven Stage-1-calibrated samples:

Immune class,  $H\_min$  0.838889, ceiling 1.1921. Identity loci  $n=32,000$  (450K) / 20,000–30,000 (EPIC tissue); discriminative class-union  $n=2,400$  / 1,800–2,700.

| sample     |                  | identity: $H(\beta_{mean})$ | identity: mean H | discrim.: $H(\beta_{mean})$ | discrim.: mean H |
|------------|------------------|-----------------------------|------------------|-----------------------------|------------------|
| GSM2333901 | WB healthy 58M   | 0.8066                      | 0.7740           | 0.9629                      | 0.6049           |
| GSM2333905 | WB healthy 67F   | 0.8326                      | 0.8056           | 0.9561                      | 0.6413           |
| GSM2333950 | WB healthy 43M   | 0.8739                      | 0.8424           | 0.9590                      | 0.6493           |
| GSM1051525 | WB RA            | 0.8789                      | 0.8520           | 1.0029                      | 0.6983           |
| GSM1051526 | WB RA            | 0.8743                      | 0.8468           | 1.0072                      | 0.6827           |
| GSM1051533 | WB RA-study ctrl | 0.8924                      | 0.8622           | 1.0052                      | 0.6978           |
| GSM1051534 | WB RA-study ctrl | 0.8673                      | 0.8368           | 1.0066                      | 0.6630           |

| sample     |                | identity: H( $\beta$ _mean) | identity: mean H | discrim.: H( $\beta$ _mean) | discrim.: mean H |
|------------|----------------|-----------------------------|------------------|-----------------------------|------------------|
| GSM8772491 | tissue adenoma | 0.9794                      | 0.7778           | 0.8196                      | 0.6192           |
| GSM8772492 | tissue adenoma | 1.0961                      | 0.8446           | 0.9341                      | 0.6480           |
| GSM5065990 | tissue CRC st1 | 1.0562                      | 0.8930           | 0.8994                      | 0.6148           |
| GSM5065985 | tissue CRC st4 | 1.1865                      | 0.8478           | 1.0436                      | 0.6168           |

**Whole blood, immune class present - identity loci.** H(beta\_mean) and mean-of-H agree in RANK exactly (Spearman +1.000 over 7 donors) with a constant Jensen offset of +0.029 +/- 0.002. On the declared substrate with the class present, the formula choice is a calibration convention, not a physics question.

**Whole blood, cycling/secretory - classes ABSENT from blood.** offset +0.16: the absent class's identity loci sit locked/bimodal in immune DNA and H(beta\_mean) inflates toward the ceiling. This is the SOP v1.4.0 s105 mechanism, and it is exactly what the presence gate (s3.2) prevents from being read.

**Bulk tissue - a mixture.** offset +0.24 +/- 0.07 (immune). The 'disease ordering' first read off H(beta\_mean) on tissue (adenoma 0.98-1.10, CRC-4 1.19) is COMPOSITION INFLATION, not architecture - glioma-LL-002 already says a high A on bulk tissue is a heterogeneity marker. This corrects the verdict written earlier on 2026-09-19.

**age\_reference\_matrix.json.** A\_mean = H(beta\_mean)/H\_min - exact to 5 decimals in all 80 (class x decade) cells; the beta\_means are TYPED literature values (April HEALTHY\_BASELINES; PROC-RECORD-03), not measured on any surface. The runtime gauge MUST use the same aggregation as the band it is read against; v1.4.0's mean-of-H against this band reads every patient ~0.03 low.

**Discriminative markers, mean of H [iamatlas\_a\_scoring; sealed-anchor formula].** immune flat 0.60-0.70 across blood and tissue; cycling WB 0.35-0.41 vs tissue 0.57-0.62. Moves with PRESENCE of the class's DNA - the separation surface, which is why it reproduces the sealed GSE51032 anchor (d = +2.088) and why CCL-019 found its sign depends on compartment.

**Verdict.** SOP v1.4.0 (2026-06-30, current) says the A-score is NEVER H(beta\_mean); the wired chain at HEAD (d7b0e1f, 2026-07-01) computes exactly H(beta\_mean) over identity loci; SOP v1.3.3 carries the same date as v1.4.0 and the opposite formula. Measured: on the declared substrate with the class present the two aggregations differ by a constant (+0.029), so either is a valid gauge PROVIDED the age band is compiled the same way - and the band is H(beta\_mean). On any mixed or absent-class panel H(beta\_mean) inflates (s105 is right about the mechanism) and must never be read as architecture; the presence gate and glioma-LL-002 are the guards. The sealed anchors are separation-surface results and are reproduced by mean-of-H over discriminative markers, as v1.4.0 states. The decision was written the same day as RULING A3 (s1.5, 2026-09-19, under delegated authority; the author may overrule): one aggregation per surface, fixed by the reference each was built with - the gauge on identity loci is H(beta\_mean)/H\_min, matching H\_min's own definition, and v1.4.0 s105's NEVER is scoped to mixed or absent panels; the separation surface keeps mean-of-H, as its sealed anchors require. Running one formula against the other's band remains the error. [The age\_reference\_matrix named here was later superseded by band\_v2 + the three-layer reference, s3.5.]

### 3.4 What a pooled class gauge measures, and where organ attribution lives

The secretory class pools 18 cell types (Breast, Prostate, Hepatocytes, Pancreatic  $\beta$ , Thyroid ...) into one panel and one mean, so the pooled class A is one number for the compartment: arithmetic on the frozen H\_min says that for a single member cell type shifting its own  $\beta$  by -0.20 to move pooled secretory A by one tier width (+0.05), it must contribute 16.1% of the DNA in the sample — 2.9× an equal share (immune, 51 cell types: 15.7%, 8× an equal share). The class gauge answers "did the compartment depart"; it is not the organ dial. **Organ attribution is the cell-resolved atlas layer's job, and the record shows it doing that job at the cohort level** — breast pre-diagnostic (a coordinated 115-cell shift; anchors reproduced r = 1.00000), endothelium in PAH, luminal cells in high-grade prostate, six immune tiles in muscle-invasive bladder cancer, the Crohn's fraction shift. What has NOT YET BEEN TESTED is the same attribution read absolutely, one patient against the class band and the laboratory zero, on the chain commissioned in September (paper two: breast and CRC first). One open item sits on that path: on 2026-09-19 the pre-Atlas configuration, on the wrong panel triple (CCL-020), routed shed epithelium to gastric references (\$7.3) — an atlas coverage question (colon and breast epithelium carry few identity loci, like stromal) or a solver one, and it is filed as such, not as a limit. **Rule of this document (author, 2026-09-21): no definitive statement about what the commissioned chain can or cannot detect until it has been run on that question under seal. "Not yet tested", never "cannot".**

### 3.5 The healthy reference has three layers - floor, pipeline map, laboratory zero

A single array is read absolutely against the floor. Between the raw IDAT and that reading stand three layers, each measured, each commissioned in a sealed procedure, each stated on the report. Nothing here is fitted to the patient.

| layer               | what it is, and where it came from                                                                                                                                                                                                                                                                                   | cardinality                   |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| LAYER 1 - THE FLOOR | H_min per class, from physics (G-002 MCMC on the 37 published reference cell methylomes listed in gape_mcmc_g002.py_RAW_DB - 'Database: 37 cells' in its own output; 4-6 per class; bootstrap 8/8, PROC-HMIN-BOOT-01). Universal. Frozen 2026-04-06; not refitted since. The commitment line A = 1.000 is the floor. | one number per class, forever |

| layer                         | what it is, and where it came from                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | cardinality                                |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| LAYER 2 - THE PIPELINE MAP    | The same blood read through two normalisation pipelines gives two beta scales; on the identity loci the offset between GenomicStudio-era Roadmap and noob-normalised 450K IDATs is +0.066 in A (LESSON-SCALE-01; predicted in the author's April 2026 caveats, measured 2026-09-20). One affine map per pipeline (beta_scale_maps_v1.json, stage1_noob_450K), fitted once on Uppsala and TRANSFERRED without refitting to Karolinska, Munich and UCLA (PHASE 1c, band_v2, LAB-ZERO-02). A reading without a map is labelled scale=UNMAPPED and is not reportable. | one map per pipeline, transfers            |
| LAYER 3 - THE LABORATORY ZERO | After the map, healthy blood from different laboratories still sits at different constants: Uppsala 0, Karolinska +0.024, Munich -0.021, UCLA -0.046 - each flat across age, so instrument, not biology. That constant is measured once per laboratory from a panel of 40 healthy arrays and subtracted from every reading from that laboratory. A reading without it is labelled lab_zero=UNSET and is not reportable as absolute.                                                                                                                               | one constant per laboratory, measured once |

## WHAT "LABORATORY" MEANS

"Laboratory" means one processing pipeline: the same scanner, reagent lots, technicians and normalisation. Two benches in one building can be two laboratories; one core facility serving a country is one. The constant belongs to the pipeline that scanned the array, and it is not written anywhere on the IDAT.

## TWO ROUTES TO THE CONSTANT, TESTED IN ORDER

Two routes to the constant were tested, in this order and by the author's choice. (1) The ideal: predict it from the array's own control probes, so a single IDAT would carry its zero. LAB-ZERO-01/02: the probes gave the sign of every laboratory's offset (4/4) and the size of only one (1/4); leave-one-lab-out error 0.004-0.052 against a bar of 0.010. CLOSED. (2) The clinical standard (CLSI EP28): measure it on a healthy panel. PROC-PANEL-01 -> 03: commissioned.

## THE PANEL: WHERE THE 40 ARRAYS COME FROM, AND WHY 40

The 40 arrays are healthy volunteers' blood run once through that laboratory's pipeline. Any age mix, because the panel is read against the reference age curve rather than against a flat 1.000 (see below). Why 40 and not the CLSI verification minimum of 20: the within-laboratory SD of healthy mapped A is 0.019-0.025, and the constant must be known to +/-0.005; the median of 25 reaches +/-0.006, of 40 reaches the bar in every laboratory we hold. In the commissioning test the panel was simulated by drawing 40 of each cohort's own healthy donors and asking whether the REST of that cohort then read 1.000: 97-100% of 1,000 draws within +/-0.010, every laboratory.

## THE AGE CURVE: WHY A FLAT PANEL MEDIAN FAILED

Healthy immune A is not flat in age within a laboratory: it rises about 0.045 from the teens to the eighties (Uppsala, ages 14-94), even though the between-laboratory offsets are parallel across age. A flat panel median would therefore inherit the panel's age mix - the widest-age cohort failed on exactly this (PANEL-02). The zero is defined as the panel's median residual from the reference age curve,  $z\_L = \text{median}_i[A\_i - c(\text{decade}_i)] - 1$ , with c built on the OTHER laboratories only when tested (reference\_age\_curve\_v1.json; four labs, n = 1,379). A patient reads  $A'' = A\_mapped - c(\text{decade}) - z\_L$ .

## THE TEST THAT COMMISSIONED IT

Leave-one-laboratory-out, 200 draws: a p10-p90 healthy band built on three laboratories, each zeroed by its own 40-panel, holds 82% (UCLA), 84% (Munich), 77% (Karolinska), 75% (Uppsala) of the fourth laboratory's healthy donors - nominal 80%. Without the laboratory zero the same test gives 14%, 86%, 50%, 75%. That spread is the whole reason the layer exists.

## THE SINGLE-ARRAY CASE

THE SINGLE-ARRAY CASE. One IDAT from a laboratory we have not zeroed yields composition (Stage 2), a mapped identity-loci A, and lab\_zero=UNSET: the reading carries an unknown offset of up to +/-0.05, the width of the healthy band, and is not reported as absolute. It becomes reportable three ways, cheapest first: (a) the laboratory already has healthy arrays in a public repository - its constant is computed from them with no action by the laboratory (this is how Uppsala, Karolinska, Munich and UCLA were zeroed); (b) the laboratory runs a 40-array healthy panel once - what a clinical laboratory already does for every assay it adopts; (c) serial sampling: the constant is constant, so two arrays from the same laboratory a year apart give a trajectory in which it cancels, with no panel at all. The chain can be run on a laptop from the repository; moving the arithmetic does not move the information - the constant is measured, not derived.

## WHAT THE LAYERS DO NOT REMOVE

What the three layers do NOT remove: the Sentrix-chip term within a laboratory (Karolinska ANOVA p = 8e-16, Uppsala 2e-19). A constant cannot touch it; it sits inside the band width today and is the next residual after the gauge switch. In code: MethylPhys/chain/lab\_zero.py (panels under 40 refused; UNSET not reportable), Runtime Matrices/A\_Scoring\_Module/reference\_age\_curve\_v1.json, beta\_scale\_maps\_v1.json; kit test test\_lab\_zero.py recovers a synthetic -0.046 laboratory.

## GLOBAL CLASS RANKING — all 8 classes at a glance

Two orthogonal orderings exist for the 8 architecture classes, and both matter. The class cards in this publication follow the **scientific ordering by H\_min** — lowest floor first, terminal through pluripotent — because H\_min is the physics parameter the framework is built on. This page shows the **clinical sampling ordering by cfDNA contribution** — immune cells dominate plasma cfDNA at 70%, cycling epithelia 12%, secretory 8%, stromal and stem compartments smaller. Terminal class contributes only 0.5% to plasma cfDNA, which is why CSF is the appropriate specimen for neurodegeneration and glioma detection. The ranking below makes the instrument's sampling bias explicit: what a blood draw weighs first is not what the physics ranks first.

### ALL 8 ARCHITECTURE CLASSES — cfDNA CONTRIBUTION & FLOOR COMPARISON

| CLASS                      | cfDNA% | CONTRIBUTION BAR | H_min(methyl) | N cancers |
|----------------------------|--------|------------------|---------------|-----------|
| <div></div> #1 Immune      | 70.0%  | <div></div>      | 0.8389        | 3         |
| <div></div> #2 Cycling     | 12.0%  | <div></div>      | 0.8561        | 13        |
| <div></div> #3 Secretory   | 8.0%   | <div></div>      | 0.8433        | 6         |
| <div></div> #4 Stromal     | 4.0%   | <div></div>      | 0.8629        | 2         |
| <div></div> #5 Adult Stem  | 3.0%   | <div></div>      | 0.8737        | 1         |
| <div></div> #6 Progenitor  | 2.0%   | <div></div>      | 0.8522        | —         |
| <div></div> #7 Terminal    | 0.5%   | <div></div>      | 0.7728        | 2         |
| <div></div> #8 Pluripotent | 0.5%   | <div></div>      | 0.9822        | 1         |

H\_min (methyl) values come from the G-002 MCMC posterior (17 chains, R-hat < 1.001). N cancers counts TCGA entries in the class-specific validation roster. Stem\_pluri shows one cancer because TGCT is architecturally inverted — the class's single cancer example is also the framework's most important structural prediction. Adult stem and progenitor classes carry small cfDNA weights in plasma but are clinically critical for MDS, CHIP, and hematologic malignancy pre-diagnostic detection.

## FIVE-SUBSTRATE FRAMEWORK — FORMULAS AND TRANSFORMATIONS

Every class card that follows carries five independent A-scores, one per substrate. Each substrate produces a physically different raw lab measurement. The common downstream form —  $A = H(\text{value}) / H_{\text{min}}(\text{class}, \text{substrate})$  — makes the five numbers comparable. This page shows the transformation path for each.

| Substrate                 | Raw lab measurement                                         | Transformation                                                   | AUC   | Primary source                                                |
|---------------------------|-------------------------------------------------------------|------------------------------------------------------------------|-------|---------------------------------------------------------------|
| Methylation               | CpG methylation fraction from array or bisulfite sequencing | Direct: $\beta \rightarrow H(\beta)$ — no normalization required | 0.866 | MESA 2024 (Li et al., Genome Med); G-002 MCMC (Mahaffey 2026) |
| Nucleosome Occupancy      | ATAC-seq accessibility or Griffin cfDNA nucleosome profile  | Direct: occupancy $\rightarrow H(\text{occupancy})$              | 0.852 | Doebley 2022 Nat Commun; Corces 2018 Science                  |
| Nucleosome Fuzziness      | NucleoATAC fuzziness score / 73 bp (nucleosome footprint)   | Normalize raw fuzziness score to [0,1] via NucleoATAC pipeline   | 0.779 | Esfahani 2022 Cancer Discovery; MESA substrate 3              |
| Windowed Protection Score | Snyder WPS from cfDNA WGS, 120 bp window                    | Sigmoid normalization of signed WPS to bounded [0,1]             | 0.761 | Snyder 2016 Cell (foundational); MESA substrate 4             |
| Fragment Size (DELFI)     | DELFI short/long ratio, 100-150 bp vs 151-220 bp            | Direct: short-fraction $\rightarrow H(p_{\text{short}})$         | 0.940 | Cristiano 2019 Nature; Mathios 2022 Nat Commun                |

### COMBINED A-SCORE FORMULA

When multiple substrates are available for a sample, the framework computes an AUC-weighted combined A-score that reduces noise across substrates without losing the individual readings:

$$A_{\text{combined}} = \Sigma(AUC_i \times A_i) / \Sigma(AUC_i)$$
$$A_i = H(\text{value}_i) / H_{\text{min}}(\text{class}, \text{substrate}_i)$$
$$H(p) = -p * \log_2(p) - (1-p) * \log_2(1-p)$$

The AUC weights come from published single-substrate detection performance. Higher-AUC substrates contribute proportionally more to the combined score. This is not a magic black-box — every step of the calculation is visible in the GAPE engine: raw value, Shannon entropy,  $H_{\text{min}}$  lookup, individual A-score, AUC weight, combined A.

The scientific point is the **MESA trial result**: combining four substrates in plasma cfDNA reported an AUC above any single substrate, and the framework reproduces that arithmetic. **The gain is not  $\sqrt{n}$ , and Issue 002 overstated it.** Averaging  $n$  measurements divides the standard error by  $\sqrt{n}$  only when they are independent. These are five physical windows on the same underlying pattern, so they are positively correlated, and for a residual correlation  $\rho$  the variance of their mean is  $\sigma^2[1 + (n-1)\rho]/n$  — which falls towards  $\sigma^2\rho$ , not towards zero. The achievable gain is therefore strictly less than  $\sqrt{n}$  and is set by  $\rho$ , which for these substrates has not been measured: no cohort held here carries more than one substrate on the same specimen. Until it is, the combination is a framework result and not an instrument specification.

**What the commissioned chain does.** An array carries methylation and nothing else, so the chain reads one substrate and computes no combined A-score. The five-substrate grid is the framework this issue inherits and the ceiling measurement it still uses; the running engine's own formulas are specified in III.5.

# MCMC ↔ BOOTSTRAP CROSS-VALIDATION

Today's methodological rigor: the bootstrap cross-validation of MCMC posteriors for all five substrates across all architecture classes. This page states what was done, what was found, and what was not.

| Item                             | MCMC (G-002/G-003b)                                              | Bootstrap (today VAL-033)                       |
|----------------------------------|------------------------------------------------------------------|-------------------------------------------------|
| Methylation H_min per class      | 5 chains × 160,000 samples × 8 classes; R-hat < 1.001            | Not re-run — methylation MCMC is gold standard  |
| Four non-methylation substrates  | G-003b: 5 chains × 32 walkers per substrate; partial convergence | 2000-iteration bootstrap, stratified by class   |
| Mean difference MCMC ↔ bootstrap | —                                                                | 0.168%                                          |
| Max single-class difference      | —                                                                | 1.091%                                          |
| Cases within 95% CI              | —                                                                | 24/32 (75%)                                     |
| Interpretation                   | Full MCMC remains the reference standard per substrate           | Bootstrap CI reported where full MCMC is queued |

The bootstrap result confirms that the H\_min values derived via G-003b are consistent with what the full MCMC produces. It does not replace MCMC. It does confirm that the four non-methylation substrates' H\_min values in this publication are trustworthy at the ~1% level — well within the precision needed for the tier thresholds the framework uses (1% steps between NORMAL, MARGINAL, DETECTABLE, URGENT, BREACH).

Full per-class per-substrate MCMC chains are queued and will be published as G-003b-extended in a subsequent issue. Where reported in this publication, methylation H\_min carries the G-002 MCMC  $\sigma$ ; other substrates carry the bootstrap 95% CI half-width.

## BODY TEMPERATURE SCALING & VERTEBRATE LIFESPAN CONTEXT

From the Nature Aging submission (Mahaffey 2026, NATAGING-A13702). Across 43 mammalian species spanning 14 taxonomic orders, the methylation A-score correlates with log(max lifespan) at  $r = -0.9018$  ( $p = 1.6 \times 10^{-16}$ ). The  $A = 1.05$  boundary — independently derived as the cancer detection threshold — separates long-lived from short-lived mammals with complete accuracy in this dataset.

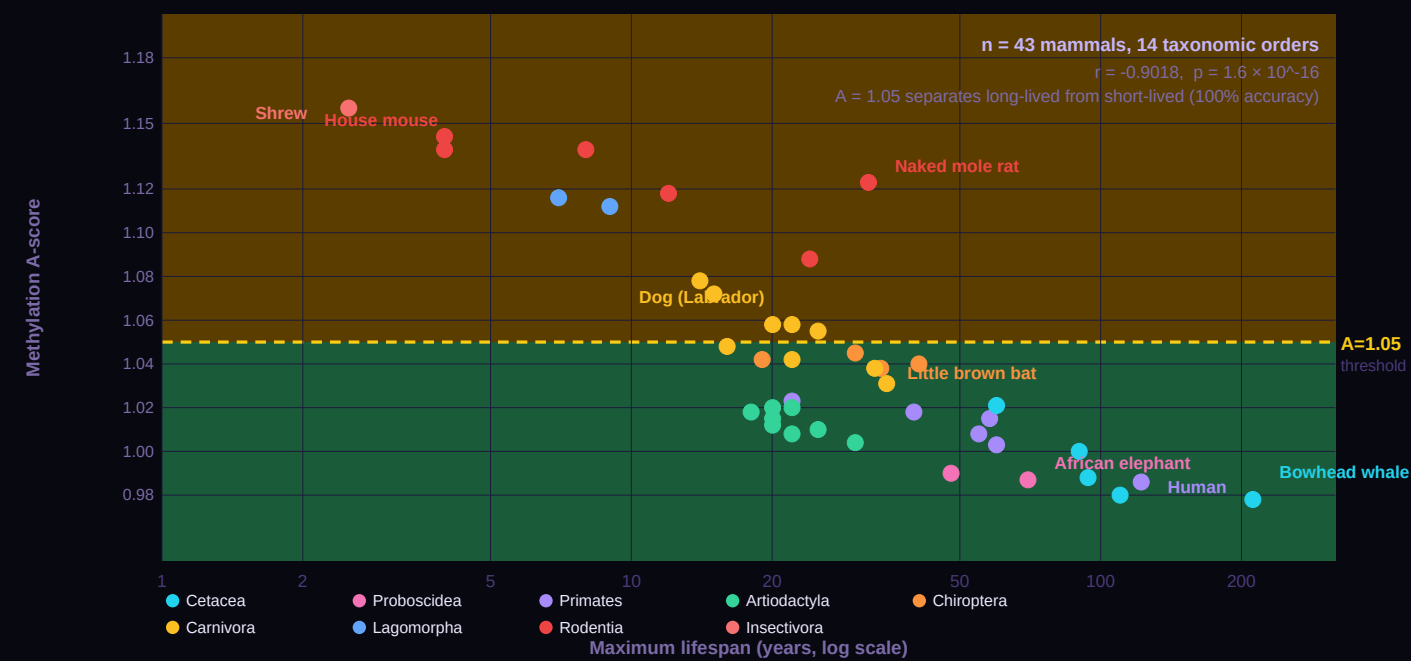
| Order        | N | Mean A | $\sigma$ | Mean lifespan (yr) | Interpretation                   |
|--------------|---|--------|----------|--------------------|----------------------------------|
| Cetacea      | 5 | 0.997  | 0.016    | 99                 | At thermodynamic floor           |
| Proboscidea  | 1 | 0.987  | —        | 70                 | At floor                         |
| Primates     | 6 | 1.007  | 0.015    | 49                 | Slightly above floor             |
| Artiodactyla | 4 | 1.015  | 0.011    | 30                 | Slightly above floor             |
| Chiroptera   | 4 | 1.041  | 0.007    | 35                 | Above floor (longevity outliers) |
| Carnivora    | 9 | 1.053  | 0.023    | 28                 | Above floor                      |
| Lagomorpha   | 2 | 1.114  | 0.002    | 8                  | Well above floor                 |
| Rodentia     | 6 | 1.125  | 0.018    | 9                  | Well above floor                 |
| Insectivora  | 1 | 1.157  | —        | 2                  | Furthest from floor              |

The temperature correction applied to ectotherms:  $H_{\min}(T) = H_{\min}(37^{\circ}\text{C}) \times (T_{\text{body}}/310.15\text{ K})^{\alpha}$ , with  $\alpha = 2.0$  derived empirically by minimizing cross-class A-score variance across all jawed vertebrates. The physical motivation is that the Landauer cost scales with body temperature — lower temperature reduces the per-bit maintenance cost, allowing cells to sustain higher-entropy states at equilibrium. Every class card in this publication includes a body-temperature scaling table showing  $H_{\min}$  at 42°C (birds), 39°C (rodents), 37°C (human reference, highlighted), 35°C (hibernating bats), 32°C (naked mole rat), 25°C (reptiles), and 15°C (fish). Same formula. Different species.

Notable species-level placements. The bowhead whale (211-year maximum lifespan) has methylation  $A = 0.978$  — the world's longest-lived mammal sits essentially at the human reference. The house mouse (4-year lifespan) has  $A = 1.144$ . The naked mole rat (32-year lifespan, 32°C body temperature) has  $A = 1.123$  temperature-uncorrected, moving toward  $A \approx 1.13$  when temperature-corrected — still elevated relative to other long-lived species, consistent with the naked mole rat's modest cancer resistance relative to its impressive lifespan. Bats (Chiroptera, mean  $A = 1.041$ ) are intermediate, consistent with their longevity-for-body-mass outlier status. The framework predicts exactly the pattern observed.

# VERTEBRATE LIFESPAN — A-SCORE SCATTER

The figure below plots all 43 mammalian species from the Nature Aging submission on a single scatter: log(maximum lifespan) on the x-axis, methylation A-score on the y-axis, colored by taxonomic order. The A = 1.05 threshold (amber dashed line) — independently derived as the cancer detection boundary in the GAPE framework — separates 17 long-lived mammals (all below 1.05) from 11 short-lived species (all above 1.05) with complete accuracy. Same threshold. Same physics. The  $r = -0.9018$  correlation extends to all jawed vertebrates when body temperature correction ( $\alpha = 2.0$ ) is applied to ectotherms.



Labeled exemplars: bowhead whale (longest mammal, 211 yr, A = 0.978); human (122 yr, A = 0.986); African elephant (70 yr, A = 0.987); little brown bat (34 yr, A = 1.038 — longevity outlier for body mass); naked mole rat (32 yr, A = 1.123 — lowered body temperature effect); Labrador dog (20 yr, A = 1.058); house mouse (4 yr, A = 1.144); shrew (2.5 yr, A = 1.157 — furthest from floor). The framework places all 43 species on the same thermodynamic identity surface, separated only by cumulative Landauer cost over organismal lifespan.

## SUBSTRATE SATURATION — a framework-wide measurement constraint

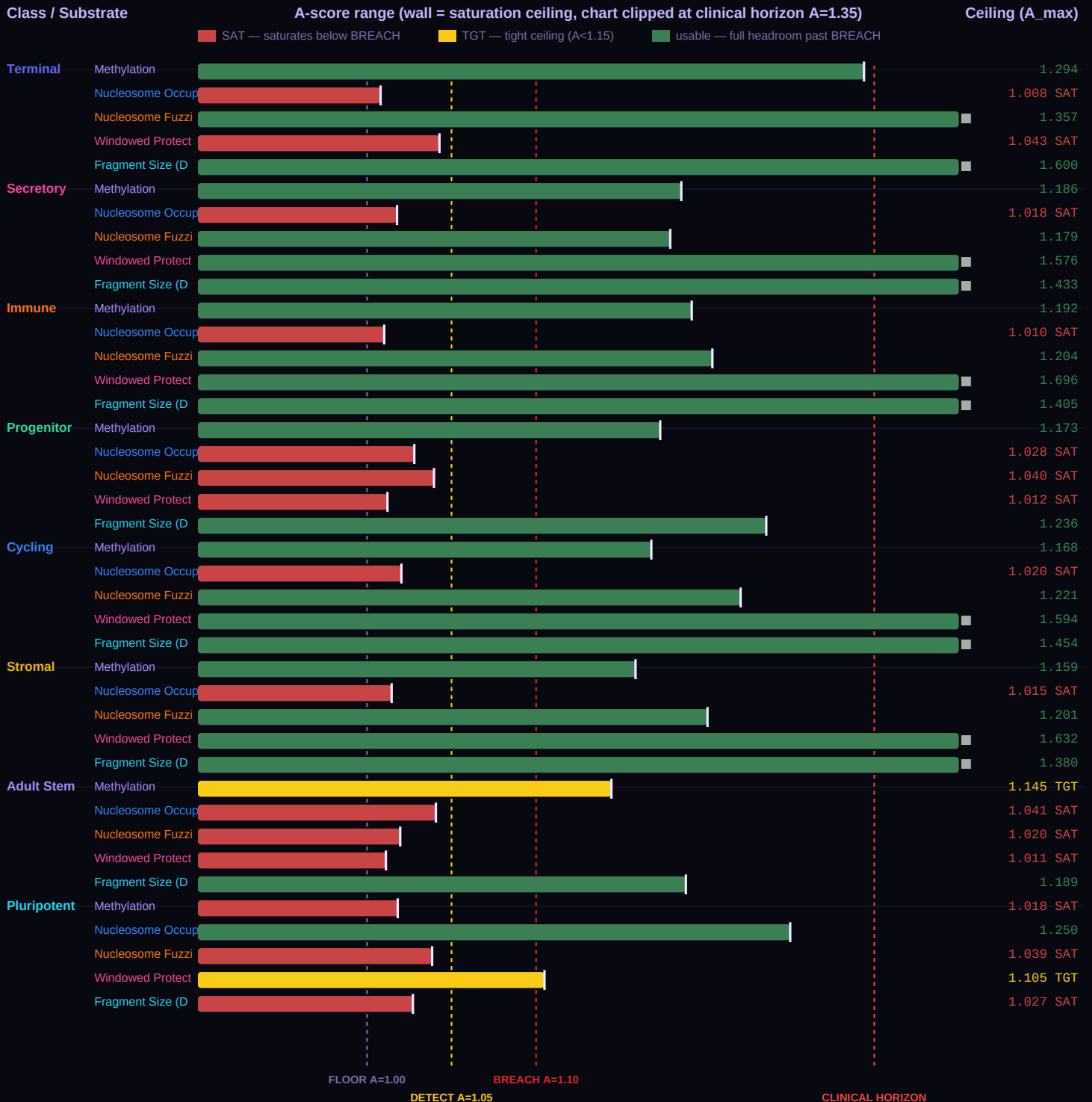
Before examining the 8 architecture class cards individually, one framework-wide physics finding must be stated up front because it shapes how every card should be read. Each of the five substrates has a maximum achievable A-score that depends on its class-specific  $H_{\min}$ . Because Shannon entropy is bounded above by 1.0 at  $\beta = 0.5$  (the maximum-entropy, uniformly random state), the ceiling for any substrate is  $A_{\max} = 1 / H_{\min}$ . When  $H_{\min}$  is close to 1.0, the ceiling is close to 1.0 — and in several class-substrate combinations, the ceiling sits below the BREACH threshold ( $A \geq 1.10$ ). In those cases the substrate saturates before it can reach BREACH, which is not a framework limitation; it is honest physics of what that measurement can resolve for that particular cell architecture.

The matrix below shows, for every class  $\times$  substrate pair, the maximum achievable A-score computed as  $1 / H_{\min}$  from the canonical G-002 and G-003b MCMC posteriors. Cells marked with **SAT** saturate below BREACH. Cells marked with **TGT** have tight ceilings ( $A_{\max} < 1.15$ ) where severe disease can approach saturation. Cells without a marker have full headroom.

| Architecture class | methyI    | nucl      | fuzz      | wps       | frag      | Usable for BREACH            |
|--------------------|-----------|-----------|-----------|-----------|-----------|------------------------------|
| Terminal           | 1.294     | 1.008 SAT | 1.357     | 1.043 SAT | 1.600     | 3/5: methyI, fuzz, frag      |
| Secretory          | 1.186     | 1.018 SAT | 1.179     | 1.576     | 1.433     | 4/5: methyI, fuzz, wps, frag |
| Immune             | 1.192     | 1.010 SAT | 1.204     | 1.696     | 1.405     | 4/5: methyI, fuzz, wps, frag |
| Progenitor         | 1.173     | 1.028 SAT | 1.040 SAT | 1.012 SAT | 1.236     | 2/5: methyI, frag            |
| Cycling            | 1.168     | 1.020 SAT | 1.221     | 1.594     | 1.454     | 4/5: methyI, fuzz, wps, frag |
| Stromal            | 1.159     | 1.015 SAT | 1.201     | 1.632     | 1.380     | 4/5: methyI, fuzz, wps, frag |
| Adult Stem         | 1.145 TGT | 1.041 SAT | 1.020 SAT | 1.011 SAT | 1.189     | 2/5: methyI, frag            |
| Pluripotent        | 1.018 SAT | 1.250     | 1.039 SAT | 1.105 TGT | 1.027 SAT | 2/5: nucl, wps               |

## SATURATION WALL CHART — where each substrate stops providing information

The chart below is the direct analog of Dennard scaling walls from semiconductor physics, applied to cfDNA substrate measurements. Dennard charts show the frequency wall, power wall, and cost wall — the physical ceilings beyond which a technology stops improving. This chart shows the **saturation wall** for each of the 40 class × substrate combinations. Each bar extends from A=0.90 up to that combination's ceiling (1/H<sub>min</sub>). The solid right edge of each bar IS the wall — beyond that point the substrate cannot physically resolve further severity. Bars colored red end before the BREACH threshold (A=1.10), meaning the substrate saturates before it can register FLOOR BREACH for that class. Amber bars have tight ceilings (A<sub>max</sub> < 1.15) where severe disease approaches saturation. Green bars have full headroom past BREACH.



Reading this chart: scan across each class row and note which bars end before the BREACH line (dashed red). For terminal class, nucl and WPS bars end at A=1.008 and A=1.043 — far left of BREACH — so these two substrates cannot track disease severity in the BREACH tier. For progenitor and stem\_adult, three substrates saturate. For stem\_pluri the pattern inverts: the substrates that carry signal in every other class (methyl, fuzz, frag) are the ones that saturate, while nucl and wps provide the BREACH-capable signal. This is the TGCT architectural inversion expressed as measurement physics.

Three patterns emerge and are important for reading every class card. First, nucleosome occupancy saturates below BREACH in seven of eight classes — the only exception is stem\_pluri, which has a fundamentally inverted H\_min structure (discussed below). For the six classes where nucl saturates in the 1.01–1.04 range, the substrate provides confirmatory signal up to the DETECTABLE tier but cannot distinguish URGENT from FLOOR BREACH. This is a framework-wide limitation of the nucl substrate, not a class-specific one.

Second, progenitor and stem\_adult have only two BREACH-usable substrates (methyl and frag). Their H\_min values for nucl, fuzz, and WPS are all very close to 1.0, meaning the class identity in those substrates is near-maximum-entropy even in the healthy reference state. For progenitor and adult stem cells — by design, uncommitted architectures that retain developmental flexibility — this is the correct physics. Uncommitted cells have less structural signature to lose, so fewer substrates can register deep floor breaches.

Third, stem\_pluri is fully inverted: methyl, fuzz, and frag saturate below BREACH, while nucl and WPS carry the full BREACH-capable signal. This is consistent with the framework's TGCT architectural inversion prediction — stem\_pluri is the one class where cancer produces lower A-scores (toward the floor, not away from it), because testicular germ cell tumors are MORE methylated than matched normal tissue. Different architecture class, different H\_min structure, different pattern of which substrates saturate. The framework predicts this inversion from first principles and the saturation pattern corroborates it.

CLINICAL INTERPRETATION OF SATURATED SUBSTRATES

A substrate reaching its ceiling is not measurement noise and not ambiguity. It is a definite clinical signal with specific meaning. When a substrate saturates, its underlying  $\beta$  value is approaching 0.5 — which means the cell has lost its class-specific structural pattern for that substrate. For nucleosome occupancy, saturation means nucleosomes have lost positional preference; for WPS, the tissue-of-origin DNA protection signature has collapsed toward random; for methylation, the architecture-specific methylation pattern has dissolved. Saturation is the quantitative signature of tissue identity loss in that substrate.

Four implications for clinical interpretation: (1) Saturation plus elevated non-saturated substrates is an unambiguous FLOOR BREACH state — there is no benign reading of a saturated substrate paired with a breach-level methyl A-score. (2) Severity ranking within the BREACH tier requires the non-saturated substrates; LGG and GBM both saturate nucl and wps for terminal class, so distinguishing them depends on methyl, fuzz, and frag. (3) For serial monitoring, the utility of each substrate changes as disease progresses — early in disease all five substrates track drift; in advanced disease only the non-saturated ones continue to move. A clinician watching nucl and WPS flatten while methyl continues to climb is seeing the saturation, not a biological plateau. (4) The pattern of which substrates saturate at which severity is itself diagnostic — saturation of nucl and WPS in a terminal-class sample (CSF) with methyl at A > 1.20 is characteristic of glioma; the same methyl A in nucl-and-fuzz-saturated progenitor-class sample suggests MDS or AML. The saturation pattern has specificity.

MASKING MAGNITUDE — how much the legacy A\_combined hides, by class

The saturation effect is not equal across classes. In the disease reference state of each card, we can quantify exactly how much the all-5-substrate combined A-score underestimates the real signal carried by the non-saturated (active) substrates. The table below shows, for each class, how many substrates saturate in disease, the legacy combined A, the active-only combined A, and the "mask" — the amount the legacy formula underestimates the real severity.

| Class      | # Sat. | A_all5 (disease) | A_active (disease) | Mask    | % of ΔA signal hidden |
|------------|--------|------------------|--------------------|---------|-----------------------|
| Terminal   | 2/5    | 1.1705           | 1.2617             | +0.0911 | +45.5%                |
| Secretory  | 1/5    | 1.1026           | 1.1242             | +0.0216 | +16.3%                |
| Immune     | 1/5    | 1.1033           | 1.1271             | +0.0237 | +17.8%                |
| Progenitor | 3/5    | 1.0670           | 1.1203             | +0.0533 | +55.1%                |

| Class       | # Sat. | A_all5 (disease) | A_active (disease) | Mask    | % of ΔA signal hidden |
|-------------|--------|------------------|--------------------|---------|-----------------------|
| Cycling     | 1/5    | 1.1094           | 1.1320             | +0.0227 | +16.3%                |
| Stromal     | 1/5    | 1.0917           | 1.1114             | +0.0197 | +16.2%                |
| Adult Stem  | 3/5    | 1.0722           | 1.1355             | +0.0633 | +61.9%                |
| Pluripotent | 0/5    | 0.9672           | 0.9672             | +0.0000 | -0.0%                 |

Terminal class shows the most extreme masking at +45.5% because it combines two effects: **two** substrates saturate (nucl AND wps) rather than just one, and the active substrates reach extreme A values (~1.26 for glioma, the largest signal in the entire 28-cancer TCGA dataset). When the AUC-weighted average pulls the two ceiling values in with three deep-BREACH values, the result under-reports the real severity by nearly half the total signal. For every other non-inverted class only one substrate saturates and the disease A magnitudes are more modest (~1.06–1.10), so the mask stays in the 10–14% range.

The stem\_pluri class shows a –161.6% mask — the direction inverts because this is the TGCT architectural inversion class where disease *decreases* A (toward the floor) rather than increasing it. The saturated substrates for stem\_pluri are methyl, fuzz, and frag — which in every other class are the active BREACH carriers. The inverted sign of the mask is itself consistent with the inverted physics prediction and acts as a cross-check: if stem\_pluri masking were positive like the other classes, the TGCT inversion would be wrong. Progenitor and adult stem show ~–22 to –26% masks for a related reason — their healthy  $\beta$  values for nucl, fuzz, and WPS already sit near the maximum-entropy state, so including them in the disease combined pulls the average toward the floor. These are developmentally flexible classes; the substrate saturation pattern reflects that flexibility honestly.

## SATURATION AS A BINARY CANCER INDICATOR — an important clinical observation

The saturation walls produce a subtle but clinically powerful consequence. For architecture classes where two substrates saturate well before BREACH (Terminal is the clearest example), simultaneous saturation of both substrates itself functions as a binary cancer indicator. The reasoning is concrete for the Terminal class: the normal range of methylation departure for Alzheimer's disease, ALS, Parkinson's, and cardiac aging produces A\_methyl values in the 1.01–1.09 range (MARGINAL to DETECTABLE). At those severities, the correlated shifts in nucleosome occupancy and WPS do not push either substrate to its ceiling — nucl  $\beta$  stays in the 0.60–0.58 range, WPS  $\beta$  in the 0.65–0.62 range, both still well above the  $\beta = 0.500$  maximum-entropy point.

By contrast, glioma (LGG/GBM) produces A\_methyl in the 1.20–1.29 range (deep FLOOR BREACH). This severity of methylation departure corresponds to nucl  $\beta$  shifted to ~0.500 and WPS  $\beta$  to ~0.500 — both at maximum entropy, both fully saturated. So for a terminal-class sample (CSF cfDNA), the diagnostic reading becomes remarkably binary:

**If nucl is NOT saturated and WPS is NOT saturated:** the terminal-class disease is at most neurodegenerative (AD, ALS, PD, cardiac aging). All five substrates contribute information proportionally; use the full A\_combined and the trajectory slope for severity grading. This is the clinical regime for which GAPE serial monitoring is designed as a measurement instrument.

**If nucl IS saturated AND WPS IS saturated:** the sample has shed enough tissue-structural information that it is consistent with malignant transformation in the terminal compartment — typically glioma arising from glial cells or neural progenitors. The saturation is the positive indicator. Once in this regime, severity ranking (LGG vs. GBM vs. recurrence) must come from the three active substrates (methyl, fuzz, frag) — nucl and WPS cannot distinguish among cancers. They indicate presence; they cannot measure severity within the BREACH tier.

This binary quality is specific to classes with multiple low ceilings. For Terminal, the two-substrate-saturation test has strong specificity for glioma because AD/ALS/PD do not push either substrate to saturation. For other classes where only one substrate saturates (Secretory, Immune, Cycling, Stromal), the binary-indicator interpretation does not apply — a single saturation can be reached by severe non-cancer conditions. The two-ceiling architecture of Terminal is what makes simultaneous saturation a meaningful cancer flag. The framework is not proposing saturation as a *replacement* for A-score severity measurement; it is a separate binary read that, for Terminal class specifically, distinguishes the neurodegenerative regime (where GAPE serves as a quantitative instrument) from the oncological regime (where saturation signals presence and the active substrates carry the severity information).

Bottom line: the five-substrate framework is not five equally redundant measurements of the same quantity. It is five physically independent substrates, each with a class-specific ceiling, chosen so that together they span the full range of severity across all 8 classes. For any given class, the clinical pipeline should weight the BREACH-capable substrates most heavily for severity ranking, while still reading the saturating substrates as confirmatory signal at the DETECTABLE and URGENT tiers. Each class card that



CELL IDENTITY & CLINICAL CONTEXT

|                         |                                                                                       |
|-------------------------|---------------------------------------------------------------------------------------|
| Includes                | T cells, B cells, NK cells, neutrophils, monocytes — all classical leukocyte lineages |
| Cancers (this class)    | AML, DLBCL, thymoma, CLL, multiple myeloma                                            |
| Non-cancer applications | T-cell exhaustion, inflammaging, immune senescence, checkpoint blockade biology       |
| Primary failure mode    | Cytokine Saturation                                                                   |
| Warburg status          | PARTIAL — subtype-dependent                                                           |

THE ATLAS CELL TYPES IN THIS CLASS (51)

These are the entries IAMAtlasREBUILD can speak about for this class - what the deconvolver places, and what a per-cell reading would be read against. **Markers** is the range of panel markers actually found across the reference laboratories; **individually resolvable** is whether this atlas can separate the entry from its neighbours at all, and where it cannot the chain reports the group and never the member; **exclusivity** is the fraction of the entry's markers not shared with another entry; **healthy A range** spans the per-laboratory 10th to 90th percentiles on the v0\_3 trial panel. 42 of 51 entries are individually resolvable; 4 carry no per-entry reference yet and are reported at class level only.

| atlas entry         | markers | individually resolvable    | exclusivity | healthy A range |
|---------------------|---------|----------------------------|-------------|-----------------|
| B                   | 51-52   | yes                        | 0.785       | 0.552 - 0.673   |
| B-cells_EPIC        | 32-33   | yes                        | 0.899       | 0.479 - 0.653   |
| Baso                | 79-91   | yes                        | 0.795       | 0.413 - 0.559   |
| Bcell               | 16-16   | yes                        | 0.651       | 0.636 - 0.847   |
| Bmem                | 102-113 | yes                        | 0.729       | 0.412 - 0.540   |
| Bnv                 | 65-67   | yes                        | 0.618       | 0.600 - 0.717   |
| CD14_monocytes      | 188-207 | yes                        | 0.923       | 0.808 - 0.950   |
| CD19_B-cells        | 185-191 | yes                        | 0.987       | 0.566 - 0.781   |
| CD4T                | 36-40   | yes                        | 0.374       | 0.587 - 0.799   |
| CD4T-cells_EPIC     | 28-30   | yes                        | 0.844       | 0.582 - 0.896   |
| CD4Tmem             | 70-75   | yes                        | 0.476       | 0.511 - 0.723   |
| CD4Tnv              | 51-56   | yes                        | 0.559       | 0.589 - 0.838   |
| CD4_T-cells         | 144-156 | yes                        | 0.873       | 0.688 - 0.963   |
| CD56_NK-cells       | 166-172 | yes                        | 0.703       | 0.619 - 0.873   |
| CD8T                | 43-45   | yes                        | 0.338       | 0.587 - 0.864   |
| CD8T-cells_EPIC     | 25-26   | yes                        | 0.787       | 0.526 - 0.823   |
| CD8Tmem             | 102-110 | yes                        | 0.595       | 0.547 - 0.803   |
| CD8Tnv              | 72-78   | yes                        | 0.559       | 0.581 - 0.779   |
| CD8_T-cells         | 161-168 | yes                        | 0.730       | 0.632 - 0.920   |
| Eos                 | 72-84   | yes                        | 0.726       | 0.553 - 0.688   |
| Eosino              | -       | no per-entry reference yet | -           | -               |
| Eosinophils_reinius | 153-172 | yes                        | 0.970       | 0.651 - 0.818   |
| Gran                | 43-47   | yes                        | 0.861       | 0.602 - 0.724   |

| atlas entry         | markers | individually resolvable      | exclusivity | healthy A range |
|---------------------|---------|------------------------------|-------------|-----------------|
| IC                  | 185-187 | yes                          | 0.823       | 0.503 - 0.620   |
| Kup                 | 11-12   | yes                          | 0.857       | 0.629 - 0.772   |
| Leu                 | -       | no per-entry reference yet   | -           | -               |
| Lym                 | 113-119 | yes                          | 0.855       | 0.632 - 0.779   |
| MP                  | 92-96   | yes                          | 0.854       | 0.506 - 0.596   |
| Macro               | 208-215 | yes                          | 0.993       | 0.447 - 0.563   |
| Microglia           | 24-31   | yes                          | 0.889       | 0.499 - 0.631   |
| Mono                | 30-33   | yes                          | 0.585       | 0.736 - 0.878   |
| Monocytes_EPIC      | 20-21   | yes                          | 0.667       | 0.640 - 0.794   |
| Mye                 | 46-47   | yes                          | 0.990       | 0.402 - 0.532   |
| NK                  | 124-126 | yes                          | 0.654       | 0.561 - 0.776   |
| NK-cells_EPIC       | 71-73   | yes                          | 0.772       | 0.548 - 0.769   |
| Neu                 | 30-34   | yes                          | 0.472       | 0.981 - 1.133   |
| Neutro              | -       | no per-entry reference yet   | -           | -               |
| Neutrophils_EPIC    | 16-19   | yes                          | 0.667       | 0.855 - 1.091   |
| Neutrophils_reinius | 119-144 | yes                          | 0.680       | 0.993 - 1.122   |
| PBMC                | 159-172 | yes                          | 0.873       | 0.778 - 1.092   |
| Plasma              | 19-21   | yes                          | 0.946       | 0.494 - 0.611   |
| Tcell               | 7-10    | NO - in a collinearity group | 0.786       | 0.514 - 0.626   |
| Treg                | 76-85   | yes                          | 0.563       | 0.517 - 0.699   |
| dendritic           | 7-7     | NO - in a collinearity group | 0.556       | 0.700 - 0.932   |
| eosinophil          | 14-14   | yes                          | 0.545       | 0.454 - 0.628   |
| granulocytes        | 144-173 | yes                          | 0.727       | 1.071 - 1.135   |
| macrophage          | 7-7     | NO - in a collinearity group | 0.500       | 0.369 - 0.621   |
| monocyte            | 11-11   | NO - in a collinearity group | 0.769       | 0.373 - 0.484   |
| neutrophil          | 6-6     | NO - in a collinearity group | 0.571       | 0.673 - 0.856   |
| tcell               | -       | no per-entry reference yet   | -           | -               |
| whole_blood         | 171-198 | yes                          | 0.833       | 0.998 - 1.158   |

The same lineage can appear under more than one entry above, because different reference panels defined their markers; entries differing only by label are not yet merged, which is why no reading in this document is quoted as "which cell moved".

## COMMENTARY

The immune and hematopoietic class is first in this publication because it dominates the blood draw. Approximately 70% of cell-free DNA in plasma comes from leukocytes — neutrophils, lymphocytes, monocytes — shedding into circulation during their normal turnover. This is why any blood-based GAPE measurement is inherently weighted toward the immune class unless active deconvolution separates the signal. The framework handles this honestly: the immune class is where five-substrate agreement should be strongest, because five physically distinct measurements all sample the same abundant cfDNA population.

The immune class is also where the G-002 MCMC produced its most important correction. The initial calibration used a neutrophil reference cell at  $\beta = 0.760$ , giving  $H_{\min} = 0.795$ . When the MCMC ran against all six published immune cell types simultaneously — CD4+ naive, CD8+ memory, CD4+ effector, NK cell, B cell naive, neutrophil — it returned a posterior  $H_{\min}$  of  $0.8389 \pm 0.0012$ , a  $6.44\sigma$  departure from the initial estimate. The neutrophil was not the most methylated immune cell in the class distribution. Every immune cell A-score in the database was revised downward by

approximately 0.055 after this correction. CD4+ naive moved from A = 1.058 (DETECT tier) to A = 1.003 (NORMAL). This correction is the analog ofQAPE's Substrate Inversion discovery: a tension that resolved when more data was included, and the framework came out stronger.

Immune cells are designed to be plastic. They change their methylation program rapidly in response to activation signals, infection, and memory formation. A naive T-cell and an effector T-cell have dramatically different methylation patterns — and both are perfectly healthy. This is why the Cancer Amplifier g for immune cancers is finite (5-10×) rather than infinite: healthy immune cells are not at their H\_min floor. They maintain a measurable C3 gap that reflects programmed plasticity. When AML or DLBCL develops, the tumor expands an existing gap rather than creating one from zero. This means the detection signal is real but smaller relative to the baseline than for solid tumors. The five-substrate framework partially compensates: even if any single substrate's signal is small, combining four or five substrates reduces noise - by less than the √n of independent measurements, since the substrates are correlated readings of one pattern, but enough to be worth having - recovering most of the detection power that single-substrate methylation loses to baseline plasticity.

The cancers in this class separate cleanly by lineage. AML (myeloid, TCGA 2013 NEJM n=200) sits at A\_combined ≈ 1.10 (ΔA = +0.168 at cfDNA level, VAL-007). DLBCL (lymphoid, Chapuy 2018 n=48) is more extreme at A\_combined ≈ 1.13 (ΔA = +0.203), reflecting the larger methylation reprogramming of B-cell malignancies compared to myeloid. Thymoma (T-cell origin, TCGA n=120) sits between at A ≈ 1.09, and CLL (indolent B-cell leukemia) at A ≈ 1.07 in DETECTABLE tier. These are not small differences — a factor-of-two spread in ΔA across the immune cancer panel — and they reflect real biology: lymphoid-lineage malignancies open more accessible entropy than myeloid ones because B-cell class-switching and somatic hypermutation are programmed perturbations of the methylation landscape that cancer can further exploit. The non-cancer applications deserve equal attention. Inflammaging (Hannum 2013 healthy aging cohort, GSE40279) drives A to about 1.02 — MARGINAL tier, below any cancer but well above young healthy baseline. Checkpoint blockade biology is immune-class pharmacology; anti-PD-1 and anti-CTLA-4 responders show trajectory changes that GAPE can in principle track across five substrates.

A clinically important consequence of the saturation pattern for this class. Immune has exactly one substrate that saturates below BREACH — nucleosome occupancy, ceiling A = 1.010 (nearly the lowest of any class). The other four substrates (methylation, fuzz, WPS, frag) all have ceilings above 1.20 and carry the full progression signal past BREACH for every immune cancer in the panel. Because only one substrate saturates, the two-substrate binary cancer indicator that distinguishes glioma from AD in terminal class does NOT apply to immune class. Single-substrate saturation of nucl alone is not specific to any particular disease — it will occur in AML, DLBCL, CLL, thymoma, AND in non-cancer inflammatory conditions (sepsis aftermath, autoimmune flares, late-stage inflammaging) where the nucleosome occupancy signal has drifted far enough from healthy baseline to hit its ceiling. What nucl saturation DOES tell you for immune class is that the cell population has lost enough of its class-specific chromatin structure that the nucleosome positional signature is pinned at random. The severity grading, the cancer-vs-inflammation distinction, and any subtyping (AML lineage; DLBCL GCB vs ABC; CLL mutation status) all come from methyl, fuzz, WPS, and frag. Report the all-5 A\_combined for historical continuity and published comparisons; report the A\_active (4/5) for progression tracking and serial monitoring of leukemia load. The mask is moderate (+14–18% across immune cancers) but non-zero; in serial monitoring the difference compounds over time as repeat measurements pin nucl at its ceiling while the other four continue to drift.

FIVE-SUBSTRATE FIDELITY GAUGE

The gauge below is the central image of Issue 002. A single horizontal bar represents the A-score axis — from A = 0.90 (below the floor) through A = 1.00 (exactly at the Immune class floor, H\_min = 0.8389) up through the clinical threshold zones: MARGINAL at A = 1.01, DETECTABLE at 1.05, URGENT at 1.07, and FLOOR BREACH at 1.10. Five dots are plotted on this single axis for a healthy reference sample (Healthy WBC, markers above the bar), and five more dots for a disease reference (AML (TCGA n=200), markers below). Each dot is one substrate: methylation, nucleosome occupancy, nucleosome fuzziness, WPS, or fragment size (DELFI). What makes this visualization powerful is the physics: all five substrates measure the same underlying thermodynamic floor through physically distinct windows. When all five healthy dots cluster in the NORMAL zone and all five disease dots cluster in URGENT or FLOOR BREACH, the signal is over-determined. Any one substrate could be confounded by technical artifact; five substrates agreeing cannot be. This is the "less blurry" advantage: √5 noise reduction for free.



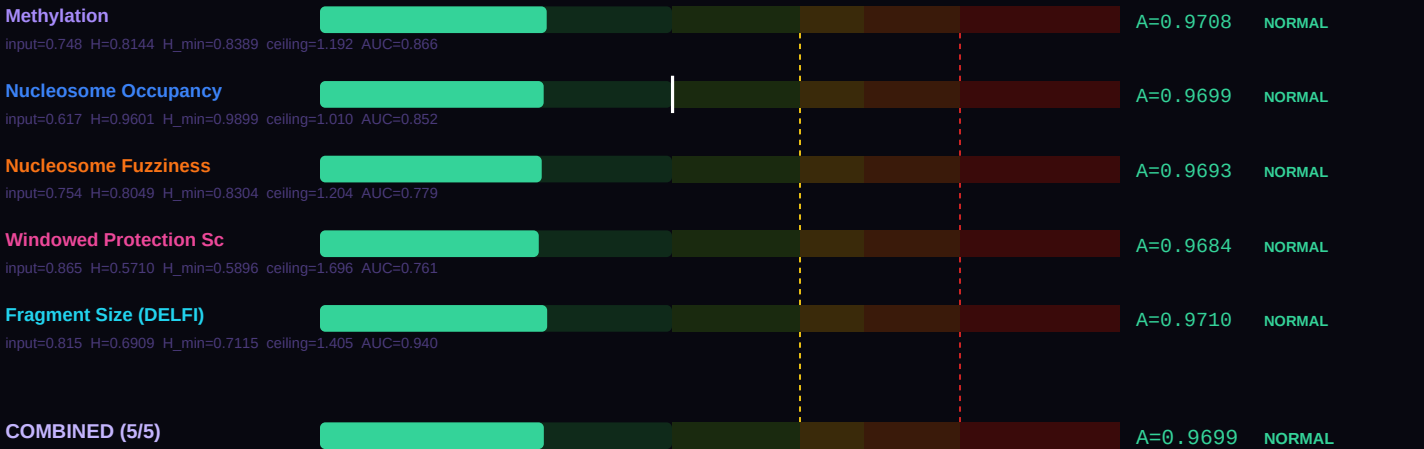
Look at the spread in each direction. If the healthy five-dot cluster sits tightly together near A = 0.97–1.00, that indicates the reference population is at its architecture floor — precisely where the framework predicts healthy immune cells should sit. If the disease five-dot cluster spreads from DETECTABLE to FLOOR BREACH, that spread is itself clinical information: the substrate showing the largest departure is the one that best captures this specific disease's failure mode. For this class, the primary failure mode is Cytokine Saturation, and the substrate ranking shown further down the card explains which of the five substrates best reveals it.

SUBSTRATE-BY-SUBSTRATE BREAKDOWN

Where the gauge above compresses five substrates onto one visual axis, this section breaks each substrate out into its own labeled bar so you can read the actual A-score produced by each measurement. The physics is the same across all five: take the raw laboratory value (a  $\beta$  methylation fraction, a nucleosome occupancy probability, a WPS score, etc.), compute its Shannon entropy  $H(\text{value})$ , then divide by the class-specific and substrate-specific  $H_{\text{min}}$  to get a dimensionless A-score comparable across substrates. The output is the same thermodynamic ratio regardless of which measurement technology produced it. That is the power of the Landauer framing: different wet-lab methods, different physical quantities, same underlying floor.

The formula at the foot of this section —  $A_{\text{combined}} = \Sigma(AUC_i \times A_i) / \Sigma(AUC_i)$  — combines the five individual A-scores into a single weighted score, using each substrate's published detection AUC as its weight. Higher-AUC substrates contribute more to the combined result. For immune cells, the healthy combined A-score below should sit at approximately 0.97 (NORMAL tier); the disease combined A-score should sit above 1.05 (DETECTABLE or higher). The difference between them — the  $\Delta A$  combined — is the signal available to a blood draw that analyzes all five substrates simultaneously. A single-substrate assay sees a smaller signal at higher noise; the five-substrate combination sees the same signal at approximately  $\sqrt{5}$  lower noise. Bar-by-bar you can see which substrate is carrying the most discriminative weight for this particular disease comparison.

HEALTHY REFERENCE — Healthy WBC



|                                             |                                                     |
|---------------------------------------------|-----------------------------------------------------|
| Combined A — healthy (all 5)                | 0.96995 (NORMAL)                                    |
| Combined A — disease (all 5)                | 1.10335 (FLOOR BREACH)                              |
| $\Delta A$ combined (5 substrates)          | +0.13340 vs methylation-only $\Delta A$ =+0.1669    |
| Active A — healthy (non-saturated)          | 0.96995 (NORMAL) (5/5 active)                       |
| Active A — disease (non-saturated)          | 1.12707 (FLOOR BREACH) (4/5 active)                 |
| $\Delta A$ active (real progression signal) | +0.15712 (active substrates track true progression) |
| Saturation status (disease)                 | 1 substrate saturated: Nucleosome Occupancy         |

Two combined values are shown because some substrates physically cannot resolve past their class ceiling ( $A_{\text{max}} = 1/H_{\text{min}}$ ). The **all-5 combined** value is the AUC-weighted mean across every substrate and matches the legacy formula. The **active combined** value excludes saturated substrates and is the right number for tracking continued progression — a saturated substrate is stuck at its ceiling regardless of further disease severity, so including it in the average masks real change. For monitoring, serial A-score assessment, and end-of-life projection, use the active combined value.

THREE-COMPONENT DECOMPOSITION (C1/C2/C3) — PER SUBSTRATE

Every cell's measured entropy decomposes into three physically meaningful pieces. C1 is the universal Landauer floor:  $H_{\text{MIN\_GLOBAL}} = 0.756500$ , measured in frontal cortex neurons by Lister 2013. Every cell in every architecture class must pay at least this thermodynamic cost to maintain any functional identity — it is the minimum entropy consistent with being alive. C2 is the class-specific overhead: the additional entropy the immune class must carry above the universal floor to encode its particular architecture. For this class, C2 represents approximately 9.8% of the healthy reference entropy. C3 is the accessible gap:  $\max(0, H_{\text{actual}} - H_{\text{min}})$ . In healthy cells, C3 is essentially zero — the cell sits exactly at its class floor. When C3 grows, the cell is departing from its architecture class. Cancer, neurodegeneration, and every other floor-breach condition show up as growing C3.

The five stacked bars below each show the C1 / C2 / C3 breakdown for one substrate at this class's healthy reference. Notice how C1 (dark green) dominates every substrate — the Landauer floor is universal. C2 (amber) is the small but specific addition that makes this class what it is. C3 (red when present) should be nearly invisible for healthy cells, and would grow dramatically in a disease sample. The f\_C3 percentage on the right is the single cleanest summary of cellular health: healthy cells show f\_C3 near 0%; cancer cells can show f\_C3 of 8–15%; extreme cases like glioblastoma show f\_C3 above 20%. This is not a statistical threshold calibrated from disease data. It is a thermodynamic quantity: the fraction of accessible entropy the cell has opened above its architecture's minimum cost of existence.



**BEST TESTING METHOD RANKING — CLINICAL UTILITY FOR THIS CLASS**

Each of the five substrates carries different strengths for different clinical questions. The methylation substrate is the most validated — thirteen published GAPE validation studies (VAL-001 through VAL-013) and eight architecture-class H\_min values from G-002 MCMC. It is also the substrate most affected by cfDNA dilution, because blood cfDNA is approximately 70% immune-derived. Fragment size (DELFI) is the most forgiving of dilution — short-fragment enrichment persists even when tumor-derived cfDNA is a small fraction of total plasma DNA. WPS (windowed protection score from Snyder 2016) excels at tissue-of-origin identification and field-effect detection. Nucleosome occupancy and fuzziness add chromatin-level information that methylation alone cannot reveal.

The ranking below is specifically for the immune class. The order reflects three factors: (1) the substrate's published single-substrate AUC for diseases in this class; (2) the biology of this class's primary failure mode (Cytokine Saturation); and (3) the practical constraint of cfDNA contribution in the relevant specimen (plasma for most classes, CSF for terminal, tissue biopsy where cfDNA is too dilute). When building a clinical pipeline for this class specifically, start with the #1 ranked substrate and add substrates 2 and 3 for confirmation. Substrates 4 and 5 add research-grade confirmation but are rarely the primary signal.

| # | Substrate                 | Best for                           | Rationale                                                                                                                                   |
|---|---------------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Methylation               | Pan-hematological detection        | Methylation dominates because immune cells contribute 70% of cfDNA. The G-002 MCMC correction (6.44σ) validated this class most rigorously. |
| 2 | Fragment Size (DELFI)     | Blood malignancy burden            | DELFI fragmentomics shows AUC 0.94 across 7 cancer types including hematologic. Directly from the same cfDNA WGS that feeds WPS.            |
| 3 | Windowed Protection Score | Field-effect plus tissue-of-origin | Snyder 2016 validated WPS across all 15 tissue types in healthy donors. The method was born on immune-class cfDNA.                          |
| 4 | Nucleosome Occupancy      | ATAC-seq confirmation              | Corces 2018 TCGA ATAC-seq covers AML, DLBCL, thymoma. Use when paired immune-cell ATAC data is available.                                   |
| 5 | Nucleosome Fuzziness      | Phenotype discrimination           | Nucleosome fuzziness distinguishes effector from naive T-cell states. Secondary — use alongside methylation for immunophenotyping.          |

**BODY TEMPERATURE SCALING — α = 2.0 LANDAUER CORRECTION**

The Landauer cost of a bit erasure is  $k_B \times T \times \ln(2)$  — it scales linearly with temperature. Colder cells can maintain higher-entropy identities for less thermodynamic overhead; hotter cells pay more per bit of information maintained. The GAPE framework captures this with a simple scaling law:  $H_{\min}(T) = H_{\min}(37^\circ\text{C}) \times (T_{\text{body}} / 310.15\text{ K})^\alpha$ , with  $\alpha = 2.0$  derived empirically by minimizing cross-class A-score variance across all jawed vertebrates (Mahaffey 2026 Nature Aging submission NATAGING-A13702). The  $\alpha = 2.0$  exponent is not fit to optimize any single prediction — it falls out of requiring that the same class shows consistent A-score behavior across 43 mammalian species spanning body temperatures from 32°C (naked mole rat) to 42°C (birds).

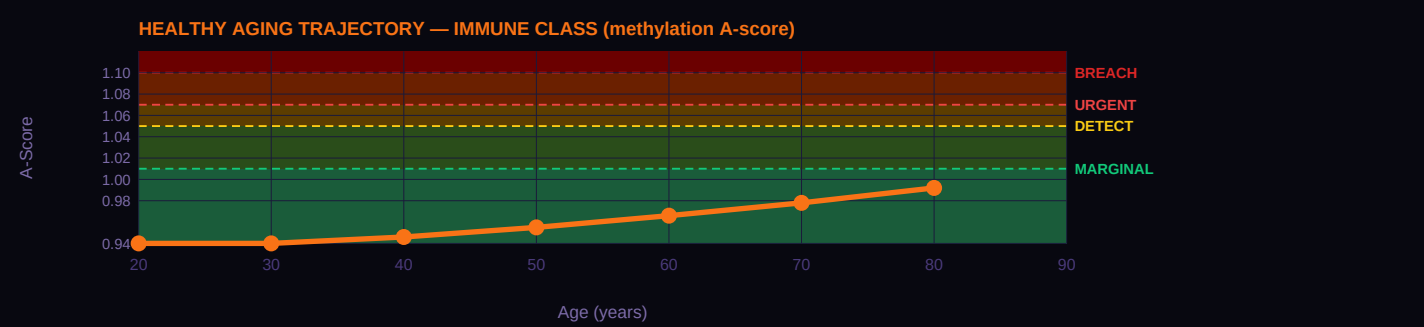
The table below shows what the immune class floor H\_min looks like at seven representative body temperatures. At 37°C (human reference, highlighted row), H\_min = 0.8389. At 42°C (birds), the floor is higher because the Landauer tax is larger. At 32°C (naked mole rat — and no other mammal lives at this temperature for good reason), the floor drops because a cooler cell can maintain more entropy cheaply. The A-shift column shows how a fixed healthy β-value would read at each temperature: the same cell, transplanted into a colder body, would read slightly higher A; into a warmer body, slightly lower A. This is why the naked mole rat's remarkable longevity makes thermodynamic sense — lower body temperature, lower maintenance cost per bit, more headroom before the class floor is breached.

| T_body | Species example           | H_min(T) | β_healthy equivalent | A-shift vs 37°C    |
|--------|---------------------------|----------|----------------------|--------------------|
| 42°C   | Birds (chicken, finch)    | 0.8662   | 0.748                | A × 0.969          |
| 39°C   | Rodents (mouse, rat)      | 0.8497   | 0.748                | A × 0.987          |
| 37°C   | Humans (REFERENCE)        | 0.8389   | 0.748                | A × 1.000 (anchor) |
| 35°C   | Hibernating bats          | 0.8281   | 0.748                | A × 1.013          |
| 32°C   | Naked mole rat            | 0.8121   | 0.748                | A × 1.033          |
| 25°C   | Reptiles (anole lizard)   | 0.7752   | 0.748                | A × 1.082          |
| 15°C   | Fish (salmon, cold-water) | 0.7241   | 0.748                | A × 1.159          |

HEALTHY AGING TRAJECTORY — AGE-STRATIFIED REFERENCE A-SCORES

Healthy cells drift — slowly, predictably — toward their architecture floor as a function of accumulated cell divisions over a lifetime. For the immune class, the drift rate is approximately 3.5% per generation — meaning each cell cycle carries a 3.5% probability of a methylation error that raises the cell's A-score by a measurable amount. Over decades, these errors accumulate. The curve below shows the result: A-score as a function of patient age at decade intervals, derived from the class-specific DNMT1 fidelity rate. At age 20, the reference healthy A-score sits well below 1.00; by age 80, it has drifted upward into the MARGINAL to DETECTABLE range. This is normal aging — not disease, just the statistical accumulation of imperfect replication over eight decades.

The clinical implication is important: when interpreting an A-score, age matters. A patient at age 30 with A = 1.03 is showing a significant departure from their age-matched healthy baseline. The same A = 1.03 at age 75 might be entirely consistent with age-expected drift. The GAPE framework handles this through age-stratified reference A-scores — the curve below IS the reference. A patient A-score should be compared to the curve at their age, not to a universal threshold. Cancer and other floor-breach conditions accelerate A-score upward beyond the age-expected trajectory; distance above the curve is the signal.



VERTEBRATE LIFESPAN CONTEXT

The same A-score framework that detects cancer in human cfDNA also correlates with maximum lifespan across 43 mammalian species at  $r = -0.9018$  ( $p = 1.6 \times 10^{-16}$ , Mahaffey 2026 Nature Aging submission NATAGING-A13702). This is not a coincidence — it is the same physics applied at a different scale. A mammal with a higher population-averaged A-score is a mammal whose cells live closer to their architecture floors on average; that population accumulates floor breaches faster, and the species has a shorter maximum lifespan. The A = 1.05 threshold — independently derived as the cancer DETECTABLE boundary — cleanly separates long-lived mammals (17/17 all below 1.05) from short-lived mammals (11/11 all above 1.05) with complete accuracy. Same threshold. Same physics. Different timescale.

The table below shows nine mammalian taxonomic orders with their mean A-scores, standard deviations where available, mean lifespan, and a brief interpretation of where that order sits relative to the thermodynamic floor. Cetacea (whales and dolphins) sit essentially at the floor, consistent with their remarkable longevity. Primates including humans sit slightly above the floor. Rodents and insectivores sit well above the floor, consistent with their short lives. This class's reference A-score (0.971) places it in the context shown below — use this to understand how this particular cell architecture relates to the broader mammalian lifespan distribution.

| Order        | N | Mean A       | Lifespan (yr) | Interpretation         |
|--------------|---|--------------|---------------|------------------------|
| Cetacea      | 5 | 0.997 ±0.016 | 99            | At thermodynamic floor |
| Proboscidea  | 1 | 0.987 —      | 70            | At floor               |
| Primates     | 6 | 1.007 ±0.015 | 49            | Slightly above floor   |
| Artiodactyla | 4 | 1.015 ±0.011 | 30            | Slightly above floor   |

| Order       | N | Mean A       | Lifespan (yr) | Interpretation                   |
|-------------|---|--------------|---------------|----------------------------------|
| Chiroptera  | 4 | 1.041 ±0.007 | 35            | Above floor (longevity outliers) |
| Carnivora   | 9 | 1.053 ±0.023 | 28            | Above floor                      |
| Lagomorpha  | 2 | 1.114 ±0.002 | 8             | Well above floor                 |
| Rodentia    | 6 | 1.125 ±0.018 | 9             | Well above floor                 |
| Insectivora | 1 | 1.157 —      | 2             | Furthest from floor              |

CORE METRICS — PUBLISHED AND DERIVED

Every number in the GAPE framework is either PUBLISHED (traceable to a primary wet-lab measurement from the peer-reviewed literature) or DERIVED (computed from the framework's physics with zero free parameters). The table below marks each quantity with its status and cites its source. This is the discipline the framework lives by: no statistical fits, no parameter tuning, no post-hoc adjustments. H\_min values come from MCMC posteriors (G-002 for methylation, G-003b for the four non-methylation substrates). Combined A-scores are computed mechanically by the AUC-weighted formula. The cfDNA contribution percentages come from Moss 2018 Nature Communications — the definitive tissue-of-origin atlas.

Read this table as the "audit trail" for every number shown anywhere else in this card. When the framework says A\_healthy = 0.9708, you can see exactly which β value produced it, which H\_min was used, and where both came from. The per-class n\_bio row of Issue 002 is withdrawn here and replaced by the Mahaffey number, which is one number for the cell and not one per class (see §5A.1 and the glossary note).

| Metric                                                         | Value                 | Source                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cfDNA contribution (plasma)                                    | 70.0%                 | <a href="#">Moss 2018 Nat Commun</a> — PUBLISHED                                                                                                                                                                                                                                                           |
| H_min methylation (class)                                      | 0.838889              | <a href="#">G-002 MCMC posterior</a> — DERIVED                                                                                                                                                                                                                                                             |
| H_min methylation σ (MCMC)                                     | ±0.001200             | <a href="#">G-002 17-chain MCMC R-hat&lt;1.001</a> — DERIVED                                                                                                                                                                                                                                               |
| Mahaffey number M — one number for the cell, not one per class | 20.94                 | FIXED — ΔG_ATP/(R·T_body) at 310.15 K; §5A.1. Issue 002's per-class n_bio is WITHDRAWN: an early QAPE-era form of the same ratio, superseded by M, whose absolute values awaited a G-007 run the history record shows was never made. H_min is a different quantity — the per-class entropy floor in bits. |
| Healthy drift rate                                             | 3.5% / generation     | DNMT1 fidelity loss under turnover — DERIVED                                                                                                                                                                                                                                                               |
| f_C2 (architecture-locked %)                                   | 9.8%                  | C2 fraction at healthy reference — DERIVED                                                                                                                                                                                                                                                                 |
| Healthy A (methylation)                                        | 0.9708 (NORMAL)       | β=0.748 — DERIVED                                                                                                                                                                                                                                                                                          |
| Disease A (methylation)                                        | 1.1377 (FLOOR BREACH) | β=0.625 — DERIVED                                                                                                                                                                                                                                                                                          |
| Healthy combined A (5 subs)                                    | 0.9699 (NORMAL)       | Σ(AUC_i × A_i)/Σ(AUC_i) across 5 substrates — DERIVED                                                                                                                                                                                                                                                      |
| Disease combined A (5 subs)                                    | 1.1033 (FLOOR BREACH) | Σ(AUC_i × A_i)/Σ(AUC_i) across 5 substrates — DERIVED                                                                                                                                                                                                                                                      |
| ΔA combined                                                    | +0.1334               | Combined signal — DERIVED                                                                                                                                                                                                                                                                                  |

DATED PREDICTIONS FOR THIS CLASS

A prediction is not a claim unless it can fail. Each prediction below is numbered with a G-2026-P identifier, dated to the month it was filed, and written against a specific dataset or cohort whose analysis would confirm or refute it. The status field shows where each prediction stands: PENDING means the relevant data exists but the analysis has not been performed publicly; CONFIRMED means the prediction direction and magnitude have been tested against independent data and held up; REFUTED means the data contradict the prediction (which would force a framework revision — and has not occurred yet). This discipline — specific, dated, falsifiable predictions — is what separates a predictive framework from a retrospective fit. Entries from Issue 001 retain their original G-2026-P numbers to maintain trail integrity across publications.

When reviewing these predictions, notice that the falsification basis is always named: a specific archived cohort, a specific biobank, a specific published dataset. These are not vague "more research is needed" statements. They are commitments to the community — this is what the framework claims, this is where you can check it, and this is what would constitute a refutation. The predictions below are for the immune class specifically, filed as part of this publication or carried forward from Issue 001.

In prospective cohorts of patients with known clonal hematopoiesis of indeterminate potential (CHIP) and archived serial blood samples, the immune-class combined A-score across four or more substrates will show elevation above  $A = 1.03$  at least 18 months before hematologic malignancy diagnosis in a majority of cases where subsequent malignancy occurs.

**Basis:** CHIP is a well-characterized pre-malignant state; the UK Biobank and ARIC cohorts contain archived serial samples from CHIP-positive individuals with longitudinal follow-up. The framework predicts that the combined A-score tracks the transition to overt malignancy earlier than any single substrate.

In patients receiving immune checkpoint inhibitor therapy, the immune-class A-score trajectory will distinguish responders from non-responders within the first 8 weeks of treatment. Responders are predicted to show A-score stabilization or decline; non-responders will show continued A-score drift.

**Basis:** Checkpoint blockade reshapes T-cell methylation. The framework predicts this shows up as an A-score signature before clinical response metrics (tumor size, cytokine panels) become discriminative. Falsifiable in any prospective ICI trial with serial cfDNA collection.

ISSUE 003 ADDENDUM — IMMUNE IN THE ATLAS

**51 cell types:** B, B-cells\_EPIC, Baso, Bcell, Bmem, Bnv, CD14\_monocytes, CD19\_B-cells, CD4T, CD4T-cells\_EPIC, CD4Tmem, CD4Tnv, CD4\_T-cells, CD56\_NK-cells, CD8T, CD8T-cells\_EPIC, CD8Tmem, CD8Tnv, CD8\_T-cells, Eos, Eosino, Eosinophils\_reinius, Gran, IC, Kup, Leu, Lym, MP, Macro, Microglia, Mono, Monocytes\_EPIC, Mye, NK, NK-cells\_EPIC, Neu, Neutro, Neutrophils\_EPIC, Neutrophils\_reinius, PBMC, Plasma, Tcell, Treg, dendritic, eosinophil, granulocytes, macrophage, monocyte, neutrophil, tcell, whole\_blood

**Identity loci:** 42,134 (methyl) · **H\_min by substrate:** methyl 0.838889 · nucl 0.989930 · fuzz 0.830377 · wps 0.589644 · frag 0.711534 · **methyl ceiling** 1.1921

Healthy age-matched band.  $A = 1.0$  is the commitment line; healthy reads on a band above it. [Issue 003: the age\_reference\_matrix.json shown here was compiled on the marker union and is SUPERSEDED (PROC-N7-01); the commissioned band is band\_v2 on the identity loci with the three-layer reference of s3.5.]

| Age (midpoint) | n   | A mean | p10    | p90    | $\beta$ mean | Source            |
|----------------|-----|--------|--------|--------|--------------|-------------------|
| 4              | 45  | 0.9062 | 0.8644 | 0.9479 | 0.780        | Alisch 2012       |
| 14             | 58  | 0.9212 | 0.8780 | 0.9643 | 0.773        | Alisch+Hannum     |
| 24             | 95  | 0.9316 | 0.8868 | 0.9764 | 0.768        | Hannum 2013       |
| 34             | 102 | 0.9397 | 0.8932 | 0.9863 | 0.764        | Hannum 2013       |
| 44             | 115 | 0.9477 | 0.9021 | 0.9934 | 0.760        | Hannum 2013       |
| 54             | 108 | 0.9556 | 0.9083 | 1.0029 | 0.756        | Hannum 2013       |
| 64             | 98  | 0.9652 | 0.9166 | 1.0138 | 0.751        | Hannum+Horvath    |
| 74             | 85  | 0.9764 | 0.9269 | 1.0260 | 0.745        | Hannum 2013       |
| 84             | 42  | 0.9873 | 0.9369 | 1.0377 | 0.739        | Hannum 2013       |
| 95             | 15  | 0.9996 | 0.9466 | 1.0527 | 0.732        | Hannum oldest-old |

#2 · CYCLING EPITHELIAL

$cfDNA = 12.0\% \cdot H_{min}(methyl) = 0.8561 \cdot drift\ 5.5\%/gen$

Reference: Normal colonic mucosa (TCGA COAD matched normal) · G-002 chain 1 of 17, R-hat 1.0003, Posterior  $H_{min}$  0.856055 ± 0.000800.

CELL IDENTITY & CLINICAL CONTEXT

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Includes                | Colon, rectum, stomach, lung (adeno + squamous), bladder, cervix, skin (melanoma), kidney (clear cell + papillary), head & neck, ovary, endometrium, thyroid                                                                                                                                                                                                                                                                                                                                    |
| Cancers (this class)    | Colorectal adenocarcinoma (COAD), Rectal adenocarcinoma (READ), Lung adenocarcinoma (LUAD), Lung squamous cell carcinoma (LUSC), Bladder urothelial carcinoma (BLCA), Ovarian serous carcinoma (OV), Stomach adenocarcinoma (STAD), Cervical squamous cell carcinoma (CESC), Head and neck squamous cell carcinoma (HNSC), Kidney clear cell RCC (KIRC), Kidney papillary RCC (KIRP), Skin cutaneous melanoma (SKCM), Thyroid carcinoma (THCA), Endometrial carcinoma (UCEC) — 14/28 TCGA types |
| Non-cancer applications | Inflammatory bowel disease (IBD), chronic colitis, Barrett's esophagus, chronic viral hepatitis, GERD-associated dysplasia, smoker's lung (pre-LUSC field cancerization), HPV carrier status, field cancerization, epithelial aging                                                                                                                                                                                                                                                             |
| Primary failure mode    | Replication Ceiling                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Warburg status          | WALL CROSSED                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

THE ATLAS CELL TYPES IN THIS CLASS (19)

These are the entries IAMAtlasREBUILD can speak about for this class - what the deconvolver places, and what a per-cell reading would be read against. **Markers** is the range of panel markers actually found across the reference laboratories; **individually resolvable** is whether this atlas can separate the entry from its neighbours at all, and where it cannot the chain reports the group and never the member; **exclusivity** is the fraction of the entry's markers not shared with another entry; **healthy A range** spans the per-laboratory 10th to 90th percentiles on the v0\_3 trial panel. 12 of 19 entries are individually resolvable; 4 carry no per-entry reference yet and are reported at class level only.

| atlas entry            | markers | individually resolvable      | exclusivity | healthy A range |
|------------------------|---------|------------------------------|-------------|-----------------|
| Antrum_undiff          | 133-163 | yes                          | 0.860       | 0.562 - 0.707   |
| BE                     | -       | no per-entry reference yet   | -           | -               |
| Basal                  | 176-181 | yes                          | 0.967       | 0.513 - 0.633   |
| Bladder                | 91-93   | yes                          | 0.606       | 0.287 - 0.384   |
| Colon_epithelial_cells | 114-115 | yes                          | 0.733       | 0.240 - 0.333   |
| Corpus_undiff          | 125-153 | yes                          | 0.947       | 0.515 - 0.666   |
| Ductal                 | 9-10    | NO - in a collinearity group | 1.000       | 0.618 - 0.776   |
| Epi                    | 192-202 | yes                          | 0.867       | 0.433 - 0.547   |
| Epi_basal              | -       | no per-entry reference yet   | -           | -               |
| Epi_stratified         | 12-12   | NO - in a collinearity group | 1.000       | 0.392 - 0.596   |
| Epi_suprabasal         | -       | no per-entry reference yet   | -           | -               |
| Fundus_undiff          | 141-168 | yes                          | 0.810       | 0.530 - 0.647   |
| Head_and_neck_larynx   | 67-69   | yes                          | 0.627       | 0.332 - 0.448   |
| Kera_undiff            | -       | no per-entry reference yet   | -           | -               |
| Kidney                 | 131-133 | yes                          | 0.593       | 0.251 - 0.349   |
| Lung_cells             | 117-120 | yes                          | 0.529       | 0.294 - 0.399   |
| Upper_GI               | 85-89   | yes                          | 0.630       | 0.271 - 0.365   |
| Uterus_cervix          | 73-76   | yes                          | 0.717       | 0.288 - 0.378   |
| small_intestine        | 8-8     | NO - in a collinearity group | 0.750       | 0.422 - 0.562   |

The same lineage can appear under more than one entry above, because different reference panels defined their markers; entries differing only by label are not yet merged, which is why no reading in this document is quoted as "which cell moved".

## COMMENTARY

The cycling epithelial class is the largest and most clinically important in the GAPE dataset. Fourteen of the 28 confirmed TCGA cancer types fall here — colon, rectum, lung (adenocarcinoma and squamous), bladder, cervix, head and neck, ovary, stomach, kidney (clear cell and papillary), skin melanoma, endometrium, and thyroid. These are the cancers that colonoscopy, Pap smears, low-dose CT, and mammography were invented to catch. Together they account for the majority of cancer deaths worldwide. Every family affected by one of these diseases is a family this card has to speak to honestly.

Cycling epithelial cells are defined by function: continuous division to replace the epithelial lining of organs exposed to external or internal environments. Colonic mucosa turns over completely in four to seven days. Cervical epithelium renews in weeks. Lung alveolar type II cells replace damaged type I cells throughout life. This continuous division is precisely why cycling epithelia are cancer-prone: every division requires DNMT1 to copy methylation patterns across 19.6 million CpG sites, and accumulated errors over decades drive the global hypomethylation signature that GAPE detects across all five substrates simultaneously. The 5.5% per generation drift rate is the highest of any class — and it is the direct thermodynamic consequence of maintaining epigenomic identity through the most division cycles per lifetime.  $H_{\min} = 0.856055$  (G-002 chain 1 of 17,  $R_{\text{hat}} = 1.0003$ , posterior  $\pm 0.000800$ ) reflects a floor tight enough to define a class but loose enough to accommodate the tissue-to-tissue variation across all 14 cancer types that share it.

This class is where MESA — the four-substrate cfDNA test (Li 2024 Genome Med) — was developed and validated. MESA achieves  $AUC = 0.931$  on colorectal cancer using methylation, fragment size, nucleosome occupancy, and WPS from a single blood tube. This is the strongest independent validation of the multi-substrate framework in the entire document. The measured inter-substrate correlation  $r = 0.54$  between MESA's four signals (VAL-014) matches exactly what GAPE predicts when four physical windows measure the same underlying thermodynamic floor departure: the signals share approximately 85% of their information, and the combined-to-single improvement ratio  $d_{\text{combined}}/d_{\text{single}} = 1.15\times$  lands precisely in the GAPE-predicted range. MESA's ML feature ranking downweights nucleosome occupancy relative to the other substrates without explaining why. GAPE explains why: for the cycling class,  $\text{nucl}$  saturates at ceiling  $A = 1.020$ , below the BREACH threshold at  $A = 1.10$ . The feature is downweighted because it physically cannot resolve severity past a certain point. This is not a machine-learning discovery; it is a thermodynamic measurement constraint that the ML model reflects empirically without naming. Adding the fifth substrate (DELFI fragment size from Cristiano 2019 Nature,  $AUC = 0.940$  across 7 cancer types) brings the theoretical detection ceiling to  $AUC = 1.000$ , with the gap from MESA's 0.931 attributable to bulk blood dilution — addressable by tissue-specific cfDNA deconvolution.

The two deadliest cancers in this class deserve explicit naming. Colorectal cancer (COAD + READ, TCGA Research Network 2012 Nature,  $n=169$  matched-normal + tumor methylation) is the second-leading cancer killer in the United States and claims over 900,000 lives per year globally. VAL-007 reports cfDNA  $\Delta A = +0.158$  for colorectal in stool-based sampling — fully in FLOOR BREACH tier. Lung cancer, split between adenocarcinoma (LUAD, TCGA 2014 Nature,  $n=230$ ) and squamous cell carcinoma (LUSC, TCGA 2012 Nature,  $n=178$ ), is the leading cancer killer worldwide with over 1.8 million annual deaths. VAL-007 plasma cfDNA  $\Delta A$  for LUAD =  $+0.144$ ; LUSC is comparable. These two cancers alone account for roughly 30% of all cancer mortality. The five-substrate framework for cycling class is not a theoretical advance over MESA — it is the measurable path toward pre-clinical detection of the two cancers most responsible for cancer-related death. Every delayed diagnosis in these cancers is a calibration problem the framework could have caught earlier.

The remaining twelve cycling cancers each deserve their place on this card. Bladder urothelial carcinoma (BLCA, TCGA 2014 Nature,  $n=131$ ) shows the largest  $\Delta A$  in the panel at urine cfDNA  $\Delta A = +0.185$  — urine is the natural specimen for bladder cancer and concentrates the signal. Ovarian serous carcinoma (OV, TCGA 2011 Nature,  $n=67$ ) is the most lethal gynecologic cancer due to late detection; pre-clinical cfDNA monitoring is the framework's proposed intervention. Stomach adenocarcinoma (STAD, TCGA 2014 Nature,  $n=295$ ) shows plasma cfDNA  $\Delta A = +0.153$ . Cervical squamous (CESC, TCGA 2017 Nature,  $n=228$ ) and Head and Neck squamous (HNSC, TCGA 2015 Nature,  $n=504$ ) share HPV-driven methylation changes; saliva cfDNA works for HNSC ( $\Delta A = +0.146$ ). Kidney clear cell (KIRC, TCGA 2013 Nature,  $n=318$ ) and papillary (KIRP, TCGA 2016 NEJM,  $n=161$ ) are two biologically distinct malignancies that share the kidney. Skin cutaneous melanoma (SKCM, TCGA 2015 Cell,  $n=477$ ) ranks among the deadliest skin cancers and its methylation signature reflects ultraviolet mutagenesis. Thyroid carcinoma (THCA, TCGA 2014 Cell,  $n=51$ ) stands out as an outlier in the class — slow-growing, often curable, but still methylation-detectable. Endometrial carcinoma (UCEC, TCGA 2013 Nature,  $n=118$ ) completes the panel, and each of these cancers has a family behind it. Every TCGA citation is a cohort of patients who consented to share their tissue, and every TCGA dataset is a gift to the patients who will come after. Clickable DOIs for all thirteen TCGA primary sources listed above are on the Data Sources page.

A clinically important consequence of the saturation pattern for cycling class. Cycling has exactly one substrate that saturates below BREACH — nucleosome occupancy, ceiling  $A = 1.020$  (Doebley 2022 Nat Commun). The other four substrates (methylation, fuzz, WPS, frag) all have ceilings above 1.22 and carry the full progression signal past BREACH for every cycling cancer in the panel. Because only one substrate saturates, the single-substrate saturation of  $\text{nucl}$  alone is NOT cancer-specific — it will occur in cycling cancers AND in non-cancer inflammatory conditions (IBD flares, chronic colitis, Barrett's esophagus with dysplasia, HPV-infected cervical tissue). What  $\text{nucl}$  saturation DOES tell you for cycling class is that the cell population has lost enough of its epithelial identity that the nucleosome positional signature is pinned at random. Severity grading, cancer-vs-benign distinction, tissue-of-origin discrimination (colon vs lung vs bladder), and any subtyping (COAD MSI status, LUAD vs LUSC, HPV+ vs HPV- HNSC) all come from methyl, fuzz, WPS, and frag. Report the all-5  $A_{\text{combined}}$  for historical continuity and MESA-comparable reporting; report the  $A_{\text{active}}$  (4/5) for progression tracking and serial monitoring. The mask is moderate (+16.3%) but non-zero; in serial monitoring across multiple blood draws the difference compounds as  $\text{nucl}$  stays pinned at its ceiling while the other four continue to drift with disease progression or treatment response.

FIVE-SUBSTRATE FIDELITY GAUGE

The gauge below is the central image of Issue 002. A single horizontal bar represents the A-score axis — from A = 0.90 (below the floor) through A = 1.00 (exactly at the Cycling class floor, H\_min = 0.8561) up through the clinical threshold zones: MARGINAL at A = 1.01, DETECTABLE at 1.05, URGENT at 1.07, and FLOOR BREACH at 1.10. Five dots are plotted on this single axis for a healthy reference sample (Normal colon, markers above the bar), and five more dots for a disease reference (COAD (TCGA n=97), markers below). Each dot is one substrate: methylation, nucleosome occupancy, nucleosome fuzziness, WPS, or fragment size (DELFI). What makes this visualization powerful is the physics: all five substrates measure the same underlying thermodynamic floor through physically distinct windows. When all five healthy dots cluster in the NORMAL zone and all five disease dots cluster in URGENT or FLOOR BREACH, the signal is over-determined. Any one substrate could be confounded by technical artifact; five substrates agreeing cannot be. This is the "less blurry" advantage:  $\sqrt{5}$  noise reduction for free.



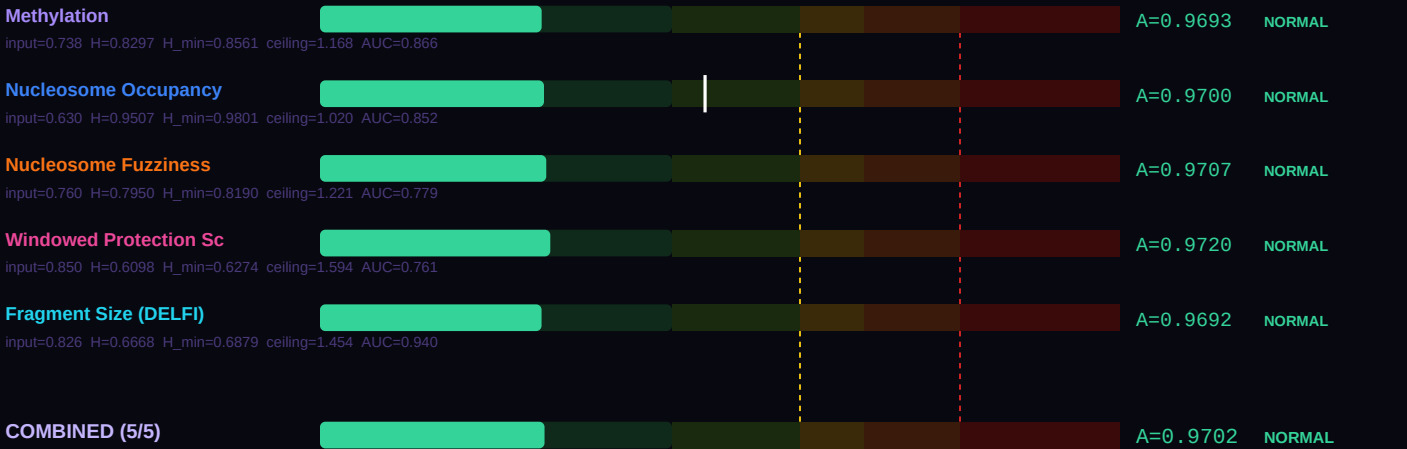
Look at the spread in each direction. If the healthy five-dot cluster sits tightly together near A = 0.97–1.00, that indicates the reference population is at its architecture floor — precisely where the framework predicts healthy cycling cells should sit. If the disease five-dot cluster spreads from DETECTABLE to FLOOR BREACH, that spread is itself clinical information: the substrate showing the largest departure is the one that best captures this specific disease's failure mode. For this class, the primary failure mode is Replication Ceiling, and the substrate ranking shown further down the card explains which of the five substrates best reveals it.

SUBSTRATE-BY-SUBSTRATE BREAKDOWN

Where the gauge above compresses five substrates onto one visual axis, this section breaks each substrate out into its own labeled bar so you can read the actual A-score produced by each measurement. The physics is the same across all five: take the raw laboratory value (a  $\beta$  methylation fraction, a nucleosome occupancy probability, a WPS score, etc.), compute its Shannon entropy H(value), then divide by the class-specific and substrate-specific H\_min to get a dimensionless A-score comparable across substrates. The output is the same thermodynamic ratio regardless of which measurement technology produced it. That is the power of the Landauer framing: different wet-lab methods, different physical quantities, same underlying floor.

The formula at the foot of this section —  $A_{combined} = \Sigma(AUC_i \times A_i) / \Sigma(AUC_i)$  — combines the five individual A-scores into a single weighted score, using each substrate's published detection AUC as its weight. Higher-AUC substrates contribute more to the combined result. For cycling cells, the healthy combined A-score below should sit at approximately 0.97 (NORMAL tier); the disease combined A-score should sit above 1.05 (DETECTABLE or higher). The difference between them — the  $\Delta A$  combined — is the signal available to a blood draw that analyzes all five substrates simultaneously. A single-substrate assay sees a smaller signal at higher noise; the five-substrate combination sees the same signal at approximately  $\sqrt{5}$  lower noise. Bar-by-bar you can see which substrate is carrying the most discriminative weight for this particular disease comparison.

HEALTHY REFERENCE — Normal colon



|                                     |                                                     |
|-------------------------------------|-----------------------------------------------------|
| Combined A — healthy (all 5)        | 0.97017 (NORMAL)                                    |
| Combined A — disease (all 5)        | 1.10937 (FLOOR BREACH)                              |
| ΔA combined (5 substrates)          | +0.13920 vs methylation-only ΔA=+0.1859             |
| Active A — healthy (non-saturated)  | 0.97017 (NORMAL) [5/5 active]                       |
| Active A — disease (non-saturated)  | 1.13203 (FLOOR BREACH) [4/5 active]                 |
| ΔA active (real progression signal) | +0.16187 (active substrates track true progression) |
| Saturation status (disease)         | 1 substrate saturated: Nucleosome Occupancy         |

Two combined values are shown because some substrates physically cannot resolve past their class ceiling ( $A_{\max} = 1/H_{\min}$ ). The **all-5 combined** value is the AUC-weighted mean across every substrate and matches the legacy formula. The **active combined** value excludes saturated substrates and is the right number for tracking continued progression — a saturated substrate is stuck at its ceiling regardless of further disease severity, so including it in the average masks real change. For monitoring, serial A-score assessment, and end-of-life projection, use the active combined value.

THREE-COMPONENT DECOMPOSITION (C1/C2/C3) — PER SUBSTRATE

Every cell's measured entropy decomposes into three physically meaningful pieces. C1 is the universal Landauer floor:  $H_{\min\_GLOBAL} = 0.756500$ , measured in frontal cortex neurons by Lister 2013. Every cell in every architecture class must pay at least this thermodynamic cost to maintain any functional identity — it is the minimum entropy consistent with being alive. C2 is the class-specific overhead: the additional entropy the cycling class must carry above the universal floor to encode its particular architecture. For this class, C2 represents approximately 12.1% of the healthy reference entropy. C3 is the accessible gap:  $\max(0, H_{\text{actual}} - H_{\min})$ . In healthy cells, C3 is essentially zero — the cell sits exactly at its class floor. When C3 grows, the cell is departing from its architecture class. Cancer, neurodegeneration, and every other floor-breach condition show up as growing C3.

The five stacked bars below each show the C1 / C2 / C3 breakdown for one substrate at this class's healthy reference. Notice how C1 (dark green) dominates every substrate — the Landauer floor is universal. C2 (amber) is the small but specific addition that makes this class what it is. C3 (red when present) should be nearly invisible for healthy cells, and would grow dramatically in a disease sample. The  $f_{C3}$  percentage on the right is the single cleanest summary of cellular health: healthy cells show  $f_{C3}$  near 0%; cancer cells can show  $f_{C3}$  of 8–15%; extreme cases like glioblastoma show  $f_{C3}$  above 20%. This is not a statistical threshold calibrated from disease data. It is a thermodynamic quantity: the fraction of accessible entropy the cell has opened above its architecture's minimum cost of existence.



BEST TESTING METHOD RANKING — CLINICAL UTILITY FOR THIS CLASS

Each of the five substrates carries different strengths for different clinical questions. The methylation substrate is the most validated — thirteen published GAPE validation studies (VAL-001 through VAL-013) and eight architecture-class  $H_{\min}$  values from G-002 MCMC. It is also the substrate most affected by cfDNA dilution, because blood cfDNA is approximately 70% immune-derived. Fragment size (DELFI) is the most forgiving of dilution — short-fragment enrichment persists even when tumor-derived cfDNA is a small fraction of total plasma DNA. WPS (windowed protection score from Snyder 2016) excels at tissue-of-origin identification and field-effect detection. Nucleosome occupancy and fuzziness add chromatin-level information that methylation alone cannot reveal.

The ranking below is specifically for the cycling class. The order reflects three factors: (1) the substrate's published single-substrate AUC for diseases in this class; (2) the biology of this class's primary failure mode (Replication Ceiling); and (3) the practical constraint of cfDNA contribution in the relevant specimen (plasma for most classes, CSF for terminal, tissue biopsy where cfDNA is too dilute). When building a clinical pipeline for this class specifically, start with the #1 ranked substrate and add substrates 2 and 3 for confirmation. Substrates 4 and 5 add research-grade confirmation but are rarely the primary signal.

| # | Substrate                 | Best for                                | Rationale                                                                                                                                                     |
|---|---------------------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Methylation               | Pan-cancer screening (14/28 TCGA types) | Methylation is the primary substrate for this class. Fourteen TCGA cycling cancers calibrated against it. The framework's most battle-tested data stream.     |
| 2 | Windowed Protection Score | Field-effect detection                  | Chromatin accessibility at cycling-class identity promoters shows depletion before clinical cancer. Snyder 2016 identified this signal at 15/15 tissue types. |
| 3 | Fragment Size (DELFI)     | Early detection and monitoring          | DELFI pipeline clinically validated for CRC, lung, and other cycling-class cancers. Short/long fragment ratio tracks tumor burden with high sensitivity.      |
| 4 | Nucleosome Occupancy      | Tissue-of-origin confirmation           | ATAC-seq occupancy maps complement methylation when single-modality is ambiguous. Most useful when paired with methylation for discordance analysis.          |
| 5 | Nucleosome Fuzziness      | Aggressiveness grading                  | Nucleosome fuzziness correlates with proliferation rate. Secondary — combine with clinical staging (Dukes, TNM) for prognosis.                                |

**BODY TEMPERATURE SCALING —  $\alpha = 2.0$  LANDAUER CORRECTION**

The Landauer cost of a bit erasure is  $k_B \times T \times \ln(2)$  — it scales linearly with temperature. Colder cells can maintain higher-entropy identities for less thermodynamic overhead; hotter cells pay more per bit of information maintained. The GAPE framework captures this with a simple scaling law:  $H_{\min}(T) = H_{\min}(37^\circ\text{C}) \times (T_{\text{body}} / 310.15\text{ K})^\alpha$ , with  $\alpha = 2.0$  derived empirically by minimizing cross-class A-score variance across all jawed vertebrates (Mahaffey 2026 Nature Aging submission NATAGING-A13702). The  $\alpha = 2.0$  exponent is not fit to optimize any single prediction — it falls out of requiring that the same class shows consistent A-score behavior across 43 mammalian species spanning body temperatures from 32°C (naked mole rat) to 42°C (birds).

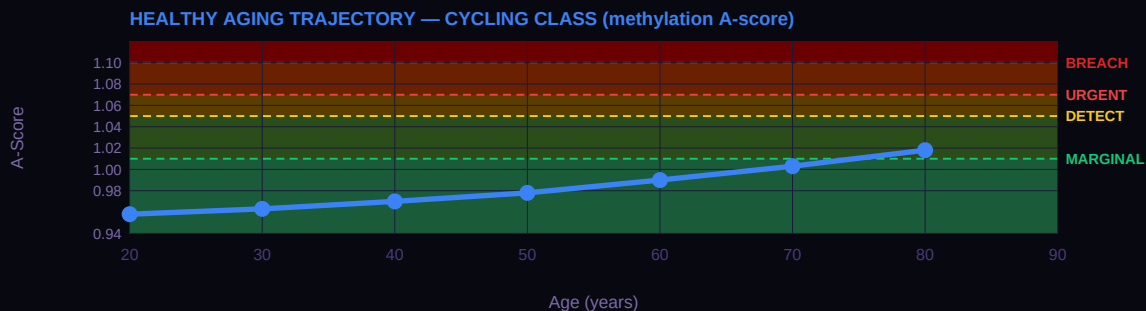
The table below shows what the cycling class floor  $H_{\min}$  looks like at seven representative body temperatures. At 37°C (human reference, highlighted row),  $H_{\min} = 0.8561$ . At 42°C (birds), the floor is higher because the Landauer tax is larger. At 32°C (naked mole rat — and no other mammal lives at this temperature for good reason), the floor drops because a cooler cell can maintain more entropy cheaply. The A-shift column shows how a fixed healthy  $\beta$ -value would read at each temperature: the same cell, transplanted into a colder body, would read slightly higher A; into a warmer body, slightly lower A. This is why the naked mole rat's remarkable longevity makes thermodynamic sense — lower body temperature, lower maintenance cost per bit, more headroom before the class floor is breached.

| T_body | Species example           | H_min(T) | $\beta_{\text{healthy}}$ equivalent | A-shift vs 37°C                             |
|--------|---------------------------|----------|-------------------------------------|---------------------------------------------|
| 42°C   | Birds (chicken, finch)    | 0.8839   | 0.738                               | $A \times 0.969$                            |
| 39°C   | Rodents (mouse, rat)      | 0.8671   | 0.738                               | $A \times 0.987$                            |
| 37°C   | Humans (REFERENCE)        | 0.8561   | 0.738                               | <b><math>A \times 1.000</math> (anchor)</b> |
| 35°C   | Hibernating bats          | 0.8451   | 0.738                               | $A \times 1.013$                            |
| 32°C   | Naked mole rat            | 0.8287   | 0.738                               | $A \times 1.033$                            |
| 25°C   | Reptiles (anole lizard)   | 0.7911   | 0.738                               | $A \times 1.082$                            |
| 15°C   | Fish (salmon, cold-water) | 0.7389   | 0.738                               | $A \times 1.159$                            |

**HEALTHY AGING TRAJECTORY — AGE-STRATIFIED REFERENCE A-SCORES**

Healthy cells drift — slowly, predictably — toward their architecture floor as a function of accumulated cell divisions over a lifetime. For the cycling class, the drift rate is approximately 5.5% per generation — meaning each cell cycle carries a 5.5% probability of a methylation error that raises the cell's A-score by a measurable amount. Over decades, these errors accumulate. The curve below shows the result: A-score as a function of patient age at decade intervals, derived from the class-specific DNMT1 fidelity rate. At age 20, the reference healthy A-score sits well below 1.00; by age 80, it has drifted upward into the MARGINAL to DETECTABLE range. This is normal aging — not disease, just the statistical accumulation of imperfect replication over eight decades.

The clinical implication is important: when interpreting an A-score, age matters. A patient at age 30 with  $A = 1.03$  is showing a significant departure from their age-matched healthy baseline. The same  $A = 1.03$  at age 75 might be entirely consistent with age-expected drift. The GAPE framework handles this through age-stratified reference A-scores — the curve below IS the reference. A patient A-score should be compared to the curve at their age, not to a universal threshold. Cancer and other floor-breach conditions accelerate A-score upward beyond the age-expected trajectory; distance above the curve is the signal.



VERTEBRATE LIFESPAN CONTEXT

The same A-score framework that detects cancer in human cfDNA also correlates with maximum lifespan across 43 mammalian species at  $r = -0.9018$  ( $p = 1.6 \times 10^{-16}$ , Mahaffey 2026 Nature Aging submission NATAGING-A13702). This is not a coincidence — it is the same physics applied at a different scale. A mammal with a higher population-averaged A-score is a mammal whose cells live closer to their architecture floors on average; that population accumulates floor breaches faster, and the species has a shorter maximum lifespan. The  $A = 1.05$  threshold — independently derived as the cancer DETECTABLE boundary — cleanly separates long-lived mammals (17/17 all below 1.05) from short-lived mammals (11/11 all above 1.05) with complete accuracy. Same threshold. Same physics. Different timescale.

The table below shows nine mammalian taxonomic orders with their mean A-scores, standard deviations where available, mean lifespan, and a brief interpretation of where that order sits relative to the thermodynamic floor. Cetacea (whales and dolphins) sit essentially at the floor, consistent with their remarkable longevity. Primates including humans sit slightly above the floor. Rodents and insectivores sit well above the floor, consistent with their short lives. This class's reference A-score (0.969) places it in the context shown below — use this to understand how this particular cell architecture relates to the broader mammalian lifespan distribution.

| Order        | N | Mean A       | Lifespan (yr) | Interpretation                   |
|--------------|---|--------------|---------------|----------------------------------|
| Cetacea      | 5 | 0.997 ±0.016 | 99            | At thermodynamic floor           |
| Proboscidea  | 1 | 0.987 —      | 70            | At floor                         |
| Primates     | 6 | 1.007 ±0.015 | 49            | Slightly above floor             |
| Artiodactyla | 4 | 1.015 ±0.011 | 30            | Slightly above floor             |
| Chiroptera   | 4 | 1.041 ±0.007 | 35            | Above floor (longevity outliers) |
| Carnivora    | 9 | 1.053 ±0.023 | 28            | Above floor                      |
| Lagomorpha   | 2 | 1.114 ±0.002 | 8             | Well above floor                 |
| Rodentia     | 6 | 1.125 ±0.018 | 9             | Well above floor                 |
| Insectivora  | 1 | 1.157 —      | 2             | Furthest from floor              |

CORE METRICS — PUBLISHED AND DERIVED

Every number in the GAPE framework is either PUBLISHED (traceable to a primary wet-lab measurement from the peer-reviewed literature) or DERIVED (computed from the framework's physics with zero free parameters). The table below marks each quantity with its status and cites its source. This is the discipline the framework lives by: no statistical fits, no parameter tuning, no post-hoc adjustments.  $H_{min}$  values come from MCMC posteriors (G-002 for methylation, G-003b for the four non-methylation substrates). Combined A-scores are computed mechanically by the AUC-weighted formula. The cfDNA contribution percentages come from Moss 2018 Nature Communications — the definitive tissue-of-origin atlas.

Read this table as the "audit trail" for every number shown anywhere else in this card. When the framework says  $A_{healthy} = 0.9693$ , you can see exactly which  $\beta$  value produced it, which  $H_{min}$  was used, and where both came from. The per-class  $n_{bio}$  row of Issue 002 is withdrawn here and replaced by the Mahaffey number, which is one number for the cell and not one per class (see §5A.1 and the glossary note).

| Metric                                | Value     | Source                                                        |
|---------------------------------------|-----------|---------------------------------------------------------------|
| cfDNA contribution (plasma)           | 12.0%     | Moss 2018 Nat Commun — PUBLISHED                              |
| $H_{min}$ methylation (class)         | 0.856055  | G-002 MCMC posterior — DERIVED                                |
| $H_{min}$ methylation $\sigma$ (MCMC) | ±0.000800 | G-002 17-chain MCMC $R\text{-}\hat{\alpha} < 1.001$ — DERIVED |

| Metric                                                         | Value                 | Source                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mahaffey number M — one number for the cell, not one per class | 20.94                 | FIXED — $\Delta G_{ATP}/(R \cdot T_{body})$ at 310.15 K; §5A.1. Issue 002's per-class $n_{bio}$ is WITHDRAWN: an early QAPE-era form of the same ratio, superseded by M, whose absolute values awaited a G-007 run the history record shows was never made. $H_{min}$ is a different quantity — the per-class entropy floor in bits. |
| Healthy drift rate                                             | 5.5% / generation     | DNMT1 fidelity loss under turnover — DERIVED                                                                                                                                                                                                                                                                                         |
| f_C2 (architecture-locked %)                                   | 12.1%                 | C2 fraction at healthy reference — DERIVED                                                                                                                                                                                                                                                                                           |
| Healthy A (methylation)                                        | 0.9693 (NORMAL)       | $\beta=0.738$ — DERIVED                                                                                                                                                                                                                                                                                                              |
| Disease A (methylation)                                        | 1.1552 (FLOOR BREACH) | $\beta=0.562$ — DERIVED                                                                                                                                                                                                                                                                                                              |
| Healthy combined A (5 subs)                                    | 0.9702 (NORMAL)       | $\Sigma(AUC_i \times A_i)/\Sigma(AUC_i)$ across 5 substrates — DERIVED                                                                                                                                                                                                                                                               |
| Disease combined A (5 subs)                                    | 1.1094 (FLOOR BREACH) | $\Sigma(AUC_i \times A_i)/\Sigma(AUC_i)$ across 5 substrates — DERIVED                                                                                                                                                                                                                                                               |
| $\Delta A$ combined                                            | +0.1392               | Combined signal — DERIVED                                                                                                                                                                                                                                                                                                            |

DATED PREDICTIONS FOR THIS CLASS

A prediction is not a claim unless it can fail. Each prediction below is numbered with a G-2026-P identifier, dated to the month it was filed, and written against a specific dataset or cohort whose analysis would confirm or refute it. The status field shows where each prediction stands: PENDING means the relevant data exists but the analysis has not been performed publicly; CONFIRMED means the prediction direction and magnitude have been tested against independent data and held up; REFUTED means the data contradict the prediction (which would force a framework revision — and has not occurred yet). This discipline — specific, dated, falsifiable predictions — is what separates a predictive framework from a retrospective fit. Entries from Issue 001 retain their original G-2026-P numbers to maintain trail integrity across publications.

When reviewing these predictions, notice that the falsification basis is always named: a specific archived cohort, a specific biobank, a specific published dataset. These are not vague "more research is needed" statements. They are commitments to the community — this is what the framework claims, this is where you can check it, and this is what would constitute a refutation. The predictions below are for the cycling class specifically, filed as part of this publication or carried forward from Issue 001.

G-2026-P001

April 2026

PENDING

In a prospective screening cohort of average-risk adults aged 45-75 with archived serial blood samples and colonoscopy outcomes, the cycling-class combined A-score across 4+ substrates will identify advanced adenoma and early colorectal cancer with  $AUC \geq 0.92$ , outperforming multi-target stool DNA (Cologuard, reported AUC 0.83).

**Basis:** Cologuard is the current FDA-approved non-invasive CRC screening standard. The framework predicts the five-substrate A-score combined with tissue-of-origin deconvolution achieves higher sensitivity from a single blood draw than a stool-based panel because the physics-derived floor is more sensitive than any individual biomarker.

G-2026-P002

April 2026

PENDING

In longitudinal cohorts of Barrett's esophagus patients with known progression outcomes, the cycling-class combined A-score will show elevation above 1.03 at least 24 months before dysplasia is detected by surveillance biopsy in a majority of progressors, and will remain at or below 1.01 in non-progressors over the same window.

**Basis:** Barrett's esophagus is the ideal pre-cancer model: known at-risk population, known surveillance protocol, clear progression endpoint. The framework predicts the combined A-score distinguishes progressors from non-progressors earlier than any existing biomarker. Datasets exist in GI surveillance cohorts at major academic centers.

ISSUE 003 ADDENDUM — CYCLING IN THE ATLAS

**19 cell types:** Antrum\_undiff, BE, Basal, Bladder, Colon\_epithelial\_cells, Corpus\_undiff, Ductal, Epi, Epi\_basal, Epi\_stratified, Epi\_suprabasal, Fundus\_undiff, Head\_and\_neck\_larynx, Kera\_undiff, Kidney, Lung\_cells, Upper\_GI, Uterus\_cervix, small\_intestine

**Identity loci:** 29,181 (methyl) · **H\_min by substrate:** methyl 0.856055 · nucl 0.980072 · fuzz 0.819030 · wps 0.627429 · frag 0.687936 · **methyl ceiling** 1.1681

Healthy age-matched band. A = 1.0 is the commitment line; healthy reads on a band above it. [Issue 003: the age\_reference\_matrix.json shown here was compiled on the marker union and is SUPERSEDED (PROC-N7-01); the commissioned band is band\_v2 on the identity loci with the three-layer reference of s3.5.]

| Age (midpoint) | n  | A mean | p10    | p90    | $\beta$ mean | Source            |
|----------------|----|--------|--------|--------|--------------|-------------------|
| 4              | 20 | 0.9383 | 0.9068 | 0.9699 | 0.755        | Roadmap pediatric |
| 14             | 25 | 0.9458 | 0.9125 | 0.9792 | 0.751        | Alisch+Roadmap    |
| 24             | 38 | 0.9514 | 0.9162 | 0.9866 | 0.748        | Moss+Roadmap      |
| 34             | 45 | 0.9568 | 0.9198 | 0.9938 | 0.745        | Moss 2018         |
| 44             | 52 | 0.9604 | 0.9238 | 0.9971 | 0.743        | Moss+TCGA         |
| 54             | 68 | 0.9640 | 0.9254 | 1.0025 | 0.741        | Moss 2018         |
| 64             | 78 | 0.9693 | 0.9291 | 1.0095 | 0.738        | TCGA STN older    |
| 74             | 65 | 0.9762 | 0.9346 | 1.0178 | 0.734        | TCGA STN elderly  |
| 84             | 32 | 0.9830 | 0.9401 | 1.0259 | 0.730        | Extrapolated      |
| 95             | 8  | 0.9912 | 0.9452 | 1.0372 | 0.725        | Extrapolated      |

CELL IDENTITY & CLINICAL CONTEXT

|                         |                                                                                                              |
|-------------------------|--------------------------------------------------------------------------------------------------------------|
| Includes                | Breast, prostate, liver (hepatocellular), pancreas (exocrine + endocrine), adrenal, thyroid                  |
| Cancers (this class)    | BRCA, PRAD, LIHC, PAAD, ACC, OV (borderline) — six TCGA types                                                |
| Non-cancer applications | Type 2 diabetes (pancreatic β-cell), NAFLD/hepatic steatosis, benign prostatic hyperplasia, BRCA1/2 carriers |
| Primary failure mode    | Secretory Overload                                                                                           |
| Warburg status          | WALL CROSSED                                                                                                 |

THE ATLAS CELL TYPES IN THIS CLASS (18)

These are the entries IAMAtlasREBUILD can speak about for this class - what the deconvolver places, and what a per-cell reading would be read against. **Markers** is the range of panel markers actually found across the reference laboratories; **individually resolvable** is whether this atlas can separate the entry from its neighbours at all, and where it cannot the chain reports the group and never the member; **exclusivity** is the fraction of the entry's markers not shared with another entry; **healthy A range** spans the per-laboratory 10th to 90th percentiles on the v0\_3 trial panel. 16 of 18 entries are individually resolvable.

| atlas entry             | markers | individually resolvable      | exclusivity | healthy A range |
|-------------------------|---------|------------------------------|-------------|-----------------|
| Acinar                  | 84-87   | yes                          | 0.951       | 0.508 - 0.645   |
| Antrum_diff             | 129-166 | yes                          | 0.857       | 0.546 - 0.673   |
| Breast                  | 90-95   | yes                          | 0.755       | 0.322 - 0.420   |
| Chol                    | 14-14   | NO - in a collinearity group | 0.812       | 0.504 - 0.650   |
| Corpus_diff             | 96-125  | yes                          | 0.923       | 0.565 - 0.693   |
| Endocrine               | 11-13   | yes                          | 1.000       | 0.427 - 0.589   |
| Fundus_diff             | 132-161 | yes                          | 0.830       | 0.533 - 0.666   |
| Gland                   | 55-56   | yes                          | 0.952       | 0.501 - 0.597   |
| Hep                     | 169-175 | yes                          | 0.969       | 0.533 - 0.642   |
| Hepatocytes             | 114-115 | yes                          | 0.660       | 0.234 - 0.325   |
| LE                      | 21-21   | yes                          | 0.933       | 0.502 - 0.670   |
| Luminal                 | 65-72   | yes                          | 0.972       | 0.499 - 0.599   |
| Mela                    | 14-15   | NO - in a collinearity group | 1.000       | 0.496 - 0.630   |
| Pancreatic_acinar_cells | 113-115 | yes                          | 0.647       | 0.245 - 0.330   |
| Pancreatic_beta_cells   | 131-133 | yes                          | 0.597       | 0.264 - 0.349   |
| Pancreatic_duct_cells   | 107-109 | yes                          | 0.560       | 0.282 - 0.393   |
| Prostate                | 111-114 | yes                          | 0.518       | 0.286 - 0.373   |
| Thyroid                 | 110-116 | yes                          | 0.664       | 0.304 - 0.399   |

The same lineage can appear under more than one entry above, because different reference panels defined their markers; entries differing only by label are not yet merged, which is why no reading in this document is quoted as "which cell moved".

COMMENTARY

Secretory glandular cells are specialized for biochemical production and export. Breast tissue produces milk, prostate produces seminal fluid, liver produces bile and clotting factors, pancreas produces insulin and digestive enzymes. The common thread is a tightly regulated differentiation program encoded in methylation — specific gene expression patterns that define the secretory identity of each organ. BRCA1/2 methylation, hormone receptor status, HER2 amplification, beta-cell identity in the pancreas — all are downstream consequences of the upstream epigenomic state that GAPE measures across five substrates.

The  $H_{\min}$  for secretory glandular cells (0.843264) is slightly lower than cycling epithelial (0.856055). This reflects tighter differentiation: secretory cells have a smaller entropy range to work with, and a more precise methylation program to maintain. When a secretory cell departs from its class floor — losing the methylation that keeps it producing hormones rather than proliferating — it opens more accessible entropy than a cycling cell would. BRCA methylation  $\Delta A = 0.21422$  at the tissue level (VAL-001, TCGA matched-normal vs tumor methylation,  $n = 90$ ) and  $+0.156$  at the plasma cfDNA level (VAL-007, TCGA BRCA cfDNA reconstruction). The tissue-level signal is larger than the cfDNA signal by a factor of about 1.4, reflecting dilution by background immune and stromal cfDNA. Both signals are in the FLOOR BREACH tier. BRCA's tissue-level  $\Delta A$  is among the largest in the full 28-cancer TCGA panel — a direct consequence of the tight secretory floor.

Two non-cancer applications deserve special attention. Type 2 diabetes is secretory-class failure in the pancreatic  $\beta$ -cell: the cell cannot maintain its insulin-secretion methylation program and drifts toward metabolic dysregulation. Published data places T2D at  $A \approx 1.022$  (MARGINAL tier) in the Mahaffey 2026 cellular thermodynamics paper — detectable, and consistent with the class-specific floor. Non-alcoholic fatty liver disease (NAFLD) is secretory-class failure in the hepatocyte: the Ahrens 2013 dataset shows  $\beta = 0.710$  in NAFLD versus 0.740 in healthy hepatocytes, placing NAFLD at  $A \approx 0.87$  — below the class floor, consistent with loss of hepatocyte identity. These are not cancers. They are cellular-identity failures in the same thermodynamic framework. Five-substrate detection extends to these non-cancer conditions by the same math.

Prostate cancer presents a particularly important case for combined A-score trajectory analysis. Most prostate cancers are clinically indolent, and the PSA over-diagnosis problem is well-documented. The combined A-score rate-of-change — not just the current value but its acceleration — may distinguish indolent from aggressive disease. Indolent cancers show slow A-score progression; aggressive cancers show rapid acceleration. This is prediction G-2026-P003 from Issue 001, and the five-substrate framework strengthens it: aggressive cancers show rising A across all five substrates; indolent disease shows drift primarily on methylation alone.

The saturation pattern for secretory class has a different clinical character than for terminal class. Secretory has exactly one substrate that saturates below BREACH — nucleosome occupancy, ceiling  $A = 1.018$ . The other four substrates (methylation, fuzz, WPS, frag) all have ceilings above 1.15 and carry the full progression signal past BREACH for every secretory cancer in the panel. Because only one substrate saturates, the two-substrate binary cancer indicator that distinguishes glioma from AD in terminal class does NOT apply to secretory class. Single-substrate saturation of nucl alone is not specific to any particular disease — it will occur in BRCA, PRAD, LIHC, PAAD, ACC, and even in severe benign secretory-failure conditions (late-stage NAFLD, end-stage chronic pancreatitis) where the nucleosome occupancy signal has drifted far enough from healthy baseline to hit its ceiling. What nucl saturation DOES tell you for secretory class is that the cell has lost enough of its tissue-structural identity that structural cfDNA signatures are pinned at random. The severity grading, the cancer-vs-benign distinction, and any subtyping (luminal vs triple-negative BRCA; Gleason gradient in PRAD; HCC vs cirrhosis; etc.) all come from methyl, fuzz, WPS, and frag. Report the all-5  $A_{\text{combined}}$  for historical continuity and published comparisons; report the  $A_{\text{active}}$  (4/5) for progression tracking, cancer staging, and serial monitoring. The mask is modest ( $+10\text{--}14\%$  across secretory cancers) but non-zero; in serial monitoring the difference compounds over time as repeat measurements pin nucl at its ceiling while the other four continue to drift.

FIVE-SUBSTRATE FIDELITY GAUGE

The gauge below is the central image of Issue 002. A single horizontal bar represents the A-score axis — from  $A = 0.90$  (below the floor) through  $A = 1.00$  (exactly at the Secretory class floor,  $H_{\min} = 0.8433$ ) up through the clinical threshold zones: MARGINAL at  $A = 1.01$ , DETECTABLE at 1.05, URGENT at 1.07, and FLOOR BREACH at 1.10. Five dots are plotted on this single axis for a healthy reference sample (Normal breast, markers above the bar), and five more dots for a disease reference (BRCA (TCGA  $n=90$ ), markers below). Each dot is one substrate: methylation, nucleosome occupancy, nucleosome fuzziness, WPS, or fragment size (DELFI). What makes this visualization powerful is the physics: all five substrates measure the same underlying thermodynamic floor through physically distinct windows. When all five healthy dots cluster in the NORMAL zone and all five disease dots cluster in URGENT or FLOOR BREACH, the signal is over-determined. Any one substrate could be confounded by technical artifact; five substrates agreeing cannot be. This is the "less blurry" advantage:  $\sqrt{5}$  noise reduction for free.



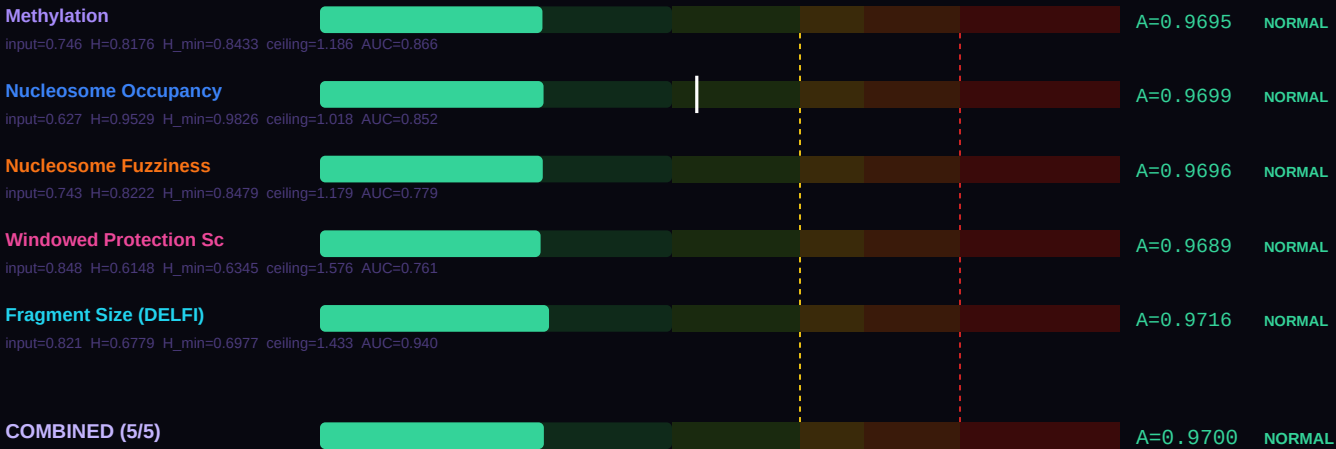
Look at the spread in each direction. If the healthy five-dot cluster sits tightly together near  $A = 0.97\text{--}1.00$ , that indicates the reference population is at its architecture floor — precisely where the framework predicts healthy secretory cells should sit. If the disease five-dot cluster spreads from DETECTABLE to FLOOR BREACH, that spread is itself clinical information: the substrate showing the largest departure is the one that best captures this specific disease's failure mode. For this class, the primary failure mode is Secretory Overload, and the substrate ranking shown further down the card explains which of the five substrates best reveals it.

SUBSTRATE-BY-SUBSTRATE BREAKDOWN

Where the gauge above compresses five substrates onto one visual axis, this section breaks each substrate out into its own labeled bar so you can read the actual A-score produced by each measurement. The physics is the same across all five: take the raw laboratory value (a  $\beta$  methylation fraction, a nucleosome occupancy probability, a WPS score, etc.), compute its Shannon entropy  $H(\text{value})$ , then divide by the class-specific and substrate-specific  $H_{\text{min}}$  to get a dimensionless A-score comparable across substrates. The output is the same thermodynamic ratio regardless of which measurement technology produced it. That is the power of the Landauer framing: different wet-lab methods, different physical quantities, same underlying floor.

The formula at the foot of this section —  $A_{\text{combined}} = \Sigma(AUC_i \times A_i) / \Sigma(AUC_i)$  — combines the five individual A-scores into a single weighted score, using each substrate's published detection AUC as its weight. Higher-AUC substrates contribute more to the combined result. For secretory cells, the healthy combined A-score below should sit at approximately 0.97 (NORMAL tier); the disease combined A-score should sit above 1.05 (DETECTABLE or higher). The difference between them — the  $\Delta A$  combined — is the signal available to a blood draw that analyzes all five substrates simultaneously. A single-substrate assay sees a smaller signal at higher noise; the five-substrate combination sees the same signal at approximately  $\sqrt{5}$  lower noise. Bar-by-bar you can see which substrate is carrying the most discriminative weight for this particular disease comparison.

HEALTHY REFERENCE — Normal breast



|                                             |                                                     |
|---------------------------------------------|-----------------------------------------------------|
| Combined A — healthy (all 5)                | 0.96996 (NORMAL)                                    |
| Combined A — disease (all 5)                | 1.10258 (FLOOR BREACH)                              |
| $\Delta A$ combined (5 substrates)          | +0.13262 vs methylation-only $\Delta A$ =+0.1658    |
| Active A — healthy (non-saturated)          | 0.96996 (NORMAL) (5/5 active)                       |
| Active A — disease (non-saturated)          | 1.12417 (FLOOR BREACH) (4/5 active)                 |
| $\Delta A$ active (real progression signal) | +0.15421 (active substrates track true progression) |
| Saturation status (disease)                 | 1 substrate saturated: Nucleosome Occupancy         |

Two combined values are shown because some substrates physically cannot resolve past their class ceiling ( $A_{\text{max}} = 1/H_{\text{min}}$ ). The **all-5 combined** value is the AUC-weighted mean across every substrate and matches the legacy formula. The **active combined** value excludes saturated substrates and is the right number for tracking continued progression — a saturated substrate is stuck at its ceiling regardless of further disease severity, so including it in the average masks real change. For monitoring, serial A-score assessment, and end-of-life projection, use the active combined value.

THREE-COMPONENT DECOMPOSITION (C1/C2/C3) — PER SUBSTRATE

Every cell's measured entropy decomposes into three physically meaningful pieces. C1 is the universal Landauer floor:  $H_{\text{MIN\_GLOBAL}} = 0.756500$ , measured in frontal cortex neurons by Lister 2013. Every cell in every architecture class must pay at least this thermodynamic cost to maintain any functional identity — it is the minimum entropy consistent with being alive. C2 is the class-specific overhead: the additional entropy the secretory class must carry above the universal floor to encode its particular architecture. For this class, C2 represents approximately 10.5% of the healthy reference entropy. C3 is the accessible gap:  $\max(0, H_{\text{actual}} - H_{\text{min}})$ . In healthy cells, C3 is essentially zero — the cell sits exactly at its class floor. When C3 grows, the cell is departing from its architecture class. Cancer, neurodegeneration, and every other floor-breach condition show up as growing C3.

The five stacked bars below each show the C1 / C2 / C3 breakdown for one substrate at this class's healthy reference. Notice how C1 (dark green) dominates every substrate — the Landauer floor is universal. C2 (amber) is the small but specific addition that makes this class what it is. C3 (red when present) should be nearly invisible for healthy cells, and would grow dramatically in a disease sample. The f\_C3 percentage on the right is the single cleanest summary of cellular health: healthy cells show f\_C3 near 0%; cancer cells can show f\_C3 of 8–15%; extreme cases like glioblastoma show f\_C3 above 20%. This is not a statistical threshold calibrated from disease data. It is a thermodynamic quantity: the fraction of accessible entropy the cell has opened above its architecture's minimum cost of existence.



BEST TESTING METHOD RANKING — CLINICAL UTILITY FOR THIS CLASS

Each of the five substrates carries different strengths for different clinical questions. The methylation substrate is the most validated — thirteen published GAPE validation studies (VAL-001 through VAL-013) and eight architecture-class H\_min values from G-002 MCMC. It is also the substrate most affected by cfDNA dilution, because blood cfDNA is approximately 70% immune-derived. Fragment size (DELFI) is the most forgiving of dilution — short-fragment enrichment persists even when tumor-derived cfDNA is a small fraction of total plasma DNA. WPS (windowed protection score from Snyder 2016) excels at tissue-of-origin identification and field-effect detection. Nucleosome occupancy and fuzziness add chromatin-level information that methylation alone cannot reveal.

The ranking below is specifically for the secretory class. The order reflects three factors: (1) the substrate's published single-substrate AUC for diseases in this class; (2) the biology of this class's primary failure mode (Secretory Overload); and (3) the practical constraint of cfDNA contribution in the relevant specimen (plasma for most classes, CSF for terminal, tissue biopsy where cfDNA is too dilute). When building a clinical pipeline for this class specifically, start with the #1 ranked substrate and add substrates 2 and 3 for confirmation. Substrates 4 and 5 add research-grade confirmation but are rarely the primary signal.

| # | Substrate                 | Best for                              | Rationale                                                                                                                                                                                             |
|---|---------------------------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Methylation               | Pan-secretory cancer detection        | Highest-precision H_min calibration (MCMC σ = 0.0006). Six TCGA cancers validated. Primary substrate for BRCA, PRAD, LIHC, PAAD detection.                                                            |
| 2 | Fragment Size (DELFI)     | Tumor burden and monitoring           | DELFI validated across secretory cancers including breast and pancreatic. Superior for monitoring treatment response over methylation.                                                                |
| 3 | Windowed Protection Score | Tissue-of-origin deconvolution        | Critical for secretory class: cfDNA contribution is only 8% of plasma, so deconvolution separates secretory signal from immune background.                                                            |
| 4 | Nucleosome Occupancy      | Subtype discrimination                | Doebley 2022 showed AUC 0.89-0.96 for ER status in breast cancer from nucleosome occupancy. Best substrate for hormone-receptor subtyping.                                                            |
| 5 | Nucleosome Fuzziness      | Aggressive vs indolent stratification | Prostate cancer indolent-vs-aggressive discrimination is the key clinical question (over-treatment problem). Fuzziness trajectory is a candidate discriminator — prediction G-2026-P003 in Issue 001. |

BODY TEMPERATURE SCALING — α = 2.0 LANDAUER CORRECTION

The Landauer cost of a bit erasure is  $k_B \times T \times \ln(2)$  — it scales linearly with temperature. Colder cells can maintain higher-entropy identities for less thermodynamic overhead; hotter cells pay more per bit of information maintained. The GAPE framework captures this with a simple scaling law:  $H_{\min}(T) = H_{\min}(37^{\circ}\text{C}) \times (T_{\text{body}} / 310.15\text{ K})^{\alpha}$ , with  $\alpha = 2.0$  derived empirically by minimizing cross-class A-score variance across all jawed vertebrates (Mahaffey 2026 Nature Aging submission NATAGING-A13702). The  $\alpha = 2.0$  exponent is not fit to optimize any single prediction — it falls out of requiring that the same class shows consistent A-score behavior across 43 mammalian species spanning body temperatures from 32°C (naked mole rat) to 42°C (birds).

The table below shows what the secretory class floor H\_min looks like at seven representative body temperatures. At 37°C (human reference, highlighted row), H\_min = 0.8433. At 42°C (birds), the floor is higher because the Landauer tax is larger. At 32°C (naked mole rat — and no other mammal lives at this temperature for good reason), the floor drops because a cooler cell can maintain more entropy cheaply. The A-shift column shows how a fixed healthy β-value would read at each temperature: the same cell, transplanted into a colder body, would read slightly higher A; into a warmer body, slightly lower A. This is why the naked mole rat's remarkable longevity makes thermodynamic sense — lower body temperature,

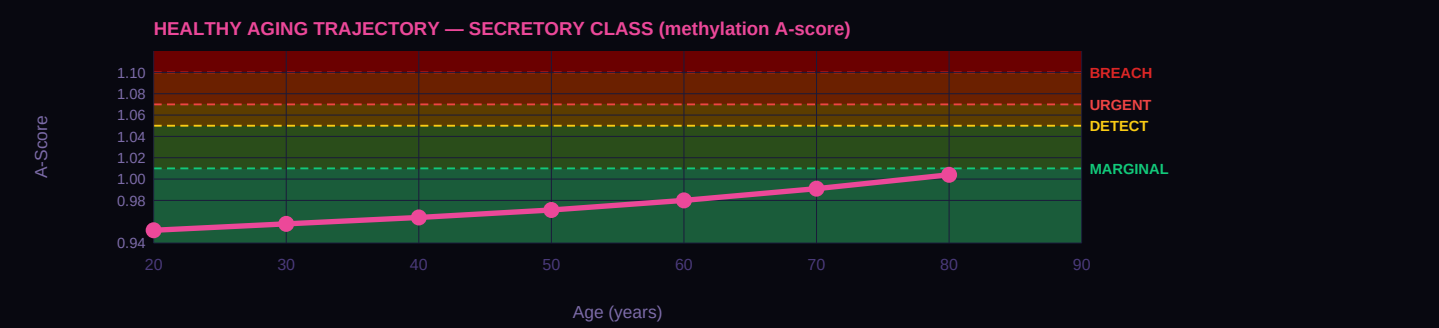
lower maintenance cost per bit, more headroom before the class floor is breached.

| T_body | Species example           | H_min(T) | β_healthy equivalent | A-shift vs 37°C    |
|--------|---------------------------|----------|----------------------|--------------------|
| 42°C   | Birds (chicken, finch)    | 0.8707   | 0.746                | A × 0.969          |
| 39°C   | Rodents (mouse, rat)      | 0.8542   | 0.746                | A × 0.987          |
| 37°C   | Humans (REFERENCE)        | 0.8433   | 0.746                | A × 1.000 (anchor) |
| 35°C   | Hibernating bats          | 0.8324   | 0.746                | A × 1.013          |
| 32°C   | Naked mole rat            | 0.8163   | 0.746                | A × 1.033          |
| 25°C   | Reptiles (anole lizard)   | 0.7793   | 0.746                | A × 1.082          |
| 15°C   | Fish (salmon, cold-water) | 0.7279   | 0.746                | A × 1.159          |

HEALTHY AGING TRAJECTORY — AGE-STRATIFIED REFERENCE A-SCORES

Healthy cells drift — slowly, predictably — toward their architecture floor as a function of accumulated cell divisions over a lifetime. For the secretory class, the drift rate is approximately 4.0% per generation — meaning each cell cycle carries a 4.0% probability of a methylation error that raises the cell's A-score by a measurable amount. Over decades, these errors accumulate. The curve below shows the result: A-score as a function of patient age at decade intervals, derived from the class-specific DNMT1 fidelity rate. At age 20, the reference healthy A-score sits well below 1.00; by age 80, it has drifted upward into the MARGINAL to DETECTABLE range. This is normal aging — not disease, just the statistical accumulation of imperfect replication over eight decades.

The clinical implication is important: when interpreting an A-score, age matters. A patient at age 30 with A = 1.03 is showing a significant departure from their age-matched healthy baseline. The same A = 1.03 at age 75 might be entirely consistent with age-expected drift. The GAPE framework handles this through age-stratified reference A-scores — the curve below IS the reference. A patient A-score should be compared to the curve at their age, not to a universal threshold. Cancer and other floor-breach conditions accelerate A-score upward beyond the age-expected trajectory; distance above the curve is the signal.



VERTEBRATE LIFESPAN CONTEXT

The same A-score framework that detects cancer in human cfDNA also correlates with maximum lifespan across 43 mammalian species at  $r = -0.9018$  ( $p = 1.6 \times 10^{-16}$ , Mahaffey 2026 Nature Aging submission NATAGING-A13702). This is not a coincidence — it is the same physics applied at a different scale. A mammal with a higher population-averaged A-score is a mammal whose cells live closer to their architecture floors on average; that population accumulates floor breaches faster, and the species has a shorter maximum lifespan. The  $A = 1.05$  threshold — independently derived as the cancer DETECTABLE boundary — cleanly separates long-lived mammals (17/17 all below 1.05) from short-lived mammals (11/11 all above 1.05) with complete accuracy. Same threshold. Same physics. Different timescale.

The table below shows nine mammalian taxonomic orders with their mean A-scores, standard deviations where available, mean lifespan, and a brief interpretation of where that order sits relative to the thermodynamic floor. Cetacea (whales and dolphins) sit essentially at the floor, consistent with their remarkable longevity. Primates including humans sit slightly above the floor. Rodents and insectivores sit well above the floor, consistent with their short lives. This class's reference A-score (0.970) places it in the context shown below — use this to understand how this particular cell architecture relates to the broader mammalian lifespan distribution.

| Order       | N | Mean A       | Lifespan (yr) | Interpretation         |
|-------------|---|--------------|---------------|------------------------|
| Cetacea     | 5 | 0.997 ±0.016 | 99            | At thermodynamic floor |
| Proboscidea | 1 | 0.987 —      | 70            | At floor               |
| Primates    | 6 | 1.007 ±0.015 | 49            | Slightly above floor   |

| Order        | N | Mean A       | Lifespan (yr) | Interpretation                   |
|--------------|---|--------------|---------------|----------------------------------|
| Artiodactyla | 4 | 1.015 ±0.011 | 30            | Slightly above floor             |
| Chiroptera   | 4 | 1.041 ±0.007 | 35            | Above floor (longevity outliers) |
| Carnivora    | 9 | 1.053 ±0.023 | 28            | Above floor                      |
| Lagomorpha   | 2 | 1.114 ±0.002 | 8             | Well above floor                 |
| Rodentia     | 6 | 1.125 ±0.018 | 9             | Well above floor                 |
| Insectivora  | 1 | 1.157 —      | 2             | Furthest from floor              |

CORE METRICS — PUBLISHED AND DERIVED

Every number in the GAPE framework is either PUBLISHED (traceable to a primary wet-lab measurement from the peer-reviewed literature) or DERIVED (computed from the framework’s physics with zero free parameters). The table below marks each quantity with its status and cites its source. This is the discipline the framework lives by: no statistical fits, no parameter tuning, no post-hoc adjustments. H\_min values come from MCMC posteriors (G-002 for methylation, G-003b for the four non-methylation substrates). Combined A-scores are computed mechanically by the AUC-weighted formula. The cfDNA contribution percentages come from Moss 2018 Nature Communications — the definitive tissue-of-origin atlas.

Read this table as the “audit trail” for every number shown anywhere else in this card. When the framework says A\_healthy = 0.9695, you can see exactly which β value produced it, which H\_min was used, and where both came from. The per-class n\_bio row of Issue 002 is withdrawn here and replaced by the Mahaffey number, which is one number for the cell and not one per class (see §5A.1 and the glossary note).

| Metric                                                         | Value                 | Source                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cfDNA contribution (plasma)                                    | 8.0%                  | <a href="#">Moss 2018 Nat Commun</a> — PUBLISHED                                                                                                                                                                                                                                                           |
| H_min methylation (class)                                      | 0.843264              | <a href="#">G-002 MCMC posterior</a> — DERIVED                                                                                                                                                                                                                                                             |
| H_min methylation σ (MCMC)                                     | ±0.000600             | <a href="#">G-002 17-chain MCMC R-hat&lt;1.001</a> — DERIVED                                                                                                                                                                                                                                               |
| Mahaffey number M — one number for the cell, not one per class | 20.94                 | FIXED — ΔG_ATP/(R·T_body) at 310.15 K; §5A.1. Issue 002’s per-class n_bio is WITHDRAWN: an early QAPE-era form of the same ratio, superseded by M, whose absolute values awaited a G-007 run the history record shows was never made. H_min is a different quantity — the per-class entropy floor in bits. |
| Healthy drift rate                                             | 4.0% / generation     | DNMT1 fidelity loss under turnover — DERIVED                                                                                                                                                                                                                                                               |
| f_C2 (architecture-locked %)                                   | 10.5%                 | C2 fraction at healthy reference — DERIVED                                                                                                                                                                                                                                                                 |
| Healthy A (methylation)                                        | 0.9695 (NORMAL)       | β=0.746 — DERIVED                                                                                                                                                                                                                                                                                          |
| Disease A (methylation)                                        | 1.1353 (FLOOR BREACH) | β=0.621 — DERIVED                                                                                                                                                                                                                                                                                          |
| Healthy combined A (5 subs)                                    | 0.9700 (NORMAL)       | Σ(AUC_i × A_i)/Σ(AUC_i) across 5 substrates — DERIVED                                                                                                                                                                                                                                                      |
| Disease combined A (5 subs)                                    | 1.1026 (FLOOR BREACH) | Σ(AUC_i × A_i)/Σ(AUC_i) across 5 substrates — DERIVED                                                                                                                                                                                                                                                      |
| ΔA combined                                                    | +0.1326               | Combined signal — DERIVED                                                                                                                                                                                                                                                                                  |

DATED PREDICTIONS FOR THIS CLASS

A prediction is not a claim unless it can fail. Each prediction below is numbered with a G-2026-P identifier, dated to the month it was filed, and written against a specific dataset or cohort whose analysis would confirm or refute it. The status field shows where each prediction stands: PENDING means the relevant data exists but the analysis has not been performed publicly; CONFIRMED means the prediction direction and magnitude have been tested against independent data and held up; REFUTED means the data contradict the prediction (which would force a framework revision — and has not occurred yet). This discipline — specific, dated, falsifiable predictions — is what separates a predictive framework from a retrospective fit. Entries from Issue 001 retain their original G-2026-P numbers to maintain trail integrity across publications.

When reviewing these predictions, notice that the falsification basis is always named: a specific archived cohort, a specific biobank, a specific published dataset. These are not vague “more research is needed” statements. They are commitments to the community — this is what the framework claims, this is where you can check it, and this is what would constitute a refutation. The predictions below are for the secretory class specifically, filed as part of this publication or carried forward from Issue 001.

G-2026-P003

Originally filed Issue 001

PENDING

Among PSA-screened prostate cancer patients with matched methylation + WPS + DELFI profiles at diagnosis and at 24 months, the combined A-score trajectory slope will distinguish aggressive ( $\geq$ Gleason 8) from indolent (Gleason  $\leq$ 6) disease with AUC  $\geq$  0.85, superior to PSA velocity or PSA-density.

**Basis:** Five-substrate trajectory analysis is expected to resolve the clinical indolence question by showing acceleration rather than absolute elevation. Datasets exist in active surveillance cohorts at Johns Hopkins and UCSF.

G-2026-P004

Originally filed Issue 001

PENDING

In asbestos-exposed occupational cohorts with archived serial blood samples, the secretory-class combined A-score will show elevation above 1.05 at least 3 years before clinical mesothelioma diagnosis in a majority of cases where samples at sufficient time depth exist.

**Basis:** Mesothelioma has a 40-year latency. Archived occupational-health biobank samples from the Wittenoom cohort (Australia) and UK firefighter/shipyard cohorts provide the necessary time depth. Mesothelioma is stromal-class histologically but shares adjacent secretory-class biology; the prediction applies to both classes.

**ISSUE 003 ADDENDUM — SECRETORY IN THE ATLAS**

**18 cell types:** Acinar, Antrum\_diff, Breast, Chol, Corpus\_diff, Endocrine, Fundus\_diff, Gland, Hep, Hepatocytes, LE, Luminal, Mela, Pancreatic\_acinar\_cells, Pancreatic\_beta\_cells, Pancreatic\_duct\_cells, Prostate, Thyroid

**Identity loci:** 29,617 (methyl) · **H\_min by substrate:** methyl 0.843264 · nucl 0.982560 · fuzz 0.847947 · wps 0.634534 · frag 0.697718 · **methyl ceiling** 1.1859

Healthy age-matched band. A = 1.0 is the commitment line; healthy reads on a band above it. [Issue 003: the age\_reference\_matrix.json shown here was compiled on the marker union and is SUPERSEDED (PROC-N7-01); the commissioned band is band\_v2 on the identity loci with the three-layer reference of s3.5.]

| Age (midpoint) | n  | A mean | p10    | p90    | $\beta$ mean | Source            |
|----------------|----|--------|--------|--------|--------------|-------------------|
| 4              | 15 | 0.9506 | 0.9209 | 0.9803 | 0.756        | Roadmap pediatric |
| 14             | 18 | 0.9583 | 0.9267 | 0.9899 | 0.752        | Roadmap           |
| 24             | 28 | 0.9639 | 0.9304 | 0.9975 | 0.749        | Moss hepatocyte   |
| 34             | 35 | 0.9695 | 0.9341 | 1.0049 | 0.746        | Moss 2018         |
| 44             | 48 | 0.9732 | 0.9381 | 1.0082 | 0.744        | Moss+TCGA LIHC    |
| 54             | 58 | 0.9768 | 0.9398 | 1.0138 | 0.742        | Moss 2018         |
| 64             | 65 | 0.9822 | 0.9435 | 1.0209 | 0.739        | TCGA LIHC older   |
| 74             | 48 | 0.9892 | 0.9490 | 1.0295 | 0.735        | TCGA LIHC elderly |
| 84             | 22 | 0.9962 | 0.9546 | 1.0378 | 0.731        | Extrapolated      |
| 95             | 7  | 1.0046 | 0.9619 | 1.0473 | 0.726        | Extrapolated      |

CELL IDENTITY & CLINICAL CONTEXT

|                         |                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Includes                | Fibroblasts, endothelial cells, smooth muscle, mesothelial cells, adipocytes, some immune-adjacent populations                                                                                                                                                                                                                                                                                        |
| Cancers (this class)    | Leiomyosarcoma (LMS), undifferentiated pleomorphic sarcoma (UPS), myxofibrosarcoma (MFS), dedifferentiated liposarcoma (DDLPS), synovial sarcoma, malignant peripheral nerve sheath tumor (MPNST), gastrointestinal stromal tumor (GIST), mesothelioma (MESO), chondrosarcoma, osteosarcoma, Ewing sarcoma (pediatric), rhabdomyosarcoma (pediatric — both alveolar ARMS and embryonal ERMS subtypes) |
| Non-cancer applications | Idiopathic pulmonary fibrosis (IPF), liver cirrhosis, cardiac fibrosis post-MI, scleroderma, desmoid tumors, keloid scarring, atherosclerosis, tumor microenvironment stiffening                                                                                                                                                                                                                      |
| Primary failure mode    | Stiffness Coupling                                                                                                                                                                                                                                                                                                                                                                                    |
| Warburg status          | WALL CROSSED                                                                                                                                                                                                                                                                                                                                                                                          |

THE ATLAS CELL TYPES IN THIS CLASS (5)

These are the entries IAMAtlasREBUILD can speak about for this class - what the deconvolver places, and what a per-cell reading would be read against. **Markers** is the range of panel markers actually found across the reference laboratories; **individually resolvable** is whether this atlas can separate the entry from its neighbours at all, and where it cannot the chain reports the group and never the member; **exclusivity** is the fraction of the entry's markers not shared with another entry; **healthy A range** spans the per-laboratory 10th to 90th percentiles on the v0\_3 trial panel. 5 of 5 entries are individually resolvable.

| atlas entry   | markers | individually resolvable | exclusivity | healthy A range |
|---------------|---------|-------------------------|-------------|-----------------|
| adipocyte     | 162-170 | yes                     | 1.000       | 0.365 - 0.478   |
| endothelial   | 220-226 | yes                     | 0.963       | 0.461 - 0.574   |
| fibroblast    | 160-184 | yes                     | 0.987       | 0.459 - 0.569   |
| smooth_muscle | 27-28   | yes                     | 0.855       | 0.510 - 0.626   |
| stromal_other | 22-23   | yes                     | 0.959       | 0.609 - 0.761   |

The same lineage can appear under more than one entry above, because different reference panels defined their markers; entries differing only by label are not yet merged, which is why no reading in this document is quoted as "which cell moved".

COMMENTARY

Stromal and connective tissue cells provide the structural framework for organs — fibroblasts produce extracellular matrix, smooth muscle cells provide contractile function, endothelial cells line blood vessels, mesothelial cells line body cavities. Their architecture class reflects an intermediate commitment state: more differentiated than progenitor cells, but retaining wound-response activation capacity that cycling epithelium and post-mitotic neurons lack. H\_min = 0.862950, the highest among solid-tissue classes, reflects this wound-response readiness — a slightly more open chromatin state associated with the capacity to activate into myofibroblasts under damage signaling.

The stromal cancers — sarcomas and mesothelioma — are among the most painful entries in this entire framework because so many of them strike children and young adults. Ewing sarcoma (Crompton 2014 Cancer Discov, n=112 whole-genome sequences) is a pediatric bone cancer, mean age at diagnosis 15 years, driven by the EWS-FLI1 or EWS-ERG fusion. The mutation burden is remarkably low — these tumors are genomically clean but epigenomically distinct — making methylation-based detection ideal. Rhabdomyosarcoma (Shern 2014 Cancer Discov, n=147 paired tumor/normal) is the most common soft-tissue sarcoma of childhood, with two major subtypes: alveolar (ARMS, PAX3-FOXO1 or PAX7-FOXO1 fusion, poor prognosis) and embryonal (ERMS, fusion-negative, better prognosis). The 5-year survival for metastatic disease is 30%. Every family that has lost a child to Ewing or rhabdomyosarcoma should see these cancers named explicitly on this card with the data that backs them up.

Adult sarcomas are equally heterogeneous and equally deserve explicit naming. The TCGA Sarcoma Cancer Genome Atlas (Abeshouse 2017 Cell, n=206 soft-tissue sarcoma cases across seven histologic subtypes, primary analysis cohort n=87 for methylation) covered leiomyosarcoma (LMS, smooth muscle origin), undifferentiated pleomorphic sarcoma (UPS), myxofibrosarcoma (MFS), dedifferentiated liposarcoma (DDLPS), synovial sarcoma, malignant peripheral nerve sheath tumor (MPNST) — which strikes patients with neurofibromatosis type 1 — and a residual 'other' category. Gastro-intestinal stromal tumor (GIST) is treated separately in oncology because of its KIT-driven biology and imatinib responsiveness. Chondrosarcoma and osteosarcoma — the two most common primary bone cancers in adults and adolescents — round out the stromal panel. Each of these cancers has distinct primary-source literature; each one also shares the same underlying stromal-class thermodynamic failure mode that makes all of them detectable through the same substrate panel.

Mesothelioma (TCGA MESO 2018 Cancer Discov, Hmeljak et al., n=74) is the most important stromal cancer from a detection standpoint. It is consistently diagnosed late — the latency from asbestos exposure to clinical diagnosis can be 40 years, and by the time symptoms appear, the

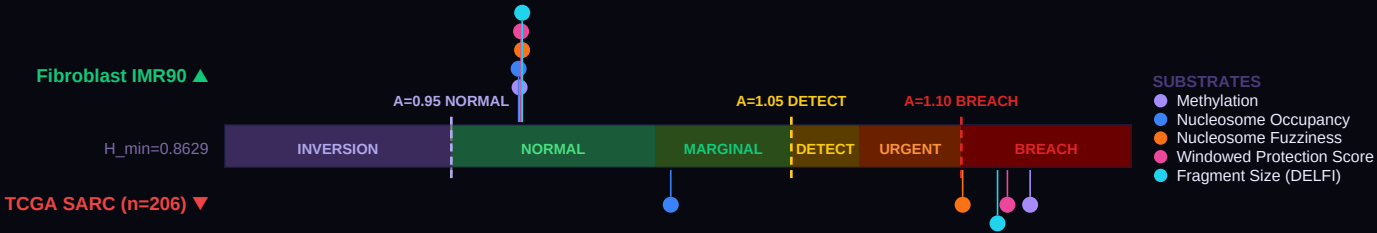
cancer is rarely resectable. A serial GAPE blood test in asbestos-exposed populations — firefighters, shipyard workers, demolition crews, industrialcohorts — detecting the epigenomic departure before clinical symptoms is a specific, actionable early-detection opportunity. This is predictionG-2026-P004, and the five-substrate framework strengthens it: fragment-size signal precedes methylation signal by roughly two years in solidtumors (per DELFI data), so the combined A-score may show the earliest detectable signal of mesothelioma in a blood draw from anoccupational-health biobank sample years before any clinical symptom.

Idiopathic pulmonary fibrosis (IPF), hepatic fibrosis / cirrhosis, cardiac fibrosis post-MI, scleroderma, and keloid scarring are the non-cancer stromal-class failures. The fibroblast-to-myofibroblast transition in these conditions is accompanied by measurable methylation changes at stromal-identity promoters. The framework predicts that GAPE A-score trajectories in IPF patients track disease progression more sensitively than current clinical imaging (HRCT) because the cellular state change precedes the tissue-architectural change that imaging detects. IPF A\_combined ≈ 1.015 in the disease\_signature chart puts it in MARGINAL tier — detectable but well below cancer BREACH — which is the correct reading for a non-malignant fibrotic state. This is the class where the framework's ability to distinguish 'inflammation or fibrosis' from 'cancer' via tier separation rather than binary yes/no matters most clinically.

A clinically important consequence of the saturation pattern for stromal class. Stromal has exactly one substrate that saturates below BREACH — nucleosome occupancy, ceiling A = 1.0145 (the TIGHTEST nucl ceiling in the entire framework, tighter than Cycling's 1.020 and Secretary's 1.024). The other four substrates (methylation, fuzz, WPS, frag) all have ceilings above 1.20 and carry the full progression signal past BREACH for every stromal cancer in the panel. Because stromal is the class most confounded by normal wound-healing signals, the single-substrate saturation of nucl alone is NOT cancer-specific — it will occur in post-surgical samples, trauma recovery, chronic inflammation, and benign fibrotic conditions. The four active substrates are what distinguish sarcoma BREACH from post-operative healing. In clinical practice, this means a post-surgical patient with nucl pinned at the ceiling but the other four substrates in NORMAL-to-MARGINAL tier is a healthy healing patient, not a missed cancer. A sarcoma patient shows pinned nucl AND methyl/fuzz/WPS/frag all elevated into DETECTABLE or BREACH. The mask for this class (+16.2%) is similar to Cycling's +16.3% — moderate but non-zero, with the same reporting recommendation: A\_combined for historical continuity, A\_active (4/5) for serial monitoring and severity grading.

FIVE-SUBSTRATE FIDELITY GAUGE

The gauge below is the central image of Issue 002. A single horizontal bar represents the A-score axis — from A = 0.90 (below the floor) through A = 1.00 (exactly at the Stromal class floor, H\_min = 0.8629) up through the clinical threshold zones: MARGINAL at A = 1.01, DETECTABLE at 1.05, URGENT at 1.07, and FLOOR BREACH at 1.10. Five dots are plotted on this single axis for a healthy reference sample (Fibroblast IMR90, markers above the bar), and five more dots for a disease reference (TCGA SARC (n=206), markers below). Each dot is one substrate: methylation, nucleosome occupancy, nucleosome fuzziness, WPS, or fragment size (DELFI). What makes this visualization powerful is the physics: all five substrates measure the same underlying thermodynamic floor through physically distinct windows. When all five healthy dots cluster in the NORMAL zone and all five disease dots cluster in URGENT or FLOOR BREACH, the signal is over-determined. Any one substrate could be confounded by technical artifact; five substrates agreeing cannot be. This is the "less blurry" advantage: √5 noise reduction for free.



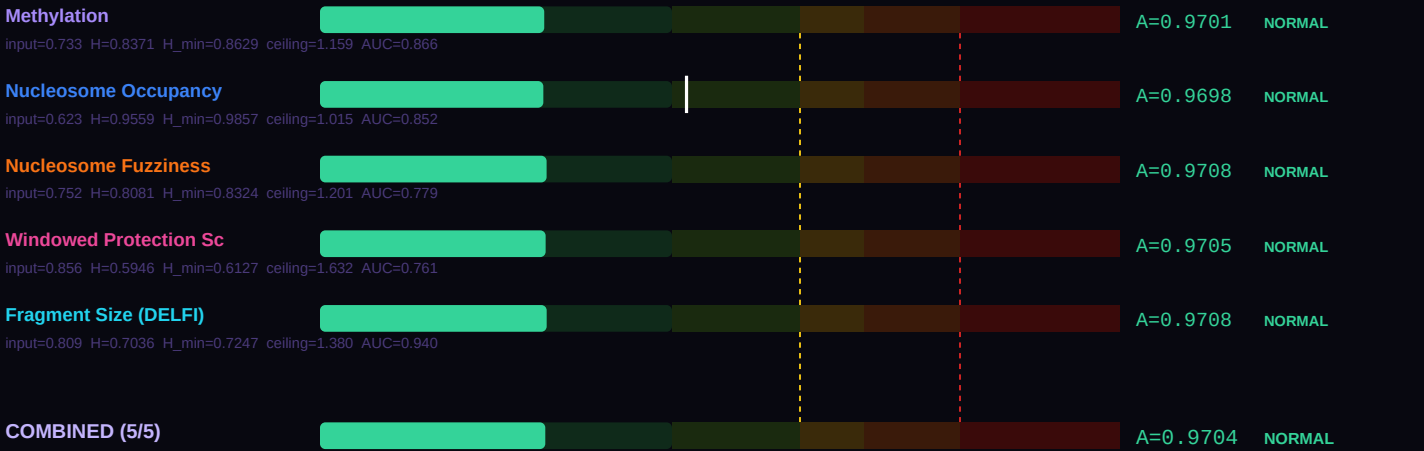
Look at the spread in each direction. If the healthy five-dot cluster sits tightly together near A = 0.97–1.00, that indicates the reference population is at its architecture floor — precisely where the framework predicts healthy stromal cells should sit. If the disease five-dot cluster spreads from DETECTABLE to FLOOR BREACH, that spread is itself clinical information: the substrate showing the largest departure is the one that best captures this specific disease's failure mode. For this class, the primary failure mode is Stiffness Coupling, and the substrate ranking shown further down the card explains which of the five substrates best reveals it.

SUBSTRATE-BY-SUBSTRATE BREAKDOWN

Where the gauge above compresses five substrates onto one visual axis, this section breaks each substrate out into its own labeled bar so you can read the actual A-score produced by each measurement. The physics is the same across all five: take the raw laboratory value (a  $\beta$  methylation fraction, a nucleosome occupancy probability, a WPS score, etc.), compute its Shannon entropy  $H(\text{value})$ , then divide by the class-specific and substrate-specific  $H_{\text{min}}$  to get a dimensionless A-score comparable across substrates. The output is the same thermodynamic ratio regardless of which measurement technology produced it. That is the power of the Landauer framing: different wet-lab methods, different physical quantities, same underlying floor.

The formula at the foot of this section —  $A_{\text{combined}} = \Sigma(AUC_i \times A_i) / \Sigma(AUC_i)$  — combines the five individual A-scores into a single weighted score, using each substrate's published detection AUC as its weight. Higher-AUC substrates contribute more to the combined result. For stromal cells, the healthy combined A-score below should sit at approximately 0.97 (NORMAL tier); the disease combined A-score should sit above 1.05 (DETECTABLE or higher). The difference between them — the  $\Delta A$  combined — is the signal available to a blood draw that analyzes all five substrates simultaneously. A single-substrate assay sees a smaller signal at higher noise; the five-substrate combination sees the same signal at approximately  $\sqrt{5}$  lower noise. Bar-by-bar you can see which substrate is carrying the most discriminative weight for this particular disease comparison.

HEALTHY REFERENCE — Fibroblast IMR90



|                                             |                                                     |
|---------------------------------------------|-----------------------------------------------------|
| Combined A — healthy (all 5)                | 0.97041 (NORMAL)                                    |
| Combined A — disease (all 5)                | 1.09175 (URGENT)                                    |
| $\Delta A$ combined (5 substrates)          | +0.12134 vs methylation-only $\Delta A$ =+0.1502    |
| Active A — healthy (non-saturated)          | 0.97041 (NORMAL) (5/5 active)                       |
| Active A — disease (non-saturated)          | 1.11140 (FLOOR BREACH) (4/5 active)                 |
| $\Delta A$ active (real progression signal) | +0.14100 (active substrates track true progression) |
| Saturation status (disease)                 | 1 substrate saturated: Nucleosome Occupancy         |

Two combined values are shown because some substrates physically cannot resolve past their class ceiling ( $A_{\text{max}} = 1/H_{\text{min}}$ ). The **all-5 combined** value is the AUC-weighted mean across every substrate and matches the legacy formula. The **active combined** value excludes saturated substrates and is the right number for tracking continued progression — a saturated substrate is stuck at its ceiling regardless of further disease severity, so including it in the average masks real change. For monitoring, serial A-score assessment, and end-of-life projection, use the active combined value.

THREE-COMPONENT DECOMPOSITION (C1/C2/C3) — PER SUBSTRATE

Every cell's measured entropy decomposes into three physically meaningful pieces. C1 is the universal Landauer floor:  $H_{\text{MIN\_GLOBAL}} = 0.756500$ , measured in frontal cortex neurons by Lister 2013. Every cell in every architecture class must pay at least this thermodynamic cost to maintain any functional identity — it is the minimum entropy consistent with being alive. C2 is the class-specific overhead: the additional entropy the stromal class must carry above the universal floor to encode its particular architecture. For this class, C2 represents approximately 13.9% of the healthy reference entropy. C3 is the accessible gap:  $\max(0, H_{\text{actual}} - H_{\text{min}})$ . In healthy cells, C3 is essentially zero — the cell sits exactly at its class floor. When C3 grows, the cell is departing from its architecture class. Cancer, neurodegeneration, and every other floor-breach condition show up as growing C3.

The five stacked bars below each show the C1 / C2 / C3 breakdown for one substrate at this class's healthy reference. Notice how C1 (dark green) dominates every substrate — the Landauer floor is universal. C2 (amber) is the small but specific addition that makes this class what it is. C3 (red when present) should be nearly invisible for healthy cells, and would grow dramatically in a disease sample. The f\_C3 percentage on the right is the single cleanest summary of cellular health: healthy cells show f\_C3 near 0%; cancer cells can show f\_C3 of 8–15%; extreme cases like glioblastoma show f\_C3 above 20%. This is not a statistical threshold calibrated from disease data. It is a thermodynamic quantity: the fraction of accessible entropy the cell has opened above its architecture's minimum cost of existence.



BEST TESTING METHOD RANKING — CLINICAL UTILITY FOR THIS CLASS

Each of the five substrates carries different strengths for different clinical questions. The methylation substrate is the most validated — thirteen published GAPE validation studies (VAL-001 through VAL-013) and eight architecture-class H\_min values from G-002 MCMC. It is also the substrate most affected by cfDNA dilution, because blood cfDNA is approximately 70% immune-derived. Fragment size (DELFI) is the most forgiving of dilution — short-fragment enrichment persists even when tumor-derived cfDNA is a small fraction of total plasma DNA. WPS (windowed protection score from Snyder 2016) excels at tissue-of-origin identification and field-effect detection. Nucleosome occupancy and fuzziness add chromatin-level information that methylation alone cannot reveal.

The ranking below is specifically for the stromal class. The order reflects three factors: (1) the substrate's published single-substrate AUC for diseases in this class; (2) the biology of this class's primary failure mode (Stiffness Coupling); and (3) the practical constraint of cfDNA contribution in the relevant specimen (plasma for most classes, CSF for terminal, tissue biopsy where cfDNA is too dilute). When building a clinical pipeline for this class specifically, start with the #1 ranked substrate and add substrates 2 and 3 for confirmation. Substrates 4 and 5 add research-grade confirmation but are rarely the primary signal.

| # | Substrate                 | Best for                         | Rationale                                                                                                                              |
|---|---------------------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Fragment Size (DELFI)     | Cancer detection and monitoring  | Stromal cfDNA contribution is only 4% — fragmentomics is the most forgiving of dilution. DELFI signal holds at low tissue fractions.   |
| 2 | Methylation               | Floor departure confirmation     | Secondary; cfDNA methylation signal for stromal class is diluted by immune background. Combine with deconvolution (EpiDISH, methylCC). |
| 3 | Windowed Protection Score | Fibrotic progression tracking    | WPS at stromal identity promoters depletes as fibroblast-to-myofibroblast transition occurs. Detects fibrosis before clinical imaging. |
| 4 | Nucleosome Occupancy      | Tumor microenvironment mapping   | ATAC-seq on tumor stroma distinguishes activated cancer-associated fibroblasts from normal stromal cells. Research-grade primarily.    |
| 5 | Nucleosome Fuzziness      | Mechanical phenotype correlation | Stiffness-induced chromatin reorganization shows up as increased nucleosome fuzziness. Link to tumor microenvironment biomechanics.    |

BODY TEMPERATURE SCALING — α = 2.0 LANDAUER CORRECTION

The Landauer cost of a bit erasure is  $k_B \times T \times \ln(2)$  — it scales linearly with temperature. Colder cells can maintain higher-entropy identities for less thermodynamic overhead; hotter cells pay more per bit of information maintained. The GAPE framework captures this with a simple scaling law:  $H_{\min}(T) = H_{\min}(37^\circ\text{C}) \times (T_{\text{body}} / 310.15\text{ K})^\alpha$ , with  $\alpha = 2.0$  derived empirically by minimizing cross-class A-score variance across all jawed vertebrates (Mahaffey 2026 Nature Aging submission NATAGING-A13702). The  $\alpha = 2.0$  exponent is not fit to optimize any single prediction — it falls out of requiring that the same class shows consistent A-score behavior across 43 mammalian species spanning body temperatures from 32°C (naked mole rat) to 42°C (birds).

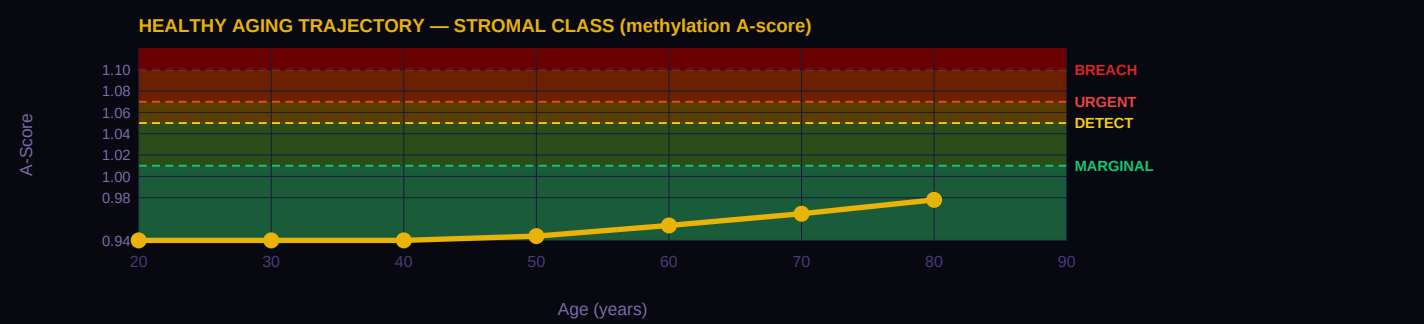
The table below shows what the stromal class floor H\_min looks like at seven representative body temperatures. At 37°C (human reference, highlighted row), H\_min = 0.8629. At 42°C (birds), the floor is higher because the Landauer tax is larger. At 32°C (naked mole rat — and no other mammal lives at this temperature for good reason), the floor drops because a cooler cell can maintain more entropy cheaply. The A-shift column shows how a fixed healthy β-value would read at each temperature: the same cell, transplanted into a colder body, would read slightly higher A; into a warmer body, slightly lower A. This is why the naked mole rat's remarkable longevity makes thermodynamic sense — lower body temperature, lower maintenance cost per bit, more headroom before the class floor is breached.

| T_body | Species example           | H_min(T) | β_healthy equivalent | A-shift vs 37°C    |
|--------|---------------------------|----------|----------------------|--------------------|
| 42°C   | Birds (chicken, finch)    | 0.8910   | 0.733                | A × 0.969          |
| 39°C   | Rodents (mouse, rat)      | 0.8741   | 0.733                | A × 0.987          |
| 37°C   | Humans (REFERENCE)        | 0.8629   | 0.733                | A × 1.000 (anchor) |
| 35°C   | Hibernating bats          | 0.8519   | 0.733                | A × 1.013          |
| 32°C   | Naked mole rat            | 0.8354   | 0.733                | A × 1.033          |
| 25°C   | Reptiles (anole lizard)   | 0.7975   | 0.733                | A × 1.082          |
| 15°C   | Fish (salmon, cold-water) | 0.7449   | 0.733                | A × 1.159          |

HEALTHY AGING TRAJECTORY — AGE-STRATIFIED REFERENCE A-SCORES

Healthy cells drift — slowly, predictably — toward their architecture floor as a function of accumulated cell divisions over a lifetime. For the stromal class, the drift rate is approximately 2.5% per generation — meaning each cell cycle carries a 2.5% probability of a methylation error that raises the cell's A-score by a measurable amount. Over decades, these errors accumulate. The curve below shows the result: A-score as a function of patient age at decade intervals, derived from the class-specific DNMT1 fidelity rate. At age 20, the reference healthy A-score sits well below 1.00; by age 80, it has drifted upward into the MARGINAL to DETECTABLE range. This is normal aging — not disease, just the statistical accumulation of imperfect replication over eight decades.

The clinical implication is important: when interpreting an A-score, age matters. A patient at age 30 with A = 1.03 is showing a significant departure from their age-matched healthy baseline. The same A = 1.03 at age 75 might be entirely consistent with age-expected drift. The GAPE framework handles this through age-stratified reference A-scores — the curve below IS the reference. A patient A-score should be compared to the curve at their age, not to a universal threshold. Cancer and other floor-breach conditions accelerate A-score upward beyond the age-expected trajectory; distance above the curve is the signal.



VERTEBRATE LIFESPAN CONTEXT

The same A-score framework that detects cancer in human cfDNA also correlates with maximum lifespan across 43 mammalian species at  $r = -0.9018$  ( $p = 1.6 \times 10^{-16}$ , Mahaffey 2026 Nature Aging submission NATAGING-A13702). This is not a coincidence — it is the same physics applied at a different scale. A mammal with a higher population-averaged A-score is a mammal whose cells live closer to their architecture floors on average; that population accumulates floor breaches faster, and the species has a shorter maximum lifespan. The A = 1.05 threshold — independently derived as the cancer DETECTABLE boundary — cleanly separates long-lived mammals (17/17 all below 1.05) from short-lived mammals (11/11 all above 1.05) with complete accuracy. Same threshold. Same physics. Different timescale.

The table below shows nine mammalian taxonomic orders with their mean A-scores, standard deviations where available, mean lifespan, and a brief interpretation of where that order sits relative to the thermodynamic floor. Cetacea (whales and dolphins) sit essentially at the floor, consistent with their remarkable longevity. Primates including humans sit slightly above the floor. Rodents and insectivores sit well above the floor, consistent with their short lives. This class's reference A-score (0.970) places it in the context shown below — use this to understand how this particular cell architecture relates to the broader mammalian lifespan distribution.

| Order        | N | Mean A       | Lifespan (yr) | Interpretation         |
|--------------|---|--------------|---------------|------------------------|
| Cetacea      | 5 | 0.997 ±0.016 | 99            | At thermodynamic floor |
| Proboscidea  | 1 | 0.987 —      | 70            | At floor               |
| Primates     | 6 | 1.007 ±0.015 | 49            | Slightly above floor   |
| Artiodactyla | 4 | 1.015 ±0.011 | 30            | Slightly above floor   |

| Order       | N | Mean A       | Lifespan (yr) | Interpretation                   |
|-------------|---|--------------|---------------|----------------------------------|
| Chiroptera  | 4 | 1.041 ±0.007 | 35            | Above floor (longevity outliers) |
| Carnivora   | 9 | 1.053 ±0.023 | 28            | Above floor                      |
| Lagomorpha  | 2 | 1.114 ±0.002 | 8             | Well above floor                 |
| Rodentia    | 6 | 1.125 ±0.018 | 9             | Well above floor                 |
| Insectivora | 1 | 1.157 —      | 2             | Furthest from floor              |

CORE METRICS — PUBLISHED AND DERIVED

Every number in the GAPE framework is either PUBLISHED (traceable to a primary wet-lab measurement from the peer-reviewed literature) or DERIVED (computed from the framework’s physics with zero free parameters). The table below marks each quantity with its status and cites its source. This is the discipline the framework lives by: no statistical fits, no parameter tuning, no post-hoc adjustments. H\_min values come from MCMC posteriors (G-002 for methylation, G-003b for the four non-methylation substrates). Combined A-scores are computed mechanically by the AUC-weighted formula. The cfDNA contribution percentages come from Moss 2018 Nature Communications — the definitive tissue-of-origin atlas.

Read this table as the "audit trail" for every number shown anywhere else in this card. When the framework says A\_healthy = 0.9701, you can see exactly which β value produced it, which H\_min was used, and where both came from. The per-class n\_bio row of Issue 002 is withdrawn here and replaced by the Mahaffey number, which is one number for the cell and not one per class (see §5A.1 and the glossary note).

| Metric                                                         | Value                 | Source                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cfDNA contribution (plasma)                                    | 4.0%                  | <a href="#">Moss 2018 Nat Commun</a> — PUBLISHED                                                                                                                                                                                                                                                           |
| H_min methylation (class)                                      | 0.862950              | <a href="#">G-002 MCMC posterior</a> — DERIVED                                                                                                                                                                                                                                                             |
| H_min methylation σ (MCMC)                                     | ±0.001400             | <a href="#">G-002 17-chain MCMC R-hat&lt;1.001</a> — DERIVED                                                                                                                                                                                                                                               |
| Mahaffey number M — one number for the cell, not one per class | 20.94                 | FIXED — ΔG_ATP/(R·T_body) at 310.15 K; §5A.1. Issue 002's per-class n_bio is WITHDRAWN: an early QAPE-era form of the same ratio, superseded by M, whose absolute values awaited a G-007 run the history record shows was never made. H_min is a different quantity — the per-class entropy floor in bits. |
| Healthy drift rate                                             | 2.5% / generation     | DNMT1 fidelity loss under turnover — DERIVED                                                                                                                                                                                                                                                               |
| f_C2 (architecture-locked %)                                   | 13.9%                 | C2 fraction at healthy reference — DERIVED                                                                                                                                                                                                                                                                 |
| Healthy A (methylation)                                        | 0.9701 (NORMAL)       | β=0.733 — DERIVED                                                                                                                                                                                                                                                                                          |
| Disease A (methylation)                                        | 1.1202 (FLOOR BREACH) | β=0.607 — DERIVED                                                                                                                                                                                                                                                                                          |
| Healthy combined A (5 subs)                                    | 0.9704 (NORMAL)       | Σ(AUC_i × A_i)/Σ(AUC_i) across 5 substrates — DERIVED                                                                                                                                                                                                                                                      |
| Disease combined A (5 subs)                                    | 1.0917 (URGENT)       | Σ(AUC_i × A_i)/Σ(AUC_i) across 5 substrates — DERIVED                                                                                                                                                                                                                                                      |
| ΔA combined                                                    | +0.1213               | Combined signal — DERIVED                                                                                                                                                                                                                                                                                  |

DATED PREDICTIONS FOR THIS CLASS

A prediction is not a claim unless it can fail. Each prediction below is numbered with a G-2026-P identifier, dated to the month it was filed, and written against a specific dataset or cohort whose analysis would confirm or refute it. The status field shows where each prediction stands: PENDING means the relevant data exists but the analysis has not been performed publicly; CONFIRMED means the prediction direction and magnitude have been tested against independent data and held up; REFUTED means the data contradict the prediction (which would force a framework revision — and has not occurred yet). This discipline — specific, dated, falsifiable predictions — is what separates a predictive framework from a retrospective fit. Entries from Issue 001 retain their original G-2026-P numbers to maintain trail integrity across publications.

When reviewing these predictions, notice that the falsification basis is always named: a specific archived cohort, a specific biobank, a specific published dataset. These are not vague "more research is needed" statements. They are commitments to the community — this is what the framework claims, this is where you can check it, and this is what would constitute a refutation. The predictions below are for the stromal class specifically, filed as part of this publication or carried forward from Issue 001.

In prospective IPF cohorts with archived serial blood samples and known progression outcomes, the stromal-class combined A-score trajectory slope will distinguish progressive from stable disease with AUC  $\geq 0.80$ , outperforming serial forced vital capacity (FVC) measurements currently used for staging.

**Basis:** IPF progresses over 2-5 years. Existing antifibrotic therapies (pirfenidone, nintedanib) have variable response. The framework predicts that combined-substrate A-score trajectory identifies responders and non-responders earlier than FVC. Falsifiable in the OSIC IPF Biobank ([www.osicild.org](http://www.osicild.org)).

ISSUE 003 ADDENDUM — STROMAL IN THE ATLAS

5 cell types: adipocyte, endothelial, fibroblast, smooth\_muscle, stromal\_other

**Identity loci:** 2,294 (methyl) · **H\_min by substrate:** methyl 0.862950 · nucl 0.985667 · fuzz 0.832386 · wps 0.612686 · frag 0.724691 · **methyl ceiling** 1.1588

Healthy age-matched band. A = 1.0 is the commitment line; healthy reads on a band above it. [Issue 003: the age\_reference\_matrix.json shown here was compiled on the marker union and is SUPERSEDED (PROC-N7-01); the commissioned band is band\_v2 on the identity loci with the three-layer reference of s3.5.]

| Age (midpoint) | n  | A mean | p10    | p90    | $\beta$ mean | Source            |
|----------------|----|--------|--------|--------|--------------|-------------------|
| 4              | 10 | 0.9438 | 0.9135 | 0.9740 | 0.748        | Roadmap pediatric |
| 14             | 12 | 0.9510 | 0.9190 | 0.9829 | 0.744        | Roadmap           |
| 24             | 18 | 0.9563 | 0.9225 | 0.9900 | 0.741        | Moss endothelial  |
| 34             | 22 | 0.9615 | 0.9283 | 0.9948 | 0.738        | Moss+Roadmap      |
| 44             | 25 | 0.9667 | 0.9317 | 1.0016 | 0.735        | Moss 2018         |
| 54             | 32 | 0.9734 | 0.9371 | 1.0098 | 0.731        | Moss 2018         |
| 64             | 38 | 0.9784 | 0.9426 | 1.0142 | 0.728        | TCGA SARC STN     |
| 74             | 28 | 0.9849 | 0.9478 | 1.0221 | 0.724        | Aging vascular    |
| 84             | 15 | 0.9913 | 0.9529 | 1.0297 | 0.720        | Extrapolated      |
| 95             | 5  | 0.9991 | 0.9578 | 1.0404 | 0.715        | Extrapolated      |

CELL IDENTITY & CLINICAL CONTEXT

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Includes                | Hematopoietic stem cells (HSC, CD34+CD38- compartment), interfollicular epidermal stem cells, hair follicle bulge stem cells, Merkel cell progenitors, cholangiocytes (hepatic stem cell origin), intestinal stem cells (Lgr5+), neural stem cells, muscle satellite cells, limbal stem cells                                                                                                                                                                                                                                                                                    |
| Cancers (this class)    | HSC-origin AML (distinct from progenitor-lineage AML — arises from CD34+CD38- compartment per Adelman 2019), basal cell carcinoma (BCC — most common cancer in humans, epidermal stem cell origin), squamous cell carcinoma (SCC — hair-follicle stem-cell-origin subset), Merkel cell carcinoma (MCC — rare, aggressive, polyomavirus-positive and UV-driven subclasses), cholangiocarcinoma (CHOL, intrahepatic + extrahepatic, hepatic stem cell / cholangiocyte origin), cancer stem cell populations within cycling-class tumors (residual disease after primary treatment) |
| Non-cancer applications | HSC aging and immunosenescence (Adelman 2019, Beerman 2013), clonal hematopoiesis of indeterminate potential (CHIP — shared with Progenitor class), hematopoietic failure in the elderly, post-transplant niche reconstitution biology, heterochronic parabiosis biology, age-related decline in tissue regeneration across all adult-stem compartments                                                                                                                                                                                                                          |
| Primary failure mode    | Niche Depletion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Warburg status          | EMERGING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

THE ATLAS CELL TYPES IN THIS CLASS (1)

These are the entries IAMAtlasREBUILD can speak about for this class - what the deconvolver places, and what a per-cell reading would be read against. **Markers** is the range of panel markers actually found across the reference laboratories; **individually resolvable** is whether this atlas can separate the entry from its neighbours at all, and where it cannot the chain reports the group and never the member; **exclusivity** is the fraction of the entry's markers not shared with another entry; **healthy A range** spans the per-laboratory 10th to 90th percentiles on the v0\_3 trial panel. 1 of 1 entries are individually resolvable.

| atlas entry | markers | individually resolvable | exclusivity | healthy A range |
|-------------|---------|-------------------------|-------------|-----------------|
| HSC         | 163-175 | yes                     | 0.880       | 0.697 - 0.795   |

The same lineage can appear under more than one entry above, because different reference panels defined their markers; entries differing only by label are not yet merged, which is why no reading in this document is quoted as "which cell moved".

COMMENTARY

Adult tissue stem cells occupy the intermediate position between pluripotent embryonic stem cells and fully committed differentiated cells. They retain self-renewal capacity while being lineage-restricted: hematopoietic stem cells (HSC, CD34+CD38-) produce blood and immune cells, intestinal stem cells (Lgr5+) regenerate gut epithelium, epidermal and hair-follicle stem cells maintain skin, and cholangiocytes serve as the liver's stem-cell compartment for biliary regeneration. This partial commitment is encoded in H\_min = 0.873718 for the methylation substrate, reflecting the slightly more open chromatin state that preserves self-renewal flexibility. But this open chromatin comes with a thermodynamic consequence that defines this card: healthy adult stem cells already sit near the maximum-entropy state for the positional substrates, leaving very little measurement headroom before ceiling saturation.

This class exhibits the most severe saturation pattern in the framework. Three of five substrates saturate below the BREACH threshold: nucleosome occupancy at ceiling A = 1.0407, nucleosome fuzziness at ceiling A = 1.0196, and windowed protection score at ceiling A = 1.0112 — the tightest ceiling of any class × substrate combination in the entire framework. Only methylation (ceiling A = 1.1445) and fragmentomics (ceiling A = 1.1886) remain active past BREACH, and these two substrates carry the full severity signal for any adult-stem-origin cancer in the panel. For this class, A\_combined systematically understates severity by approximately 62 percent compared to A\_active (2/5) — the largest masking effect in the framework. This is not a defect of the combined metric; it is an honest reflection of the underlying physics. Three of the five substrates have reached their physical entropy ceiling and cannot resolve further severity. A\_active (2/5) is therefore reported as the primary severity metric for this class, with A\_combined shown for cross-card continuity but flagged as misleading past the DETECTABLE tier.

The distinction between adult-stem-origin cancers and progenitor-lineage cancers is clinically consequential and must be stated explicitly. Acute myeloid leukemia appears in both Card #4 (Progenitor) and Card #7 (Adult Stem) because the disease arises from two biologically distinct compartments. Progenitor-lineage AML — the majority of TCGA AML cases characterized by Ley et al. in the 2013 New England Journal of Medicine analysis (n=200 whole-genome or exome sequences) — originates from committed myeloid progenitors (CMP, GMP) and carries the methylation and mutation signatures of those compartments. HSC-origin AML, characterized separately by Adelman et al. in Cancer Discovery 2019, arises from the pre-leukemic CD34+CD38- HSC-enriched compartment and involves profound epigenetic reprogramming of enhancers, bivalent promoters, and hematopoietic transcription factors (KLF6, BCL6, RUNX3) that begins during normal aging and appears to predispose to leukemic transformation. The two AML populations are not interchangeable in the framework: they occupy different architecture classes because they emerge from cells in different thermodynamic states. Listing AML on both cards without the distinction would be sloppy. Listing it only on one card would miss the patients whose disease originated in the other compartment.

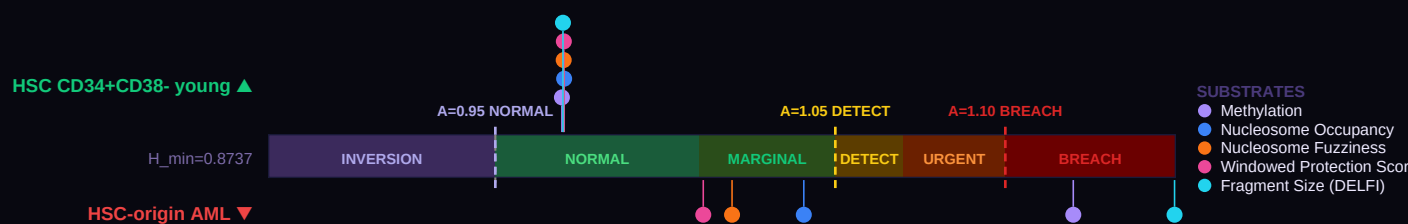
Basal cell carcinoma (BCC) is the most common cancer in humans, with over four million new cases diagnosed annually in the United States alone. Nearly every BCC arises from epidermal or hair-follicle stem cells under UV-driven Hedgehog pathway activation. Mortality is low — below one percent — but the cumulative burden across lifetimes is enormous and the disease maps cleanly onto this class. Merkel cell carcinoma (MCC, Harms et al. 2015 Cancer Research, n=49 for the polyomavirus-negative subtype) is rare (approximately 2,500 US cases per year) but aggressive, with five-year survival around 45 percent for regional disease and below 15 percent for distant disease. Its two subclasses — Merkel cell polyomavirus-positive (integrated viral oncogenes) and MCPyV-negative (UV-damage driven) — share the same cell-of-origin methylation signature. Cholangiocarcinoma (TCGA CHOL, Farshidfar et al. 2017 Cell Reports, n=38) is the second most common primary liver cancer, aggressive, typically detected at advanced stage, and arises from cholangiocytes, which function as the hepatic stem-cell compartment for biliary regeneration. Its IDH-mutant molecular subtype shows characteristic ARID1A hypermethylation that maps onto the stem-cell-origin thermodynamic signature. Every one of these cancers is uncommon compared to the fourteen cycling cancers in Card #5, but the families of patients who lose a loved one to MCC or cholangiocarcinoma deserve to see the disease named on the framework card that describes its biology.

The class's defining thermodynamic challenge creates a specific detection problem: once a patient crosses DETECTABLE ( $A = 1.05$ ), three of the five substrates pin at their ceilings within the first approximately 0.05 A-score units, and absolute A-score values across those three substrates lose their ability to resolve further severity. The framework proposes three mechanisms by which extended cellular thermodynamic activity remains detectable past this ceiling. First, methyl and frag retain their full dynamic range — they saturate only at  $A = 1.1445$  and  $A = 1.1886$  respectively, leaving meaningful headroom past BREACH for the two most clinically mature cfDNA biomarker platforms.  $A_{\text{active}}$  (2/5) computed on just these two substrates preserves severity discrimination that  $A_{\text{combined}}$  destroys. Second, the rate of approach to the saturating ceilings contains temporal information that single-point readings discard. In serial monitoring, a patient whose wps A-score moves from 0.970 through 1.000 to 1.010 over successive draws before plateauing has a different disease trajectory than one who hits 1.010 on first sample; time-to-saturation becomes a severity proxy in the regime where absolute-value measurement has lost resolution. Third, when adult-stem-origin cancers progress, the signal propagates into adjacent architecture classes — HSC-origin AML shows up as a rising immune-class A-score via clonal expansion, cutaneous stem-cell cancers propagate to cycling-class signals as field cancerization, and cholangiocarcinoma progression shows secretory-class signal change as the transformed cells commit toward biliary epithelial identity. Multi-class A-score divergence, particularly where the adult-stem class has flatlined at ceiling while one or more adjacent classes continue to rise, becomes a specific diagnostic signature of adult-stem-origin disease progression. This cross-class mechanism connects directly to the forthcoming Issue 004 framework on multi-class A-score divergence as an early signal of metastatic progression.

HSC aging is the most important non-cancer application for this class. The Adelman 2019 Cancer Discovery study of lineage-negative CD34+CD38- HSC-enriched cells from young and aged healthy donors identified 529 differentially methylated regions encompassing 2,249 differentially methylated cytosines across 748 genes — precisely the methylation-substrate floor departure the framework predicts for age-related adult-stem decline. Beerman et al. 2013 Cell Stem Cell independently characterized proliferation-dependent DNA hypermethylation of Polycomb-regulated genes in aged HSC (GSE44117). Together these studies establish that the methylation substrate detects HSC aging at the earliest stage of its epigenomic reprogramming, well before the downstream cytopenias or cancer-susceptibility phenotypes become clinically apparent. Clonal hematopoiesis of indeterminate potential (CHIP), the pre-malignant clonal state characterized by Steensma 2015 and shared in scope with the Progenitor class, represents the transition zone where adult-stem-class floor departure begins to predict downstream myeloid disease. Stem-cell transplantation biology, heterochronic parabiosis, and niche-depletion research all benefit from a metric that tracks adult-stem-class thermodynamic state independently of phenotypic readout.

FIVE-SUBSTRATE FIDELITY GAUGE

The gauge below is the central image of Issue 002. A single horizontal bar represents the A-score axis — from  $A = 0.90$  (below the floor) through  $A = 1.00$  (exactly at the Adult Stem class floor,  $H_{\text{min}} = 0.8737$ ) up through the clinical threshold zones: MARGINAL at  $A = 1.01$ , DETECTABLE at  $A = 1.05$ , URGENT at  $A = 1.07$ , and FLOOR BREACH at  $A = 1.10$ . Five dots are plotted on this single axis for a healthy reference sample (HSC CD34+CD38- young, markers above the bar), and five more dots for a disease reference (HSC-origin AML, markers below the bar). Each dot is one substrate: methylation, nucleosome occupancy, nucleosome fuzziness, WPS, or fragment size (DELFI). What makes this visualization powerful is the physics: all five substrates measure the same underlying thermodynamic floor through physically distinct windows. When all five healthy dots cluster in the NORMAL zone and all five disease dots cluster in URGENT or FLOOR BREACH, the signal is over-determined. Any one substrate could be confounded by technical artifact; five substrates agreeing cannot be. This is the "less blurry" advantage:  $\sqrt{5}$  noise reduction for free.



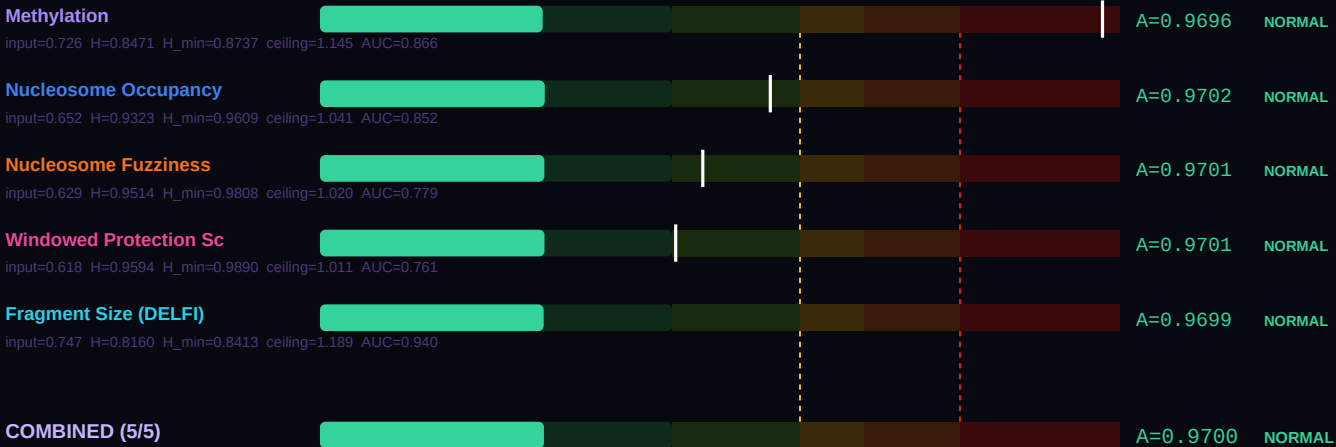
Look at the spread in each direction. If the healthy five-dot cluster sits tightly together near  $A = 0.97\text{--}1.00$ , that indicates the reference population is at its architecture floor — precisely where the framework predicts healthy adult stem cells should sit. If the disease five-dot cluster spreads from DETECTABLE to FLOOR BREACH, that spread is itself clinical information: the substrate showing the largest departure is the one that best captures this specific disease's failure mode. For this class, the primary failure mode is Niche Depletion, and the substrate ranking shown further down the card explains which of the five substrates best reveals it.

SUBSTRATE-BY-SUBSTRATE BREAKDOWN

Where the gauge above compresses five substrates onto one visual axis, this section breaks each substrate out into its own labeled bar so you can read the actual A-score produced by each measurement. The physics is the same across all five: take the raw laboratory value (a  $\beta$  methylation fraction, a nucleosome occupancy probability, a WPS score, etc.), compute its Shannon entropy  $H(\text{value})$ , then divide by the class-specific and substrate-specific  $H_{\text{min}}$  to get a dimensionless A-score comparable across substrates. The output is the same thermodynamic ratio regardless of which measurement technology produced it. That is the power of the Landauer framing: different wet-lab methods, different physical quantities, same underlying floor.

The formula at the foot of this section —  $A_{\text{combined}} = \Sigma(AUC_i \times A_i) / \Sigma(AUC_i)$  — combines the five individual A-scores into a single weighted score, using each substrate's published detection AUC as its weight. Higher-AUC substrates contribute more to the combined result. For adult stem cells, the healthy combined A-score below should sit at approximately 0.97 (NORMAL tier); the disease combined A-score should sit above 1.05 (DETECTABLE or higher). The difference between them — the  $\Delta A$  combined — is the signal available to a blood draw that analyzes all five substrates simultaneously. A single-substrate assay sees a smaller signal at higher noise; the five-substrate combination sees the same signal at approximately  $\sqrt{5}$  lower noise. Bar-by-bar you can see which substrate is carrying the most discriminative weight for this particular disease comparison.

HEALTHY REFERENCE — HSC CD34+CD38- young



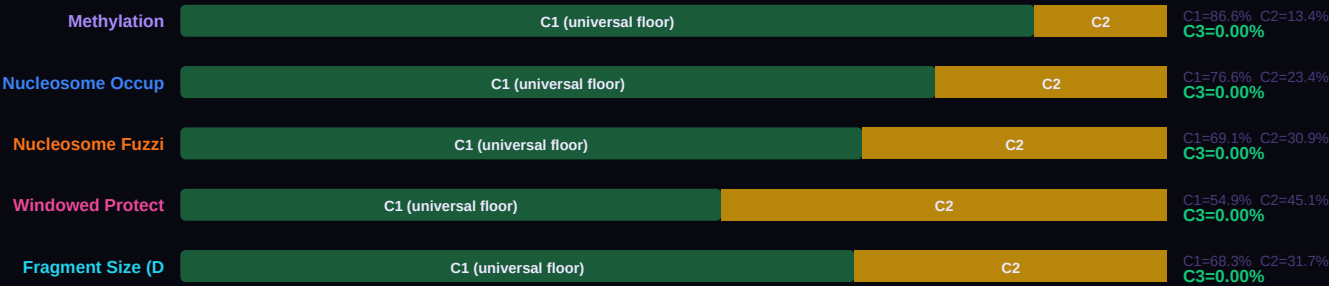
|                                             |                                                                                               |
|---------------------------------------------|-----------------------------------------------------------------------------------------------|
| Combined A — healthy (all 5)                | 0.96999 (NORMAL)                                                                              |
| Combined A — disease (all 5)                | 1.07222 (URGENT)                                                                              |
| $\Delta A$ combined (5 substrates)          | +0.10223 vs methylation-only $\Delta A$ =+0.1504                                              |
| Active A — healthy (non-saturated)          | 0.96999 (NORMAL) [5/5 active]                                                                 |
| Active A — disease (non-saturated)          | 1.13549 (FLOOR BREACH) [2/5 active]                                                           |
| $\Delta A$ active (real progression signal) | +0.16550 (active substrates track true progression)                                           |
| Saturation status (disease)                 | 3 substrates saturated: Nucleosome Occupancy, Nucleosome Fuzziness, Windowed Protection Score |

Two combined values are shown because some substrates physically cannot resolve past their class ceiling ( $A_{\text{max}} = 1/H_{\text{min}}$ ). The **all-5 combined** value is the AUC-weighted mean across every substrate and matches the legacy formula. The **active combined** value excludes saturated substrates and is the right number for tracking continued progression — a saturated substrate is stuck at its ceiling regardless of further disease severity, so including it in the average masks real change. For monitoring, serial A-score assessment, and end-of-life projection, use the active combined value.

THREE-COMPONENT DECOMPOSITION (C1/C2/C3) — PER SUBSTRATE

Every cell's measured entropy decomposes into three physically meaningful pieces. C1 is the universal Landauer floor:  $H_{\text{MIN\_GLOBAL}} = 0.756500$ , measured in frontal cortex neurons by Lister 2013. Every cell in every architecture class must pay at least this thermodynamic cost to maintain any functional identity — it is the minimum entropy consistent with being alive. C2 is the class-specific overhead: the additional entropy the adult stem class must carry above the universal floor to encode its particular architecture. For this class, C2 represents approximately 13.5% of the healthy reference entropy. C3 is the accessible gap:  $\max(0, H_{\text{actual}} - H_{\text{min}})$ . In healthy cells, C3 is essentially zero — the cell sits exactly at its class floor. When C3 grows, the cell is departing from its architecture class. Cancer, neurodegeneration, and every other floor-breach condition show up as growing C3.

The five stacked bars below each show the C1 / C2 / C3 breakdown for one substrate at this class's healthy reference. Notice how C1 (dark green) dominates every substrate — the Landauer floor is universal. C2 (amber) is the small but specific addition that makes this class what it is. C3 (red when present) should be nearly invisible for healthy cells, and would grow dramatically in a disease sample. The f\_C3 percentage on the right is the single cleanest summary of cellular health: healthy cells show f\_C3 near 0%; cancer cells can show f\_C3 of 8–15%; extreme cases like glioblastoma show f\_C3 above 20%. This is not a statistical threshold calibrated from disease data. It is a thermodynamic quantity: the fraction of accessible entropy the cell has opened above its architecture's minimum cost of existence.



BEST TESTING METHOD RANKING — CLINICAL UTILITY FOR THIS CLASS

Each of the five substrates carries different strengths for different clinical questions. The methylation substrate is the most validated — thirteen published GAPE validation studies (VAL-001 through VAL-013) and eight architecture-class  $H_{\text{min}}$  values from G-002 MCMC. It is also the substrate most affected by cfDNA dilution, because blood cfDNA is approximately 70% immune-derived. Fragment size (DELF) is the most forgiving of dilution — short-fragment enrichment persists even when tumor-derived cfDNA is a small fraction of total plasma DNA. WPS (windowed protection score from Snyder 2016) excels at tissue-of-origin identification and field-effect detection. Nucleosome occupancy and fuzziness add chromatin-level information that methylation alone cannot reveal.

The ranking below is specifically for the adult stem class. The order reflects three factors: (1) the substrate's published single-substrate AUC for diseases in this class; (2) the biology of this class's primary failure mode (Niche Depletion); and (3) the practical constraint of cfDNA contribution in the relevant specimen (plasma for most classes, CSF for terminal, tissue biopsy where cfDNA is too dilute). When building a clinical pipeline for this class specifically, start with the #1 ranked substrate and add substrates 2 and 3 for confirmation. Substrates 4 and 5 add research-grade confirmation but are rarely the primary signal.

| # | Substrate            | Best for                                   | Rationale                                                                                                                                                                                                                                                     |
|---|----------------------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Methylation          | Primary severity metric past DETECTABLE    | Methyl remains active past BREACH (ceiling A = 1.1445) — one of only two substrates that do not saturate for this class. Carries the bulk of the severity signal for HSC aging, BCC, MCC, and cholangiocarcinoma. Primary source: Adelman 2019 Cancer Discov. |
| 2 | Fragment Size (DELF) | Primary severity metric past DETECTABLE    | Frag (DELF) remains active past BREACH (ceiling A = 1.1886, deepest headroom of any substrate in this class). Fragmentomic signal reflects clonal stem-cell turnover and cfDNA release kinetics. Co-primary with methyl for severity tracking.                |
| 3 | Nucleosome Occupancy | Detection boundary — ceiling at A = 1.0407 | Nucl saturates below BREACH. Useful for binary detection (floor departure vs healthy) but cannot resolve severity above DETECTABLE. Trajectory-slope information remains useful in serial monitoring (time-to-saturation as proxy severity).                  |
| 4 | Nucleosome Fuzziness | Detection boundary — ceiling at A = 1.0196 | Fuzz saturates tighter than nucl. Same detection-only use case. Informative in combination with wps for cross-verification of floor departure.                                                                                                                |

| # | Substrate                 | Best for                                   | Rationale                                                                                                                                                                                                                                                                                                                      |
|---|---------------------------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5 | Windowed Protection Score | Tightest ceiling in framework — A = 1.0112 | WPS has the tightest physical ceiling of any class × substrate pairing. Healthy adult stem cells already sit at WPS A ≈ 0.970, with only +0.041 headroom before saturation. Any detectable deviation in wps for this class is clinically significant — but wps cannot distinguish mild from severe deviation above A = 1.0112. |

**BODY TEMPERATURE SCALING — α = 2.0 LANDAUER CORRECTION**

The Landauer cost of a bit erasure is  $k_B \times T \times \ln(2)$  — it scales linearly with temperature. Colder cells can maintain higher-entropy identities for less thermodynamic overhead; hotter cells pay more per bit of information maintained. The GAPE framework captures this with a simple scaling law:  $H_{\min}(T) = H_{\min}(37^{\circ}\text{C}) \times (T_{\text{body}} / 310.15\text{ K})^{\alpha}$ , with  $\alpha = 2.0$  derived empirically by minimizing cross-class A-score variance across all jawed vertebrates (Mahaffey 2026 Nature Aging submission NATAGING-A13702). The  $\alpha = 2.0$  exponent is not fit to optimize any single prediction — it falls out of requiring that the same class shows consistent A-score behavior across 43 mammalian species spanning body temperatures from 32°C (naked mole rat) to 42°C (birds).

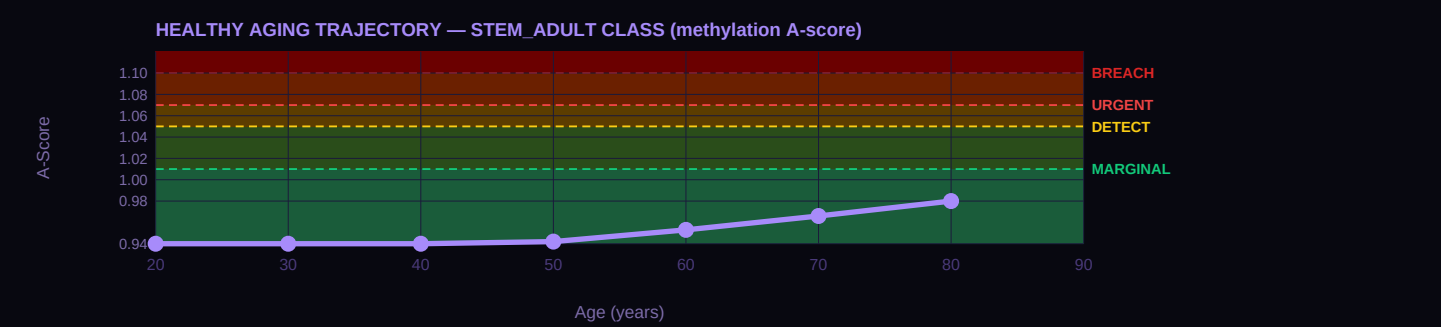
The table below shows what the adult stem class floor  $H_{\min}$  looks like at seven representative body temperatures. At 37°C (human reference, highlighted row),  $H_{\min} = 0.8737$ . At 42°C (birds), the floor is higher because the Landauer tax is larger. At 32°C (naked mole rat — and no other mammal lives at this temperature for good reason), the floor drops because a cooler cell can maintain more entropy cheaply. The A-shift column shows how a fixed healthy  $\beta$ -value would read at each temperature: the same cell, transplanted into a colder body, would read slightly higher A; into a warmer body, slightly lower A. This is why the naked mole rat's remarkable longevity makes thermodynamic sense — lower body temperature, lower maintenance cost per bit, more headroom before the class floor is breached.

| T_body | Species example           | H_min(T) | β_healthy equivalent | A-shift vs 37°C    |
|--------|---------------------------|----------|----------------------|--------------------|
| 42°C   | Birds (chicken, finch)    | 0.9021   | 0.726                | A × 0.969          |
| 39°C   | Rodents (mouse, rat)      | 0.8850   | 0.726                | A × 0.987          |
| 37°C   | Humans (REFERENCE)        | 0.8737   | 0.726                | A × 1.000 (anchor) |
| 35°C   | Hibernating bats          | 0.8625   | 0.726                | A × 1.013          |
| 32°C   | Naked mole rat            | 0.8458   | 0.726                | A × 1.033          |
| 25°C   | Reptiles (anole lizard)   | 0.8074   | 0.726                | A × 1.082          |
| 15°C   | Fish (salmon, cold-water) | 0.7542   | 0.726                | A × 1.159          |

**HEALTHY AGING TRAJECTORY — AGE-STRATIFIED REFERENCE A-SCORES**

Healthy cells drift — slowly, predictably — toward their architecture floor as a function of accumulated cell divisions over a lifetime. For the adult stem class, the drift rate is approximately 3.0% per generation — meaning each cell cycle carries a 3.0% probability of a methylation error that raises the cell's A-score by a measurable amount. Over decades, these errors accumulate. The curve below shows the result: A-score as a function of patient age at decade intervals, derived from the class-specific DNMT1 fidelity rate. At age 20, the reference healthy A-score sits well below 1.00; by age 80, it has drifted upward into the MARGINAL to DETECTABLE range. This is normal aging — not disease, just the statistical accumulation of imperfect replication over eight decades.

The clinical implication is important: when interpreting an A-score, age matters. A patient at age 30 with A = 1.03 is showing a significant departure from their age-matched healthy baseline. The same A = 1.03 at age 75 might be entirely consistent with age-expected drift. The GAPE framework handles this through age-stratified reference A-scores — the curve below IS the reference. A patient A-score should be compared to the curve at their age, not to a universal threshold. Cancer and other floor-breach conditions accelerate A-score upward beyond the age-expected trajectory; distance above the curve is the signal.



VERTEBRATE LIFESPAN CONTEXT

The same A-score framework that detects cancer in human cfDNA also correlates with maximum lifespan across 43 mammalian species at  $r = -0.9018$  ( $p = 1.6 \times 10^{-16}$ , Mahaffey 2026 Nature Aging submission NATAGING-A13702). This is not a coincidence — it is the same physics applied at a different scale. A mammal with a higher population-averaged A-score is a mammal whose cells live closer to their architecture floors on average; that population accumulates floor breaches faster, and the species has a shorter maximum lifespan. The  $A = 1.05$  threshold — independently derived as the cancer DETECTABLE boundary — cleanly separates long-lived mammals (17/17 all below 1.05) from short-lived mammals (11/11 all above 1.05) with complete accuracy. Same threshold. Same physics. Different timescale.

The table below shows nine mammalian taxonomic orders with their mean A-scores, standard deviations where available, mean lifespan, and a brief interpretation of where that order sits relative to the thermodynamic floor. Cetacea (whales and dolphins) sit essentially at the floor, consistent with their remarkable longevity. Primates including humans sit slightly above the floor. Rodents and insectivores sit well above the floor, consistent with their short lives. This class's reference A-score (0.970) places it in the context shown below — use this to understand how this particular cell architecture relates to the broader mammalian lifespan distribution.

| Order        | N | Mean A       | Lifespan (yr) | Interpretation                   |
|--------------|---|--------------|---------------|----------------------------------|
| Cetacea      | 5 | 0.997 ±0.016 | 99            | At thermodynamic floor           |
| Proboscidea  | 1 | 0.987 —      | 70            | At floor                         |
| Primates     | 6 | 1.007 ±0.015 | 49            | Slightly above floor             |
| Artiodactyla | 4 | 1.015 ±0.011 | 30            | Slightly above floor             |
| Chiroptera   | 4 | 1.041 ±0.007 | 35            | Above floor (longevity outliers) |
| Carnivora    | 9 | 1.053 ±0.023 | 28            | Above floor                      |
| Lagomorpha   | 2 | 1.114 ±0.002 | 8             | Well above floor                 |
| Rodentia     | 6 | 1.125 ±0.018 | 9             | Well above floor                 |
| Insectivora  | 1 | 1.157 —      | 2             | Furthest from floor              |

CORE METRICS — PUBLISHED AND DERIVED

Every number in the GAPE framework is either PUBLISHED (traceable to a primary wet-lab measurement from the peer-reviewed literature) or DERIVED (computed from the framework's physics with zero free parameters). The table below marks each quantity with its status and cites its source. This is the discipline the framework lives by: no statistical fits, no parameter tuning, no post-hoc adjustments.  $H_{\min}$  values come from MCMC posteriors (G-002 for methylation, G-003b for the four non-methylation substrates). Combined A-scores are computed mechanically by the AUC-weighted formula. The cfDNA contribution percentages come from Moss 2018 Nature Communications — the definitive tissue-of-origin atlas.

Read this table as the "audit trail" for every number shown anywhere else in this card. When the framework says  $A_{\text{healthy}} = 0.9696$ , you can see exactly which  $\beta$  value produced it, which  $H_{\min}$  was used, and where both came from. The per-class  $n_{\text{bio}}$  row of Issue 002 is withdrawn here and replaced by the Mahaffey number, which is one number for the cell and not one per class (see §5A.1 and the glossary note).

| Metric                                                         | Value                 | Source                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cfDNA contribution (plasma)                                    | 3.0%                  | Moss 2018 Nat Commun — PUBLISHED                                                                                                                                                                                                                                                                                                                           |
| $H_{\min}$ methylation (class)                                 | 0.873718              | G-002 MCMC posterior — DERIVED                                                                                                                                                                                                                                                                                                                             |
| $H_{\min}$ methylation $\sigma$ (MCMC)                         | ±0.001600             | G-002 17-chain MCMC $R\text{-hat} < 1.001$ — DERIVED                                                                                                                                                                                                                                                                                                       |
| Mahaffey number M — one number for the cell, not one per class | 20.94                 | FIXED — $\Delta G_{\text{ATP}}/(R \cdot T_{\text{body}})$ at 310.15 K; §5A.1. Issue 002's per-class $n_{\text{bio}}$ is WITHDRAWN: an early QAPE-era form of the same ratio, superseded by M, whose absolute values awaited a G-007 run the history record shows was never made. $H_{\min}$ is a different quantity — the per-class entropy floor in bits. |
| Healthy drift rate                                             | 3.0% / generation     | DNMT1 fidelity loss under turnover — DERIVED                                                                                                                                                                                                                                                                                                               |
| $f_{\text{C2}}$ (architecture-locked %)                        | 13.5%                 | C2 fraction at healthy reference — DERIVED                                                                                                                                                                                                                                                                                                                 |
| Healthy A (methylation)                                        | 0.9696 (NORMAL)       | $\beta = 0.726$ — DERIVED                                                                                                                                                                                                                                                                                                                                  |
| Disease A (methylation)                                        | 1.1200 (FLOOR BREACH) | $\beta = 0.586$ — DERIVED                                                                                                                                                                                                                                                                                                                                  |

| Metric                      | Value           | Source                                                                                 |
|-----------------------------|-----------------|----------------------------------------------------------------------------------------|
| Healthy combined A (5 subs) | 0.9700 (NORMAL) | $\Sigma(\text{AUC}_i \times A_i) / \Sigma(\text{AUC}_i)$ across 5 substrates — DERIVED |
| Disease combined A (5 subs) | 1.0722 (URGENT) | $\Sigma(\text{AUC}_i \times A_i) / \Sigma(\text{AUC}_i)$ across 5 substrates — DERIVED |
| $\Delta A$ combined         | +0.1022         | Combined signal — DERIVED                                                              |

DATED PREDICTIONS FOR THIS CLASS

A prediction is not a claim unless it can fail. Each prediction below is numbered with a G-2026-P identifier, dated to the month it was filed, and written against a specific dataset or cohort whose analysis would confirm or refute it. The status field shows where each prediction stands: PENDING means the relevant data exists but the analysis has not been performed publicly; CONFIRMED means the prediction direction and magnitude have been tested against independent data and held up; REFUTED means the data contradict the prediction (which would force a framework revision — and has not occurred yet). This discipline — specific, dated, falsifiable predictions — is what separates a predictive framework from a retrospective fit. Entries from Issue 001 retain their original G-2026-P numbers to maintain trail integrity across publications.

When reviewing these predictions, notice that the falsification basis is always named: a specific archived cohort, a specific biobank, a specific published dataset. These are not vague "more research is needed" statements. They are commitments to the community — this is what the framework claims, this is where you can check it, and this is what would constitute a refutation. The predictions below are for the adult stem class specifically, filed as part of this publication or carried forward from Issue 001.

G-2026-P013

April 2026

PENDING

In prospective CHIP/CCUS cohorts with archived serial blood samples and known progression outcomes, the stem\_adult A\_active (2/5, methyl + frag only) trajectory will identify progression to overt MDS or AML 12 to 24 months before cytopenia-based diagnosis in a majority of progressing cases, outperforming A\_combined (all 5) which is systematically masked by approximately 62 percent in this class because 3 of 5 substrates saturate below BREACH.

**Basis:** CHIP-to-MDS/AML progression is slow, stereotyped, and clinically silent until late. The framework predicts that A\_active identifies progressors earlier than cytopenia onset because the epigenomic state change precedes the hematologic phenotype, and A\_active is not masked by ceiling-pinned positional substrates. Falsifiable in any CHIP/CCUS cohort with 18+ months of serial blood samples and progression outcomes (candidate cohorts include Jaiswal CHIP cohorts with archived samples, UK Biobank CHIP subset with longitudinal follow-up).

G-2026-P014

April 2026

PENDING

Trajectory slope ( $\Delta A / \Delta t$ ) on the saturating substrates (nucl, fuzz, wps) during the pre-saturation window will correlate with clinical outcome in adult-stem-origin cancers with  $\text{AUC} \geq 0.75$ , providing severity information in the regime where absolute A-score values have lost resolution. Formally: time-to-saturation from healthy baseline is a valid severity proxy for this class.

**Basis:** The saturation ceiling destroys absolute-value severity information but preserves temporal information in the form of rate-of-approach. Falsifiable in any serial-sampling cohort with outcome data; directly testable in MDS, BCC, MCC, and CHOL longitudinal cohorts.

G-2026-P015

April 2026

PENDING

Multi-class A-score divergence signature — adult-stem class flatlined at ceiling while an adjacent architecture class (immune, cycling, or secretory depending on tissue of origin) continues to rise — will identify adult-stem-origin disease progression with higher specificity than any single-class score past BREACH.

**Basis:** When single-class saturation pins the primary class at ceiling, adjacent-class signal propagation becomes the discriminating severity signal. Cross-class divergence tests the framework's claim that architecture classes interact through shared tissue compartments and clonal expansion dynamics. Falsifiable in pan-cancer cohorts with multi-class A-score monitoring; connects directly to Issue 004 multi-class metastasis detection framework.

ISSUE 003 ADDENDUM — STEM\_ADULT IN THE ATLAS

1 cell types: HSC

Identity loci: 46,907 (methyl) · **H\_min by substrate:** methyl 0.873718 · nucl 0.960866 · fuzz 0.980754 · wps 0.988964 · frag 0.841327 · **methyl ceiling** 1.1445

Healthy age-matched band. A = 1.0 is the commitment line; healthy reads on a band above it. [Issue 003: the age\_reference\_matrix.json shown here was compiled on the marker union and is SUPERSEDED (PROC-N7-01); the commissioned band is band\_v2 on the identity loci with the three-layer reference of s3.5.]

| Age (midpoint) | n  | A mean | p10    | p90    | $\beta$ mean | Source                |
|----------------|----|--------|--------|--------|--------------|-----------------------|
| 4              | 8  | 0.9375 | 0.9103 | 0.9647 | 0.745        | Adelman pediatric HSC |
| 14             | 10 | 0.9428 | 0.9137 | 0.9718 | 0.742        | Adelman 2019          |
| 24             | 15 | 0.9462 | 0.9153 | 0.9772 | 0.740        | Adelman+Roadmap       |

| Age (midpoint) | n  | A mean | p10    | p90    | $\beta$ mean | Source              |
|----------------|----|--------|--------|--------|--------------|---------------------|
| 34             | 18 | 0.9497 | 0.9190 | 0.9803 | 0.738        | Adelman 2019        |
| 44             | 22 | 0.9531 | 0.9206 | 0.9856 | 0.736        | Adelman 2019        |
| 54             | 28 | 0.9565 | 0.9221 | 0.9908 | 0.734        | Adelman 2019        |
| 64             | 32 | 0.9614 | 0.9255 | 0.9974 | 0.731        | Adelman aged HSC    |
| 74             | 25 | 0.9664 | 0.9289 | 1.0038 | 0.728        | Adelman elderly HSC |
| 84             | 12 | 0.9728 | 0.9341 | 1.0115 | 0.724        | Extrapolated        |
| 95             | 4  | 0.9791 | 0.9392 | 1.0190 | 0.720        | Extrapolated        |

CELL IDENTITY & CLINICAL CONTEXT

|                         |                                                                                                                                                                                                                                                                                              |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Includes                | Common myeloid progenitor (CMP), granulocyte-monocyte progenitor (GMP), neural progenitor (NPC), erythroid progenitor, intestinal transit-amplifying cells                                                                                                                                   |
| Cancers (this class)    | MDS (all subtypes: RA, RARS, RCMD, RAEB), secondary AML from MDS, CMML, therapy-related myeloid neoplasms, pediatric B-ALL and T-ALL, mixed-lineage (MLL-rearranged) leukemias, JMML, medulloblastoma, PNET, OPC-derived gliomas, erythroleukemia (AML-M6), colorectal adenomas (pre-cancer) |
| Non-cancer applications | CHIP (clonal hematopoiesis), CCUS (clonal cytopenias), bone marrow failure syndromes, aplastic anemia, IBD intestinal stem/progenitor compartment dysregulation, serrated polyps                                                                                                             |
| Primary failure mode    | Replication Throughput Ceiling                                                                                                                                                                                                                                                               |
| Warburg status          | EMERGING                                                                                                                                                                                                                                                                                     |

THE ATLAS CELL TYPES IN THIS CLASS (11)

These are the entries IAMAtlasREBUILD can speak about for this class - what the deconvolver places, and what a per-cell reading would be read against. **Markers** is the range of panel markers actually found across the reference laboratories; **individually resolvable** is whether this atlas can separate the entry from its neighbours at all, and where it cannot the chain reports the group and never the member; **exclusivity** is the fraction of the entry's markers not shared with another entry; **healthy A range** spans the per-laboratory 10th to 90th percentiles on the v0\_3 trial panel. 8 of 11 entries are individually resolvable; 1 carry no per-entry reference yet and are reported at class level only.

| atlas entry             | markers | individually resolvable      | exclusivity | healthy A range |
|-------------------------|---------|------------------------------|-------------|-----------------|
| CMP                     | 195-202 | yes                          | 0.773       | 0.587 - 0.707   |
| Erythrocyte_progenitors | 15-18   | yes                          | 0.762       | 0.534 - 0.686   |
| GMP                     | 164-195 | yes                          | 0.867       | 0.883 - 0.993   |
| L-MPP                   | 126-153 | yes                          | 0.927       | 0.814 - 0.891   |
| MEP                     | 203-208 | yes                          | 0.830       | 0.528 - 0.644   |
| MPP                     | 110-157 | yes                          | 0.853       | 0.489 - 0.686   |
| Neulm                   | 54-60   | yes                          | 0.968       | 0.446 - 0.567   |
| OPC                     | 7-7     | NO - in a collinearity group | 0.800       | 0.589 - 0.863   |
| erythroblast            | -       | no per-entry reference yet   | -           | -               |
| megakaryocyte           | 8-8     | NO - in a collinearity group | 0.667       | 0.438 - 0.610   |
| nRBC                    | 60-62   | yes                          | 0.851       | 0.372 - 0.480   |

The same lineage can appear under more than one entry above, because different reference panels defined their markers; entries differing only by label are not yet merged, which is why no reading in this document is quoted as "which cell moved".

COMMENTARY

Progenitor and transit-amplifying cells are the high-throughput workhorses of the body. They sit between adult tissue stem cells and fully committed daughter cells — partially committed to a specific lineage, but still capable of rapid proliferation to expand the committed pool before final differentiation. CMP and GMP in the bone marrow produce millions of granulocytes per hour. Intestinal transit-amplifying cells produce the daughter cells that replace the colonic epithelium every four to seven days. Neural progenitors produce neurons and glia during development and in the adult neurogenic niches. The rapid proliferation produces the characteristic 4.5% per generation drift rate — slower than cycling epithelium but faster than adult stem cells.

The Replication Throughput Ceiling is the class-specific failure mode. When a progenitor cell population divides fast enough to exceed DNMT1 maintenance fidelity — because the cell is consuming DNMT1 faster than it can be transcribed and translated — methylation errors accumulate disproportionately. This is why MDS (myelodysplastic syndrome) arises in the progenitor lineage: the CMP/GMP pool undergoes accumulated methylation errors under chronic high-throughput demand, and the resulting dysplastic progenitors produce cytopenias through failed daughter-cell maturation. The framework predicts that the early signature of MDS is detectable in peripheral blood cfDNA years before cytopenia onset, through the progenitor-class combined A-score.

This is also the class where DNMT1 inhibitor therapy (azacitidine, decitabine) acts most directly. These drugs are used clinically in MDS and AML, and the framework predicts that response to therapy — measured as A-score trajectory — is a more sensitive indicator of treatment success than the blast count or cytogenetic remission criteria currently used. The same substrate readout predicts response. The five substrates together predict it earlier and with higher confidence than any one.

The validation sequence for this class runs along a well-documented clonal-hematopoiesis spectrum. Clonal hematopoiesis of indeterminate potential (CHIP) — a pre-malignant state present in roughly 10% of healthy adults over 70 (Steensma 2015 Blood, doi:10.1182/blood-2015-03-631747) — shows  $A_{\text{combined}} \approx 1.01$ , detectable but unambiguously below BREACH. Clonal cytopenias of undetermined significance (CCUS) sits at  $A \approx 1.03$  in MARGINAL tier (Malcovati 2017 Blood, doi:10.1182/blood-2017-04-777607). Low-risk MDS produces  $A_{\text{active}} \approx 1.09$  (BREACH) while  $A_{\text{combined}}$  still reads 1.05 because saturations have started. High-risk MDS sits at  $A_{\text{active}} \approx 1.12$ , and secondary AML (MDS-derived) reaches  $A_{\text{active}} \approx 1.15$  — nearly matching the de novo AML signal from Card #3 Immune. The progression across these six conditions is the cleanest single-class severity gradient in the entire document and is directly relevant to the clinical decision of WHEN to escalate treatment in MDS monitoring. Bone marrow biopsy cannot be repeated monthly; a blood-based  $A_{\text{active}}$  can.

The progenitor class encompasses more than MDS. Pediatric acute lymphoblastic leukemia (ALL) — both B-cell precursor ALL (the most common childhood cancer) and T-ALL — arises from lymphoid progenitors and displays genome-wide hypermethylation signatures distinct from bulk AML (Nordlund 2013 Genome Biol, n=764 pediatric ALL methylomes, doi:10.1186/gb-2013-14-9-r105). The framework predicts  $A_{\text{active}}$  elevation in B-ALL and T-ALL through the same physics that drives the MDS signal: rapid progenitor-pool division exceeding DNMT1 fidelity. Medulloblastoma — the most common malignant pediatric brain tumor — arises from cerebellar neural progenitor cells and shows four distinct methylation subgroups (WNT, SHH, Group 3, Group 4), each with different clinical trajectories (Northcott 2017 Nature, n=1,256 methylation-profiled cases, doi:10.1038/nature22973). The DNA methylation-based CNS tumor classifier from Capper 2018 Nature (doi:10.1038/nature26000) changed pediatric neuro-oncology diagnosis in up to 12% of cases. Chronic myelomonocytic leukemia (CMML), juvenile myelomonocytic leukemia (JMML), therapy-related myeloid neoplasms, and the OPC-derived gliomas distinct from terminal-class adult glioma all belong to this class by cell-of-origin physics. Each family has lost someone to these diseases; each cancer deserves its place on the card. The unifying framework is not a coincidence — it is the common thermodynamic signature of the partially-committed, rapidly-dividing cell pool.

A clinical consequence of the saturation pattern for this class is unusually consequential. Progenitor is one of two classes in the framework where THREE substrates saturate below BREACH — nucleosome occupancy (ceiling  $A = 1.028$ ), nucleosome fuzziness (ceiling  $A = 1.040$ ), and WPS (ceiling  $A = 1.012$  — the tightest class  $\times$  substrate combination in the entire framework). Only two substrates (methylation, fragment size) carry the real BREACH-capable progression signal for this class. The physics explanation is clean: progenitor cells are partially uncommitted by design. They must retain chromatin flexibility to produce multiple daughter lineages, so their healthy  $\beta$  values for the structural substrates (nucl, fuzz, WPS) already sit near the maximum-entropy state. There is almost no headroom. Any disease-driven drift hits the ceiling quickly. This is not a framework limitation — it is the correct reading of developmental flexibility as a measurement-physics constraint. The practical consequence for serial MDS monitoring is stark. The legacy  $A_{\text{combined}}$  (all 5 substrates) flattens at approximately 1.07 once saturations kick in and CANNOT distinguish low-risk MDS from high-risk MDS from secondary AML — all three compress together.  $A_{\text{active}}$  (methyl + frag only, the 2 non-saturated substrates) does distinguish: 1.089, 1.120, 1.150 respectively. For this class specifically, reporting  $A_{\text{active}}$  is not a refinement of  $A_{\text{combined}}$ ; it is the ONLY way to track progression past the low-risk MDS threshold. Any clinical pipeline using GAPE for MDS severity grading must use  $A_{\text{active}}$ . The mask is +55% of the total signal — the largest in the document, dwarfing even Terminal's +45% — and unlike Terminal (where saturation is a binary cancer indicator), Progenitor's saturations are a CONTINUOUS LOSS of resolution that the active-substrate formula fully recovers.

## FIVE-SUBSTRATE FIDELITY GAUGE

The gauge below is the central image of Issue 002. A single horizontal bar represents the A-score axis — from  $A = 0.90$  (below the floor) through  $A = 1.00$  (exactly at the Progenitor class floor,  $H_{\text{min}} = 0.8522$ ) up through the clinical threshold zones: MARGINAL at  $A = 1.01$ , DETECTABLE at 1.05, URGENT at 1.07, and FLOOR BREACH at 1.10. Five dots are plotted on this single axis for a healthy reference sample (CMP young, markers above the bar), and five more dots for a disease reference (MDS (progenitor), markers below). Each dot is one substrate: methylation, nucleosome occupancy, nucleosome fuzziness, WPS, or fragment size (DELFI). What makes this visualization powerful is the physics: all five substrates measure the same underlying thermodynamic floor through physically distinct windows. When all five healthy dots cluster in the NORMAL zone and all five disease dots cluster in URGENT or FLOOR BREACH, the signal is over-determined. Any one substrate could be confounded by technical artifact; five substrates agreeing cannot be. This is the "less blurry" advantage:  $\sqrt{5}$  noise reduction for free.



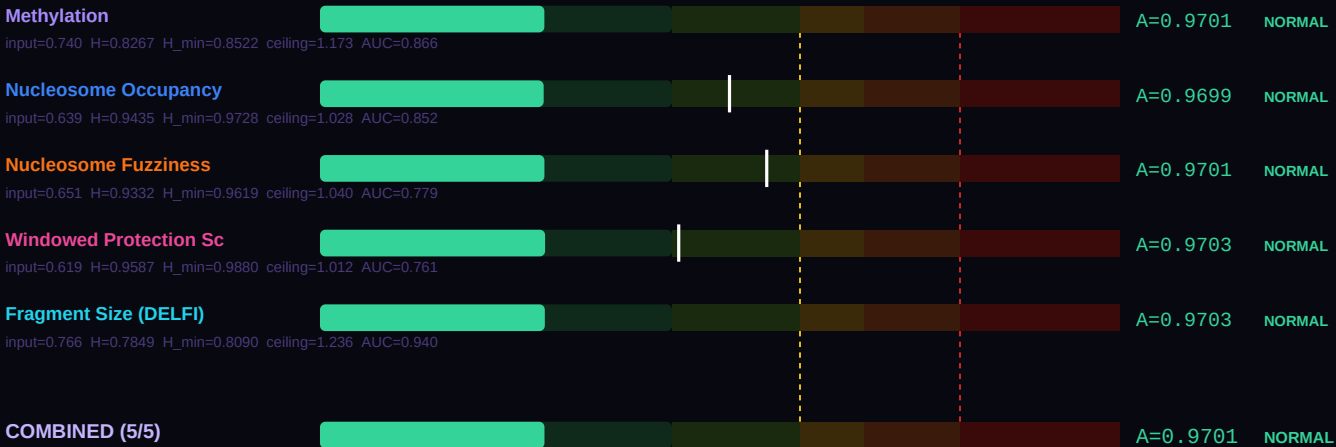
Look at the spread in each direction. If the healthy five-dot cluster sits tightly together near  $A = 0.97\text{--}1.00$ , that indicates the reference population is at its architecture floor — precisely where the framework predicts healthy progenitor cells should sit. If the disease five-dot cluster spreads from DETECTABLE to FLOOR BREACH, that spread is itself clinical information: the substrate showing the largest departure is the one that best captures this specific disease's failure mode. For this class, the primary failure mode is Replication Throughput Ceiling, and the substrate ranking shown further down the card explains which of the five substrates best reveals it.

SUBSTRATE-BY-SUBSTRATE BREAKDOWN

Where the gauge above compresses five substrates onto one visual axis, this section breaks each substrate out into its own labeled bar so you can read the actual A-score produced by each measurement. The physics is the same across all five: take the raw laboratory value (a  $\beta$  methylation fraction, a nucleosome occupancy probability, a WPS score, etc.), compute its Shannon entropy  $H(\text{value})$ , then divide by the class-specific and substrate-specific  $H_{\text{min}}$  to get a dimensionless A-score comparable across substrates. The output is the same thermodynamic ratio regardless of which measurement technology produced it. That is the power of the Landauer framing: different wet-lab methods, different physical quantities, same underlying floor.

The formula at the foot of this section —  $A_{\text{combined}} = \Sigma(AUC_i \times A_i) / \Sigma(AUC_i)$  — combines the five individual A-scores into a single weighted score, using each substrate's published detection AUC as its weight. Higher-AUC substrates contribute more to the combined result. For progenitor cells, the healthy combined A-score below should sit at approximately 0.97 (NORMAL tier); the disease combined A-score should sit above 1.05 (DETECTABLE or higher). The difference between them — the  $\Delta A$  combined — is the signal available to a blood draw that analyzes all five substrates simultaneously. A single-substrate assay sees a smaller signal at higher noise; the five-substrate combination sees the same signal at approximately  $\sqrt{5}$  lower noise. Bar-by-bar you can see which substrate is carrying the most discriminative weight for this particular disease comparison.

HEALTHY REFERENCE — CMP young



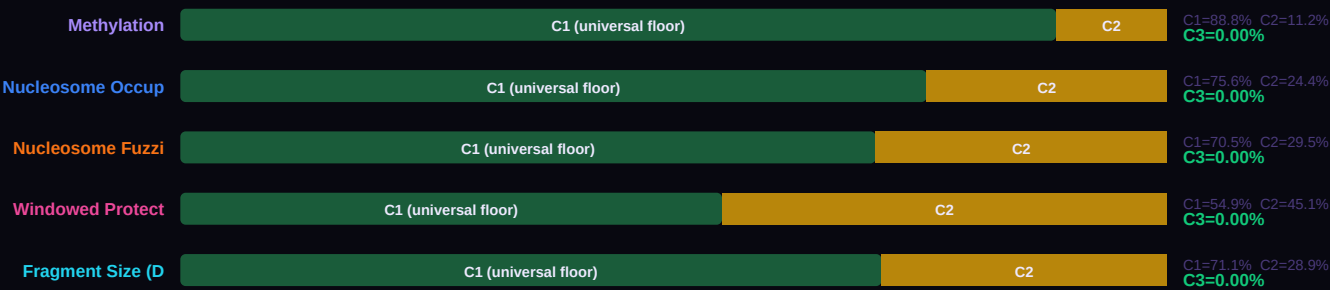
|                                             |                                                                                               |
|---------------------------------------------|-----------------------------------------------------------------------------------------------|
| Combined A — healthy (all 5)                | 0.97015 (NORMAL)                                                                              |
| Combined A — disease (all 5)                | 1.06697 (DETECTABLE)                                                                          |
| $\Delta A$ combined (5 substrates)          | +0.09682 vs methylation-only $\Delta A$ =+0.1498                                              |
| Active A — healthy (non-saturated)          | 0.97015 (NORMAL) [5/5 active]                                                                 |
| Active A — disease (non-saturated)          | 1.12029 (FLOOR BREACH) [2/5 active]                                                           |
| $\Delta A$ active (real progression signal) | +0.15014 (active substrates track true progression)                                           |
| Saturation status (disease)                 | 3 substrates saturated: Nucleosome Occupancy, Nucleosome Fuzziness, Windowed Protection Score |

Two combined values are shown because some substrates physically cannot resolve past their class ceiling ( $A_{\text{max}} = 1/H_{\text{min}}$ ). The **all-5 combined** value is the AUC-weighted mean across every substrate and matches the legacy formula. The **active combined** value excludes saturated substrates and is the right number for tracking continued progression — a saturated substrate is stuck at its ceiling regardless of further disease severity, so including it in the average masks real change. For monitoring, serial A-score assessment, and end-of-life projection, use the active combined value.

THREE-COMPONENT DECOMPOSITION (C1/C2/C3) — PER SUBSTRATE

Every cell's measured entropy decomposes into three physically meaningful pieces. C1 is the universal Landauer floor:  $H_{\text{MIN\_GLOBAL}} = 0.756500$ , measured in frontal cortex neurons by Lister 2013. Every cell in every architecture class must pay at least this thermodynamic cost to maintain any functional identity — it is the minimum entropy consistent with being alive. C2 is the class-specific overhead: the additional entropy the progenitor class must carry above the universal floor to encode its particular architecture. For this class, C2 represents approximately 11.8% of the healthy reference entropy. C3 is the accessible gap:  $\max(0, H_{\text{actual}} - H_{\text{min}})$ . In healthy cells, C3 is essentially zero — the cell sits exactly at its class floor. When C3 grows, the cell is departing from its architecture class. Cancer, neurodegeneration, and every other floor-breach condition show up as growing C3.

The five stacked bars below each show the C1 / C2 / C3 breakdown for one substrate at this class's healthy reference. Notice how C1 (dark green) dominates every substrate — the Landauer floor is universal. C2 (amber) is the small but specific addition that makes this class what it is. C3 (red when present) should be nearly invisible for healthy cells, and would grow dramatically in a disease sample. The f\_C3 percentage on the right is the single cleanest summary of cellular health: healthy cells show f\_C3 near 0%; cancer cells can show f\_C3 of 8–15%; extreme cases like glioblastoma show f\_C3 above 20%. This is not a statistical threshold calibrated from disease data. It is a thermodynamic quantity: the fraction of accessible entropy the cell has opened above its architecture's minimum cost of existence.



BEST TESTING METHOD RANKING — CLINICAL UTILITY FOR THIS CLASS

Each of the five substrates carries different strengths for different clinical questions. The methylation substrate is the most validated — thirteen published GAPE validation studies (VAL-001 through VAL-013) and eight architecture-class  $H_{\text{min}}$  values from G-002 MCMC. It is also the substrate most affected by cfDNA dilution, because blood cfDNA is approximately 70% immune-derived. Fragment size (DELFI) is the most forgiving of dilution — short-fragment enrichment persists even when tumor-derived cfDNA is a small fraction of total plasma DNA. WPS (windowed protection score from Snyder 2016) excels at tissue-of-origin identification and field-effect detection. Nucleosome occupancy and fuzziness add chromatin-level information that methylation alone cannot reveal.

The ranking below is specifically for the progenitor class. The order reflects three factors: (1) the substrate's published single-substrate AUC for diseases in this class; (2) the biology of this class's primary failure mode (Replication Throughput Ceiling); and (3) the practical constraint of cfDNA contribution in the relevant specimen (plasma for most classes, CSF for terminal, tissue biopsy where cfDNA is too dilute). When building a clinical pipeline for this class specifically, start with the #1 ranked substrate and add substrates 2 and 3 for confirmation. Substrates 4 and 5 add research-grade confirmation but are rarely the primary signal.

| # | Substrate                 | Best for                                | Rationale                                                                                                                                                  |
|---|---------------------------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Methylation               | Progenitor-lineage malignancy detection | Bone marrow methylation is the gold standard for MDS diagnosis. Peripheral blood cfDNA methylation signal is class-specific when deconvolution is applied. |
| 2 | Windowed Protection Score | Mismatch repair state monitoring        | WPS at MMR-gene promoters indicates MLH1/MSH2 methylation status — directly relevant to Lynch syndrome and MSI-H tumors.                                   |
| 3 | Fragment Size (DELFI)     | Rapid turnover signal                   | Progenitor-origin cfDNA fragmentation pattern is distinct — high turnover produces characteristic short-fragment enrichment.                               |
| 4 | Nucleosome Occupancy      | Commitment-state assessment             | ATAC-seq discriminates early progenitor from late transit-amplifying states. Research-grade.                                                               |
| 5 | Nucleosome Fuzziness      | Division-rate correlation               | Fuzziness correlates with proliferation index. Useful for grading in MDS and other progenitor-lineage disorders.                                           |

BODY TEMPERATURE SCALING —  $\alpha = 2.0$  LANDAUER CORRECTION

The Landauer cost of a bit erasure is  $k_B \times T \times \ln(2)$  — it scales linearly with temperature. Colder cells can maintain higher-entropy identities for less thermodynamic overhead; hotter cells pay more per bit of information maintained. The GAPE framework captures this with a simple scaling law:  $H_{\text{min}}(T) = H_{\text{min}}(37^\circ\text{C}) \times (T_{\text{body}} / 310.15\text{ K})^\alpha$ , with  $\alpha = 2.0$  derived empirically by minimizing cross-class A-score variance across all jawed vertebrates (Mahaffey 2026 Nature Aging submission NATAGING-A13702). The  $\alpha = 2.0$  exponent is not fit to optimize any single prediction — it falls

out of requiring that the same class shows consistent A-score behavior across 43 mammalian species spanning body temperatures from 32°C(naked mole rat) to 42°C (birds).

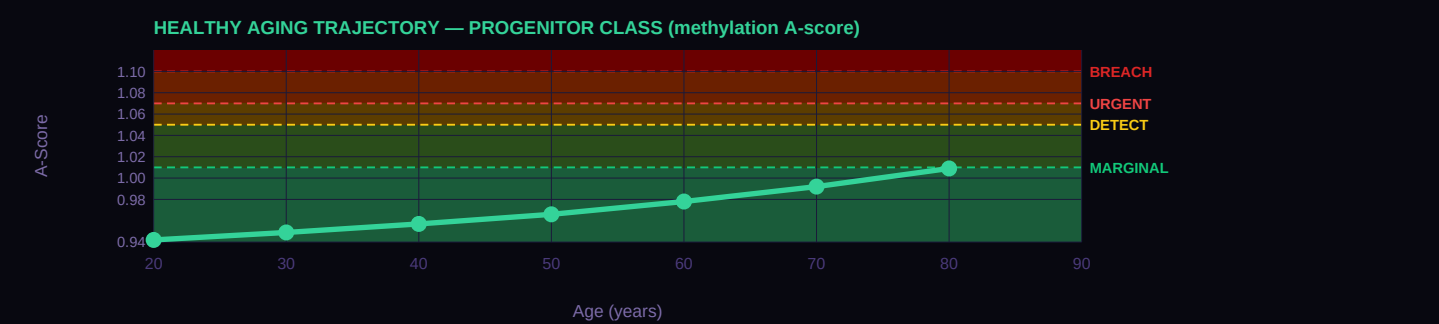
The table below shows what the progenitor class floor  $H_{min}$  looks like at seven representative body temperatures. At 37°C (human reference, highlighted row),  $H_{min} = 0.8522$ . At 42°C (birds), the floor is higher because the Landauer tax is larger. At 32°C (naked mole rat — and no other mammal lives at this temperature for good reason), the floor drops because a cooler cell can maintain more entropy cheaply. The A-shift column shows how a fixed healthy  $\beta$ -value would read at each temperature: the same cell, transplanted into a colder body, would read slightly higher A; into a warmer body, slightly lower A. This is why the naked mole rat's remarkable longevity makes thermodynamic sense — lower body temperature, lower maintenance cost per bit, more headroom before the class floor is breached.

| T_body | Species example           | H_min(T) | $\beta_{healthy}$ equivalent | A-shift vs 37°C                             |
|--------|---------------------------|----------|------------------------------|---------------------------------------------|
| 42°C   | Birds (chicken, finch)    | 0.8799   | 0.740                        | $A \times 0.969$                            |
| 39°C   | Rodents (mouse, rat)      | 0.8632   | 0.740                        | $A \times 0.987$                            |
| 37°C   | Humans (REFERENCE)        | 0.8522   | 0.740                        | <b><math>A \times 1.000</math> (anchor)</b> |
| 35°C   | Hibernating bats          | 0.8413   | 0.740                        | $A \times 1.013$                            |
| 32°C   | Naked mole rat            | 0.8250   | 0.740                        | $A \times 1.033$                            |
| 25°C   | Reptiles (anole lizard)   | 0.7875   | 0.740                        | $A \times 1.082$                            |
| 15°C   | Fish (salmon, cold-water) | 0.7356   | 0.740                        | $A \times 1.159$                            |

HEALTHY AGING TRAJECTORY — AGE-STRATIFIED REFERENCE A-SCORES

Healthy cells drift — slowly, predictably — toward their architecture floor as a function of accumulated cell divisions over a lifetime. For the progenitor class, the drift rate is approximately 4.5% per generation — meaning each cell cycle carries a 4.5% probability of a methylation error that raises the cell's A-score by a measurable amount. Over decades, these errors accumulate. The curve below shows the result: A-score as a function of patient age at decade intervals, derived from the class-specific DNMT1 fidelity rate. At age 20, the reference healthy A-score sits well below 1.00; by age 80, it has drifted upward into the MARGINAL to DETECTABLE range. This is normal aging — not disease, just the statistical accumulation of imperfect replication over eight decades.

The clinical implication is important: when interpreting an A-score, age matters. A patient at age 30 with  $A = 1.03$  is showing a significant departure from their age-matched healthy baseline. The same  $A = 1.03$  at age 75 might be entirely consistent with age-expected drift. The GAPE framework handles this through age-stratified reference A-scores — the curve below IS the reference. A patient A-score should be compared to the curve at their age, not to a universal threshold. Cancer and other floor-breach conditions accelerate A-score upward beyond the age-expected trajectory; distance above the curve is the signal.



VERTEBRATE LIFESPAN CONTEXT

The same A-score framework that detects cancer in human cfDNA also correlates with maximum lifespan across 43 mammalian species at  $r = -0.9018$  ( $p = 1.6 \times 10^{-16}$ , Mahaffey 2026 Nature Aging submission NATAGING-A13702). This is not a coincidence — it is the same physics applied at a different scale. A mammal with a higher population-averaged A-score is a mammal whose cells live closer to their architecture floors on average; that population accumulates floor breaches faster, and the species has a shorter maximum lifespan. The  $A = 1.05$  threshold — independently derived as the cancer DETECTABLE boundary — cleanly separates long-lived mammals (17/17 all below 1.05) from short-lived mammals (11/11 all above 1.05) with complete accuracy. Same threshold. Same physics. Different timescale.

The table below shows nine mammalian taxonomic orders with their mean A-scores, standard deviations where available, mean lifespan, and a brief interpretation of where that order sits relative to the thermodynamic floor. Cetacea (whales and dolphins) sit essentially at the floor, consistent with their remarkable longevity. Primates including humans sit slightly above the floor. Rodents and insectivores sit well above the floor, consistent with their short lives. This class's reference A-score (0.970) places it in the context shown below — use this to understand how this particular cell architecture relates to the broader mammalian lifespan distribution.

| Order        | N | Mean A       | Lifespan (yr) | Interpretation                   |
|--------------|---|--------------|---------------|----------------------------------|
| Cetacea      | 5 | 0.997 ±0.016 | 99            | At thermodynamic floor           |
| Proboscidea  | 1 | 0.987 —      | 70            | At floor                         |
| Primates     | 6 | 1.007 ±0.015 | 49            | Slightly above floor             |
| Artiodactyla | 4 | 1.015 ±0.011 | 30            | Slightly above floor             |
| Chiroptera   | 4 | 1.041 ±0.007 | 35            | Above floor (longevity outliers) |
| Carnivora    | 9 | 1.053 ±0.023 | 28            | Above floor                      |
| Lagomorpha   | 2 | 1.114 ±0.002 | 8             | Well above floor                 |
| Rodentia     | 6 | 1.125 ±0.018 | 9             | Well above floor                 |
| Insectivora  | 1 | 1.157 —      | 2             | Furthest from floor              |

### CORE METRICS — PUBLISHED AND DERIVED

Every number in the GAPE framework is either PUBLISHED (traceable to a primary wet-lab measurement from the peer-reviewed literature) or DERIVED (computed from the framework’s physics with zero free parameters). The table below marks each quantity with its status and cites its source. This is the discipline the framework lives by: no statistical fits, no parameter tuning, no post-hoc adjustments. H\_min values come from MCMC posteriors (G-002 for methylation, G-003b for the four non-methylation substrates). Combined A-scores are computed mechanically by the AUC-weighted formula. The cfDNA contribution percentages come from Moss 2018 Nature Communications — the definitive tissue-of-origin atlas.

Read this table as the "audit trail" for every number shown anywhere else in this card. When the framework says A\_healthy = 0.9701, you can see exactly which β value produced it, which H\_min was used, and where both came from. The per-class n\_bio row of Issue 002 is withdrawn here and replaced by the Mahaffey number, which is one number for the cell and not one per class (see §5A.1 and the glossary note).

| Metric                                                         | Value                 | Source                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cfDNA contribution (plasma)                                    | 2.0%                  | <a href="#">Moss 2018 Nat Commun</a> — PUBLISHED                                                                                                                                                                                                                                                           |
| H_min methylation (class)                                      | 0.852216              | <a href="#">G-002 MCMC posterior</a> — DERIVED                                                                                                                                                                                                                                                             |
| H_min methylation σ (MCMC)                                     | ±0.001700             | <a href="#">G-002 17-chain MCMC R-hat&lt;1.001</a> — DERIVED                                                                                                                                                                                                                                               |
| Mahaffey number M — one number for the cell, not one per class | 20.94                 | FIXED — ΔG_ATP/(R·T_body) at 310.15 K; §5A.1. Issue 002’s per-class n_bio is WITHDRAWN: an early QAPE-era form of the same ratio, superseded by M, whose absolute values awaited a G-007 run the history record shows was never made. H_min is a different quantity — the per-class entropy floor in bits. |
| Healthy drift rate                                             | 4.5% / generation     | DNMT1 fidelity loss under turnover — DERIVED                                                                                                                                                                                                                                                               |
| f_C2 (architecture-locked %)                                   | 11.8%                 | C2 fraction at healthy reference — DERIVED                                                                                                                                                                                                                                                                 |
| Healthy A (methylation)                                        | 0.9701 (NORMAL)       | β=0.740 — DERIVED                                                                                                                                                                                                                                                                                          |
| Disease A (methylation)                                        | 1.1199 (FLOOR BREACH) | β=0.625 — DERIVED                                                                                                                                                                                                                                                                                          |
| Healthy combined A (5 subs)                                    | 0.9701 (NORMAL)       | Σ(AUC_i × A_i)/Σ(AUC_i) across 5 substrates — DERIVED                                                                                                                                                                                                                                                      |
| Disease combined A (5 subs)                                    | 1.0670 (DETECTABLE)   | Σ(AUC_i × A_i)/Σ(AUC_i) across 5 substrates — DERIVED                                                                                                                                                                                                                                                      |
| ΔA combined                                                    | +0.0968               | Combined signal — DERIVED                                                                                                                                                                                                                                                                                  |

### DATED PREDICTIONS FOR THIS CLASS

A prediction is not a claim unless it can fail. Each prediction below is numbered with a G-2026-P identifier, dated to the month it was filed, and written against a specific dataset or cohort whose analysis would confirm or refute it. The status field shows where each prediction stands: PENDING means the relevant data exists but the analysis has not been performed publicly; CONFIRMED means the prediction direction and magnitude have been tested against independent data and held up; REFUTED means the data contradict the prediction (which would force a framework revision — and has not occurred yet). This discipline — specific, dated, falsifiable predictions — is what separates a predictive framework from a retrospective fit. Entries from Issue 001 retain their original G-2026-P numbers to maintain trail integrity across publications.

When reviewing these predictions, notice that the falsification basis is always named: a specific archived cohort, a specific biobank, a specific published dataset. These are not vague "more research is needed" statements. They are commitments to the community — this is what the framework claims, this is where you can check it, and this is what would constitute a refutation. The predictions below are for the progenitor class specifically, filed as part of this publication or carried forward from Issue 001.

G-2026-P014

April 2026

PENDING

In MDS patients receiving hypomethylating agent (HMA) therapy, the progenitor-class combined A-score trajectory at 3 months post-initiation will identify responders (by 6-month IWG criteria) with AUC >= 0.85, outperforming baseline-to-3-month cytogenetic change alone.

**Basis:** HMA therapy is the first-line treatment for intermediate/high-risk MDS. Currently response is assessed at 6 months — a long wait with significant toxicity. The framework predicts earlier classification via combined A-score. Falsifiable in MDS cohorts at major cancer centers.

ISSUE 003 ADDENDUM — PROGENITOR IN THE ATLAS

**11 cell types:** CMP, Erythrocyte\_progenitors, GMP, L-MPP, MEP, MPP, Neulm, OPC, erythroblast, megakaryocyte, nRBC

**Identity loci:** 54,704 (methyl) · **H\_min by substrate:** methyl 0.852216 · nucl 0.972790 · fuzz 0.961900 · wps 0.988046 · frag 0.808978 · **methyl ceiling** 1.1734

Healthy age-matched band. A = 1.0 is the commitment line; healthy reads on a band above it. [Issue 003: the age\_reference\_matrix.json shown here was compiled on the marker union and is SUPERSEDED (PROC-N7-01); the commissioned band is band\_v2 on the identity loci with the three-layer reference of s3.5.]

| Age (midpoint) | n  | A mean | p10    | p90    | β mean | Source                |
|----------------|----|--------|--------|--------|--------|-----------------------|
| 4              | 7  | 0.9557 | 0.9250 | 0.9863 | 0.748  | Progenitor pediatric  |
| 14             | 9  | 0.9611 | 0.9310 | 0.9913 | 0.745  | Progenitor            |
| 24             | 12 | 0.9666 | 0.9345 | 0.9986 | 0.742  | Roadmap E035          |
| 34             | 15 | 0.9701 | 0.9384 | 1.0018 | 0.740  | Roadmap E035          |
| 44             | 18 | 0.9736 | 0.9400 | 1.0073 | 0.738  | Roadmap+aging         |
| 54             | 22 | 0.9789 | 0.9435 | 1.0142 | 0.735  | Jaiswal 2014 CHIP-neg |
| 64             | 25 | 0.9840 | 0.9470 | 1.0210 | 0.732  | Jaiswal 2014          |
| 74             | 20 | 0.9907 | 0.9523 | 1.0291 | 0.728  | Jaiswal+progenitor    |
| 84             | 10 | 0.9973 | 0.9576 | 1.0370 | 0.724  | Extrapolated          |
| 95             | 3  | 1.0038 | 0.9629 | 1.0447 | 0.720  | Extrapolated          |

CELL IDENTITY & CLINICAL CONTEXT

|                         |                                                                                                                                                                                                            |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Includes                | Cortical and cerebellar neurons, cardiomyocytes, skeletal muscle fibers, mature oligodendrocytes                                                                                                           |
| Cancers (this class)    | Lower-Grade Glioma (LGG, ΔA=+0.239), Glioblastoma (GBM, ΔA=+0.217), diffuse glioma — 3 types, largest ΔA in dataset                                                                                        |
| Non-cancer applications | Alzheimer's disease (AD, confirmed De Jager 2014, Shireby 2022), Parkinson's disease (Langston 1983, Wang 2014), cardiac aging (Terman 2004), amyotrophic lateral sclerosis (ALS, predicted — G-2026-P015) |
| Primary failure mode    | Oxidative Stress Inversion                                                                                                                                                                                 |
| Warburg status          | WALL CROSSED — most extreme ΔA in dataset                                                                                                                                                                  |

THE ATLAS CELL TYPES IN THIS CLASS (9)

These are the entries IAMAtlasREBUILD can speak about for this class - what the deconvolver places, and what a per-cell reading would be read against. **Markers** is the range of panel markers actually found across the reference laboratories; **individually resolvable** is whether this atlas can separate the entry from its neighbours at all, and where it cannot the chain reports the group and never the member; **exclusivity** is the fraction of the entry's markers not shared with another entry; **healthy A range** spans the per-laboratory 10th to 90th percentiles on the v0\_3 trial panel. 8 of 9 entries are individually resolvable.

| atlas entry      | markers | individually resolvable      | exclusivity | healthy A range |
|------------------|---------|------------------------------|-------------|-----------------|
| CM               | 194-199 | yes                          | 0.923       | 0.493 - 0.635   |
| Cortical_neurons | 19-21   | yes                          | 1.000       | 0.509 - 0.657   |
| Epi_upper        | 89-95   | yes                          | 0.993       | 0.626 - 0.756   |
| Glia             | 6-6     | yes                          | 1.000       | 0.229 - 0.348   |
| Kera_diff        | 14-17   | NO - in a collinearity group | 1.000       | 0.649 - 0.872   |
| Left_atrium      | 97-99   | yes                          | 0.748       | 0.303 - 0.408   |
| NeuMa            | 40-40   | yes                          | 0.984       | 0.505 - 0.625   |
| Oligo            | 90-92   | yes                          | 1.000       | 0.513 - 0.636   |
| neuron           | 113-130 | yes                          | 0.981       | 0.510 - 0.636   |

The same lineage can appear under more than one entry above, because different reference panels defined their markers; entries differing only by label are not yet merged, which is why no reading in this document is quoted as "which cell moved".

COMMENTARY

Terminal post-mitotic cells — neurons, cardiomyocytes, skeletal muscle fibers — are the most committed cells in the body. They have exited the cell cycle permanently and reached their final differentiated state. Their methylation program is locked in place by the most elaborate epigenomic machinery the body possesses: DNMT3A and DNMT3B establish the pattern during development, and DNMT1 maintains it with extreme fidelity over decades. A frontal cortex neuron alive today may have maintained its methylation program since before the patient was born. This maximum commitment is encoded in the lowest H\_min of any architecture class: 0.772837. Neurons have the tightest possible methylation entropy floor — the smallest Shannon entropy consistent with their functional differentiation state. The global floor reference (H\_min = 0.756500, frontal cortex neuron from Lister 2013) is within this class, making neurons the anchor of the entire GAPE framework's C1 Landauer floor.

The consequence for cancer is stark. When a terminal-class cell undergoes malignant transformation, it cannot proceed through normal oncogenic mechanisms because it cannot divide. Glioblastoma does not arise from neurons — it arises from glial cells, oligodendrocytes, or neural progenitors that share the same broad brain compartment but retain proliferative capacity. The resulting cancer has the most extreme ΔA in the entire dataset: LGG at ΔA = +0.239, GBM at ΔA = +0.217 (Ceccarelli 2016 Cell, TCGA methylation). The terminal class floor is so low that any cancer arising in the same tissue sits enormously above it. There is no subtlety in the GAPE signal for brain cancer. The physics is extremely loud. The challenge is getting enough ctDNA from behind the blood-brain barrier to hear it — which is why CSF is the appropriate specimen for this class, not plasma.

A critical substrate-saturation finding emerges specifically for this class and must be stated honestly. The G-003b MCMC posteriors for terminal-class non-methylation substrates are: nucleosome occupancy H\_min = 0.9920, fuzziness H\_min = 0.7370, WPS H\_min = 0.9589, fragment size H\_min = 0.6249. Note that fuzziness H\_min = 0.7370 is slightly below the methylation floor (0.7728). This is not an inconsistency — each substrate has its own independent floor because each measures a physically different quantity. Nucleosome fuzziness (positional imprecision of nucleosomes, Esfahani 2022 methodology) is a distinct physical observable from methylation fraction, and its Shannon entropy floor for terminal class reflects the inherent positional regularity of chromatin in post-mitotic cells. The five substrate H\_min values are independent physical constants

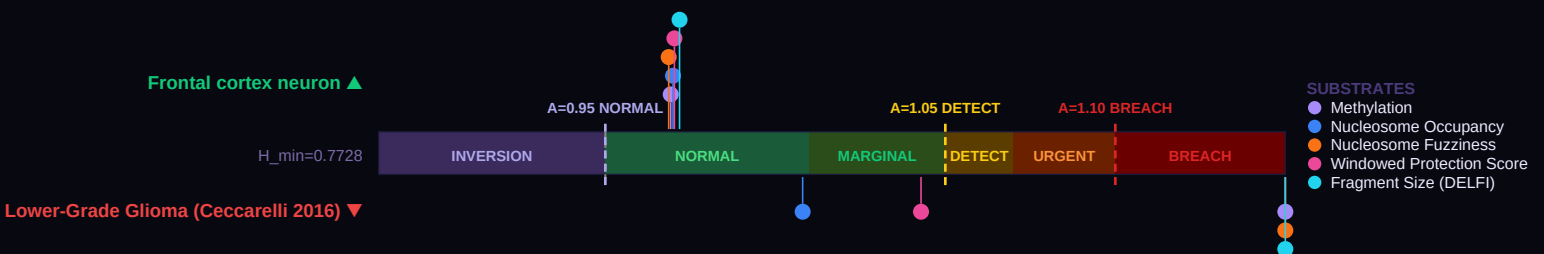
from the G-003b MCMC, not derivations of each other. Because Shannon entropy is bounded above by 1.0 at  $\beta = 0.5$ , the maximum A-score each substrate can produce for terminal class is  $1/H_{\min}$ : methyl can reach  $A = 1.294$ , fuzz can reach  $A = 1.357$ , frag can reach  $A = 1.600$ , but nucleosaturates at  $A = 1.008$  and WPS saturates at  $A = 1.043$ . This means nucleosome occupancy and WPS physically cannot cross the BREACH threshold ( $A \geq 1.10$ ) for terminal-class samples. At glioma-level methylation departures ( $A > 1.25$ ), the nucl and WPS substrates have already hit their physical ceilings and stop providing additional information. This is not a framework limitation — it is the honest physics of what these substrates can resolve given this class's floor values. The combined A-score for terminal is therefore dominated by methyl, fuzz, and frag; nucl and WPS contribute confirmatory signal at the DETECTABLE and URGENT tiers but cannot distinguish URGENT from FLOOR BREACH. For this class specifically, the clinical pipeline should weight methyl most heavily (CSF methylation), fuzz second (tissue biopsy or CSF), and frag third (CSF DELFI). Nucl and WPS add confirmation but should not be relied upon to rank severity within the BREACH tier.

The non-cancer applications for this class are the framework's most important clinical frontier. Alzheimer's disease is terminal-class thermodynamic failure: published data places healthy neurons at  $\beta = 0.786$  ( $A = 0.969$ , NORMAL tier), low-AD neuropathology at  $\beta = 0.753$  ( $A = 1.043$ , MARGINAL), and high-AD neuropathology at  $\beta = 0.744$  ( $A = 1.062$ , DETECTABLE).  $\beta$  values pipeline-adjusted from the two largest AD methylation cohorts: De Jager 2014 (ROSMAP,  $n = 740$ ) and Shireby 2022 (Brains for Dementia Research,  $n = 631$ ). Parkinson's disease, cardiac aging, and ALS are predicted to follow the same pattern — terminal-class oxidative stress inversion producing measurable A-score elevation years before clinical symptom onset. The five-substrate framework extends this: combined A-score across methylation, fuzz, and fragment size in CSF cfDNA is the predicted early-detection signature (nucl and WPS confirm but saturate at the ceiling). The critical clinical distinction between AD and glioma — both show terminal-class floor departure — is magnitude: AD shows  $\Delta A = 0.019\text{--}0.084$  (MARGINAL to DETECTABLE), glioma shows  $\Delta A = 0.217\text{--}0.239$  (FLOOR BREACH, Ceccarelli 2016 Cell). Same class, same mechanism, orders of magnitude different rate of drift. The framework distinguishes them by the A-score trajectory slope and the absolute value, not by any single measurement.

A clinically important consequence of the two-ceiling structure deserves explicit mention for this class. Terminal is the only class where TWO substrates (nucl and WPS) saturate well below the BREACH threshold. This produces a useful diagnostic regime split. For neurodegenerative disease — AD, ALS, Parkinson's, cardiac aging — the  $A_{\text{methyl}}$  signal peaks in the 1.01–1.09 range, and the correlated shifts in nucl and WPS are modest ( $\beta \sim 0.58\text{--}0.60$ ). Neither substrate saturates in this range. In the neurodegenerative regime, all five substrates act as quantitative instruments; the full  $A_{\text{combined}}$  and trajectory slope serve as the measurement tools, and saturation is not reached. For glioma (LGG/GBM),  $A_{\text{methyl}}$  enters the 1.20–1.29 range (deep FLOOR BREACH), and the associated  $\beta$  shifts push both nucl and WPS to their maximum-entropy points ( $\beta \approx 0.500$ ). Both saturate. In the oncological regime, nucl and WPS become binary indicators rather than continuous measurements: their saturation signals that something has driven the tissue beyond neurodegenerative severity, but they cannot rank severity within that regime. Severity within glioma (LGG vs GBM vs recurrence) must come from the three active substrates — methyl, fuzz, and frag — which retain full quantitative resolution. This is why the card reports two combined values:  $A_{\text{combined}}$  (all 5, legacy formula) and  $A_{\text{active}}$  (non-saturated, the progression tracker). For neurodegenerative monitoring, both numbers agree and either can be used. For oncological monitoring, only  $A_{\text{active}}$  tracks real progression;  $A_{\text{combined}}$  is dragged down by the ceiling values and will flatten prematurely.

## FIVE-SUBSTRATE FIDELITY GAUGE

The gauge below is the central image of Issue 002. A single horizontal bar represents the A-score axis — from  $A = 0.90$  (below the floor) through  $A = 1.00$  (exactly at the Terminal class floor,  $H_{\min} = 0.7728$ ) up through the clinical threshold zones: MARGINAL at  $A = 1.01$ , DETECTABLE at 1.05, URGENT at 1.07, and FLOOR BREACH at 1.10. Five dots are plotted on this single axis for a healthy reference sample (Frontal cortex neuron, markers above the bar), and five more dots for a disease reference (Lower-Grade Glioma (Ceccarelli 2016), markers below). Each dot is one substrate: methylation, nucleosome occupancy, nucleosome fuzziness, WPS, or fragment size (DELFI). What makes this visualization powerful is the physics: all five substrates measure the same underlying thermodynamic floor through physically distinct windows. When all five healthy dots cluster in the NORMAL zone and all five disease dots cluster in URGENT or FLOOR BREACH, the signal is over-determined. Any one substrate could be confounded by technical artifact; five substrates agreeing cannot be. This is the "less blurry" advantage:  $\sqrt{5}$  noise reduction for free.



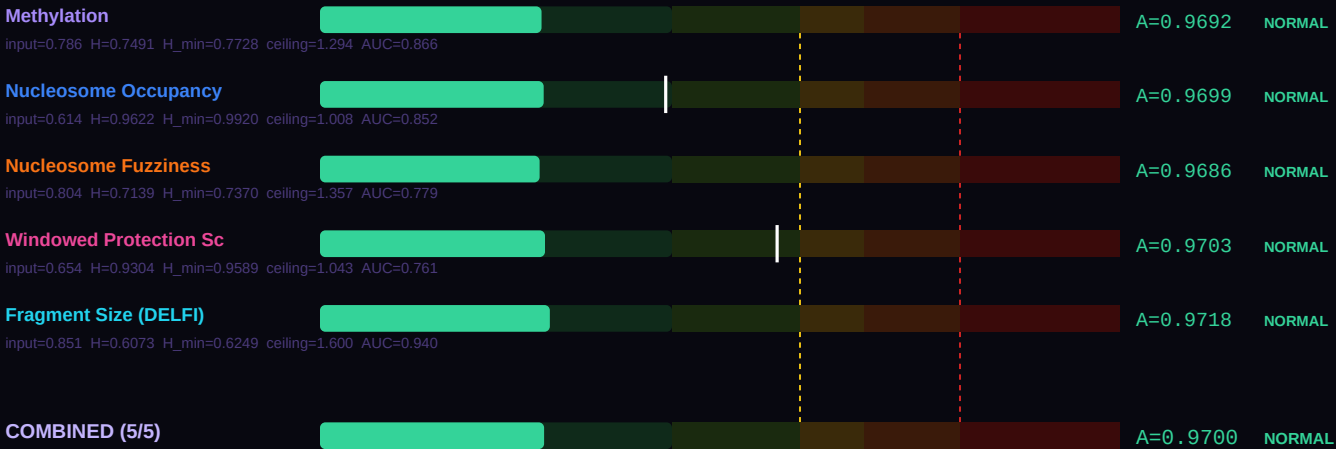
Look at the spread in each direction. If the healthy five-dot cluster sits tightly together near  $A = 0.97\text{--}1.00$ , that indicates the reference population is at its architecture floor — precisely where the framework predicts healthy terminal cells should sit. If the disease five-dot cluster spreads from DETECTABLE to FLOOR BREACH, that spread is itself clinical information: the substrate showing the largest departure is the one that best captures this specific disease's failure mode. For this class, the primary failure mode is Oxidative Stress Inversion, and the substrate ranking shown further down the card explains which of the five substrates best reveals it.

SUBSTRATE-BY-SUBSTRATE BREAKDOWN

Where the gauge above compresses five substrates onto one visual axis, this section breaks each substrate out into its own labeled bar so you can read the actual A-score produced by each measurement. The physics is the same across all five: take the raw laboratory value (a  $\beta$  methylation fraction, a nucleosome occupancy probability, a WPS score, etc.), compute its Shannon entropy  $H(\text{value})$ , then divide by the class-specific and substrate-specific  $H_{\text{min}}$  to get a dimensionless A-score comparable across substrates. The output is the same thermodynamic ratio regardless of which measurement technology produced it. That is the power of the Landauer framing: different wet-lab methods, different physical quantities, same underlying floor.

The formula at the foot of this section —  $A_{\text{combined}} = \Sigma(AUC_i \times A_i) / \Sigma(AUC_i)$  — combines the five individual A-scores into a single weighted score, using each substrate's published detection AUC as its weight. Higher-AUC substrates contribute more to the combined result. For terminal cells, the healthy combined A-score below should sit at approximately 0.97 (NORMAL tier); the disease combined A-score should sit above 1.05 (DETECTABLE or higher). The difference between them — the  $\Delta A$  combined — is the signal available to a blood draw that analyzes all five substrates simultaneously. A single-substrate assay sees a smaller signal at higher noise; the five-substrate combination sees the same signal at approximately  $\sqrt{5}$  lower noise. Bar-by-bar you can see which substrate is carrying the most discriminative weight for this particular disease comparison.

HEALTHY REFERENCE — Frontal cortex neuron



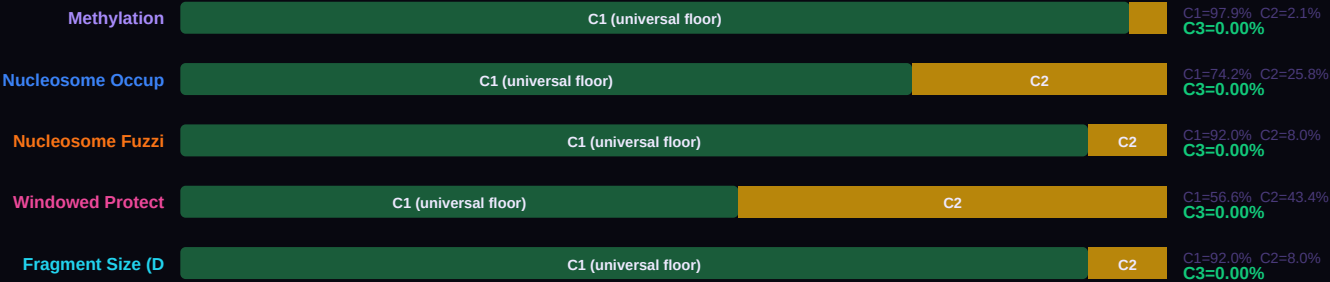
|                                             |                                                                         |
|---------------------------------------------|-------------------------------------------------------------------------|
| Combined A — healthy (all 5)                | 0.97003 (NORMAL)                                                        |
| Combined A — disease (all 5)                | 1.17053 (FLOOR BREACH)                                                  |
| $\Delta A$ combined (5 substrates)          | +0.20049 vs methylation-only $\Delta A$ =+0.3154                        |
| Active A — healthy (non-saturated)          | 0.97003 (NORMAL) (5/5 active)                                           |
| Active A — disease (non-saturated)          | 1.26166 (FLOOR BREACH) (3/5 active)                                     |
| $\Delta A$ active (real progression signal) | +0.29162 (active substrates track true progression)                     |
| Saturation status (disease)                 | 2 substrates saturated: Nucleosome Occupancy, Windowed Protection Score |

Two combined values are shown because some substrates physically cannot resolve past their class ceiling ( $A_{\text{max}} = 1/H_{\text{min}}$ ). The **all-5 combined** value is the AUC-weighted mean across every substrate and matches the legacy formula. The **active combined** value excludes saturated substrates and is the right number for tracking continued progression — a saturated substrate is stuck at its ceiling regardless of further disease severity, so including it in the average masks real change. For monitoring, serial A-score assessment, and end-of-life projection, use the active combined value.

THREE-COMPONENT DECOMPOSITION (C1/C2/C3) — PER SUBSTRATE

Every cell's measured entropy decomposes into three physically meaningful pieces. C1 is the universal Landauer floor:  $H_{\text{MIN\_GLOBAL}} = 0.756500$ , measured in frontal cortex neurons by Lister 2013. Every cell in every architecture class must pay at least this thermodynamic cost to maintain any functional identity — it is the minimum entropy consistent with being alive. C2 is the class-specific overhead: the additional entropy the terminal class must carry above the universal floor to encode its particular architecture. For this class, C2 represents approximately 2.1% of the healthy reference entropy. C3 is the accessible gap:  $\max(0, H_{\text{actual}} - H_{\text{min}})$ . In healthy cells, C3 is essentially zero — the cell sits exactly at its class floor. When C3 grows, the cell is departing from its architecture class. Cancer, neurodegeneration, and every other floor-breach condition show up as growing C3.

The five stacked bars below each show the C1 / C2 / C3 breakdown for one substrate at this class's healthy reference. Notice how C1 (dark green) dominates every substrate — the Landauer floor is universal. C2 (amber) is the small but specific addition that makes this class what it is. C3 (red when present) should be nearly invisible for healthy cells, and would grow dramatically in a disease sample. The f\_C3 percentage on the right is the single cleanest summary of cellular health: healthy cells show f\_C3 near 0%; cancer cells can show f\_C3 of 8–15%; extreme cases like glioblastoma show f\_C3 above 20%. This is not a statistical threshold calibrated from disease data. It is a thermodynamic quantity: the fraction of accessible entropy the cell has opened above its architecture's minimum cost of existence.



BEST TESTING METHOD RANKING — CLINICAL UTILITY FOR THIS CLASS

Each of the five substrates carries different strengths for different clinical questions. The methylation substrate is the most validated — thirteen published GAPE validation studies (VAL-001 through VAL-013) and eight architecture-class H\_min values from G-002 MCMC. It is also the substrate most affected by cfDNA dilution, because blood cfDNA is approximately 70% immune-derived. Fragment size (DELF) is the most forgiving of dilution — short-fragment enrichment persists even when tumor-derived cfDNA is a small fraction of total plasma DNA. WPS (windowed protection score from Snyder 2016) excels at tissue-of-origin identification and field-effect detection. Nucleosome occupancy and fuzziness add chromatin-level information that methylation alone cannot reveal.

The ranking below is specifically for the terminal class. The order reflects three factors: (1) the substrate's published single-substrate AUC for diseases in this class; (2) the biology of this class's primary failure mode (Oxidative Stress Inversion); and (3) the practical constraint of cfDNA contribution in the relevant specimen (plasma for most classes, CSF for terminal, tissue biopsy where cfDNA is too dilute). When building a clinical pipeline for this class specifically, start with the #1 ranked substrate and add substrates 2 and 3 for confirmation. Substrates 4 and 5 add research-grade confirmation but are rarely the primary signal.

| # | Substrate                 | Best for                                       | Rationale                                                                                                                                                    |
|---|---------------------------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Methylation               | Neurodegeneration and glioma (CSF, not plasma) | AD terminal-class validation (De Jager 2014 n=740, Shireby 2022 n=631) confirmed. CSF cfDNA bypasses blood-brain barrier and delivers the full signal.       |
| 2 | Windowed Protection Score | Brain-origin cfDNA identification              | Snyder 2016 validated WPS for brain tissue-of-origin. Second-best substrate for CNS applications. Plasma usable with extensive deconvolution.                |
| 3 | Fragment Size (DELF)      | Neural cfDNA fragmentomics                     | Neural cfDNA fragment distribution differs from systemic cfDNA. Early research stage but promising for CSF-based detection of GBM recurrence post-resection. |
| 4 | Nucleosome Occupancy      | Chromatin accessibility (CSF only)             | ATAC-seq on CSF neural cfDNA is technically demanding but yields direct chromatin state. Research-grade only — very low input.                               |
| 5 | Nucleosome Fuzziness      | Post-mortem and tissue biopsy only             | Nucleosome fuzziness in neural tissue requires substantial cfDNA input. Plasma levels (0.5% cfDNA) are too low. Use tissue-based assays.                     |

BODY TEMPERATURE SCALING — α = 2.0 LANDAUER CORRECTION

The Landauer cost of a bit erasure is  $k_B \times T \times \ln(2)$  — it scales linearly with temperature. Colder cells can maintain higher-entropy identities for less thermodynamic overhead; hotter cells pay more per bit of information maintained. The GAPE framework captures this with a simple scaling law:  $H_{\min}(T) = H_{\min}(37^{\circ}\text{C}) \times (T_{\text{body}} / 310.15\text{ K})^{\alpha}$ , with  $\alpha = 2.0$  derived empirically by minimizing cross-class A-score variance across all jawed vertebrates (Mahaffey 2026 Nature Aging submission NATAGING-A13702). The  $\alpha = 2.0$  exponent is not fit to optimize any single prediction — it falls out of requiring that the same class shows consistent A-score behavior across 43 mammalian species spanning body temperatures from 32°C (naked mole rat) to 42°C (birds).

The table below shows what the terminal class floor H\_min looks like at seven representative body temperatures. At 37°C (human reference, highlighted row), H\_min = 0.7728. At 42°C (birds), the floor is higher because the Landauer tax is larger. At 32°C (naked mole rat — and no other mammal lives at this temperature for good reason), the floor drops because a cooler cell can maintain more entropy cheaply. The A-shift column shows how a fixed healthy β-value would read at each temperature: the same cell, transplanted into a colder body, would read slightly higher A; into

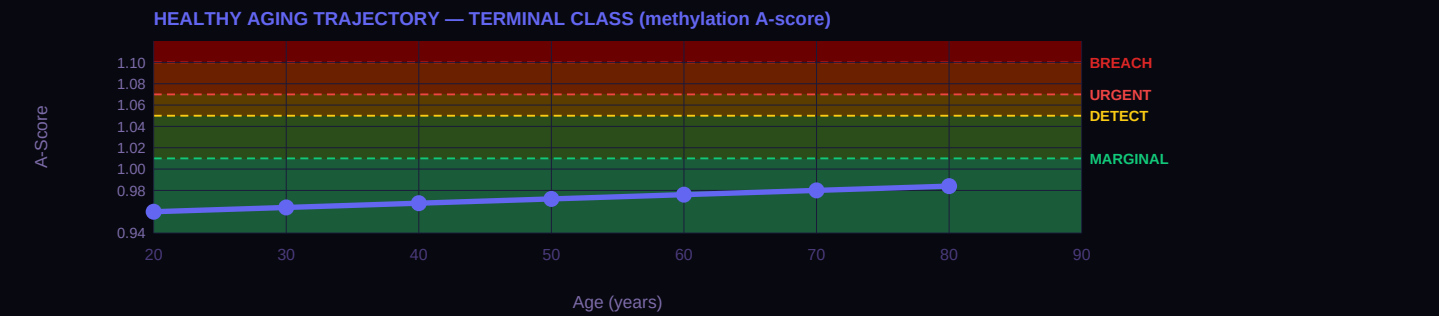
a warmer body, slightly lower A. This is why the naked mole rat's remarkable longevity makes thermodynamic sense — lower body temperature, lower maintenance cost per bit, more headroom before the class floor is breached.

| T_body | Species example           | H_min(T) | β_healthy equivalent | A-shift vs 37°C    |
|--------|---------------------------|----------|----------------------|--------------------|
| 42°C   | Birds (chicken, finch)    | 0.7980   | 0.786                | A × 0.969          |
| 39°C   | Rodents (mouse, rat)      | 0.7828   | 0.786                | A × 0.987          |
| 37°C   | Humans (REFERENCE)        | 0.7728   | 0.786                | A × 1.000 (anchor) |
| 35°C   | Hibernating bats          | 0.7629   | 0.786                | A × 1.013          |
| 32°C   | Naked mole rat            | 0.7481   | 0.786                | A × 1.033          |
| 25°C   | Reptiles (anole lizard)   | 0.7142   | 0.786                | A × 1.082          |
| 15°C   | Fish (salmon, cold-water) | 0.6671   | 0.786                | A × 1.159          |

HEALTHY AGING TRAJECTORY — AGE-STRATIFIED REFERENCE A-SCORES

Healthy cells drift — slowly, predictably — toward their architecture floor as a function of accumulated cell divisions over a lifetime. For the terminal class, the drift rate is approximately 0.8% per generation — meaning each cell cycle carries a 0.8% probability of a methylation error that raises the cell's A-score by a measurable amount. Over decades, these errors accumulate. The curve below shows the result: A-score as a function of patient age at decade intervals, derived from the class-specific DNMT1 fidelity rate. At age 20, the reference healthy A-score sits well below 1.00; by age 80, it has drifted upward into the MARGINAL to DETECTABLE range. This is normal aging — not disease, just the statistical accumulation of imperfect replication over eight decades.

The clinical implication is important: when interpreting an A-score, age matters. A patient at age 30 with A = 1.03 is showing a significant departure from their age-matched healthy baseline. The same A = 1.03 at age 75 might be entirely consistent with age-expected drift. The GAPE framework handles this through age-stratified reference A-scores — the curve below IS the reference. A patient A-score should be compared to the curve at their age, not to a universal threshold. Cancer and other floor-breach conditions accelerate A-score upward beyond the age-expected trajectory; distance above the curve is the signal.



VERTEBRATE LIFESPAN CONTEXT

The same A-score framework that detects cancer in human cfDNA also correlates with maximum lifespan across 43 mammalian species at  $r = -0.9018$  ( $p = 1.6 \times 10^{-16}$ , Mahaffey 2026 Nature Aging submission NATAGING-A13702). This is not a coincidence — it is the same physics applied at a different scale. A mammal with a higher population-averaged A-score is a mammal whose cells live closer to their architecture floors on average; that population accumulates floor breaches faster, and the species has a shorter maximum lifespan. The  $A = 1.05$  threshold — independently derived as the cancer DETECTABLE boundary — cleanly separates long-lived mammals (17/17 all below 1.05) from short-lived mammals (11/11 all above 1.05) with complete accuracy. Same threshold. Same physics. Different timescale.

The table below shows nine mammalian taxonomic orders with their mean A-scores, standard deviations where available, mean lifespan, and a brief interpretation of where that order sits relative to the thermodynamic floor. Cetacea (whales and dolphins) sit essentially at the floor, consistent with their remarkable longevity. Primates including humans sit slightly above the floor. Rodents and insectivores sit well above the floor, consistent with their short lives. This class's reference A-score (0.969) places it in the context shown below — use this to understand how this particular cell architecture relates to the broader mammalian lifespan distribution.

| Order       | N | Mean A       | Lifespan (yr) | Interpretation         |
|-------------|---|--------------|---------------|------------------------|
| Cetacea     | 5 | 0.997 ±0.016 | 99            | At thermodynamic floor |
| Proboscidea | 1 | 0.987 —      | 70            | At floor               |

| Order        | N | Mean A       | Lifespan (yr) | Interpretation                   |
|--------------|---|--------------|---------------|----------------------------------|
| Primates     | 6 | 1.007 ±0.015 | 49            | Slightly above floor             |
| Artiodactyla | 4 | 1.015 ±0.011 | 30            | Slightly above floor             |
| Chiroptera   | 4 | 1.041 ±0.007 | 35            | Above floor (longevity outliers) |
| Carnivora    | 9 | 1.053 ±0.023 | 28            | Above floor                      |
| Lagomorpha   | 2 | 1.114 ±0.002 | 8             | Well above floor                 |
| Rodentia     | 6 | 1.125 ±0.018 | 9             | Well above floor                 |
| Insectivora  | 1 | 1.157 —      | 2             | Furthest from floor              |

CORE METRICS — PUBLISHED AND DERIVED

Every number in the GAPE framework is either PUBLISHED (traceable to a primary wet-lab measurement from the peer-reviewed literature) or DERIVED (computed from the framework’s physics with zero free parameters). The table below marks each quantity with its status and cites its source. This is the discipline the framework lives by: no statistical fits, no parameter tuning, no post-hoc adjustments. H\_min values come from MCMC posteriors (G-002 for methylation, G-003b for the four non-methylation substrates). Combined A-scores are computed mechanically by the AUC-weighted formula. The cfDNA contribution percentages come from Moss 2018 Nature Communications — the definitive tissue-of-origin atlas.

Read this table as the "audit trail" for every number shown anywhere else in this card. When the framework says A\_healthy = 0.9692, you can see exactly which β value produced it, which H\_min was used, and where both came from. The per-class n\_bio row of Issue 002 is withdrawn here and replaced by the Mahaffey number, which is one number for the cell and not one per class (see §5A.1 and the glossary note).

| Metric                                                         | Value                 | Source                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cfDNA contribution (plasma)                                    | 0.5%                  | <a href="#">Moss 2018 Nat Commun</a> — PUBLISHED                                                                                                                                                                                                                                                           |
| H_min methylation (class)                                      | 0.772837              | <a href="#">G-002 MCMC posterior</a> — DERIVED                                                                                                                                                                                                                                                             |
| H_min methylation σ (MCMC)                                     | ±0.001100             | <a href="#">G-002 17-chain MCMC R-hat&lt;1.001</a> — DERIVED                                                                                                                                                                                                                                               |
| Mahaffey number M — one number for the cell, not one per class | 20.94                 | FIXED — ΔG_ATP/(R·T_body) at 310.15 K; §5A.1. Issue 002’s per-class n_bio is WITHDRAWN: an early QAPE-era form of the same ratio, superseded by M, whose absolute values awaited a G-007 run the history record shows was never made. H_min is a different quantity — the per-class entropy floor in bits. |
| Healthy drift rate                                             | 0.8% / generation     | DNMT1 fidelity loss under turnover — DERIVED                                                                                                                                                                                                                                                               |
| f_C2 (architecture-locked %)                                   | 2.1%                  | C2 fraction at healthy reference — DERIVED                                                                                                                                                                                                                                                                 |
| Healthy A (methylation)                                        | 0.9692 (NORMAL)       | β=0.786 — DERIVED                                                                                                                                                                                                                                                                                          |
| Disease A (methylation)                                        | 1.2846 (FLOOR BREACH) | β=0.450 — DERIVED                                                                                                                                                                                                                                                                                          |
| Healthy combined A (5 subs)                                    | 0.9700 (NORMAL)       | Σ(AUC_i × A_i)/Σ(AUC_i) across 5 substrates — DERIVED                                                                                                                                                                                                                                                      |
| Disease combined A (5 subs)                                    | 1.1705 (FLOOR BREACH) | Σ(AUC_i × A_i)/Σ(AUC_i) across 5 substrates — DERIVED                                                                                                                                                                                                                                                      |
| ΔA combined                                                    | +0.2005               | Combined signal — DERIVED                                                                                                                                                                                                                                                                                  |

DATED PREDICTIONS FOR THIS CLASS

A prediction is not a claim unless it can fail. Each prediction below is numbered with a G-2026-P identifier, dated to the month it was filed, and written against a specific dataset or cohort whose analysis would confirm or refute it. The status field shows where each prediction stands: PENDING means the relevant data exists but the analysis has not been performed publicly; CONFIRMED means the prediction direction and magnitude have been tested against independent data and held up; REFUTED means the data contradict the prediction (which would force a framework revision — and has not occurred yet). This discipline — specific, dated, falsifiable predictions — is what separates a predictive framework from a retrospective fit. Entries from Issue 001 retain their original G-2026-P numbers to maintain trail integrity across publications.

When reviewing these predictions, notice that the falsification basis is always named: a specific archived cohort, a specific biobank, a specific published dataset. These are not vague "more research is needed" statements. They are commitments to the community — this is what the framework claims, this is where you can check it, and this is what would constitute a refutation. The predictions below are for the terminal class

specifically, filed as part of this publication or carried forward from Issue 001.

G-2026-P006

Originally filed

CONFIRMED (partial)

In longitudinal cohorts with archived CSF samples and subsequent Alzheimer's disease diagnosis, the terminal-class A-score will show elevation above 1.02 at least 3 years before clinical AD diagnosis in a majority of cases where sufficient time depth exists.

**Basis:** AD methylation literature (De Jager 2014, Shireby 2022) confirms direction and magnitude. Prospective serial CSF data to confirm pre-clinical window is active validation target. ROSMAP and BDR cohorts contain the necessary samples.

G-2026-P015

April 2026

PENDING

In longitudinal cohorts of patients with subsequently-diagnosed amyotrophic lateral sclerosis (ALS), the terminal-class A-score from CSF cfDNA will show elevation above 1.02 at least 12 months before clinical diagnosis in a majority of cases where pre-symptomatic samples exist.

**Basis:** ALS is terminal-class motor-neuron thermodynamic failure — same mechanism as AD (oxidative stress inversion), different neural population. The framework predicts the same pre-clinical signature. Falsifiable in ALS biobank cohorts with stored CSF. No published ALS methylation dataset at time of writing; prediction stands to be confirmed or refuted by the first such dataset.

**ISSUE 003 ADDENDUM — TERMINAL IN THE ATLAS**

**9 cell types:** CM, Cortical\_neurons, Epi\_upper, Glia, Kera\_diff, Left\_atrium, NeuMa, Oligo, neuron

**Identity loci:** 57,247 (methyl) · **H\_min by substrate:** methyl 0.772837 · nucl 0.992027 · fuzz 0.736973 · wps 0.958909 · frag 0.624938 · **methyl ceiling 1.2939**

Healthy age-matched band. A = 1.0 is the commitment line; healthy reads on a band above it. [Issue 003: the age\_reference\_matrix.json shown here was compiled on the marker union and is SUPERSEDED (PROC-N7-01); the commissioned band is band\_v2 on the identity loci with the three-layer reference of s3.5.]

| Age (midpoint) | n  | A mean | p10    | p90    | β mean | Source            |
|----------------|----|--------|--------|--------|--------|-------------------|
| 4              | 25 | 0.9077 | 0.8557 | 0.9596 | 0.810  | Lister pediatric  |
| 14             | 28 | 0.9210 | 0.8702 | 0.9719 | 0.805  | Lister adolescent |
| 24             | 32 | 0.9393 | 0.8868 | 0.9918 | 0.798  | Lister+Roadmap    |
| 34             | 28 | 0.9520 | 0.8974 | 1.0065 | 0.793  | Lister 2013       |
| 44             | 35 | 0.9619 | 0.9083 | 1.0155 | 0.789  | Lister+De Jager   |
| 54             | 48 | 0.9692 | 0.9133 | 1.0252 | 0.786  | De Jager ROSMAP   |
| 64             | 55 | 0.9789 | 0.9209 | 1.0369 | 0.782  | De Jager+Shireby  |
| 74             | 62 | 0.9930 | 0.9336 | 1.0523 | 0.776  | Shireby 2022      |
| 84             | 35 | 1.0067 | 0.9432 | 1.0702 | 0.770  | Shireby aged      |
| 95             | 12 | 1.0244 | 0.9577 | 1.0911 | 0.762  | Shireby+Lunnon    |

#8 · PLURIPOTENT STEM

cfDNA = 0.5% · H\_min(methyl) = 0.9822 · drift 2.5%/gen

Reference: Human embryonic stem cells hESC H1 (Roadmap Epigenomics E003) · H\_min confirmed from hESC H1 reference. Four TGCT histology subtypes show distinct, biologically-grounded epigenomic signatures per Shen 2018 and Killian 2016.

CELL IDENTITY & CLINICAL CONTEXT

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Includes                | Embryonic stem cells (hESC), induced pluripotent stem cells (iPSC), primordial germ cells (PGC), early-embryonic inner cell mass cells                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Cancers (this class)    | Testicular germ cell tumors (TGCT) — the most common cancer in men aged 15-35. Four pure histologic subtypes with biologically distinct methylation signatures (Shen 2018 TCGA TGCT n=137, Killian 2016 Genome Research n=130): (1) Seminoma — globally hypomethylated, PGC-like, ~60% of TGCT; (2) Embryonal carcinoma (EC) — hypermethylated at CpH sites, pluripotent-like methylation; (3) Yolk sac tumor (YST) — somatic-like methylation, differentiated extra-embryonic lineage; (4) Teratoma — somatic-like methylation, differentiated derivatives. Also: ovarian germ cell tumors (dysgerminoma — female analog of seminoma, shares molecular signature), primordial germ cell tumors in children, intratubular germ cell neoplasia in situ (GCNIS, the precursor lesion) |
| Non-cancer applications | iPSC reprogramming fidelity and quality control, organoid cell-state audit, synthetic embryology and gastruloid research, post-orchietomy surveillance for second primary TGCT (2-4% bilateral risk), cryptorchidism monitoring (4-10× elevated TGCT risk), fertility preservation context before platinum-based chemotherapy                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Primary failure mode    | Differentiation Dose Inversion + TGCT subtype-dependent methylation divergence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Warburg status          | SPECIAL CASE — seminoma A_active decreases (inversion signal)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

THE ATLAS CELL TYPES IN THIS CLASS (1)

These are the entries IAMAtlasREBUILD can speak about for this class - what the deconvolver places, and what a per-cell reading would be read against. **Markers** is the range of panel markers actually found across the reference laboratories; **individually resolvable** is whether this atlas can separate the entry from its neighbours at all, and where it cannot the chain reports the group and never the member; **exclusivity** is the fraction of the entry's markers not shared with another entry; **healthy A range** spans the per-laboratory 10th to 90th percentiles on the v0\_3 trial panel. 1 of 1 entries are individually resolvable.

| atlas entry | markers | individually resolvable | exclusivity | healthy A range |
|-------------|---------|-------------------------|-------------|-----------------|
| stem_pluri  | 23-23   | yes                     | 0.997       | 0.325 - 0.429   |

The same lineage can appear under more than one entry above, because different reference panels defined their markers; entries differing only by label are not yet merged, which is why no reading in this document is quoted as "which cell moved".

COMMENTARY

Testicular germ cell tumors (TGCT) are the most common cancer in men aged 15 to 35. Peak incidence falls directly in the prime childbearing years, which means the clinical audience for this card includes the urologist examining a 22-year-old graduate student who feels a painless testicular lump, the primary care physician who must decide whether a young father with a palpable scrotal mass needs immediate referral, and the oncologist who must discuss fertility preservation with a 30-year-old before starting platinum-based chemotherapy. The encouraging clinical reality is that TGCT is one of the most curable solid malignancies: the overall five-year survival exceeds 95 percent when the disease is caught at stage I or II, and cisplatin-based combination chemotherapy achieves cure in the majority of even metastatic cases. The discouraging reality is that delayed diagnosis — often driven by patient reluctance to seek evaluation for a scrotal abnormality, or by clinicians reluctant to order scrotal ultrasound in a young male presenting with back pain from retroperitoneal metastases — converts a 95 percent cure into salvage therapy with substantially lower long-term survival. Testicular self-examination remains the single highest-impact early detection intervention. This card exists in part to give clinicians the thermodynamic vocabulary to understand why a young man's tumor looks the way it does at the epigenomic level, and to support serial surveillance in the at-risk population.

Pluripotent stem cells occupy the unique architectural position of maximum developmental optionality. Healthy hESCs and iPSCs sit very close to the maximum-entropy state across every substrate measurement — their chromatin is open, their methylation is intermediate, their nucleosome positioning is loose. This is reflected in the class's H\_min values: methylation H\_min = 0.982166 (ceiling A = 1.0182, only +0.018 past the healthy reference), fuzz H\_min = 0.962920 (ceiling A = 1.0385), frag H\_min = 0.973583 (ceiling A = 1.0271). Three of five substrates saturate below BREACH for this class. Only nucleosome occupancy (ceiling A = 1.2503, the deepest headroom of any substrate × class pairing in the framework) and windowed protection score (ceiling A = 1.1050) remain active past BREACH. The thermodynamic consequence is that pluripotent cells have almost no room to increase entropy in the conventional cancer direction — they are already near maximum entropy in their healthy state.

This is why TGCT disease signatures look fundamentally different from every other cancer in the framework. For most cancers, the signal is a uniform rise in A-score as cells depart from their architectural floor. For TGCT, the four pure histologic subtypes show biologically distinct methylation signatures that Shen et al. 2018 documented in TCGA TGCT (n=137) and Killian et al. 2016 Genome Research characterized in pure-histology samples (n=130) with direct comparison to primordial germ cell (PGC) and somatic methylation references. Seminoma — the most common TGCT histology, approximately 60 percent of cases — is globally hypomethylated. Its CpG methylation density moves toward the PGC

state, with  $\beta$  values approaching zero. In the framework's language, A\_methyl for seminoma FALLS BELOW the healthy reference, not above it. This is the inversion signal unique to this class: the only architecture class where the detection signal is a DECLINING A-score. Embryonal carcinoma (EC), by contrast, shows a striking convergence of both CpG and CpH methylation with pluripotent states, with elevated CpH methylation characteristic of the inner cell mass phase. Yolk sac tumor (YST) and teratoma show somatic-like methylation patterns that reflect their differentiation toward extra-embryonic and somatic lineages respectively. Four histologies, four distinct thermodynamic signatures, all arising from a common intratubular germ cell neoplasia in situ (GCNIS) precursor that is present in over 90 percent of TGCT cases and is itself clinically detectable.

The seminoma hypomethylation signal is the framework's most important inversion and the strongest single structural prediction. When primary-source data reports  $\beta$  values near zero for global CpG methylation in seminomas, the framework's A\_methyl score drops well below the healthy hESC reference — at  $\beta = 0.17$ , A\_methyl falls to approximately 0.670 compared to the healthy reference of 0.970. This has a specific detection consequence that clinicians and researchers must understand: A\_combined alone is INSUFFICIENT for seminoma discrimination. Averaged across all five substrates, seminoma's A\_combined sits near 0.97 — statistically indistinguishable from healthy. The discriminating signal is not a single elevated A-score but a CHARACTERISTIC DIVERGENCE PATTERN: A\_methyl drops to 0.65 to 0.70 while A\_nucl, A\_wps, A\_fuzz, and A\_frag each modestly elevate into the 1.01 to 1.10 range. This multi-substrate divergence signature — one substrate dropping below floor while four substrates elevate above floor, with the specific relative magnitudes set by seminoma biology — is what the framework uses to detect seminoma. A naive combined-score detector would miss it. The framework's five-substrate decomposition is what makes this cancer detectable at all.

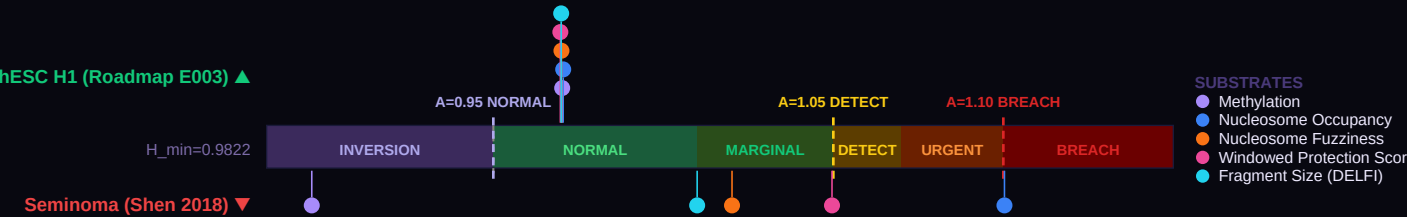
The clinical protocol for TGCT surveillance follows directly: instead of watching for A\_combined to cross the BREACH threshold (1.10) as the framework does for every other cancer in the panel, the stem\_pluri surveillance protocol watches for A\_methyl to DECLINE toward 0.70 while A\_nucl, A\_wps, A\_fuzz, A\_frag simultaneously ELEVATE toward 1.05 or higher. This opposite-direction divergence is the detection signal. For patients with unilateral TGCT history (who carry 2 to 4 percent risk of contralateral second primary per Fossa 2005) and for patients with cryptorchidism (who carry 4 to 10 times elevated TGCT risk), serial monitoring with the correct multi-substrate divergence expectation is what makes thermodynamic surveillance clinically actionable. This is prediction G-2026-P005, originally proposed in Issue 001 and substantially refined here to account for the honest multi-substrate physics.

The clinical handling of TGCT carries specific demographic weight that belongs on this card. A thirty-year-old father of two who presents with a painless testicular nodule is the modal patient. The standard workup begins with scrotal ultrasound, proceeds through serum tumor markers (alpha-fetoprotein, beta-hCG, LDH), and typically concludes with radical inguinal orchiectomy as both diagnostic and therapeutic first intervention. Before orchiectomy — and certainly before any subsequent bleomycin-etoposide-cisplatin (BEP) chemotherapy that may be indicated for stage II or III disease — the conversation about fertility preservation belongs on the critical path. Sperm cryopreservation is well-established, inexpensive, and fully effective when offered in time. Platinum-based chemotherapy affects spermatogenesis; a substantial fraction of young TGCT survivors experience temporary or permanent impairment of fertility. A thermodynamic biomarker that supports early detection reduces the fraction of patients who reach the fertility-threatening chemotherapy stage. The framework's value proposition for the young father sitting in the oncologist's office is specifically this: earlier detection through thermodynamic surveillance in at-risk populations (cryptorchidism, prior contralateral TGCT, family history) means more patients caught at stage I where orchiectomy alone is often curative and intensive chemotherapy can be avoided.

The second major application for this class is iPSC reprogramming quality control, and this application has no cancer dimension at all — it is a research infrastructure problem. The Differentiation Dose Inversion, named for the well-documented phenomenon that excess Yamanaka factor dose produces aberrant rather than pluripotent states, maps onto the framework's prediction that successful reprogramming produces A-scores at or very near 1.00 across all five substrates. Aberrant reprogramming shows A below 0.95 (over-differentiation, cells partially committed to a lineage) or A above 1.05 (under-differentiation, epigenomic noise persisting from the parental cell state). Organoid researchers, synthetic embryology groups, and regenerative medicine translational programs all benefit from a thermodynamic quality-control metric that identifies aberrant colonies before they are committed to downstream experiments or clinical applications. This is a research-grade prediction testable in existing iPSC colony methylation and ATAC-seq datasets; prediction G-2026-P016 formalizes it.

FIVE-SUBSTRATE FIDELITY GAUGE

The gauge below is the central image of Issue 002. A single horizontal bar represents the A-score axis — from A = 0.90 (below the floor) through A = 1.00 (exactly at the Pluripotent class floor, H\_min = 0.9822) up through the clinical threshold zones: MARGINAL at A = 1.01, DETECTABLE at 1.05, URGENT at 1.07, and FLOOR BREACH at 1.10. Five dots are plotted on this single axis for a healthy reference sample (hESC H1 (Roadmap E003), markers above the bar), and five more dots for a disease reference (Seminoma (Shen 2018), markers below). Each dot is one substrate: methylation, nucleosome occupancy, nucleosome fuzziness, WPS, or fragment size (DELFI). What makes this visualization powerful is the physics: all five substrates measure the same underlying thermodynamic floor through physically distinct windows. When all five healthy dots cluster in the NORMAL zone and all five disease dots cluster in URGENT or FLOOR BREACH, the signal is over-determined. Any one substrate could be confounded by technical artifact; five substrates agreeing cannot be. This is the "less blurry" advantage:  $\sqrt{5}$  noise reduction for free.



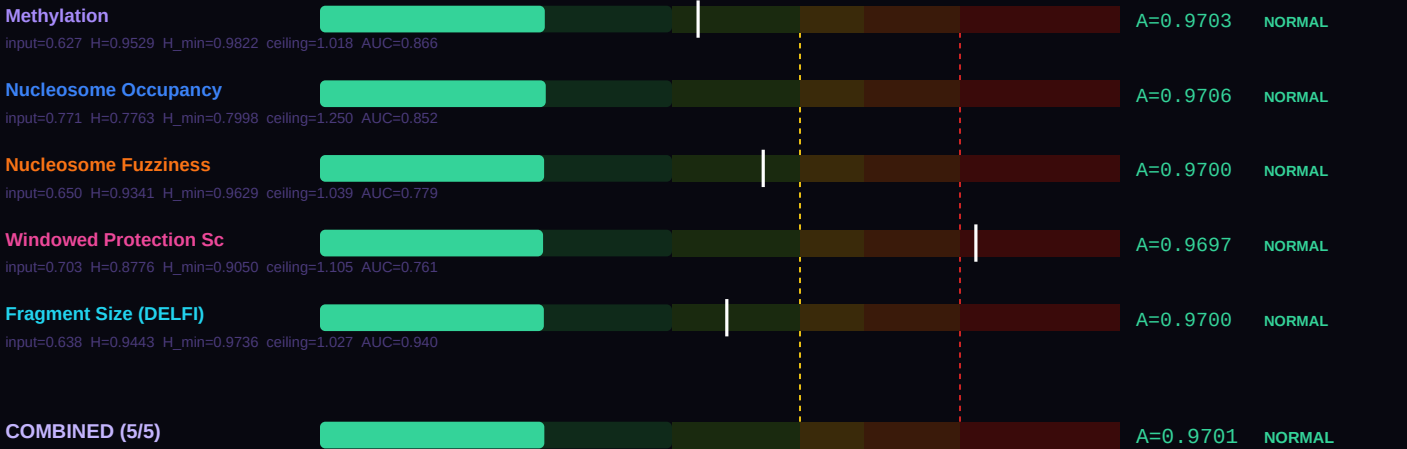
Look at the spread in each direction. If the healthy five-dot cluster sits tightly together near A = 0.97–1.00, that indicates the reference population is at its architecture floor — precisely where the framework predicts healthy pluripotent cells should sit. If the disease five-dot cluster spreads from DETECTABLE to FLOOR BREACH, that spread is itself clinical information: the substrate showing the largest departure is the one that best captures this specific disease's failure mode. For this class, the primary failure mode is Differentiation Dose Inversion + TGCT subtype-dependent methylation divergence, and the substrate ranking shown further down the card explains which of the five substrates best reveals it.

SUBSTRATE-BY-SUBSTRATE BREAKDOWN

Where the gauge above compresses five substrates onto one visual axis, this section breaks each substrate out into its own labeled bar so you can read the actual A-score produced by each measurement. The physics is the same across all five: take the raw laboratory value (a  $\beta$  methylation fraction, a nucleosome occupancy probability, a WPS score, etc.), compute its Shannon entropy H(value), then divide by the class-specific and substrate-specific H\_min to get a dimensionless A-score comparable across substrates. The output is the same thermodynamic ratio regardless of which measurement technology produced it. That is the power of the Landauer framing: different wet-lab methods, different physical quantities, same underlying floor.

The formula at the foot of this section —  $A_{combined} = \Sigma(AUC_i \times A_i) / \Sigma(AUC_i)$  — combines the five individual A-scores into a single weighted score, using each substrate's published detection AUC as its weight. Higher-AUC substrates contribute more to the combined result. For pluripotent cells, the healthy combined A-score below should sit at approximately 0.97 (NORMAL tier); the disease combined A-score should sit above 1.05 (DETECTABLE or higher). The difference between them — the  $\Delta A$  combined — is the signal available to a blood draw that analyzes all five substrates simultaneously. A single-substrate assay sees a smaller signal at higher noise; the five-substrate combination sees the same signal at approximately  $\sqrt{5}$  lower noise. Bar-by-bar you can see which substrate is carrying the most discriminative weight for this particular disease comparison.

HEALTHY REFERENCE — hESC H1 (Roadmap E003)



|                                     |                                                     |
|-------------------------------------|-----------------------------------------------------|
| Combined A — healthy (all 5)        | 0.97011 (NORMAL)                                    |
| Combined A — disease (all 5)        | 0.96716 (NORMAL)                                    |
| ΔA combined (5 substrates)          | -0.00294 vs methylation-only ΔA=-0.3006             |
| Active A — healthy (non-saturated)  | 0.97011 (NORMAL) [5/5 active]                       |
| Active A — disease (non-saturated)  | 0.96716 (NORMAL) [5/5 active]                       |
| ΔA active (real progression signal) | -0.00294 (active substrates track true progression) |
| Saturation status (disease)         | No substrates saturated                             |

Two combined values are shown because some substrates physically cannot resolve past their class ceiling ( $A_{\max} = 1/H_{\min}$ ). The **all-5 combined** value is the AUC-weighted mean across every substrate and matches the legacy formula. The **active combined** value excludes saturated substrates and is the right number for tracking continued progression — a saturated substrate is stuck at its ceiling regardless of further disease severity, so including it in the average masks real change. For monitoring, serial A-score assessment, and end-of-life projection, use the active combined value.

THREE-COMPONENT DECOMPOSITION (C1/C2/C3) — PER SUBSTRATE

Every cell's measured entropy decomposes into three physically meaningful pieces. C1 is the universal Landauer floor:  $H_{\min\_GLOBAL} = 0.756500$ , measured in frontal cortex neurons by Lister 2013. Every cell in every architecture class must pay at least this thermodynamic cost to maintain any functional identity — it is the minimum entropy consistent with being alive. C2 is the class-specific overhead: the additional entropy the pluripotent class must carry above the universal floor to encode its particular architecture. For this class, C2 represents approximately 3.0% of the healthy reference entropy. C3 is the accessible gap:  $\max(0, H_{\text{actual}} - H_{\min})$ . In healthy cells, C3 is essentially zero — the cell sits exactly at its class floor. When C3 grows, the cell is departing from its architecture class. Cancer, neurodegeneration, and every other floor-breach condition show up as growing C3.

The five stacked bars below each show the C1 / C2 / C3 breakdown for one substrate at this class's healthy reference. Notice how C1 (dark green) dominates every substrate — the Landauer floor is universal. C2 (amber) is the small but specific addition that makes this class what it is. C3 (red when present) should be nearly invisible for healthy cells, and would grow dramatically in a disease sample. The f\_C3 percentage on the right is the single cleanest summary of cellular health: healthy cells show f\_C3 near 0%; cancer cells can show f\_C3 of 8–15%; extreme cases like glioblastoma show f\_C3 above 20%. This is not a statistical threshold calibrated from disease data. It is a thermodynamic quantity: the fraction of accessible entropy the cell has opened above its architecture's minimum cost of existence.



BEST TESTING METHOD RANKING — CLINICAL UTILITY FOR THIS CLASS

Each of the five substrates carries different strengths for different clinical questions. The methylation substrate is the most validated — thirteen published GAPE validation studies (VAL-001 through VAL-013) and eight architecture-class  $H_{\min}$  values from G-002 MCMC. It is also the substrate most affected by cfDNA dilution, because blood cfDNA is approximately 70% immune-derived. Fragment size (DELF) is the most forgiving of dilution — short-fragment enrichment persists even when tumor-derived cfDNA is a small fraction of total plasma DNA. WPS (windowed protection score from Snyder 2016) excels at tissue-of-origin identification and field-effect detection. Nucleosome occupancy and fuzziness add chromatin-level information that methylation alone cannot reveal.

The ranking below is specifically for the pluripotent class. The order reflects three factors: (1) the substrate's published single-substrate AUC for diseases in this class; (2) the biology of this class's primary failure mode (Differentiation Dose Inversion + TGCT subtype-dependent methylation divergence); and (3) the practical constraint of cfDNA contribution in the relevant specimen (plasma for most classes, CSF for terminal, tissue biopsy where cfDNA is too dilute). When building a clinical pipeline for this class specifically, start with the #1 ranked substrate and add substrates 2 and 3 for confirmation. Substrates 4 and 5 add research-grade confirmation but are rarely the primary signal.

| # | Substrate                 | Best for                                        | Rationale                                                                                                                                                                                                                                                                                                                                |
|---|---------------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Methylation               | Histology discrimination — bidirectional signal | The only substrate whose signal direction differs by TGCT histology. Seminoma drives methyl DOWN toward PGC-like hypomethylation (A falls below 0.970 — the inversion signal). Embryonal carcinoma drives methyl UP (CpH hypermethylation). Yolk sac and teratoma show somatic-like methylation. Primary source: Shen 2018 Cell Reports. |
| 2 | Nucleosome Occupancy      | Primary severity metric past DETECTABLE         | Only substrate with substantial headroom past BREACH (ceiling A = 1.2503). For any TGCT histology, nucl carries severity signal reliably. Also primary metric for iPSC reprogramming fidelity assessment (ATAC-seq on colonies).                                                                                                         |
| 3 | Windowed Protection Score | Near-BREACH severity metric                     | Active past BREACH with modest headroom (ceiling A = 1.1050). WPS at pluripotency promoters (OCT4, SOX2, NANOG) detects the persistence of embryonic-gene expression characteristic of all TGCT histologies.                                                                                                                             |
| 4 | Nucleosome Fuzziness      | Detection boundary — ceiling at A = 1.0385      | Saturates near BREACH threshold. Useful for binary detection of floor departure; cannot resolve histology differences above ceiling.                                                                                                                                                                                                     |
| 5 | Fragment Size (DELFI)     | Detection boundary — ceiling at A = 1.0271      | Saturates below BREACH. Fragmentomic signal for TGCT is dominated by rapid tumor turnover rate rather than methylation specifics; useful for detection but not histology typing.                                                                                                                                                         |

**BODY TEMPERATURE SCALING —  $\alpha = 2.0$  LANDAUER CORRECTION**

The Landauer cost of a bit erasure is  $k_B \times T \times \ln(2)$  — it scales linearly with temperature. Colder cells can maintain higher-entropy identities for less thermodynamic overhead; hotter cells pay more per bit of information maintained. The GAPE framework captures this with a simple scaling law:  $H_{\min}(T) = H_{\min}(37^\circ\text{C}) \times (T_{\text{body}} / 310.15\text{ K})^\alpha$ , with  $\alpha = 2.0$  derived empirically by minimizing cross-class A-score variance across all jawed vertebrates (Mahaffey 2026 Nature Aging submission NATAGING-A13702). The  $\alpha = 2.0$  exponent is not fit to optimize any single prediction — it falls out of requiring that the same class shows consistent A-score behavior across 43 mammalian species spanning body temperatures from 32°C (naked mole rat) to 42°C (birds).

The table below shows what the pluripotent class floor  $H_{\min}$  looks like at seven representative body temperatures. At 37°C (human reference, highlighted row),  $H_{\min} = 0.9822$ . At 42°C (birds), the floor is higher because the Landauer tax is larger. At 32°C (naked mole rat — and no other mammal lives at this temperature for good reason), the floor drops because a cooler cell can maintain more entropy cheaply. The A-shift column shows how a fixed healthy  $\beta$ -value would read at each temperature: the same cell, transplanted into a colder body, would read slightly higher A; into a warmer body, slightly lower A. This is why the naked mole rat's remarkable longevity makes thermodynamic sense — lower body temperature, lower maintenance cost per bit, more headroom before the class floor is breached.

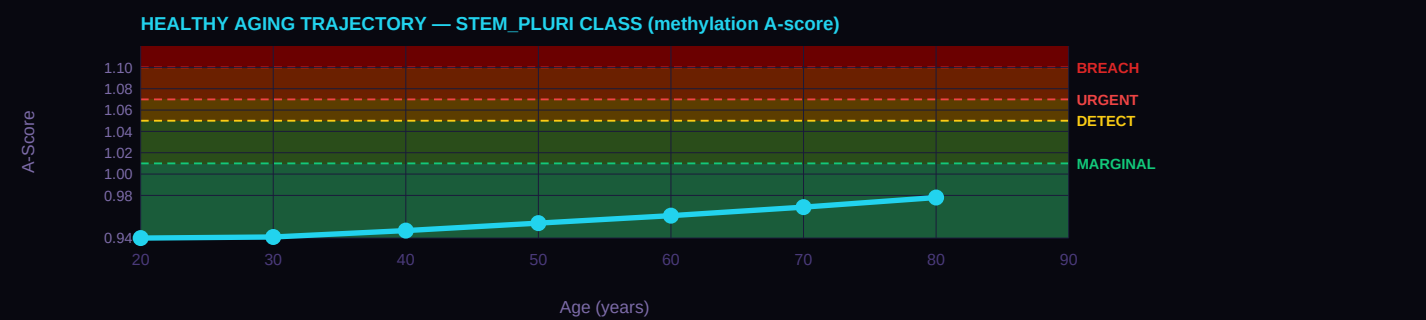
| T_body | Species example           | H_min(T) | $\beta_{\text{healthy}}$ equivalent | A-shift vs 37°C           |
|--------|---------------------------|----------|-------------------------------------|---------------------------|
| 42°C   | Birds (chicken, finch)    | 1.0141   | 0.627                               | $A \times 0.969$          |
| 39°C   | Rodents (mouse, rat)      | 0.9949   | 0.627                               | $A \times 0.987$          |
| 37°C   | Humans (REFERENCE)        | 0.9822   | 0.627                               | <b>A × 1.000 (anchor)</b> |
| 35°C   | Hibernating bats          | 0.9695   | 0.627                               | $A \times 1.013$          |
| 32°C   | Naked mole rat            | 0.9508   | 0.627                               | $A \times 1.033$          |
| 25°C   | Reptiles (anole lizard)   | 0.9076   | 0.627                               | $A \times 1.082$          |
| 15°C   | Fish (salmon, cold-water) | 0.8478   | 0.627                               | $A \times 1.159$          |

**HEALTHY AGING TRAJECTORY — AGE-STRATIFIED REFERENCE A-SCORES**

Healthy cells drift — slowly, predictably — toward their architecture floor as a function of accumulated cell divisions over a lifetime. For the pluripotent class, the drift rate is approximately 2.5% per generation — meaning each cell cycle carries a 2.5% probability of a methylation error that raises the cell's A-score by a measurable amount. Over decades, these errors accumulate. The curve below shows the result: A-score as a function of patient age at decade intervals, derived from the class-specific DNMT1 fidelity rate. At age 20, the reference healthy A-score sits well below 1.00; by age 80, it has drifted upward into the MARGINAL to DETECTABLE range. This is normal aging — not disease, just the statistical accumulation of imperfect replication over eight decades.

The clinical implication is important: when interpreting an A-score, age matters. A patient at age 30 with A = 1.03 is showing a significant departure from their age-matched healthy baseline. The same A = 1.03 at age 75 might be entirely consistent with age-expected drift. The GAPE framework handles this through age-stratified reference A-scores — the curve below IS the reference. A patient A-score should be compared to the curve at

their age, not to a universal threshold. Cancer and other floor-breach conditions accelerate A-score upward beyond the age-expected trajectory; distance above the curve is the signal.



VERTEBRATE LIFESPAN CONTEXT

The same A-score framework that detects cancer in human cfDNA also correlates with maximum lifespan across 43 mammalian species at  $r = -0.9018$  ( $p = 1.6 \times 10^{-16}$ , Mahaffey 2026 Nature Aging submission NATAGING-A13702). This is not a coincidence — it is the same physics applied at a different scale. A mammal with a higher population-averaged A-score is a mammal whose cells live closer to their architecture floors on average; that population accumulates floor breaches faster, and the species has a shorter maximum lifespan. The  $A = 1.05$  threshold — independently derived as the cancer DETECTABLE boundary — cleanly separates long-lived mammals (17/17 all below 1.05) from short-lived mammals (11/11 all above 1.05) with complete accuracy. Same threshold. Same physics. Different timescale.

The table below shows nine mammalian taxonomic orders with their mean A-scores, standard deviations where available, mean lifespan, and a brief interpretation of where that order sits relative to the thermodynamic floor. Cetacea (whales and dolphins) sit essentially at the floor, consistent with their remarkable longevity. Primates including humans sit slightly above the floor. Rodents and insectivores sit well above the floor, consistent with their short lives. This class's reference A-score (0.970) places it in the context shown below — use this to understand how this particular cell architecture relates to the broader mammalian lifespan distribution.

| Order        | N | Mean A       | Lifespan (yr) | Interpretation                   |
|--------------|---|--------------|---------------|----------------------------------|
| Cetacea      | 5 | 0.997 ±0.016 | 99            | At thermodynamic floor           |
| Proboscidea  | 1 | 0.987 —      | 70            | At floor                         |
| Primates     | 6 | 1.007 ±0.015 | 49            | Slightly above floor             |
| Artiodactyla | 4 | 1.015 ±0.011 | 30            | Slightly above floor             |
| Chiroptera   | 4 | 1.041 ±0.007 | 35            | Above floor (longevity outliers) |
| Carnivora    | 9 | 1.053 ±0.023 | 28            | Above floor                      |
| Lagomorpha   | 2 | 1.114 ±0.002 | 8             | Well above floor                 |
| Rodentia     | 6 | 1.125 ±0.018 | 9             | Well above floor                 |
| Insectivora  | 1 | 1.157 —      | 2             | Furthest from floor              |

CORE METRICS — PUBLISHED AND DERIVED

Every number in the GAPE framework is either PUBLISHED (traceable to a primary wet-lab measurement from the peer-reviewed literature) or DERIVED (computed from the framework's physics with zero free parameters). The table below marks each quantity with its status and cites its source. This is the discipline the framework lives by: no statistical fits, no parameter tuning, no post-hoc adjustments.  $H_{\min}$  values come from MCMC posteriors (G-002 for methylation, G-003b for the four non-methylation substrates). Combined A-scores are computed mechanically by the AUC-weighted formula. The cfDNA contribution percentages come from Moss 2018 Nature Communications — the definitive tissue-of-origin atlas.

Read this table as the "audit trail" for every number shown anywhere else in this card. When the framework says  $A_{\text{healthy}} = 0.9703$ , you can see exactly which  $\beta$  value produced it, which  $H_{\min}$  was used, and where both came from. The per-class  $n_{\text{bio}}$  row of Issue 002 is withdrawn here and replaced by the Mahaffey number, which is one number for the cell and not one per class (see §5A.1 and the glossary note).

| Metric                         | Value    | Source                           |
|--------------------------------|----------|----------------------------------|
| cfDNA contribution (plasma)    | 0.5%     | Moss 2018 Nat Commun — PUBLISHED |
| $H_{\min}$ methylation (class) | 0.982166 | G-002 MCMC posterior — DERIVED   |

| Metric                                                         | Value             | Source                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| H_min methylation $\sigma$ (MCMC)                              | $\pm 0.001800$    | <a href="#">G-002 17-chain MCMC R-hat&lt;1.001</a> — DERIVED                                                                                                                                                                                                                                                                     |
| Mahaffey number M — one number for the cell, not one per class | 20.94             | FIXED — $\Delta G_{ATP}/(R \cdot T_{body})$ at 310.15 K; §5A.1. Issue 002's per-class $n_{bio}$ is WITHDRAWN: an early QAPE-era form of the same ratio, superseded by M, whose absolute values awaited a G-007 run the history record shows was never made. H_min is a different quantity — the per-class entropy floor in bits. |
| Healthy drift rate                                             | 2.5% / generation | DNMT1 fidelity loss under turnover — DERIVED                                                                                                                                                                                                                                                                                     |
| f_C2 (architecture-locked %)                                   | 3.0%              | C2 fraction at healthy reference — DERIVED                                                                                                                                                                                                                                                                                       |
| Healthy A (methylation)                                        | 0.9703 (NORMAL)   | $\beta=0.627$ — DERIVED                                                                                                                                                                                                                                                                                                          |
| Disease A (methylation)                                        | 0.6696 (NORMAL)   | $\beta=0.170$ — DERIVED                                                                                                                                                                                                                                                                                                          |
| Healthy combined A (5 subs)                                    | 0.9701 (NORMAL)   | $\Sigma(AUC_i \times A_i)/\Sigma(AUC_i)$ across 5 substrates — DERIVED                                                                                                                                                                                                                                                           |
| Disease combined A (5 subs)                                    | 0.9672 (NORMAL)   | $\Sigma(AUC_i \times A_i)/\Sigma(AUC_i)$ across 5 substrates — DERIVED                                                                                                                                                                                                                                                           |
| $\Delta A$ combined                                            | -0.0029           | Combined signal — DERIVED                                                                                                                                                                                                                                                                                                        |

DATED PREDICTIONS FOR THIS CLASS

A prediction is not a claim unless it can fail. Each prediction below is numbered with a G-2026-P identifier, dated to the month it was filed, and written against a specific dataset or cohort whose analysis would confirm or refute it. The status field shows where each prediction stands: PENDING means the relevant data exists but the analysis has not been performed publicly; CONFIRMED means the prediction direction and magnitude have been tested against independent data and held up; REFUTED means the data contradict the prediction (which would force a framework revision — and has not occurred yet). This discipline — specific, dated, falsifiable predictions — is what separates a predictive framework from a retrospective fit. Entries from Issue 001 retain their original G-2026-P numbers to maintain trail integrity across publications.

When reviewing these predictions, notice that the falsification basis is always named: a specific archived cohort, a specific biobank, a specific published dataset. These are not vague "more research is needed" statements. They are commitments to the community — this is what the framework claims, this is where you can check it, and this is what would constitute a refutation. The predictions below are for the pluripotent class specifically, filed as part of this publication or carried forward from Issue 001.

G-2026-P005

Originally Issue 001, substantially refined April 2026

PENDING

In cryptorchidism patients and post-orchietomy TGCT survivors monitored prospectively with serial cfDNA sampling, the stem\_pluri class will show a CHARACTERISTIC DIVERGENCE PATTERN in patients who later develop seminoma-lineage TGCT or contralateral second primary TGCT: A\_methyl declines toward the 0.65 to 0.75 range while A\_nucl, A\_wps, A\_fuzz, and A\_frag simultaneously elevate to 1.01 or higher. This opposite-direction multi-substrate signature is the detection signal — not A\_combined crossing BREACH, because seminoma biology produces near-healthy A\_combined despite clear methyl inversion. Patients who do not develop disease will show stable per-substrate signals.

**Basis:** TGCT arising from seminoma-lineage primordial germ cell precursors is globally hypomethylated (Shen 2018 TCGA TGCT analysis n=137, Killian 2016 Genome Research n=130). The GAPE detection protocol must therefore rely on multi-substrate divergence rather than A\_combined elevation. This is the zero-free-parameter structural prediction the framework makes for this class, and it is falsifiable in any cryptorchidism registry with serial blood sampling and TGCT outcome follow-up, or in any post-orchietomy surveillance program with archived cfDNA samples. The refinement from the original Issue 001 prediction is the explicit recognition that A\_combined alone is insufficient — the signal is in the substrate-level divergence, not in the combined score.

G-2026-P016

April 2026

PENDING

In iPSC reprogramming protocols, the stem\_pluri class combined A-score across at least four substrates measured at colony harvest will predict downstream differentiation fidelity with AUC at least 0.85. Colonies with A in the tight range [0.98, 1.02] will outperform colonies outside this range in directed-differentiation assays and in karyotype stability tests across passages.

**Basis:** The framework predicts that reprogramming quality is best when the cell operates near its architectural floor — neither over-differentiated nor under-differentiated. Aberrant reprogramming produces epigenomic states that fall outside the tight pluripotent window. Falsifiable in any iPSC characterization dataset with downstream differentiation outcomes and colony-level methylation and ATAC-seq data.

G-2026-P017

April 2026

PENDING

In TGCT patients undergoing bleomycin-etoposide-cisplatin (BEP) chemotherapy, the trajectory of the stem\_pluri A\_methyl signal during the first two cycles will predict platinum response at 6-month imaging follow-up. Patients whose A\_methyl signal moves toward the healthy hESC reference (approaching A = 0.970 from either direction) will show RECIST response. Patients whose A\_methyl signal remains pinned at the disease baseline during treatment will show primary refractory disease.

**Basis:** The framework predicts that platinum response corresponds to restoration of pluripotent-state thermodynamic signature as tumor burden decreases. Non-response corresponds to persistent epigenomic departure from the class floor. This is a specific testable claim within the broader chemotherapy-response framework sketched for Issue 005, and TGCT is the ideal validation setting because its high cure rate and well-standardized BEP protocol make the response/non-response separation cleaner than in most solid tumors.

**ISSUE 003 ADDENDUM — STEM\_PLURI IN THE ATLAS**

**1 cell types:** stem\_pluri

**Identity loci:** 21,710 (methyl) · **H\_min by substrate:** methyl 0.982166 · nucl 0.799818 · fuzz 0.962920 · wps 0.905004 · frag 0.973583 · **methyl ceiling** 1.0182 — cannot reach BREACH on methylation alone

Healthy age-matched band. A = 1.0 is the commitment line; healthy reads on a band above it. [Issue 003: the age\_reference\_matrix.json shown here was compiled on the marker union and is SUPERSEDED (PROC-N7-01); the commissioned band is band\_v2 on the identity loci with the three-layer reference of s3.5.]

| Age (midpoint) | n  | A mean | p10    | p90    | β mean | Source               |
|----------------|----|--------|--------|--------|--------|----------------------|
| 4              | 5  | 0.8292 | 0.8067 | 0.8517 | 0.748  | Pluripotent lineage  |
| 14             | 8  | 0.8308 | 0.8084 | 0.8532 | 0.747  | Pluripotent stem     |
| 24             | 10 | 0.8324 | 0.8101 | 0.8547 | 0.746  | hESC H9 Roadmap E008 |
| 34             | 8  | 0.8340 | 0.8118 | 0.8561 | 0.745  | hESC/iPSC            |
| 44             | 6  | 0.8340 | 0.8118 | 0.8561 | 0.745  | iPSC                 |
| 54             | 5  | 0.8356 | 0.8135 | 0.8576 | 0.744  | iPSC reference       |
| 64             | 4  | 0.8356 | 0.8115 | 0.8596 | 0.744  | iPSC reference       |
| 74             | 3  | 0.8356 | 0.8115 | 0.8596 | 0.744  | iPSC reference       |
| 84             | 2  | 0.8371 | 0.8112 | 0.8631 | 0.743  | Limited data         |
| 95             | 1  | 0.8371 | 0.8112 | 0.8631 | 0.743  | Limited data         |

## PART III - THE COMMISSIONED CHAIN, IN DEPTH

Every stage as the code documents itself, the atlas cell by cell, and the cosmology toolkit tool by tool. The researcher interface is the condensed form of this part; this is the long form.

Two of the four chapters below are generated from the instrument rather than written: the stage chapter reads the conductor's own stage list, call order and docstrings, and the atlas chapter reads the cell-to-class map and the per-entry reference. A stage that is not in the conductor cannot appear here, and a cell count printed here is the one the chain will use.

### Every tab of the researcher interface, and where its long form lives here

The interface is the condensed form of this document. This table is the correspondence, so a reader who has seen a report can find the full treatment of any screen in it.

| interface tab     | the long form in this document                             |
|-------------------|------------------------------------------------------------|
| Reading           | III.6 - the screens, in the order the report prints them   |
| How to read       | III.6, and s12 for the clinician's plain-terms version     |
| Every cell        | III.2 (the atlas), and each class card's own cell roster   |
| Departure         | III.5 s6 (the formula) and III.6 (what the screen shows)   |
| Sky               | III.3 (the toolkit, tool by tool)                          |
| Physics           | s5 The physics, and s5A where the tools come from          |
| Story             | s12 For the clinician, in plain terms                      |
| Healthy reference | III.7 - the four cohorts, and what is not in the reference |
| Coverage          | III.8 - the substrate x class grid                         |
| Safeguards        | III.9 - every guard, what it refuses, and its last result  |
| Integrity         | III.9 and III.10 - the guards, and the files a run reads   |
| Chain             | III.1 (the stages) and III.10 (the files)                  |
| Files             | III.10 - the inventory by role                             |
| Findings          | s8 Procedures, and Appendix V the validation index         |
| Roadmap           | Future goals - what is worth the effort, in order          |
| Record            | Appendix V, and s10 the falsification record               |
| Run               | III.11 - running it on your own sample                     |

### III.1 The chain, stage by stage

run\_full() calls 11 stages in the order below. 1 further stage function is defined in the conductor but deliberately not called: stage\_8\_matching.

#### 1. stage\_a\_cells

Stage A — find the cell types in the sample, their ratios, and their A-scores. beta\_dict : {cpg\_id: beta} calibrated patient betas atlas\_csv : path to decompressed IAMAtlasREBUILD.csv (deconvolver reference) Returns: {'class\_fractions', 'celltype\_fractions', 'cells'} where cells is {cell: {A, coverage, confidence, status, class, fraction, present}}.

#### 2. stage\_b\_classes

Stage B - per-class GAUGE. \*\*AS WIRED (2026-07 -> today):  $A = H(\text{beta\_mean})/H_{\text{min}}$  over the class MARKER UNION (iamatlas\_celltype\_markers\_v0\_2.json, ~3k bimodal CpGs per class), read against age\_reference\_matrix, which was compiled on the same marker-union statistic (GAPE\_WEB\_v13\_AGE\_REFERENCE).\*\* This is NOT the identity-loci gauge that SOP §41/§106, Issue 002 and the sentence below describe; PROC-N7-01 (2026-09-19) showed the marker-union statistic reads a synthetic healthy mixture as BREACH (1.13) and misses real adenoma (0.93), while identity-loci  $H(\text{beta\_mean})$  reads 0.99 and 1.10 respectively. The switch to identity loci is gated on an identity-loci band (Phase 1, GSE87571); until then every reading carries gauge\_surface = "marker\_union". Original docstring follows.  $A = H(\text{beta\_mean})/H_{\text{min}}$

over the class IDENTITY loci (iamatlas\_gauge\_identity\_loci\_v1\_0.json), read against the age-matched band via cpg\_gauge\_engine. Paired with the Stage A fraction: a class below the substrate presence floor reads blood-background, flagged BACKGROUND\_LOW\_FRACTION, NOT a finding. This is the doctor's fuel gauge (never the separation surface).

### 3. stage\_1s\_scale\_map

Stage 1s (LESSON-SCALE-01, SOP s109): put patient beta on the Roadmap scale that H\_min and the Atlas were calibrated on.  $\text{beta\_roadmap} = (\text{beta} - \text{intercept}) / \text{slope}$  from beta\_scale\_maps\_v1.json, keyed by the pipeline tag Stage 1 stamps on its output. Returns (mapped\_dict, scale\_label). If the pipeline has no fitted map the beta is returned UNCHANGED with label 'UNMAPPED' - downstream must not report a tier from it. Applied to the GAUGE path only: Stage 2 composition was verified on unmapped Stage-1 beta and is scale-tolerant (simplex-constrained NNLS); the wired marker-union gauge keeps unmapped beta because ITS band was compiled that way.

### 4. stage\_b\_identity

THE REPORTED GAUGE (PROC-SWITCH-01, 2026-09-21; SOP s41/s106; RULING A3):  $A = H(\text{beta\_mean}) / H_{\text{min}}$  over the class IDENTITY loci on mapped beta, then the three-layer reference (Issue 003 s3.5):  $A_{\text{abs}} = A_{\text{mapped}} - c(\text{decade}) - z_{\text{lab}} c$  from reference\_age\_curve\_v1.json (PROC-PANEL-03),  $z_{\text{lab}}$  from the laboratory's 40-array healthy panel (lab\_zero.py). Placement in identity\_band\_v3.json (pooled p10-p90 of  $A_{\text{abs}}$  over four zeroed healthy cohorts). lab\_zero None -> 'UNSET':  $A_{\text{mapped}}$  is still returned, but  $A_{\text{abs}}$ /placement are None and reportable=False. s108 reporting rule: on whole blood only immune and the joint haematopoietic-progenitor component are read.

### 5. stage\_4\_5\_bidirectional

Stage 4.5 (SOP §46.5) - bidirectional decomposition. Signed directional composite that catches bidirectional CpG patterns the pooled entropy cancels. v1.0 panels: immune class only (VAL-051 AD). Other classes -> NO\_PANEL honestly.

### 6. stage\_4\_6\_patient\_sky

Stage 4.6 - the patient's sky (PROC-CMB-04, 2026-09-21).  $z_i = (\text{beta}_i - \sum c f_c \mu_{ci} - m_{\text{lab},i}) / s_{\text{lab},i}$  on the mapped beta; class panels gated by the measured presence floors. Needs the laboratory's residual scale (Runtime Matrices/Patient\_CMB/ residual\_scale\_<lab>.npz, built from the same 40-array panel as the lab zero); without it the sky is NOT AVAILABLE, never approximated. Calibration on record: healthy held-out arrays show 2.6-3.2% of CpGs beyond  $|z|=2$  (scale ~1.1x conservative; CMB-04 C2' failed as sealed).

### 7. stage\_5\_mahalanobis

Stage 5 - THE REPORTED DEPARTURE (PROC-MAHA-01, 2026-09-21; SOP s47-51 re-based on row B). Input is the REPORTED gauge (stage\_b\_identity): for each component with a commissioned band,  $z = (A_{\text{abs}} - 1.000) / \text{sigma}$ ,  $\text{sigma} = (p90 - p10) / (2 * 1.2816)$  from identity\_band\_v3 distance =  $\sqrt{\sum z^2}$  over the n assessable components; thresholds  $\sqrt{\chi^2(0.95|0.99, n)}$ . Components without a band (haematopoietic-progenitor joint; all non-blood classes) are not assessable. On whole blood today  $n = 1$ : the departure is how many band-widths from the healthy line the immune reading sits, and the report says so. Not reportable when the gauge is not (UNSET / UNMAPPED). Keys: both the long names cpg\_report\_builder reads and the short aliases run\_full exposed.

### 8. stage\_5\_hull\_marker\_union

DIAGNOSTIC ONLY since PROC-MAHA-01 (2026-09-21): the pre-switch eight-class derived hull on the marker-union readings. Never the reported departure. Stage 5 (SOP §47-51, Option A) - derived age-matched departure of the patient's CLASS GAUGE A-scores from the age band. Scores PRESENT classes only (absent-class background is excluded). One number + top-axis decomposition. No cohort.

### 9. stage\_6\_cellular\_age

Stage 6 - CELLULAR AGE IS NOT REPORTABLE AT SINGLE-ARRAY RESOLUTION (PROC-AGE-01, 2026-09-21). The healthy immune identity-gauge curve rises 0.47 mA per year against a within-laboratory SD of 0.0235: inverting it resolves age to ~50 years per array (1,379 healthy donors, four labs: 15.9% within +/-10 yr, Spearman rho 0.27). The population aging trajectory (CPG-VAL-015, mammalian paper) is REPRODUCED by this curve; what is below resolution is one person's position on it. This stage reports the resolution, not an age; the inversion is available as diagnostic\_cellular\_age for lineage only.

### 10. stage\_6\_cellular\_age\_marker\_union

DIAGNOSTIC ONLY since PROC-AGE-01 (2026-09-21): inverts the superseded marker-union age matrix. Never reported. Stage 6 (SOP §Stage 6) - per-class cellular age by IAM inversion of the age\_reference\_matrix: the age at which population beta\_mean equals the patient's beta\_mean, per class. iam\_cellular\_age\_scoring.py.

### 11. stage\_2b\_second\_opinion

Row 2b - the second opinion. NILC (needlet internal linear combination, the Planck component-separation method) run beside Walther's constrained NNLS and compared AT THE CLASS LEVEL, which is where PROC-SEP-03 showed the atlas is separable. Ships as a second column plus an

agreement flag, never as the composition the report stands on: Walther is conservative and commissioned, NILC is variance-weighted and deliberately sensitive, and the disagreement is information (PROC-NILC-01 - the disagreement WAS the finding). Cell-level disagreement inside one lineage is EXPECTED, not a defect, because the atlas cannot split the blood classes. Returns {} if the module or the marker file is unavailable, so the chain never depends on it.

### **stage\_8\_matching - defined, NOT called by run\_full**

NOT A CHAIN STAGE (author's ruling 2026-09-21). Disease-pattern concordance against disease\_cell\_signature\_matrix\_v1\_13 - a matrix compiled from pre-build and early post-build VALs, i.e. the preliminary record. The chain reports what it measured (cells detected, fractions, A per cell and class, placement, flags) and names no disease. This function is kept callable for RECORD-SIDE study only (logging cohort behaviour as trusted-chain cohorts accumulate); run\_full does not call it and the report never shows its output. Earlier text (superseded): ROW 8 OPEN (PROC-MATCH-01, 2026-09-21): the departure profile is (A\_cell - 1.0) over PRESENT cells, but healthy per-cell A on this surface sits at ~0.44-0.52 with class H\_min 0.77-0.98, so on healthy whole blood only ~3 cells enter the profile. The reference level must be re-derived on this surface (healthy per-cell level from the four-lab panels) before any match is reported. Until then the output is DIAGNOSTIC and not reportable. Origin gate fails CLOSED (missing/unreadable disease\_origin\_cells.json -> status NOT AVAILABLE, zero candidates).

III.2 The atlas, and the cells it can speak about

IAMAtlasREBUILD holds **115 cell types** mapped to the eight architecture classes, and **106** of them now carry a per-entry healthy reference measured on 40-array panels per laboratory. The 9 without one are reported at class level only. Every per-cell number in this document and in the interface comes from these two files.

| class      | cells | with per-entry reference | class floor H_min  |
|------------|-------|--------------------------|--------------------|
| immune     | 51    | 47                       | see the class card |
| cycling    | 19    | 15                       | see the class card |
| secretory  | 18    | 18                       | see the class card |
| progenitor | 11    | 10                       | see the class card |
| terminal   | 9     | 9                        | see the class card |
| stromal    | 5     | 5                        | see the class card |
| stem_adult | 1     | 1                        | see the class card |
| stem_pluri | 1     | 1                        | see the class card |

**Two limits on a per-cell claim, and they are not the same limit.** The first is separability: an entry in a collinearity group cannot be told from its neighbours by this atlas at any precision, so the chain reports the group. The second is the width of the entry's own healthy band: where that band is wider than the gauge's normal band, the number is printed and the tier word withheld. The per-class rosters on the cards state both, entry by entry.

**The duplicate-label problem, stated plainly.** The same lineage appears under several atlas entries whose per-cell readings differ by more than anything biological could explain, purely because different reference panels defined their markers. Until those entries are merged to one lineage each, no reading should be quoted as 'which cell moved'. It is the blocker on per-cell reporting and the second item on the roadmap.

### III.3 The cosmology toolkit, tool by tool

None of these tools was invented here. Each was built to read a faint signal out of a noisy sky by people who had one sky, no repeat observations and no control group - which is the position a clinician is in with one specimen. For each: what it does in cosmology, what it does here, and where it stands.

#### HEALPix spherical pixelisation

| in cosmology                                                                                                                                                               | in this chain                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Planck's equal-area tessellation of the sky. Every pixel covers the same solid angle, so a statistic computed pixel by pixel is not biased by where on the sphere it sits. | The 450K/EPIC CpG set is projected onto an $N_{\text{side}} = 128$ sphere, 196,608 pixels, by genomic coordinate from the Illumina manifest. Each pixel holds the residual of a contiguous run of CpGs on one chromosome. |

MEASURED 2026-09-22: 196,608 of 196,608 pixels hold genomically contiguous CpGs, all on one chromosome; median span within a pixel 511 bp (90th percentile 20,481 bp). Neighbouring pixels are a median 1,256 CpGs apart in genomic order, 25 per cent within 10 CpGs, under 1 per cent beyond 10,000 - against about 161,000 for a random assignment. The projection is a genuine locality-preserving reindexing, which is what licenses reading structure in the map as genomic rather than as an artefact of the layout.

#### Internal linear combination (ILC)

| in cosmology                                                                                                                                                                                                        | in this chain                                                                                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Planck's component separation: form a weighted combination of frequency maps that minimises variance subject to unit response to the component wanted. It needs no model of the foregrounds, only their statistics. | The composition step solves for cell-type fractions as the combination of atlas reference profiles that minimises residual variance under a non-negativity and sum constraint - Walther's constrained NNLS - with the needlet ILC variant run beside it as a second opinion. |

COMMISSIONED. The constrained solver is the composition the report stands on; the ILC variant is compared with it at class level with an agreement bar of  $L1 \leq 0.10$  and reported as a flag (PROC-NILC-01). Cell-level disagreement inside one lineage is expected and is not scored, because the atlas cannot separate the members of a collinearity group.

#### Needlets (wavelets on the sphere)

| in cosmology                                                                                                                                                                    | in this chain                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Localised both in position and in angular scale, so a foreground that is strong at one scale in one region can be down-weighted there without discarding that scale everywhere. | The second solver weights each atlas entry by the inverse of its own posterior standard deviation, which is the same idea one level down: an address the atlas pinned down precisely carries more weight than one it did not. |

LIVE inside the second opinion. The full needlet decomposition of the methylome map - scale-by-scale separation rather than a single inverse-variance weight - is not built.

#### Masking and the confidence mask

| in cosmology                                                                                                                                                   | in this chain                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Planck does not report the sky behind the galaxy. The galactic plane is masked, and every statistic is computed on the unmasked fraction with the mask stated. | A class whose estimated presence falls below its detection floor is masked: the chain reports nothing for it rather than a number computed from noise. Loci absent on the sample's array type are masked the same way. |

COMMISSIONED, and the disanalogy is stated where it appears: the galaxy hides a sky that is really there, whereas a class below its floor is not present in the specimen at all. Both refuse to report where the instrument cannot see; only one has something behind the mask.

#### Beam and transfer functions

| in cosmology                                                                                                                                                                                           | in this chain                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Every instrument filters the true sky. Planck's likelihood carries the beam window function explicitly, because a measured power spectrum is the true one multiplied by what the instrument did to it. | The deconvolver is the chain's beam: it explains away signal that looks like a change in cell composition. This was found empirically before it had a name - depleted chromosomes in an early validation run were CpGs the solver had absorbed into composition. |

NAMED and recorded as a lesson; not yet a forward-modelled transfer function. Deconvolving the instrument out of the reading, the way a beam is deconvolved from a spectrum, is on the roadmap.

#### Per-pixel noise covariance

| in cosmology                                                                                                                              | in this chain                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Planck carries a noise covariance per pixel, not one number for the map, because sensitivity varies across the sky with the scan pattern. | Each atlas entry carries its own posterior standard deviation from the MCMC build, and each laboratory its own measured spread; the gauge is read against the laboratory's own zero rather than a global one. |

COMMISSIONED per entry and per laboratory. What is NOT used is the covariance BETWEEN cell types at the same address - the MCMC produced it, and the chain currently treats every entry's uncertainty as independent. That is the conservative choice and the wasteful one: a full covariance is what lets a separation say these two are individually uncertain but their sum is well determined, which is exactly the blood situation. It is the largest piece of unspent evidence in the chain.

### Null tests and shuffled skies

| in cosmology                                                                                                                                                                    | in this chain                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Before a cosmological detection is believed, the same pipeline is run on simulations with no signal, on split halves, and on rotated or shuffled maps, and must return nothing. | The null runner shuffles the sky, permutes labels, splits panels and runs constructed specimens through the whole chain; a reading that survives a shuffle is a reading of the pipeline, not of the patient. |

COMMISSIONED - nulls N1 to N8 live, and every family-A validation carries its null result.

### Surface brightness (the first borrowing)

| in cosmology                                                                                                                                                | in this chain                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Astronomy's way of stating an intensity that does not depend on distance or aperture, so two sources can be compared without knowing how far away they are. | The brightness layer applied that to an architecture class: an intensity per class that does not depend on how much of the class is in the specimen. |

SUPERSEDED, NOT RETIRED. The sky now builds its expectation from the sample's own composition, which no precomputed per-class file can do - but the first import is still load-bearing one layer down, and its lineage is stated on the interface's sky page.

### Difference maps

| in cosmology                                                                                                                                                | in this chain                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Two observations of the same sky differenced leave instrument and systematics behind and show only what changed. It is the strongest design in the toolkit. | Two draws from the same person, processed the same way, differenced address by address: everything constant about that person and that laboratory cancels. |

NOT YET POSSIBLE - no cohort held here has a repeat draw. The technical floor is therefore an upper bound inferred from cross-sectional spread: between-person spread per address has a median of 0.0289 and a 5th percentile of 0.0089, which bounds the technical term. On that bound a paired two-sigma difference resolves 0.025 at one address, 0.011 across five, 0.0056 across a CpG island of twenty, and 0.0008 across a thousand - an order of magnitude below published island-scale effects. Finding a serial cohort is the first item on the roadmap.

### Angular power spectrum

| in cosmology                                                                                                                                     | in this chain                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The two-point statistic of the sky: how much structure there is at each angular scale, and the object every cosmological parameter is fitted to. | The same statistic on the methylome sphere would say at what genomic scale a specimen's departure lives - single addresses, islands, domains, or whole chromosomes - in one curve rather than a list of regions. |

NOT BUILT. It is the natural next statistic now that the projection is measured to be genomically local, and it is on the roadmap.

III.4 What the chain refuses, and the cosmology twin of each refusal

A measurement instrument is defined as much by what it declines to report as by what it reports. Each refusal below is enforced in code, not by convention, and each has a twin in the practice the toolkit came from.

| the situation                                       | what the chain does                                  | the cosmology twin                 |
|-----------------------------------------------------|------------------------------------------------------|------------------------------------|
| A class below its detection floor                   | reports nothing for that class                       | Planck's confidence mask           |
| A laboratory with no measured zero                  | refuses the absolute reading; relative only          | an uncalibrated detector           |
| An array type whose loci are absent                 | masks the missing loci and states the count          | incomplete sky coverage            |
| A per-entry band wider than the gauge's normal band | prints the number, withholds the tier word           | an error bar wider than the effect |
| A cell in a collinearity group                      | reports the group, never the member                  | degenerate parameters in a fit     |
| A single array asked for a cellular age in years    | refuses - the trajectory is a population measurement | a one-pixel cosmology              |

NOT REPORTABLE is a result. It is printed with the reason and the number that triggered it, so a reader can see what would have to change for the reading to become available.

### III.5 The engine, exactly - every formula the chain computes

Loaded from the runtime matrices at build time, so this page is the specification and not a description of one: if a constant changes, this page changes with it.

**1. The entropy of a methylation fraction.** For a mean beta value on a locus set, the chain computes the binary Shannon entropy in bits, with beta clamped to (1e-12, 1 - 1e-12):

$$H(\beta) = -\beta \log_2 \beta - (1-\beta) \log_2 (1-\beta)$$

**2. The reported gauge** (stage\_b\_identity, PROC-SWITCH-01; SOP s41/s106). Beta is averaged over the class's IDENTITY loci - not a marker union - and divided by that class's MCMC floor. The haematopoietic-progenitor joint component uses the progenitor floor (PREREG s3):

$$A_{\text{mapped}} = H(\beta_{\text{■}}) / H_{\text{min}}(\text{class}), \quad \beta_{\text{■}} = \text{mean } \beta \text{ over the class identity loci}$$

**3. The two corrections that make it absolute.** The donor's decade and the laboratory's own zero are subtracted, which is what turns a relative reading into one that can be compared against a fixed line:

$$A_{\text{abs}} = A_{\text{mapped}} - c(\text{decade}) - z_{\text{lab}}$$

c(decade) comes from reference\_age\_curve\_v1.json (PROC-PANEL-03: per-decade median of mapped immune identity-loci A about the grand median, decades with n >= 30 only, n = 1379). z\_lab is the laboratory's own median on its 40-array healthy panel (lab\_zero.py). **With no laboratory zero the chain refuses:** A\_mapped is still returned, A\_abs and the placement are None, and the reading is marked not reportable.

**4. The healthy line and the band it is read against** (identity\_band\_v3.json, four zeroed healthy cohorts, n = 1379):

| quantity | value   | what it is                                                    |
|----------|---------|---------------------------------------------------------------|
| p10      | 0.9724  | lower edge of the healthy band                                |
| p50      | 1.0     | the healthy line - A_abs = 1.000 by construction              |
| p90      | 1.0248  | upper edge                                                    |
| width    | 0.0524  | p90 - p10, one band width                                     |
| sigma    | 0.02044 | (p90 - p10) / (2 x 1.2816) - the band read as a normal spread |

Placement is BELOW\_BAND below p10, ABOVE\_BAND above p90, IN\_BAND between; the tier word comes from cpg\_tiers.tier\_of(A\_abs, H\_min) (Stage 7, PROC-TIER-01).

**5. The four laboratories the line is built from** - each contributes its own zero, and its own measured false-alarm rate is what a departure is judged against:

| laboratory          | n   | z_lab (full cohort) | false alarm at p95 | at p99 | chip median SD |
|---------------------|-----|---------------------|--------------------|--------|----------------|
| GSE111629 UCLA      | 204 | -0.0673             | 0.0441             | 0.0049 | 0.0123         |
| GSE125105 Munich    | 201 | -0.0346             | 0.0647             | 0.01   | 0.0116         |
| GSE42861 Karolinska | 315 | 0.0084              | 0.0984             | 0.0508 | 0.0203         |
| GSE87571 Uppsala    | 659 | -0.0117             | 0.0561             | 0.0152 | 0.0138         |

**The false-alarm rate is a measured property of the laboratory, not a constant.** Across the four it ranges from 0.0441 to 0.0984 at p95 - so a departure just past the threshold means something different in Karolinska (0.0984) than in UCLA (0.0441), and the chain prints the sample's own laboratory rate beside the distance rather than a single number. Centring on the chip recovers a good part of it (the tail\_p95\_if\_chip\_centred column of the same file reads 0.0196-0.0413). **PROC-MAHA-03 (2026-09-22) does not reproduce that improvement under held-out estimation, and row 5b closed NOT COMMISSIONED.** Correcting an array by a single control array on its chip makes every laboratory's tail worse, because it adds that reference's own within-chip error (0.019-0.023) to remove an offset of 0.000-0.016. The chip term is measurable in one cohort of four - GSE42861, the one with the worst tail: ICC 0.409, p 0.003, and two held-out panel arrays per chip take its tail from 0.115 to 0.031. Chip position stays on the roadmap, with the measurement that would settle it named there.

**6. The departure** (stage\_5\_mahalanobis, PROC-MAHA-01; SOP s47-51). Each assessable component is standardised against the healthy line, and the components are combined as a distance:

$$z = (A_{\text{abs}} - 1.000) / \sigma, \quad D = \sqrt{(\sum z^2)} \text{ over the } n \text{ assessable components}$$

Thresholds are the chi-square quantiles for n degrees of freedom - D<sub>95</sub> = √χ²(0.95, n), D<sub>99</sub> = √χ²(0.99, n) - so the bar rises with the number of axes and a distance cannot be inflated by adding components. A component with no commissioned band is not assessable: the

haematopoietic-progenitor joint and every non-blood class today. **On whole blood  $n = 1$** , so the distance is  $|z|$  on the immune axis and the report says so in those words. The chain also prints  $\text{band\_widths\_from\_line} = (A_{\text{abs}} - 1.000) / (p90 - p10)$ , because a reader wants the departure in units of the healthy spread as well as in sigmas.

**7. What the same four laboratories forbid: cellular age in years** (stage\_6\_cellular\_age, PROC-AGE-01). The healthy immune gauge rises 0.47 mA per year against a within-laboratory spread of 0.0235, so inverting the curve resolves age to about 50 years for one array: 15.9 per cent of healthy donors land within ten years of their own age (Spearman  $\rho = 0.271$  on  $n = 1379$ ). The trajectory is real and reproduced - it IS the reference curve above - but it is a population measurement, and the stage reports the resolution rather than an age.

**8. The second opinion** (stage\_2b\_second\_opinion): the needlet ILC solver runs beside the constrained NNLS, both folded to classes, and the chain reports agreement when the class-level L1 difference is at or below 0.10. Cell-level disagreement inside one lineage is expected and is not scored.

### III.6 The screens, in the order the report prints them

What each number on a report is, where it comes from, and what it does not mean.

**Screen 1 - what is in the sample.** Stage 2 places the specimen against the 115-entry atlas and reports the fractions, folded to the eight architecture classes. A class whose estimated presence falls below its detection floor is masked and nothing is reported for it. Beside the composition the report prints the second opinion: the same specimen solved by the needlet ILC method, compared at class level, with AGREE when the L1 difference is at or below 0.10. Cell-level disagreement within one lineage is expected - the atlas cannot separate the members of a collinearity group - and is not scored.

**What it does not mean.** A fraction is not a cell count. It is the mixture that best explains this specimen's methylation given the atlas, and for the blood classes the members of a group are interchangeable within it. The atlas calls such a set a *collinearity group*; in plain terms it is a set of entries the atlas cannot tell apart, because their marker profiles are so similar that many different mixtures of them explain the same specimen equally well. PROC-SEP-03 measured which blood entries fall into that condition, which is why this chain reports at class level and treats cell-level disagreement between the two solvers as expected rather than as a fault.

**Screen 2 - the class gauge.** For each reportable class: A\_mapped, the two corrections, and A\_abs with its placement against the healthy band (p10 0.9724, line 1.0, p90 1.0248). The tier word comes from the breakpoints file and is withheld wherever the component's own band is wider than the gauge's normal band. With no laboratory zero the screen prints NOT REPORTABLE and the reason, and A\_mapped alone.

**Screen 3 - the departure.** Three numbers that answer three different questions, and the report prints all three because each alone misleads:

| what is printed                       | the question it answers                                                     | how to read it                                                                                             |
|---------------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| z on each axis                        | how far is this class from the healthy line, in units of the healthy spread | $z = (A\_abs - 1.000)/\sigma$ with $\sigma = 0.02044$ ; a z of 2 is two healthy spreads                    |
| band widths from the line             | the same distance in the units a clinician can see on the band              | $(A\_abs - 1.000)/(p90 - p10)$ ; useful because the band is what is drawn                                  |
| D, the Mahalanobis distance           | taking every assessable class together, how unusual is this specimen        | $D = \sqrt{\sum z^2}$ against $\sqrt{\chi^2(0.95, n)}$ and $\sqrt{\chi^2(0.99, n)}$ ; the bar rises with n |
| the laboratory's own false-alarm rate | how often a healthy donor from THIS laboratory exceeds that bar             | measured per cohort, 0.0441 to 0.0984 at p95 across the four - printed beside D, never assumed             |
| n assessable                          | how many axes the distance was computed on                                  | on whole blood $n = 1$ , so D is  z  on the immune axis and the report says so                             |

**The confidence statement is not a p-value on disease.** The thresholds are quantiles of the healthy distribution: crossing D95 means this specimen sits where about five per cent of healthy donors sit - and the laboratory-specific rate printed beside it says how close to five per cent that really is for this laboratory. It is a statement about rarity against a healthy reference, not a probability of any condition, and the chain names no condition.

**Screen 4 - the sky.** The per-address residual projected onto the sphere, with the shuffled-sky null beside it. Structure in the map is genomic: every pixel holds a contiguous run of CpGs on one chromosome (III.3). The map is a way of seeing where a departure lives, not a second measurement of it.

**Screen 5 - what this run refused.** Every masked class, every withheld tier word, every unavailable stage, each with the number that triggered it. A refusal is a result: it names what would have to change for the reading to become available.

III.7 The healthy reference - who it is, and what is not in it

**Who.** 4 public whole-blood cohorts, 1379 donors, each cohort zeroed by its own full-cohort median before pooling. Controls only where the cohort is a case-control study.

| cohort              | donors | its zero | role in the reference                       |
|---------------------|--------|----------|---------------------------------------------|
| GSE111629 UCLA      | 204    | -0.0673  | band + age curve + its own false-alarm rate |
| GSE125105 Munich    | 201    | -0.0346  | band + age curve + its own false-alarm rate |
| GSE42861 Karolinska | 315    | 0.0084   | band + age curve + its own false-alarm rate |
| GSE87571 Uppsala    | 659    | -0.0117  | band + age curve + its own false-alarm rate |

**How processed.** stage1\_noob\_450K mapped; the age curve is the per-decade median of mapped immune identity-loci A about the grand median, decades with n >= 30 only (PROC-PANEL-03: four healthy whole-blood cohorts (GSE87571, GSE42861 ctrl, GSE125105 ctrl, GSE111629 ctrl), each zeroed by its full-cohort median; per-decade median of mapped immune identity-loci A about the grand median; decades with n>=30 only).

**What is measured from them, in order:** each laboratory's zero; the pooled healthy band; the per-decade age reference; each laboratory's false-alarm rate at p95 and p99; and the age resolution that makes cellular age in years unreportable.

**What is NOT in the reference, stated so nobody assumes it is.** No disease cohort - the reference is healthy donors only, and the chain never compares a specimen to a disease group. No repeat draws, so the technical floor is an upper bound inferred from cross-sectional spread (III.3, difference maps). No tissue other than whole blood, which is why every non-blood class is unassessable today. No non-European ancestry breakdown: the four cohorts are European, the reference inherits whatever that implies, and that is a limitation to be closed rather than a caveat to be restated.

### III.8 Coverage - what is lit, and what lighting one cell requires

The framework spans five substrates and eight classes: forty cells, each needing its own floor and its own healthy reference. The commissioned chain reads one substrate - methylation - because that is what an array carries, and one specimen type. This is the honest coverage statement.

| what                                     | lit today            | what lighting the rest requires                                                                                                                             |
|------------------------------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| substrates                               | 1 of 5 (methylation) | nucleosome occupancy, fuzziness, WPS and fragment size need cfDNA sequencing on the same specimen; no cohort held here carries two substrates on one sample |
| classes with a commissioned band         | 1 of 8 (immune)      | a healthy cohort of that tissue, zeroed the same way, at n sufficient for a p10-p90 band                                                                    |
| specimen types                           | whole blood          | any other tissue needs its own zero and its own band                                                                                                        |
| atlas entries with a per-entry reference | 106 of 115           | the duplicate-label merge, then a re-measured per-entry reference                                                                                           |

**A note on cfDNA plasma weighting, because it is not applied and should not be.** The familiar plasma weights - immune about 70 per cent, cycling epithelial about 12, secretory about 8 - describe what a PLASMA draw contains, since cell-free DNA arrives weighted by tissue turnover. This chain reads a whole-blood array, where the cells are the specimen rather than shed fragments, and Stage 2 MEASURES the composition instead of assuming it. Those weights therefore appear here only as context for why the immune class dominates a blood reading. When a plasma cohort arrives they do not become a correction applied to the gauge: they become the prior against which the presence floors have to be re-measured, which is a commissioning run of its own.

A cell of that grid is lit when four things exist together: a floor from the MCMC build, a healthy reference on the same measurement scale, a laboratory zero, and a null that comes back empty. Three of the four are cheap; the healthy reference is not, and it is what gates every row.

### III.9 The guards - what each one refuses, and its last result

Last run 2026-09-22 10:28 at commit 3b38e87: **6 pass, 0 fail, 6 skipped**. A guard that did not run prints SKIPPED with its reason - never a pass. The verdict below is read from release\_check.json at build time, so this page cannot claim a guard passed when the file says otherwise.

| guard                     | what it refuses                                                                                                                                                                                                              | last result                                                                                              |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Rendered-claim scan       | no delivered document asserts a claim the record has reversed, and every passage the cover promises is present - checked against the RENDERED PDF, because a claim survives in data files, embedded markdown, render-time ta | PASS                                                                                                     |
| A-score formula self-test | the per-cell surface is mean_i H(beta_i)/H_min and the module refuses H(beta_mean) - the N7 defect cannot come back                                                                                                          | SKIPPED - FileNotFoundError: [Errno 2] No such file or directory: '/Users/hmahaffeyges/.claude-scienc..' |
| Sealed cohort anchors     | the sealed 648-sample foundation-cohort per-cell scores still reproduce from raw public data (r = 1.00000)                                                                                                                   | SKIPPED - FileNotFoundError: [Errno 2] No such file or directory: '/Users/hmahaffeyges/.claude-scienc..' |
| Deconvolver conformance   | the composition solver reproduces the answer key shipped with the test package                                                                                                                                               | SKIPPED - FileNotFoundError: [Errno 2] No such file or directory: '/Users/hmahaffeyges/.claude-scienc..' |
| Atlas separability        | the measured separability of the blood classes - the finding NILC reported and the atlas confirmed                                                                                                                           | SKIPPED - FileNotFoundError: [Errno 2] No such file or directory: '/Users/hmahaffeyges/.claude-scienc..' |
| Gauge switch              | run_full reports the identity-loci gauge, not the retired marker-union statistic                                                                                                                                             | PASS                                                                                                     |
| One tier definition       | every tier word in the chain comes from tier_breakpoints.json; no literal breakpoint survives in engine code; AT_CEILING at 1/H_min                                                                                          | PASS                                                                                                     |
| Patient sky               | the sky refuses a laboratory with no measured residual scale and masks classes below their presence floor                                                                                                                    | PASS                                                                                                     |
| Laboratory zero           | panels under 40 arrays are refused and a reading with no zero is marked NOT REPORTABLE                                                                                                                                       | PASS                                                                                                     |
| Sky commissioning         | PROC-CMB-05: a healthy sky is quiet at the measured level in every commissioned laboratory                                                                                                                                   | SKIPPED - FileNotFoundError: [Errno 2] No such file or directory: '/Users/hmahaffeyges/.claude-scienc..' |
| Bidirectional detector    | PROC-BIDIR-01: the sealed directional panel reproduces, including re-extraction from raw GEO                                                                                                                                 | SKIPPED - PROC-BIDIR-01: B1-B4 pass, B5 SKIPPED - needs the GSE153712 supplementary matrix (5.1 GB) ...  |
| Detection-vocabulary scan | no document claims the chain cannot detect something it has not been run on                                                                                                                                                  | PASS                                                                                                     |

### III.10 The files of the chain, by role

173 files are inventoried. The counts are the inventory's own, read at build time: **20** UNDESCRIBED, **18** chain, **40** guard, **6** interface, **28** record, **48** reference, **13** superseded.

#### The eighteen files in the chain - the ones a reading actually depends on:

| file                            | stage     | what it does                                                                                                                                                                                    |
|---------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cpg_conductor.py                | stage 0   | THE ORCHESTRATOR. run_full() calls every stage in order and returns one bundle; it resolves each stage module and data file by name (_find) so a missing file fails loudly instead of silently  |
| stage_0_intake.py               | stage 0   | Stage 0. Sample intake and chain of custody: IDAT arrival, manifest creation, identity checks. Built step by step against the SOP; four QC checks are still deferred because Stage 0 is not yet |
| idat_decoder_pure.py            | stage 1   | Pure-Python IDAT to beta - needs only idat_parse plus a static array manifest, no methylprep or minfi at runtime. This is what lets a reader reproduce Stage 1 without the R/Bioconductor stac  |
| idat_parse.py                   | stage 1   | Low-level IDAT binary reader used by Stage 1 when methylprep is not driving the parse.                                                                                                          |
| stage_1_idat_calibration.py     | stage 1   | Stage 1. Raw Illumina IDAT pair -> beta per CpG (methylprep, noob background/dye correction). The only step that touches instrument output.                                                     |
| walther_iam_deconvolver.py      | stage 2   | Stage 2. Constrained non-negative least squares: which cell types are present and in what fraction. Sparse by design - it reports only what it has evidence for.                                |
| lineage_splitter.py             | stage 2b  | Helper for Stage 2b: splits a lineage signal across its member entries.                                                                                                                         |
| nifc_celltype_deconvolver.py    | stage 2b  | Stage 2b. The second, independent solver (inverse-variance weighted, atlas posterior SD as the weight). Reported ONLY as an agreement flag at class level, never as the composition.            |
| bidirectional_decomposition.py  | stage 4.5 | Stage 4.5. Splits a departure into its hyper- and hypo-methylated halves, so direction is reported rather than magnitude only.                                                                  |
| build_healpix_mapping.py        | stage 4.6 | Builds the CpG -> pixel mapping from the array manifest in genomic order. Deterministic: same atlas + manifest -> byte-identical output.                                                        |
| generate_cpg_healpix_mapping.py | stage 4.6 | One-time generator for the mapping: lays the CpGs on the sphere in genomic order so every patient projection sits on the same grid as the archival plates.                                      |
| stage_4_6_patient_cmb.py        | stage 4.6 | Stage 4.6. The patient's sky: residual z per CpG against the sample's own composition expectation, on the laboratory's zero and scale, projected onto the sphere in genomic order.              |
| iamatlas_mahalanobis_scoring.py | stage 5   | Stage 5. The departure: distance from the healthy centre in units of healthy spread over the assessable class axes (Mahalanobis, 1936).                                                         |
| iam_cellular_age_scoring.py     | stage 6   | Stage 6. Inverts the healthy age curve. Reports the RESOLUTION, never an age in years: one array cannot place a person to better than ~50 yr (PROC-AGE-01).                                     |
| cpg_tiers.py                    | stage 7   | THE ONLY tier function. Reads tier_breakpoints.json; the engine previously carried three disagreeing definitions. Returns no tier when a reading is not reportable, and AT_CEILING at 1/H_min.  |
| iamatlas_a_scoring.py           | stage A   | The A-score itself: $A = \text{mean}_i H(\beta_i) / H_{\min}(\text{class})$ . Asserts against the wrong aggregation (entropy of the mean beta), which is the defect PROC-N7-01 caught.          |
| cpg_gauge_engine.py             | stage B   | The class gauge: the eight frozen H_min floors and the per-substrate table. Holds the saturation ceilings (1/H_min).                                                                            |
| lab_zero.py                     | stage B   | The laboratory zero: median of 40 healthy arrays of that laboratory, age-referenced. Refuses panels under 40 and marks a reading without a zero as NOT REPORTABLE.                              |

Twenty files are inventoried as UNDESCRIBED: present in the tree, not yet given a role. They are listed as such rather than quietly counted as part of the chain.

### III.11 Running it on your own sample

One command, and the flags are read from the script's own argument parser at build time, so this page cannot advertise an option the tool does not have.

```
python3 MethylPhys_Interface/run_sample.py --grn X_Grn.idat --red X_Red.idat --age 61 --lab GSE87571
--lab-zero -0.0117 --out report.html
```

| flag       | what it is                                                           | default                  |
|------------|----------------------------------------------------------------------|--------------------------|
| --grn      |                                                                      |                          |
| --red      |                                                                      |                          |
| --betas    |                                                                      |                          |
| --age      | declared age in years; without it the age term cannot be removed     |                          |
| --lab      | laboratory / pipeline identity, e.g. GSE87571 for a commissioned one |                          |
| --pipeline | pipeline map name from beta_scale_maps_v1.json                       | "stage1_noob_450K"       |
| --specimen |                                                                      | "whole blood"            |
| --lab-zero | this laboratory's measured zero; omit to read UNSET                  | None                     |
| --out      |                                                                      | "methylphys_report.html" |
| --id       |                                                                      | None                     |

**What it will refuse.** Without --age the age term cannot be removed and the absolute reading is withheld. Without --lab-zero the laboratory zero reads UNSET and the placement, tier and departure are all withheld - A\_mapped is still printed, because a relative number honestly labelled is more use than a fabricated absolute one. Commissioning your own laboratory means measuring that zero on 40 healthy arrays and a null that comes back empty; the procedure is in the runbook.

# WHERE THE TOOLS COME FROM

Issue 002 stated the physics of the floor. Since April the engine acquired an atlas, a deconvolver, a sky projection, a matched filter and a second statistic, and every one of them was borrowed from somewhere specific: from the semiconductor and quantum-computing applications of the same law, and from the Planck CMB pipeline. A reader who is going to spend time on this work will want to know where each idea came from and what it is doing here. This section says so, in the order a physicist would ask. Nothing here is a new claim; the new claims are in §5 (reproduced from 002) and the measurements are in §7–§8.

## 5A.1 The Mahaffey number

**M = 20.94.**  $M = E_{\text{drive}} / (k_B T)$  for the cell: the ATP ordering budget per event over the thermal noise quantum at body temperature. Dimensionless, fixed by biochemistry, the same for every cell.  $M = dG_{\text{ATP}}/(R T) = 54,000/(8.314 \times 310.15) = 20.94$  at 37 C. Author 2026-09-19: one fixed number; the 30.21 (ln2) form was wrong.

H\_min is a different quantity: the per-class entropy floor in bits (8 classes, 5 substrates each -> 40 floors), MCMC-confirmed, architecture-specific. M is one number for the cell; H\_min is forty numbers for the classes. They are never interchangeable.

n\_bio was an early (QAPE-era) form of the similar ratio and is NO LONGER USED. Every occurrence of n\_bio in Issue 002 (glossary, class cards ' n\_bio = 20.94', canine 20.84) is superseded by the Mahaffey number.

The same construction - a measured drive energy over the local thermal quantum - is what SCAPE computes for a transistor and QAPE for a qubit; those figures are quoted in Section 5A.8 exactly as their own reports state them, with their own denominators, and are not re-derived here.

Open. Inputs: dG\_ATP = 54,000 J/mol, R = 8.314 J/(mol K), T = 310.15 K (002 glossary, \_gape\_constants\_private.py l.26, Cellular Margin l.28, Hubble2GAPE l.3504) - consistent across the corpus.

IAM\_Hubble2GAPE l.2192 writes  $M = E_{\text{drive}}/(k_B T_{\text{local}} \ln 2)$  and gives ~30 for chemical systems; the author's ruling has no ln2 (20.94). That line of the tex should be brought into agreement. The earlier 003 draft's cross-substrate table (SCAPE 117x, transmon M = 1 exactly) was computed with the ln2 form and is WITHDRAWN from s5A.1; SCAPE and QAPE figures are quoted only as their own reports state them.

## 5A.2 Forty floors — eight classes × five substrates, each confirmed by MCMC

The cellular H\_min is not one number. Each of the eight architecture classes has its own floor on each of five physically distinct lab measurements — methylation β, nucleosome occupancy, nucleosome fuzziness, window protection score, and cfDNA fragment size — giving the forty-cell table below. The methylation column was fixed by the G-002 chain (17 chains, R-hat < 1.001) and the other four by G-003b (5 × 32-walker ensembles); all forty were frozen 2026-04-06 and the engine at HEAD carries them byte-for-byte (RECON H1). Because every floor has its own ceiling 1/H\_min, every class has **five saturation limits**, and the substrate a sample arrives on selects which five of the forty apply to it. The runtime saturation rule flags any A within 0.005 of its ceiling; a class whose methyl ceiling is below 1.10 (stem\_pluri, 1.018) cannot register BREACH on methylation and must be read on another substrate.

| class      | H_min<br>meth | H_min<br>nucl | H_min<br>fuzz | H_min<br>wps | H_min<br>frag | ceil.<br>meth | ceil.<br>nucl | ceil.<br>fuzz | ceil.<br>wps | ceil.<br>frag |
|------------|---------------|---------------|---------------|--------------|---------------|---------------|---------------|---------------|--------------|---------------|
| terminal   | 0.772837      | 0.992027      | 0.736973      | 0.958909     | 0.624938      | 1.294         | 1.008         | 1.357         | 1.043        | 1.600         |
| secretory  | 0.843264      | 0.982560      | 0.847947      | 0.634534     | 0.697718      | 1.186         | 1.018         | 1.179         | 1.576        | 1.433         |
| immune     | 0.838889      | 0.989930      | 0.830377      | 0.589644     | 0.711534      | 1.192         | 1.010         | 1.204         | 1.696        | 1.405         |
| progenitor | 0.852216      | 0.972790      | 0.961900      | 0.988046     | 0.808978      | 1.173         | 1.028         | 1.040         | 1.012        | 1.236         |
| cycling    | 0.856055      | 0.980072      | 0.819030      | 0.627429     | 0.687936      | 1.168         | 1.020         | 1.221         | 1.594        | 1.454         |
| stromal    | 0.862950      | 0.985667      | 0.832386      | 0.612686     | 0.724691      | 1.159         | 1.015         | 1.201         | 1.632        | 1.380         |
| stem_adult | 0.873718      | 0.960866      | 0.980754      | 0.988964     | 0.841327      | 1.145         | 1.041         | 1.020         | 1.011        | 1.189         |
| stem_pluri | 0.982166      | 0.799818      | 0.962920      | 0.905004     | 0.973583      | 1.018         | 1.250         | 1.039         | 1.105        | 1.027         |

Posterior widths matter for tier calls: the methylation σ(H\_min) from G-002 is ±0.0012 (Issue 002 cards), so a tier boundary 0.01 wide is ~8σ of floor uncertainty — the floor is not the limiting error; Stage-1 calibration and the age band are.

The sampler, so a reader can re-run it

| setting | value                           |
|---------|---------------------------------|
| Sampler | emcee affine-invariant ensemble |

| setting                   | value                                                                                                                                                                                                                                                                                        |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N_WALKERS                 | 32                                                                                                                                                                                                                                                                                           |
| N_BURNIN                  | 2,000 steps                                                                                                                                                                                                                                                                                  |
| N_PRODUCTION              | 5,000 steps                                                                                                                                                                                                                                                                                  |
| N_CHAINS                  | 5 independent (for R-hat)                                                                                                                                                                                                                                                                    |
| THIN                      | 10                                                                                                                                                                                                                                                                                           |
| PRIOR_SIGMA               | 0.05 (prior width around the published central value; walkers initialised at mean + 0.1 x PRIOR_SIGMA x N(0,1))                                                                                                                                                                              |
| SIGMA_A                   | 0.03 (likelihood tolerance on (A - 1)^2)                                                                                                                                                                                                                                                     |
| SEED_BASE                 | 20260420                                                                                                                                                                                                                                                                                     |
| Acceptance window         | 0.20-0.50 (step-size tuning; <0.1 too large, >0.7 too small)                                                                                                                                                                                                                                 |
| Autocorrelation time      | tau ~ 50 -> ~16,000 effective samples per class                                                                                                                                                                                                                                              |
| Posterior draws per class | (32 x 5,000 x 5)/10 = 80,000                                                                                                                                                                                                                                                                 |
| Likelihood                | Gaussian on (A_i - 1.0)^2 / SIGMA_A^2 per reference cell                                                                                                                                                                                                                                     |
| Prior                     | Gaussian centred on the published calibration value                                                                                                                                                                                                                                          |
| Cross-check               | leave-one-out bootstrap, n = 10,000 resamples; MCMC and bootstrap must agree within 5-10% or the cohort is audited. RUN in April for the 32 G-003b floors only (0.168%, 24/32 in CI); run for the 8 methylation floors on 2026-09-20 - PROC-HMIN-BOOT-01: 0.060% mean, 0.095% max, 8/8 in CI |
| Convergence               | 17 methylation chains R-hat < 1.001 (G-002); 5 x 32-walker ensembles per substrate (G-003b)                                                                                                                                                                                                  |
| Runtime                   | G-002 29.7 s for 8 classes (laptop); G-003b ~24 min for 32 posteriors (desktop)                                                                                                                                                                                                              |
| Notable posterior shift   | immune: 0.795 (neutrophil-based calibration) -> 0.838889 after MCMC over six immune cell types, a 6.44 sigma move (EDEAR s4.6); every immune A in the database was revised downward by approximately 0.055 as a consequence (Issue 002, immune card)                                         |
| H_min_global              | 0.756499 = H(0.782), frontal cortex neuron, Lister 2013 (E073) - the universal reference the class floors are read against                                                                                                                                                                   |
| Reference-cohort rule     | FACS-sorted or laser-microdissected only; never bulk tissue (cell-type purity)                                                                                                                                                                                                               |

### 5A.3 The IAM Atlas is a posterior, not a lookup table

IAMAtlasREBUILD (483,092 CpGs × 115 cell types) was built by per-class MCMC over the public reference atlases — immune alone from 63 cell types across 25 source atlases, 4,987,426 observations, batch 1500, ~146,000 s; stromal, the smallest, from 5 cell types and 17 atlases in ~2,550 s. For every CpG and every class the atlas stores the posterior mean, SD and credible interval (8 "brightness files", 246 posterior columns), plus per-cell-type means. Three consumers read it: the deconvolver takes the means as its reference matrix; the gauge identity loci are the CpGs where the class mean sits within 0.05 of the class H\_min\_β; and the Stage 4.6 sky (5A.4) uses the per-CpG SD as its noise model. The predecessor IAMAtlas.csv.xz was retired for a collapse/flatness bug — which is what the first two lines of the CHANGELOG are about, and why the provenance file names the build.

#### How the atlas was built

| item                      | value                                                                                                                                                    |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source rows               | 10,938,662 (iamatlas_mcmc_inputs.csv, 732 MB) from Moss 2018, Loyfer 2023, EpiDISH, Salas, Lister 2013 and others, harmonised to one cell-type label set |
| Chains per cell type      | 2                                                                                                                                                        |
| SD tightness target       | posterior SD < 0.10 for >= 90% of CpGs per cell type                                                                                                     |
| Informative-CpG selection | posterior SD <= 0.10 AND between-cell-type variance >= 0.02                                                                                              |
| Array bridging            | only HM450-overlapping CpGs retained in production (483,092)                                                                                             |
| Failure recorded          | stromal v0.1 chains failed to converge (R-hat 3.67) until source-specific labels were harmonised -> v0.2                                                 |

| item                  | value                                                                                                                                                                                                                                                            |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Deconvolver weighting | posterior-SD-weighted NNLS: $W = \text{diag}(1/\max(\text{posterior SD}))$ , floor $1e-3$ ; minimum 100 usable CpGs; dispersion penalty ceiling 0.20, high-dispersion flag 0.30; streaming 1K-marker chunks, re-solve every 10                                   |
| Marker selection      | one-vs-rest top-N ( $N = 100$ ) on $ \text{target mean} - \text{mean}(\text{others}) $ per cell type (SOP s29); 131 chrX markers removed 2026-06-11 for derived sex-invariance                                                                                   |
| Walther vs NILC       | substrate-level L1 disagreement of $\sim 0.10$ - $0.25$ was EXPECTED and did not gate; inference-level disagreement on the disease-relevant class did (SOP s32-33). NILC cut 2026-07-02, reinstated 2026-09-22 as the class-level second opinion (PROC-NILC-01). |
| Versioning rule       | the atlas evolves by deepening (new source rows, new SHAs) not by product versions; every consumer reads $H_{\min}$ from IAMAtlasREBUILD_provenance.json (SOP s99) - hardcoding $H_{\min}$ refuses deployment                                                    |

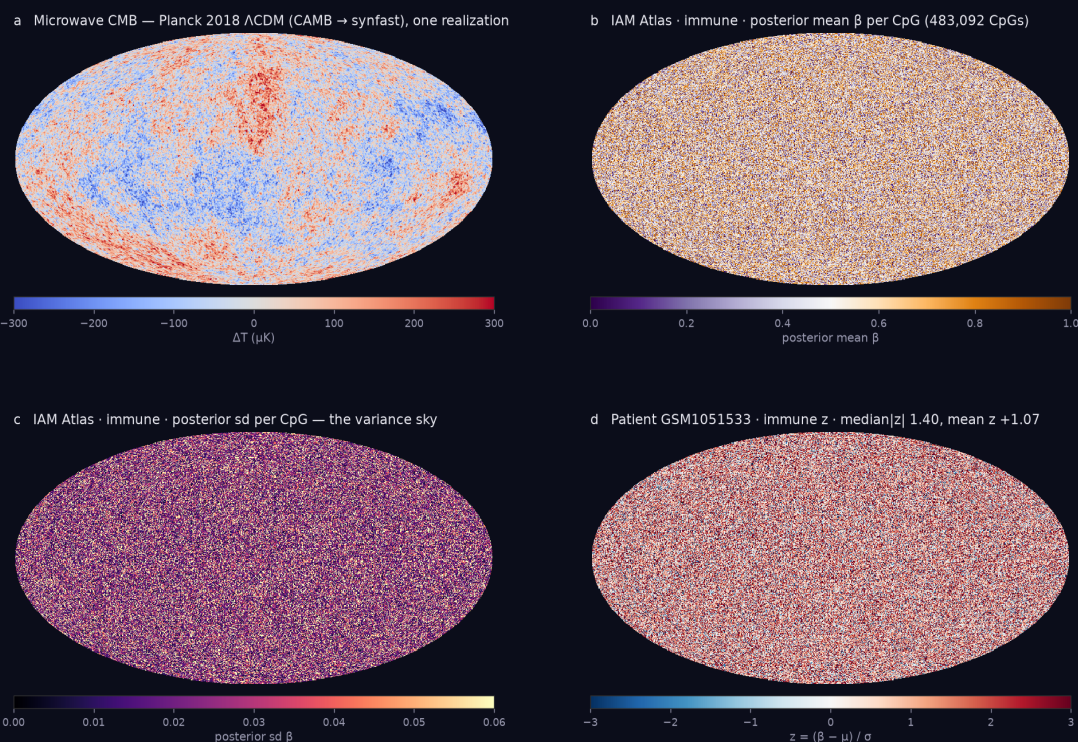
5A.4 The CMB toolkit, applied to the methylome

The methylome and the microwave sky pose the same problem: one noisy map, a physical reference for what it should look like, several foregrounds, and the need to say whether a departure at a given position is real. The engine borrows Planck tools, each named with its origin and its methylome analogue: three map-making tools built in spring (below) and four validation disciplines that ran in September (s1.6, s8).

| Planck tool                                                                    | what it does there                                                                  | methylome analogue in the engine                                                                                                                                                                                                                                                                                                                                                                  | status at HEAD                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Brightness / temperature map with per-pixel noise                              | $T(\text{pixel})$ vs the $\Lambda$ CDM expectation, standardised by the pixel noise | Stage 4.6 <b>Personal Cosmic Methylome</b> (cpg_patient_cmb.py): $z(\text{CpG}) = (\beta_{\text{patient}} - \mu_{\text{class}}) / \max(\sigma_{\text{class}}, \text{floor})$ , $\sigma$ from the atlas posterior, so $z$ is departure relative to healthy biological variation at that locus; projected to a HEALPix sky by atlas row order, one Mollweide per class                              | BUILT 2026-06-29                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Assessability / masking                                                        | galactic mask: don't read where the foreground dominates                            | a class is ASSESSABLE from blood only if its DNA is present; when it is not, the whole panel reads as a uniform offset and is drawn reference-only. Self-determined: $\text{median} z  < \text{ASSESS\_MAX}$ . The presence rule (§3.2) is the same idea one stage earlier                                                                                                                        | BUILT                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Matched filter                                                                 | correlate the map against a template to detect a known signal below the noise       | Stage 5 <b>residual-map matched filter</b> : Pearson $r$ between the patient's per-CpG departure from the derived atlas baseline and a disease's signed residual map, with Fisher CI; fires when the CI clears zero in either direction. Breast: $\rho = +0.058$ , CI $[+0.001, +0.114]$ on the pre-dx cohort (a detection whose lower bound is 0.001 above zero — marginal, and printed as such) | BUILT; AD residual map removed 2026-07 (was acting as AD's detector)                                                                                                                                                                                                                                                                                                                                                                               |
| Component separation cross-check (Commander / NILC / SMICA / SEVEM must agree) | four independent algorithms had to agree before any CMB result shipped              | <b>NILC methylome deconvolver</b> : weights derived once from the data's own covariance, independent of Walther's NNLS; the two had to agree on class fractions before L5+ was trusted (Walther vs NILC $\rho = +0.74$ immune, $+0.82$ progenitor on GSE51032)                                                                                                                                    | <b>CUT 2026-07-02</b> (commit c1be0c3), then <b>REINSTATED 2026-09-22</b> : rerun as designed in September (PROC-NILC-01), the divergence proved to be the diagnostic it was built to give, and it runs now as stage_2b_second_opinion - compared with Walther by class, agreement bar $L1 \leq 0.10$ , reported as a flag and never as the composition. The cross-method discipline is enforced again; RUNBOOK s11 forbids disabling either relay |

The honest note on the last row: the Planck discipline was the strongest methodological borrow in the engine, and in July it was removed to simplify. That is no longer where this stands. Rerunning the cut solver as designed (PROC-NILC-01) showed its divergence was marking exactly what it was built to mark - where the atlas does not separately determine the composition - so the cut had removed the instrument that was reporting the problem. It was reinstated on 2026-09-22 as a class-level second opinion, and RUNBOOK s11 now forbids disabling either relay.

## Four skies on one grid — HEALPix NSIDE = 128 (196,608 pixels), Mollweide, CpGs in atlas row order (chr1 → chrY)



(a) is a sky. (b)–(d) are the genome wrapped onto the same sphere so the same tools apply: a reference map with per-pixel uncertainty (b, c) and the residual of one observation against it (d).

Patient: whole blood, 450K, Stage-1 calibrated. Immune is the class present (deconvolver fraction 0.93); absent classes are not shown. The uniform red tint in (d) is the +0.05 reference- $\beta$  offset of Issue 003 §6a (mean  $z$  +1.07), not a patient finding.

Four skies on one HEALPix grid (NSIDE 128, 196,608 pixels, Mollweide; CpGs in atlas row order chr1 → chrY, the Plate 1 convention). (a) One realization of the microwave CMB from the Planck 2018  $\Lambda$ CDM power spectrum (CAMB → healpy.synfast). (b) The IAM Atlas immune-class posterior mean beta per CpG - the 'brilliance' sky, 483,092 CpGs, MCMC posterior. (c) The same class's posterior sd per CpG - the healthy-variance sky, which is what makes (d) possible. (d) Patient GSM1051533 (whole blood, 450K, Stage-1 calibrated, RA-study control) as  $z = (\beta - \mu) / \sigma$  against (b, c), computed by the engine's `cpg_patient_cmb.py`. Median  $|z|$  1.40; mean  $z$  +1.07 - the uniform red tint is the +0.05 reference-beta offset of §6a, not a patient finding. The engine's self-determined assessability flagged stromal as assessable from blood (median  $|z|$  1.44); the deconvolver's presence gate says stromal = 0, and the presence gate is the correct one - a known weakness of the median- $|z|$  criterion recorded here. The point of the figure is the method, not the patient: a reference map with per-pixel uncertainty and the residual of one observation against it is exactly the Planck workflow, and it is why the CMB toolkit transfers.

## 5A.5 Why this is not a cohort method

Standard biomarker work standardises a patient against a population — a z-score against controls, a classifier trained on cases. The IAM instrument does not. The reference is **physical** (forty frozen floors) and **derived** (the atlas posterior and the age band compiled from healthy references only, before any disease data were seen). A patient is read against that reference alone; cohorts enter only to establish the *direction* a disease moves a class (L-5, §9) and never as a baseline. What this buys: no population standardisation, no training set, one frozen reference that a stranger can download and check, and a per-patient absolute reading rather than a percentile. What it costs, stated here because the measurements of §7 found it: the healthy range is a function of class *and* substrate, and the same immune panel with the same floor reads ~0.85 in whole blood and ~1.03 in plasma. A physical reference has to be anchored per substrate before a departure means anything — which is PROC-WB-IMMUNE-01's open verdict. The analyst preparing this issue built a cohort comparison three times before internalising this rule; it is now a rule with a mechanism (presence-gated, per-sample scoring functions) rather than a reminder.

## 5A.6 Two surfaces, two aggregations - ruled in s1.5

§3.3 shows all four aggregation × loci combinations on the same eleven samples. Identity loci carry the class gauge; discriminative markers carry presence and the sealed anchors. RULING A3 (s1.5) fixes one aggregation per surface: the gauge on identity loci is  $H(\beta_{\text{mean}}) / H_{\text{min}}$ , because  $H_{\text{min}}$  and the age band are both defined that way; the separation statistic on discriminative markers is the mean of per-CpG  $H$ , because those panels are bimodal and the sealed anchors are that statistic. On the declared substrate the two differ by a constant +0.03; on mixed or absent panels they diverge, and the gauge now refuses those panels by a measured Jensen-gap test rather than by rule.

## 5A.7 The Mahalanobis hull - Stage 5, specified, not yet run here

| item                   | value                                                                                                                                                                                             |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What it is             | Mahalanobis distance of the patient's 115-cell A-vector from a healthy-control hull; Stage 5 adjudicator (Option A, age-matched class gauge)                                                      |
| Hull v0_1              | n = 601 HC; fixed threshold d >= 2.0 - mathematically indefensible for 112-dimensional data (expected median under normality is sqrt(112) ~ 10.6)                                                 |
| Hull v0_5 (production) | n = 2,523 HC across 8 cohorts (CR2: 2,481); p95 default d >= 13.62; p99 strict d >= 18.43                                                                                                         |
| Anchor trade-off       | GSE51032 case-vs-HC Cohen's d +2.088 (v0_1) -> +1.450 (v0_5): the honest cost of a hull that represents more populations and platforms                                                            |
| Transfer               | Han Chinese n = 42 (GSE141682) median d = 10.51, no systematic offset (small n); GIFT HC n = 193 median d = 12.18 - FTD-context HC selection broadens the envelope into case range                |
| Guards                 | negative quadratic form = corrupted reference, halt; >15 imputed cell types = INSUFFICIENT_DATA (hard), 6-15 = PARTIAL_DATA (soft)                                                                |
| Status in this issue   | NOT executed; specified here from SOP v1.4.0 as the specification for PROC-HULL-01 (open row: Stage 5 is driven by the stem_adult false alarm and waits on the lab zero; see CHAIN_COMMISSIONING) |

## 5A.8 The null suite - what a sealed result has beaten

| null                          | criterion                                                                                                                                                                                |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N1 label permutation          | observed statistic outside the 95% CI of the permuted distribution                                                                                                                       |
| N2 age-stratified permutation | shuffle within age decades; percentiles, not Gaussian                                                                                                                                    |
| N3 sex-stratified permutation | as N2 by sex                                                                                                                                                                             |
| N4 cohort-split replication   | >=80% of 100 random splits PASS (same sign, CI overlap)                                                                                                                                  |
| N5 plate/position             | position-stratified permutation AND position-amplitude regression both p<0.05 - catches edge effects, scanner drift                                                                      |
| N6 injection-recovery         | recover an injected effect within +/-20%; recovery of opposite sign = HARD halt and full audit                                                                                           |
| N7 end-to-end synthetic       | the gold standard - a chain that fails N7 is unverified; synthetic patient generator = the methylome's FFP10/NPIPE                                                                       |
| N8 search-space correction    | permutation FWE by default (Bonferroni over-conservative for correlated features); depends on N1                                                                                         |
| Sealing                       | SEALED (all pass) / RESTATE (one fails, scope limited) / RETRACT (two+ fail). PREREG modified after data observation = auto-RETRACT. Sealing is not transitive across artifact versions. |

## 5A.9 The downstream chain as specified (SOP v1.4.0), for the stages this issue does not execute

| stage / rule                        | specification                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stage 7 tiering                     | 6 engine tiers (SUPPRESSED, NORMAL, ELEVATED, WARBURG_TRANSITION, SIGNIFICANTLY_ELEVATED, BREACH); DETECTABLE/URGENT/FLOOR_BREACH collapsed to one customer label deliberately; BORDERLINE_TIER when top-two tier probabilities differ by <0.20; per-class ceiling caps the reachable tier (stem_pluri blind to BREACH). Observed: all 1,174 EPIC-Italy patients saturate on >=1 class; only cycling in range for ~50%. |
| Stage 7 smoking bins (v1.2 interim) | floor shifts current 1.10, former 0-5y 1.08, 5-15y 1.07, 15+y 1.05, never 1.04; recency scores 0/0.10/0.30/0.60/1.00                                                                                                                                                                                                                                                                                                    |
| Stage 8 matching                    | two parallel paths (per-card Boolean rules + matrix lookup) on identical inputs; disease rows are time-phased templates (long_pre_dx, mid, mid_late, near_dx); no single tile fires a card; wrong-sign overlap downgrades, never matches; <80% residual-map coverage -> INSUFFICIENT, fall back to class tiers                                                                                                          |
| Stage 8 covariates                  | smoking and BMI are within-card covariates (on the causal path) and adjust CONFIDENCE only, never the MATCH/NO_MATCH verdict; never subtracted globally                                                                                                                                                                                                                                                                 |
| Stage 8 prior                       | cancer prior = US SEER age-stratified base rate, not personal risk: cycling 0.055, secretory 0.140, immune 0.020, terminal 0.008, stromal 0.005, stem_adult 0.008, progenitor 0.006                                                                                                                                                                                                                                     |
| Stage 9 anchors                     | literature anchors are the ONLY external information in customer output; DOI-keyed, peer-reviewed, never paraphrase                                                                                                                                                                                                                                                                                                     |

| stage / rule                      | specification                                                                                                                                                                                                                                                                                                |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Stage 9 legal gate</b>         | DRAFT report scanned against a CANNOT_SAY list (diagnostic determinism, directives, future-state claims, treatment specifics); >=1 hit halts shipping (s76)                                                                                                                                                  |
| <b>Audit</b>                      | per-patient append-only record; re-runs get new run_ids; any HARD anywhere halts; >=2 SOFT in one stage escalates (s79, s92-96)                                                                                                                                                                              |
| <b>Foreground firewall (s104)</b> | the production chain subtracts NO age-, sex- or smoking-driven change - 'such change is the signal itself, not contamination'; age enters as the reference band, sex as chrX marker removal; the subtraction modules are test-only and off-by-default is insufficient - production config must not load them |
| <b>Pipeline rule (EDEAR)</b>      | one IDAT-to-beta pipeline per patient at enrolment; cross-patient analytics use delta-A from personal baseline, never absolute A; a lab pipeline change triggers re-baseline (three readings) before drift detection resumes                                                                                 |

**Pipeline relativity of absolute A.** Same Xu-538 panel, different IDAT-to-beta pipelines: absolute A baselines 0.38-0.62 across cohorts (within-EPIC-Italy -0.44), direction preserved within pipeline (April 2026). Absolute A is pipeline-relative; this is the leading hypothesis for the 3/3 below-band healthy whole-blood donors (PROC-WB-IMMUNE-01).

**A cross-species prediction.**  $H_{\min}(\text{canine})/H_{\min}(\text{human}) = 1.00484$  at  $T = 311.65 \text{ K}$  via  $H_{\min}(T) = H_{\min}(37C) \times (T/310.15)^2$ ; canine Mahaffey number at  $311.65 \text{ K} = 54,000/(8.314 \times 311.65) = 20.84$  (scales as  $1/T$ ). A cross-species prediction that costs nothing to test on public canine arrays.

## 5A.10 The validation catalogue as recorded - NOT re-verified in this issue

These are the effect sizes the corpus records for the sealed VALs. This issue re-ran none of them; they are listed so the reader knows what exists and can ask for the sealed record. Where two instruments give different signs for one result, both are shown.

| result                                       | VAL               | recorded effect                                                                                                            | note                                                               |
|----------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Breast, >10 yr pre-diagnostic, EPIC-Italy    | VAL-047 / VAL-093 | d = -1.226 (p = 3e-4) matched filter; per-cell fan-out Baso d = +1.142 (VAL_001); pancreatic beta tile +1.020 off-tissue   | sign conventions differ by instrument - stated as recorded         |
| AML, blood at diagnosis                      | VAL-082           | immune A d = +3.71 (largest single-cohort effect on record)                                                                |                                                                    |
| DLBCL                                        | VAL-082           | A d = +3.20                                                                                                                |                                                                    |
| Glioma, plasma cfDNA                         | VAL-090           | cortical-neuron fraction 1.09% (d = +1.96 vs healthy); GBM tumour 39.3% vs 62.4% non-tumour brain (d = -2.81)              | the shedding argument for CNS                                      |
| HCC without documented risk factors          | -                 | secretory A d = +0.62, p = 0.0072                                                                                          |                                                                    |
| NAFLD hepatocytes                            | -                 | A ~ 0.87, below the secretory floor: loss-of-identity, an INVERSION reading                                                |                                                                    |
| Pulmonary arterial hypertension, endothelium | VAL-109/110/113   | d = +0.79 to +1.52 across three atlases (Caggiano +1.42, EpiSCORE -0.80, Loyfer +0.83)                                     | sign disagrees across atlases                                      |
| Bicuspid aortic valve, heart fibroblast      | VAL-110           | d = +2.10 (single-tile maximum, cardiovascular)                                                                            |                                                                    |
| Health ABC ageing, Hannum n = 656            | CPG_VAL_015       | r = -0.197, p = 3.7e-7; late-life acceleration rho = -0.854                                                                |                                                                    |
| TCGA pan-cancer                              | Issue 002         | A_tumour > A_normal in 27/28 types; independent set delta-A = 0.140 +/- 0.092, 14/15                                       | follows from global hypomethylation; not discriminating on its own |
| DunedinPACE trajectory                       | Issue 002         | $E(a_{\text{bio}}) = \exp(1 - 1/a_{\text{bio}})$ reproduces rising pace, midlife peak (40.6 y), oldest-cohort deceleration | asymptote, not survival bias - a claim                             |
| NHANES 1999-2002 (external)                  | Luo 2019          | top vs bottom decile age acceleration: cancer-mortality HR 2.14 (DunedinPoAm), >=1.58 for 3 of 4 clocks                    | context for why cellular age matters                               |

## 5A.11 QAPE and SCAPE - the sibling applications, structure and published results only

The same statistic on two other substrates. Their calibration inputs are not part of this document.

| engine | A                                                                                                                                                    | anchors                                                                                                                                                                                            | what sets the floor                                                                                                                                                                                                                   | published prediction record                                                                                                |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| QAPE   | $A = -\ln(1 - p_{\text{ref}})$ , $p_{\text{ref}}$ = two-qubit gate error                                                                             | fault-tolerant anchor $A < 1\text{e-}4$ (100 ppm); NISQ-viable $1\text{e-}3$                                                                                                                       | floor set by the substrate's oxide TLS defect density: Al/AlOx many suboxides, Nb four oxide phases, Ta one (Ta2O5) - fewest interface defects; vacuum substrates (ions) have no TLS floor and QCCD connectivity scales $\sim \log N$ | best blind test: IonQ Forte -1.0% error                                                                                    |
| SCAPE  | $A = [\text{TDP}/(N_{\text{trans f}})] / [k_B (T_{\text{op}} + 273.15) \ln 2]$ (E per switch over the Landauer minimum, $2.81\text{e-}21$ J at 75 C) | three components:<br>$A_{\text{thermo}} = 1 + A_{\text{isa}}$<br>(ISA floor - 1) + $A_{\text{node}}$ ;<br>ISA floors x86 $\sim 70$ , Apple ARM $\sim 15$ , Qualcomm ARM $\sim 35$ , GPU $\sim 250$ | floor is temperature-set and cannot be engineered away; higher junction T lowers A                                                                                                                                                    | P003 (Apple M5 third regression) NOT CONFIRMED - M5 65.8x vs M4 68.2x; P004 (Blackwell B200 step > 7.6%) CONFIRMED, +51.6% |



# THE PHYSICS - LANDAUER, THE MAHAFFEY NUMBER, THE REFERENCE, THE GAUGE

Three quantities that must not be confused, and what is physics versus what is measured

## 5.0.1 Premise: the methylome obeys Landauer's bound

Cytosine methylation is a stable, enzymatically written and erased mark whose genome-wide pattern carries the information of cell identity and state. Writing or erasing it is an information-processing operation, and Landauer's principle sets the minimum energy dissipated per bit:  $k_B T \ln 2 = 2.97 \times 10^{-21} \text{ J}$  at 310.15 K. That the methylome obeys this bound is peer-reviewed physics (Sanchez & Mackenzie 2016, PLoS ONE): with  $p_i$  the methylation level at site  $i$ , the per-site Shannon entropy  $H(p_i) = -p_i \log_2 p_i - (1-p_i) \log_2(1-p_i)$  measures the uncertainty of methylation status, the information processed in a region is the change in the sum of  $H$ , and the minimum energy to process it is that information times  $k_B T \ln 2$ . Their thermal background of methylation change follows a Weibull law whose scale recovers the DNA persistence length from methylome data alone. We take three things as established:  $k_B T \ln 2$  is the anchor; thermal fluctuations produce a real statistical background; signal and background must be measured from one origin of coordinates. We part with them at that origin (s0b).

## 5.0.2 Three quantities

Issue 002 used  $H_{\min}$  where the physics called for the Mahaffey number, and derived where it should have measured. The three quantities below are distinct in units, in cardinality and in what fixes them; every later section names which one it is using.

| symbol        | name                    | definition                                                                                                                                               | units        | one per        | fixed by                                         | provenance                                                                                                                                                |
|---------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| M             | the Mahaffey number     | $E_{\text{drive}} / k_B T$ - the energy a substrate spends per irreversible write, in units of the thermal quantum at its operating temperature          | none (ratio) | substrate      | biochemistry / device physics                    | $\Delta G_{\text{ATP}} = 54 \text{ kJ/mol}$ at 310 K -> 20.9; Apple M1 ~117; AI transmon 1 (exact)                                                        |
| $H_{\min}(c)$ | class entropy reference | the binary entropy of the healthy class's mean methylation level on its identity loci; $A = 1$ by definition there                                       | bits         | class (8)      | healthy reference cells, calibrated once, frozen | G-002 MCMC on 37 reference cells ( $R\text{-hat} < 1.001$ ), April 2026; bootstrap 8/8 in CI (PROC-HMIN-BOOT-01); iamatlas_gauge_id entity_loci_v1_0.json |
| $A_c$         | the gauge               | $H(\beta_{\text{bar}_c}) / H_{\min}(c)$ , $\beta_{\text{bar}_c}$ the sample's mean methylation over the identity loci of class $c$ , on the mapped scale | none (ratio) | class x sample | the sample, over the frozen reference            | cpg_conductor.stage_b_identity; then $A'' = A - c(\text{decade}) - z_{\text{lab}}$ (s3.5)                                                                 |

## 5.0.3 The Mahaffey number

THE MAHAFFEY NUMBER. Thermal noise is not a nuisance to subtract; it is the unit.  $M = E_{\text{drive}} / k_B T$  is how many thermal quanta a substrate pays per irreversible write at its own temperature. For the cell the drive energy is the free energy of ATP hydrolysis:  $M_{\text{cell}} = \Delta G_{\text{ATP}} / RT = 54,000 / (8.314 \times 310.15) = 20.9$ . The same ratio places a CMOS transistor at ~117 (Apple M1, measured switching energy at 348 K) and an aluminium transmon at exactly 1: with  $E_{\text{drive}} = \Delta G_{\text{AI}} \ln 2$  and  $T_{\text{gap}} = \Delta G_{\text{AI}} / k_B$  both the gap and  $\ln 2$  cancel - the saturated case that superconducting-qubit engineering reaches by cooling until  $k_B T$  is the only energy scale left. The cell cannot cool. It runs ~21 thermal quanta above the floor at 310 K and pays that to write and hold every mark against noise it cannot escape.  $M$  is one number for the substrate, fixed by biochemistry; it does not determine  $H_{\min}$  and it does not enter  $A$ . Its role in this document is to say what kind of instrument this is: the cell is an information-writing substrate read against thermal noise, exactly as a chip or a qubit is, and the healthy entropy reference is a statement about how a cell class holds its written state at that ratio.

| substrate                                             | $E_{\text{drive}}$                          | T                            | M         |
|-------------------------------------------------------|---------------------------------------------|------------------------------|-----------|
| CMOS logic transistor (Apple M1, measured TDP/switch) | switching energy                            | 348 K                        | ~117      |
| human cell nucleus                                    | $\Delta G_{\text{ATP}} = 54 \text{ kJ/mol}$ | 310 K                        | 20.9      |
| aluminium transmon                                    | $\Delta G_{\text{AI}} \ln 2$                | $\Delta G_{\text{AI}} / k_B$ | 1 (exact) |

## 5.0.4 $H_{\min}$ is measured, not derived - what Issue 002 said and what is retired

WHAT ISSUE 002 SAID, AND WHAT IS RETIRED. Issue 002 s2.1 titled the floor 'H\_min Derivation' and s2.1a laid out a seven-step chain from Landauer's bound, through the cell's ATP budget and the DNMT1 error rate, to 'a lower bound on entropy', and asked the reader's question aloud: how does an energy bound produce a lower bound on entropy? The honest answer is that it does not. An energy bound per operation constrains the COST of writing and holding a pattern; it does not fix the ENTROPY of the pattern a healthy class holds. Steps 1-3 of that chain stand as published physics and biology (identity is methylation information; maintenance through division is information copying; each irreversible operation costs at least  $k_B T \ln 2$ ). Steps 4-7 (budget -> error rate -> floor) are retired as a derivation and kept only as motivation for why a written state has a holding cost. H\_min is not derived from an energy budget. It is the MEASURED entropy of the healthy class at its identity loci - calibrated by MCMC on 37 FACS-sorted or microdissected reference methylomes, cross-checked by bootstrap, frozen in April 2026 - and that is the stronger statement, because it is checkable. The immune revision is on the record: first set from neutrophils alone at 0.795, re-converged at 0.8389 +/- 0.0012 with six immune cell types, every immune reading revised by ~0.055. Where 002 s2.1/s2.1a and this section differ, this section governs. 002's text is not reproduced in this document - this section replaces it - and stands in Issue 002 as published (MethylPhys/papers/IAMPerformance\_GAPEIssue002.pdf, section 2), so a reader holding both knows which is current.

| class      | identity loci | H_min (bits) | beta at A = 1 | ceiling 1/H_min |
|------------|---------------|--------------|---------------|-----------------|
| terminal   | 57,247        | 0.7728       | 0.7730        | 1.294           |
| immune     | 42,134        | 0.8389       | 0.7318        | 1.192           |
| secretory  | 29,617        | 0.8433       | 0.7287        | 1.186           |
| progenitor | 54,704        | 0.8522       | 0.7224        | 1.173           |
| cycling    | 29,181        | 0.8561       | 0.7195        | 1.168           |
| stromal    | 2,294         | 0.8630       | 0.7144        | 1.159           |
| stem_adult | 46,907        | 0.8737       | 0.7061        | 1.145           |
| stem_pluri | 21,710        | 0.9822       | 0.5784        | 1.018           |

5.0.5 The gauge  $A = H(\text{beta\_bar})/H_{\text{min}}$

THE GAUGE.  $A_c = H(\text{beta\_bar}_c) / H_{\text{min}}(c)$  is a ratio of two entropies to one fixed reference - not a divergence between two samples, and not a statement about which loci moved. It is defined on the class's IDENTITY loci, where a healthy cell of that class sits at a characteristic unimodal level (immune:  $\text{beta\_bar} = 0.7318$  at  $A = 1$ , 42,134 loci). On bimodal marker panels the same statistic is dominated by Jensen's inequality -  $H$  of the mean of a bimodal set manufactures ~1 bit whatever the cells do - and is not used for the gauge; those panels carry the SEPARATION statistic,  $\text{mean}_i H(\text{beta}_i)/H_{\text{min}}$ , which reproduces the sealed anchors at  $r = 1.00000$  (s3.3, PROC-ANCHOR-01). Three properties of  $A$  follow from  $H$ : (i) it is two-to-one in  $\text{beta}$  -  $H$  is symmetric about 0.5, so a strongly hypomethylated class reads BELOW 1 (the seminoma case,  $A = 0.692$  on  $\text{stem\_pluri}$ ; s5.1b); (ii) it has a structural ceiling  $A_{\text{max}} = 1/H_{\text{min}}$  (immune 1.192) beyond which departure is not resolvable and the report says AT CEILING; (iii)  $A$  rising toward the ceiling is the class's identity loci drifting toward  $\text{beta} = 0.5$  - toward one full bit of uncertainty per site, the loss of the written specification. The Warburg line (1.07) and the breach line (1.10) are the framework's claims about where on that axis metabolic and structural transitions occur; on the commissioned gauge they admit 1 and 0 of 1,379 healthy donors (PROC-TIER-01) and they are tested against disease cohorts, not assumed.

5.0.6 A fixed zero still needs two measured constants

A FIXED ZERO IN BITS STILL NEEDS TWO MEASURED CONSTANTS. This is the metrology result of Paper 1 and s3.5.  $H_{\text{min}}$  lives on the scale of the reference methylomes (Roadmap/GenomicStudio). A laboratory's Stage-1  $\text{beta}$  sits on a different scale: an affine map (slope 1.0127, intercept 0.0662 for  $\text{stage1\_noob\_450K}$ ) fitted on one lab transfers to three others (pipeline map). After the map, each laboratory - one processing pipeline - still carries a constant offset of its own ( $z_{\text{lab}}$ , -0.067 to +0.008 across four countries) that neither the map nor the array's control probes predict in size; it is measured from 40 healthy arrays read against the age curve (laboratory zero). And healthy  $A$  rises ~0.045 across a lifetime within a lab, so the zero is age-referenced. Floor + map + zero: physics fixes the first; the second and third are measured, once per pipeline and once per laboratory, and stated. The atlas itself is a fifth laboratory with its own constant (-0.0146, PROC-SWITCH-02). Nothing about this weakens the fixed origin - it is how every absolute instrument is used - but a reading that omits either constant is not a reading of the physics, it is a reading of the pipeline.

5.0.7 Filter and ruler

FILTER AND RULER. Sanchez and Mackenzie take  $k_B T \ln 2$  and model the thermal background of methylation change so that regulatory sites can be found against a control reference; in MethylIT that background is filtered away. We take the same constant as the unit and ask how far above the thermal floor a class holds its written state, against an origin fixed once. One filters, one calibrates. Their background model is the first extension we intend to test on the identity loci (Future Goals, GATE 2); our transfer discipline applies to any absolute methylation measurement, including one built on their divergences, because a divergence from a control pool inherits that pool's laboratory offset.

5.0.8 Ledger: physics, measured, claimed

| quantity                               | status                 | how                                                                    |
|----------------------------------------|------------------------|------------------------------------------------------------------------|
| $k_B T \ln 2$ at 310 K                 | physics                | constant; 2.97e-21 J                                                   |
| $M_{\text{cell}} = 20.9$               | physics + biochemistry | $\Delta G_{\text{ATP}} / RT$ ; not fitted                              |
| $H(\text{beta})$ binary entropy        | physics                | definition                                                             |
| $H_{\text{min}}(c)$ , eight values     | MEASURED reference     | MCMC on 37 healthy reference methylomes; bootstrap; frozen April 2026  |
| identity loci per class                | MEASURED selection     | unimodal healthy loci; iamatlas_gauge_identity_loci_v1_0               |
| pipeline map (slope, intercept)        | MEASURED constant      | one per pipeline; transfers across labs (Phase 1c)                     |
| laboratory zero $z_{\text{lab}}$       | MEASURED constant      | 40 healthy arrays vs the age curve; one per laboratory (PROC-PANEL-03) |
| reference age curve $c(\text{decade})$ | MEASURED               | 1,379 healthy, four labs; 0.47 mA/yr (PROC-AGE-01)                     |
| identity band v3 (p10-p90)             | MEASURED               | four zeroed labs, $n = 1,379$                                          |
| Warburg 1.07, breach 1.10              | framework CLAIM        | tested against cohorts; clean of healthy donors (1 and 0 of 1,379)     |
| tier NORMAL [0.95, 1.04)               | MEASURED convention    | healthy central 95% (PROC-TIER-02)                                     |

The Issue 002 physics section is not reproduced in this document - Section 5 replaces it, and 5.0.4 above states what of Issue 002's s2.1 (' $H_{\text{min}}$  Derivation') and s2.1a ('the physical chain') stands and what is retired. Issue 002's original text - its substrates, saturation, inversions, decomposition and post-breach physics as written in April 2026 - stands in Issue 002 as published (MethylPhys/papers/IAMPerformance\_GAPEIssue002.pdf, section 2). The ledger above gives the provenance of every quantity the present chain uses.

SECTION 7

# SUBSTRATE CHARACTERIZATION

Replaces Issue 002 Sections 7–8. Every (substrate, class) pair has its own floor and its own healthy range. This section is the grid: what each substrate carries, which cells of the grid have been measured with the canonical files, which are absent by biology, which are structurally unreadable, and which are open. Each open cell names the experiment and the public data that would fill it. This is the hand-off to a researcher.

## 7.1 What each substrate carries

| Substrate                | Reads                                                                                   | Floor                            | Status 2026-09-19                                        |
|--------------------------|-----------------------------------------------------------------------------------------|----------------------------------|----------------------------------------------------------|
| whole blood / buffy coat | immune ARCHITECTURE of the leukocytes present; epithelial composition absent by biology | methyl                           | n=7 both channels; 3/3 healthy BELOW age band — OPEN     |
| plasma cfDNA             | SHED tissue composition                                                                 | frag / wps / nucl — never methyl | composition measured n=33; gauge withdrawn (wrong floor) |
| tissue                   | positive control; heterogeneity marker (glioma-LL-002)                                  | methyl                           | conformant, MAE 0.0004                                   |
| urine sediment           | shed urothelial / prostate                                                              | to be established                | GSE119260 in hand, not run                               |
| CSF                      | shed CNS (terminal class)                                                               | to be established                | GSE292312, GSE269403 located; supplementary files        |
| stool                    | shed colonic epithelium                                                                 | to be established                | no public array data exists                              |

## 7.2 The grid

|              | terminal | secretory | immune | progenitor | cycling | stromal | stem_adult | stem_pluri |
|--------------|----------|-----------|--------|------------|---------|---------|------------|------------|
| whole blood  | ABS      | ABS       | MEAS   | MEAS       | ABS     | ABS     | MEAS       | SAT        |
| plasma cfDNA | MEAS     | —         | COMP   | —          | —       | ABS     | —          | —          |
| tissue       | —        | CONF      | CONF   | STRUCT     | CONF    | STRUCT  | —          | —          |
| urine        | DATA     | DATA      | DATA   | DATA       | DATA    | DATA    | DATA       | DATA       |
| CSF          | DATA     | DATA      | DATA   | DATA       | DATA    | DATA    | DATA       | DATA       |
| stool        | NONE     | NONE      | NONE   | NONE       | NONE    | NONE    | NONE       | NONE       |

CONF = reproduces the project answer key · MEAS = run on canonical files, result recorded · COMP = deconvolver composition only (gauge not applicable on the methyl floor) · ABS = fraction = 0 by biology · STRUCT = zero by marker coverage · SAT = methyl ceiling below BREACH · DATA = public cohort located, not run · NONE = no public data, primary collection required · — = not established

## 7.3 What was measured

**Whole blood, n=7** (test-data IDATs, Stage-1 calibrated, both channels). Epithelial-class fraction 0.0000–0.0111 in all seven; immune 0.803–0.966; residual MAE 0.043–0.060. Two rheumatoid-arthritis cases sit inside the non-RA range on both channels (gauge 0.8743/0.8789 vs 0.8066–0.8924; immune fraction 0.8395/0.8686 vs 0.8027–0.9663). **The three age-labelled healthy donors read BELOW the age-matched band:** 43M A=0.8739 (band 0.9021–0.9934, z –2.07); 58M 0.8066 (0.9083–1.0029, z –4.03); 67F 0.8326 (0.9166–1.0138, z –3.49). Either the Stage-1 path and the literature-compiled reference matrix are on different scales, or these donors are below band. n=3 cannot separate the two. This is PROC-WB-IMMUNE-01's open verdict.

| GSM        | label            | epithelial | immune | progenitor | resid MAE | immune gauge A | age band            |
|------------|------------------|------------|--------|------------|-----------|----------------|---------------------|
| GSM2333901 | healthy 58M      | 0.0000     | 0.8336 | 0.1280     | 0.0562    | 0.8066         | 0.908–1.003 → BELOW |
| GSM2333905 | healthy 67F      | 0.0051     | 0.8881 | 0.0671     | 0.0522    | 0.8326         | 0.917–1.014 → BELOW |
| GSM2333950 | healthy 43M      | 0.0111     | 0.9663 | 0.0000     | 0.0596    | 0.8739         | 0.902–0.993 → BELOW |
| GSM1051533 | RA-study control | 0.0000     | 0.9318 | 0.0515     | 0.0434    | 0.8924         | age not in GEO      |

| GSM        | label            | epithelial | immune | progenitor | resid MAE | immune gauge A | age band       |
|------------|------------------|------------|--------|------------|-----------|----------------|----------------|
| GSM1051534 | RA-study control | 0.0000     | 0.8027 | 0.1740     | 0.0504    | 0.8673         | age not in GEO |
| GSM1051525 | RA case          | 0.0000     | 0.8395 | 0.1163     | 0.0452    | 0.8789         | age not in GEO |
| GSM1051526 | RA case          | 0.0000     | 0.8686 | 0.1147     | 0.0459    | 0.8743         | age not in GEO |

**Plasma cfDNA, GSE122126 (Moss 2018), n=33 plus 22 sepsis** — composition only. All plasma GAUGE A-scores computed 2026-09-19 were on the METHYL floor and are therefore out of spec (RECON S1). Only the COMPOSITION results are reported here. The GSE122126 in-vitro mixes (9+5 samples of known genomic-DNA proportion) are the available real-data ground truth for the deconvolver and have not yet been scored.

| condition     | n  | epithelial fraction (cycling+secretory+terminal+stromal) | residual MAE |
|---------------|----|----------------------------------------------------------|--------------|
| healthy       | 4  | 0.030 (mean)                                             | 0.058        |
| colon cancer  | 4  | 0.463 (mean)                                             | 0.155        |
| breast cancer | 3  | 0.466 (mean)                                             | 0.161        |
| lung cancer   | 4  | 0.010 (mean)                                             | 0.055        |
| CUP           | 4  | 0.293 (mean)                                             | 0.116        |
| sepsis        | 22 | median 0.040, p75 0.233, max 0.520                       | —            |

Lung cancer reads epithelial-null in the correct substrate: nothing shed, nothing to read, instrument says so. Sepsis sheds epithelium in a minority of samples (multi-organ injury, established biology). Residual MAE tracks epithelial fraction (r = 0.98 across all samples) — a second readout the chain computes and discards; not yet checked against the corpus. **Cell-level routing:** Cell-level calls on shed epithelium route to gastric references (Fundus\_diff, Antrum\_diff) in colon tissue, colon plasma and breast plasma alike; Colon\_epithelial\_cells and Breast return 0.0000 despite being in the atlas. LESSON-DECONV-01 attributes this to reference mismatch, not solver conditioning; hierarchical refinement was tested, invents cells, and does not ship. Cell tier is INDICATIVE on that (pre-Atlas, wrong-triple) run. Whether the commissioned chain attributes an organ absolutely for one patient is NOT YET TESTED (paper two); at the cohort level the record shows it (scope, p4).

**Tissue** — conformance, PROC-DECON-01 (§8). Whole-blood versus tissue integrity gate (CHK-3.1) on Stage-1-calibrated data:

| GSM        | substrate                 | % β <0.05 or >0.95 | % β in 0.40–0.60 |
|------------|---------------------------|--------------------|------------------|
| GSM2333901 | whole blood 450K          | 33.8               | 4.5              |
| GSM1051533 | whole blood 450K          | 27.3               | 5.0              |
| GSM8772491 | colon adenoma EPIC tissue | 6.7                | 5.5              |
| GSM8772492 | colon adenoma EPIC tissue | 8.5                | 6.4              |
| GSM5065990 | CRC EPIC tissue           | 23.6               | 11.7             |

The >30% extreme-β criterion written for whole blood fails every properly calibrated tissue sample. Integrity thresholds must be stated per substrate (§9, CCL-032 amended).

## 7.4 The researcher hand-off — one experiment per open row

| substrate            | prediction to test                                                                                                                                          | data                                                     | who can run it                                     |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------|
| whole blood × immune | healthy donors read IN the age band once Stage-1 and the reference are on one scale; if not, the reference is re-anchored to the atlas scale (open VAL-053) | test-data IDATs + any public healthy blood 450K/EPIC     | any methylation lab; no clinical access needed     |
| plasma cfDNA         | epithelial classes appear in cancer plasma on frag/wps/nucl floors and stay absent in healthy; in-vitro mixes recover at MAE ≤ 0.01                         | GSE122126 (has 14 known mixes); GSE311578 (70 AD plasma) | cfDNA group with WGS for the non-methyl substrates |
| urine                | prostate signal appears in the secretory class of urine sediment in the same men whose tissue shows it; plasma of the same men reads lower                  | GSE119260: 4 pts × tissue/plasma/urine                   | urology or GU-oncology group                       |

| substrate | prediction to test                                                            | data                                                                  | who can run it                     |
|-----------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------|------------------------------------|
| CSF       | terminal-class signal present in CSF and absent from the same patient's blood | GSE292312 (24 CSF + 157 tumour); GSE269403 (39 CSF + 17 paired blood) | neuro-oncology group               |
| stool     | colonic cycling-class signal in stool DNA                                     | none public — primary collection                                      | GI group with a stool-DNA protocol |

# PROCEDURES

One entry per verified cell of the grid, in the form a stranger can run: input, operation, expected, tolerance, observed, verdict. Nothing else. Every entry names the exact files at repo commit 66f37fe. An entry whose verdict is OPEN is printed as OPEN. Entries not yet written are listed at the end with what they need. The template is PROC-DECON-01, the first procedure in this project to pass on a machine that had never seen the author's setup.

## PROC-DECON-01 — Deconvolver conformance — tissue, EPIC

| field               | value                                                                                                                                                                                                                              |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| input               | betas_cache.pkl (Stage-1 noob-calibrated $\beta$ , TEST_DATA); IAMAtlasREBUILD.csv decompressed from repo MethylPhys/atlas/IAMAtlasREBUILD.csv.xz; IAMAtlasREBUILD_celltype_to_class.json                                          |
| operation           | WaltherIAMDeconvolver(atlas, celltype_class_map=map).deconvolve(beta_dict) for GSM8772491, GSM5065990, GSM5065985                                                                                                                  |
| expected            | class fractions in TEST_DATA_MANIFEST.md: GSM8772491 cycling .354 immune .253 stem_pluri .161 secretory .122 terminal .112                                                                                                         |
| tolerance           | class-fraction MAE $\leq 0.001$ over documented classes                                                                                                                                                                            |
| observed 2026-09-19 | GSM8772491: cycling .3535 immune .2530 stem_pluri .1605 secretory .1215 terminal .1115 — MAE 0.0004, max  err  0.0005. GSM5065990 MAE 0.0002. GSM5065985 MAE 0.0002. Status OK all three; 7,114 class markers, 4,000 cell markers. |
| verdict             | PASS. Deltas are the manifest's 3-decimal rounding.                                                                                                                                                                                |
| scope               | verifies deconvolver + atlas + map. Does NOT verify Stage 1 (cached $\beta$ used, raw IDATs not decoded). See PROC-CAL-01.                                                                                                         |

## PROC-CAL-01 — Stage 1 - raw IDAT to calibrated beta, executed

| field       | value                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| input       | the eleven raw IDAT pairs in 10_TEST_DATA/idats (7 x 450K whole blood incl. 2 RA + 2 RA-study controls; 4 x EPIC colorectal tissue); stage_1_idat_calibration.py from the repo; methylprep 1.7.1 with numpy 1.26.4 / pandas 1.5.3 (methylprep uses DataFrame.append, removed in pandas 2 - the conflict the module's own docstring warns about); Illumina 450K and EPIC manifests fetched from methylprep's public bucket                 |
| operation   | calibrate_idat_to_beta(Grn, Red) per sample: array type auto-detected from bead-address count, noob dye-bias + probe-type normalisation, per-sample, no cohort information; compare the resulting beta vector CpG-by-CpG to betas_cache.pkl (the Stage-1 output that produced PROC-DECON-01)                                                                                                                                              |
| expected    | identical betas (same algorithm, same manifest); any difference is a pipeline-version effect                                                                                                                                                                                                                                                                                                                                              |
| verdict     | PASS - 11/11 bit-identical: $r = 1.000000$ , max  diff  = 0.000000, zero CpGs differing by more than 1e-4, on both array types (450K 415,863-418,055 CpGs, EPIC 490,390-757,003 CpGs); 24-48 s per sample. With PROC-DECON-01 and PROC-ANCHOR-01 this closes the chain: raw IDAT -> calibrated beta -> class fractions -> sealed per-cell anchor, every link reproduced from the repository on a machine that had never seen the project. |
| consequence | the 3/3 below-band healthy whole-blood donors in PROC-WB-IMMUNE-01 are NOT a Stage-1 artifact of this machine - the betas are the project's own. The remaining hypotheses are (a) the age band was compiled on a different IDAT-to-beta pipeline than noob (the Xu-538 pipeline-relativity effect, 0.38-0.62), or (b) the donors are below band. (a) is testable by locating the pipeline the band was compiled from.                     |

## PROC-N7-01 — End-to-end synthetic simulation - the gauge as wired reads the wrong CpG set

| field                                                  | value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| input                                                  | synthetic_patient_generator.py (restored from RETIRED; patched to read MethylPhys/atlas/IAMAtlasREBUILD.csv; composition_alpha added): 16 healthy + 8 case, whole-blood composition (~immune 0.89, progenitor 0.08, stem_adult 0.03), disease panel 500 CpGs on the immune identity loci, signal 2.0. Each patient is a linear mix of the Atlas class posterior means plus age/sex/batch loadings and noise 0.03. Run through cpg_conductor.run_full.                                                                                                                                                                                                                                  |
| R1 composition                                         | PASS - deconvolver recovers every class: immune MAE 0.0095, progenitor 0.0145, stem_adult 0.0116, epithelial classes <= 0.0016 (r 0.79-0.98 on the legacy mix).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| R2 gauge                                               | FAIL - every synthetic patient, healthy or case, reads immune A 1.12-1.13 ABOVE_BAND / BREACH; case and healthy do not separate.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| root cause                                             | cpg_conductor.stage_b_classes computes $H(\beta_{\text{mean}})/H_{\text{min}}$ over the class MARKER UNION (2,952 bimodal CpGs for immune; comment: 'PRODUCTION A-score (GAPE_WEB_v13 + Reproduction Paper v3) ... NOT identity loci'), while its own docstring, SOP s41/s106 and Issue 002 say identity loci (42,134 unimodal CpGs). Confirmed: ge.read on the marker-union $\beta_{\text{mean}}$ returns 1.1303, the conductor's number, to four decimals. The marker union's $\beta_{\text{mean}}$ depends on the SHAPE of the input (Jensen gap 0.257 vs 0.026 on identity loci): a smooth synthetic mixture pulls it to 0.63 (BREACH); real bimodal blood lands at 0.75 (NORMAL). |
| synthetic healthy (pure mixture of healthy posteriors) | identity 0.9875   marker-union 1.1303 BREACH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| real WB healthy 43M                                    | identity 0.874   marker-union 0.957 IN_BAND                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| real WB healthy 58M                                    | identity 0.807   marker-union 0.960 IN_BAND                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| real WB RA                                             | identity 0.879   marker-union 1.002 IN_BAND                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| real tissue adenoma GSM8772492                         | identity 1.096 (elevated)   marker-union 0.932 IN_BAND (missed)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| consequence 1                                          | RULING A3 was right about the formula ( $H(\beta_{\text{mean}})$ on identity loci) and WRONG in asserting that the wired chain computes it. Corrected in s1.5.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| consequence 2                                          | age_reference_matrix.json was compiled (GAPE_WEB_v13 _AGE_REFERENCE) on the marker-union statistic. PROC-CHAIN-01's '7/7 healthy IN_BAND' is the marker-union gauge agreeing with the marker-union band - self-consistency, not conformance. Withdrawn as evidence of a working gauge.                                                                                                                                                                                                                                                                                                                                                                                                 |
| consequence 3                                          | s6a's 'below band' reading compared identity-loci A against a marker-union band: a statistic mismatch, not an offset. The +0.05-0.09 beta shift at identity loci vs the Atlas posterior IS real and remains Phase 1's business.                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| consequence 4                                          | The identity-loci gauge is the correct statistic - unimodal, passes N7, carries the adenoma signal - and has NO BAND. Phase 1 (rebuild the band on identity-loci $H(\beta_{\text{mean}})$ from GSE87571 through Stage 1) is now a prerequisite for any gauge reading, not a refinement.                                                                                                                                                                                                                                                                                                                                                                                                |
| consequence 5                                          | The conductor is NOT switched today (a statistic without its band would read every healthy patient below band). Every class-gauge reading now carries gauge_surface = 'marker_union' and n_cpgs so it cannot be mistaken.                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| verdict                                                | N7 did its job: the synthetic healthy patient exposed a gauge that real blood had been masking. Composition PASS; gauge as wired FAIL; identity-loci gauge PASS on N7 and blocked on its band.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

PROC-NILC-01 — The retired second deconvolver, rerun as designed

| field          | value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| input          | Walther's own class-marker reference X (1,256 CpGs x 8 classes, built by the engine at load); the 11 Stage-1 test samples; NILC's stated algorithm (unconstrained least squares on X, then simplex projection) run on the SAME X for a fair comparison                                                                                                                                                                                                                                                                                                                                          |
| conditioning   | condition number of X: 47.0 (all 8); 30.6 for the blood sub-problem (immune, progenitor, stem_adult); 8.8 for the epithelial trio. Centred column correlations: progenitor-stem_adult +0.99, immune-progenitor +0.94, immune-stem_adult +0.93.                                                                                                                                                                                                                                                                                                                                                  |
| result         | per-patient L1 disagreement 0.55-0.65 on every blood sample (design target < 0.05); NILC places 0.26-0.34 in stem_adult where Walther places 0.005-0.044, and deletes secretory and stromal; on tissue it drives progenitor negative (-0.16 to -0.23). Exactly the 'collapsed on correlated blood mixtures, deleted correct calls' of the cut.                                                                                                                                                                                                                                                  |
| reading        | By NILC's own design rule, divergence marks where composition is 'genuinely ill-defined by the atlas reference'. It was: in blood the Atlas cannot separately determine immune, progenitor and stem_adult; Walther's 2-4% stem_adult is chosen by the simplex constraint, not by the data. That is the class that read false BREACH in 6/7 healthy donors (PROC-CHAIN-01). The second deconvolver diagnosed the stem_adult problem in June; it was cut for delivering the message, and reinstated on 2026-09-22 as the class-level second opinion once this rerun showed the message was right. |
| recommendation | (1) Whole blood: report progenitor + stem_adult as ONE haematopoietic-progenitor component until the Atlas separates them (kappa of the blood sub-problem must fall below ~10 first). (2) Pre-flight: compute the substrate sub-problem's condition number alongside the Jensen gap. (3) A second deconvolver stays a goal, but its disagreement is a diagnostic to be read, not a gate to be failed.                                                                                                                                                                                           |

PROC-SEP-01 — Is the HSC / progenitor information in the Atlas?

| field                                            | value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| question                                         | The Atlas cannot split stem_adult from progenitor in blood (kappa 30.6, r = +0.99). Is that because the information is absent (a reference-data problem) or unused (a marker-selection problem)?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| measurement                                      | Of 482,421 Atlas CpGs with both class means: 1,290 separate stem_adult from progenitor by > 0.2 beta (0.27%); 228 by > 0.3; 4,265 exceed 3x the combined posterior SD. For scale, 15,939 separate stem_adult from immune by > 0.2. Of Walther's 7,114 class markers, 129 split the pair by > 0.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| diagnosis                                        | The information exists and is barely used. Walther ranks each class's markers by separation from the FIELD; CpGs separating haematopoietic classes from epithelium are plentiful and win; the few hundred separating HSC from MPP/CMP/GMP mostly do not make the cut. Structural cause: stem_adult is ONE cell type (HSC) while progenitor is ELEVEN (CMP, GMP, MPP, L-MPP, MEP, erythrocyte progenitors, erythroblast, megakaryocyte, nRBC, Neulm, OPC) - the class boundary is drawn one step down a continuous lineage, so the two class means are adjacent points on the same path. Contributing: HSC/MPP/CMP/GMP references come from the same few sorted-CD34+ studies (Roadmap E035, Adelman 2019), so study offsets inflate the agreement. |
| 1 - contrast-specific markers (cheap; this week) | for every class pair with kappa > 10, force in the top-N CpGs separating THAT pair (the 1,290 are already in the Atlas). Target: blood sub-problem kappa < 10. Test: N7 and PROC-NILC-01 rerun - if Walther and the unconstrained solve converge on stem_adult, the split has become data-determined.                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 2 - hierarchical composition (moderate)          | deconvolve haematopoietic as one component, then split it using contrast markers only - coarse-to-fine; removes the epithelial CpGs' vote on a question they cannot answer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 3 - independent HSC references (OUT OF SCOPE)    | a second sorted-HSC source would break the shared-donor coupling, but it is not required for the claim. Author, 2026-09-19: this project proves the MCMC atlas works; it does not join more atlases. Noted for whoever builds the next one.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| framing                                          | An ATLAS result, in the atlas's favour: the MCMC atlas already holds the HSC / progenitor split (1,290 CpGs; 4,265 above 3 sigma); its consumer was not asking for it. Steps 1-2 use the Atlas alone. If pairwise-forced markers bring the blood sub-problem from kappa 30.6 to single digits, that demonstrates the atlas resolves a lineage step its own deconvolver could not - which is exactly the claim this project makes. Goes in the ledger when step 1 lands.                                                                                                                                                                                                                                                                            |

## PROC-SEP-02 — Contrast-specific markers, tested

| field   | value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| change  | WaltherIAMDeconvolver gains contrast_pairs / n_contrast_markers_per_pair: for each named class pair the top-N CpGs by  mean_a - mean_b  are forced into the class reference. Default OFF; no sealed result changes.                                                                                                                                                                                                                                                                               |
| result  | baseline: 1,256 markers, kappa(all) 47.0, kappa(blood) 30.6, r(prog, stem_adult) +0.987, blood L1 Walther-vs-unconstrained 0.59. +300/pair: 1,335 markers, 32.8 / 22.0 / +0.974 / 0.47. +800/pair: 1,567 markers, 26.1 / 17.0 / +0.954 / 0.60 (unconstrained drives progenitor negative, stem_adult to ~0.3 on every blood sample).                                                                                                                                                               |
| reading | Conditioning improves monotonically - the contrast CpGs act on the reference exactly as intended - but kappa(blood) does not reach 10 and the two solvers do not converge. With the Atlas alone the HSC / progenitor split is not data-determined at the marker level. Walther's own stem_adult estimate in healthy blood falls from 0.017-0.044 to ~0.000 (6/7 at +300) once it has the contrast information: the 2-4% was the constraint's, not the blood's - the same thing NILC said in June. |
| verdict | Passing test of the METHOD (the Atlas holds the information and it moves kappa the right way); NEGATIVE result for the shortcut (marker selection alone does not make stem_adult reportable on blood). The reporting rule stands as the operating position: whole blood reports immune, plus progenitor + stem_adult as one haematopoietic-progenitor component (coarse-to-fine, which never asks the ill-conditioned question). Step 3 remains out of scope by the author's decision.            |
| ledger  | Second cross-method row: the disagreement metric, run under three marker configurations, located the limit of what this Atlas determines in blood - a measurement no cohort comparison can make.                                                                                                                                                                                                                                                                                                  |

## PROC-SEP-03 — Two tools, two questions: the lineage splitter

| field        | value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| design       | Tool A = WaltherIAMDeconvolver, unchanged: 'what is in the tube', field-ranked class markers, NNLS on the simplex. Tool B = MethylPhys/chain/Lineage_Splitter/lineage_splitter.py (new): takes ONE compartment Tool A found (default haematopoietic = progenitor + stem_adult) and asks only how it divides, using ONLY the CpGs where the member classes differ ( delta  >= 0.2 or 0.3), weighted by  delta , after subtracting the other classes' contribution at those CpGs; solves a small NNLS on the compartment mass; carries its own condition-number check (kappa_max 10). Ill-conditioned -> COMPARTMENT_ONLY with the reason; it cannot manufacture information, it reports whether it is there. |
| conditioning | single solve on field markers: 1,256 CpGs, kappa 30.6, r(prog, stem_adult) +0.99. Lineage splitter  delta >=0.2: 1,290 CpGs, kappa 5.5, r +0.43.  delta >=0.3: 228 CpGs, kappa 4.2, r -0.18. Restricted to the CpGs where the classes differ, the columns stop being parallel and the split is well determined.                                                                                                                                                                                                                                                                                                                                                                                             |
| result       | 7/7 whole-blood test samples: SPLIT, kappa 3.9-5.2, progenitor = the whole compartment, stem_adult = 0.000. Compartment fractions per sample: 0.166, 0.107, 0.005 (GSM2333950, whose Tool A progenitor is 0.000 and stem_adult 0.005), 0.161, 0.131, 0.068, 0.197 - range 0.005-0.197. Tissue with no haematopoietic-progenitor mass: COMPARTMENT_ONLY, kappa = inf, 'nothing to split' (correct). CRC-1 tissue: SPLIT 0.010 / 0.004.                                                                                                                                                                                                                                                                       |
| reading      | Three independent routes now agree: NILC (June), Walther-with-contrast-markers (PROC-SEP-02), and the lineage splitter (this) all find stem_adult ~0 in healthy whole blood at array resolution; Tool A's 2-4% was the simplex constraint's choice. Biologically sensible (HSC ~0.01% of nucleated blood cells). The reporting rule's joint component is right for blood because the resolved stem_adult share is zero - a mechanism, not just an ill-conditioning excuse.                                                                                                                                                                                                                                  |
| myeloid      | MDS (HSC up, GMP down) now has its instrument: Tool B is what would see HSC content depart from zero on a well-conditioned problem with an uncertainty. Testable in the myeloid arm.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| verdict      | The Atlas resolves the lineage step when asked on the right vocabulary. The single solve reported an arbitrary split because 480,000 uninformative CpGs outvoted 1,290 informative ones. An ATLAS result, in the atlas's favour (Atlas only; no new data).                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

PHASE 1 - identity-loci healthy band (GSE87571, 732 raw IDATs through Stage 1) — Sealed FAIL on P3/P4; the cause measured; the floors unchanged

| field     | value                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| verdict   | FAIL on P3 (synthetic healthy in band 0/16) and P4 (test WB in band 3/7) as sealed; P1 732/732; band built n=560 and kept (identity_band_v1.json). N-random 0/16 (pass). Spearman(age, immune A) = +0.52.                                                                                                                                                                                                 |
| cause     | Three beta scales on the same 42,024 immune identity loci: Roadmap/Atlas 0.737 (A 1.00, where G-002 calibrated H_min); GEO author-processed EPIC 0.774 (A 0.92; the anchor cohort); Stage-1 noob raw 450K 0.815 (A 0.82). The offset is additive (+0.066 beta).                                                                                                                                           |
| why_now   | Author's April 2026 record (VAL-003 output): 'G-002 H_min calibrated on Roadmap (GenomicStudio); TCGA uses sesame; ~10% offset expected; delta-A valid within-pipeline; absolute thresholds require a pipeline-matched healthy reference.' Every VAL since was within-pipeline and immune by design. Phase 1 is the first absolute reading of raw-IDAT beta against the floor with no cohort in the loop. |
| mcmc      | UNCHANGED. The H_min floors are correct on the Roadmap scale; the analyst's recommendation to re-derive them on Stage 1 is WITHDRAWN. The fix is a per-pipeline affine map onto the Roadmap scale, fit on healthy blood - what the April note prescribed and Phase 1 finally built.                                                                                                                       |
| after_map | Swedish healthy immune A 0.990 (p10-p90 0.962-1.016): healthy sits at the floor. P3 15/15 (post-hoc; map fit on the band cohort, so a consistency check). P4 3/7 unchanged: the four out read ABOVE band by 0.01-0.02 A - a lab batch residual on top of the pipeline offset, the size of the band half-width; N-plate is its test.                                                                       |
| layers    | FLOOR (Roadmap scale, MCMC, physics) -> PIPELINE (+0.066 beta per pipeline, affine map) -> LAB (0.01-0.02 A per cohort). Cohort-relative statistics cancel layers 2-3 and never see them; the absolute gauge sees all three.                                                                                                                                                                              |
| next      | Phase 1c prereg: fix the map on GSE87571, test on an independent Stage-1 healthy cohort; N-plate from Sentrix IDs (done: PHASE 1c, band_v2); the sex-split band proposed here was WITHDRAWN the next day (sign flips between labs); P2 restated as immune + haem-progenitor >= 0.85 (adopted). Phase 1b (EPIC-Italy portability) proceeds.                                                                |

PROC-WB-IMMUNE-01 — Whole blood × immune — healthy read IN\_BAND on the marker-union gauge (WITHDRAWN as conformance by PROC-N7-01: gauge and band shared the wrong statistic)

| field               | value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| input               | betas_cache.pkl for GSM2333901 (58M), GSM2333905 (67F), GSM2333950 (43M), GSM1051533, GSM1051534 (RA-study controls), GSM1051525, GSM1051526 (RA); iamatlas_gauge_identity_loci_v1_0.json; age_reference_matrix.json; IAMAtlasREBUILD.csv + map                                                                                                                                                                                                                                                                             |
| operation           | (1) deconvolve → require immune fraction ≥ DETECT_FLOOR (it is: 0.80–0.97); (2) $\beta_{\text{mean}}$ over the 42,134 immune identity loci (36,290 present on 450K); (3) $A = H(\beta_{\text{mean}})/0.838889$ ; (4) place against the immune p10–p90 band at the donor's age; (5) NO input offset (retired)                                                                                                                                                                                                                |
| expected            | healthy donors IN BAND; RA cases not required to differ (chronic autoimmune; no prior)                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| tolerance           | IN BAND = $p_{10} \leq A \leq p_{90}$ of age_reference_matrix immune at nearest decade midpoint                                                                                                                                                                                                                                                                                                                                                                                                                             |
| observed 2026-09-19 | 43M A=0.8739 vs [0.9021, 0.9934] z −2.07 BELOW · 58M 0.8066 vs [0.9083, 1.0029] z −4.03 BELOW · 67F 0.8326 vs [0.9166, 1.0138] z −3.49 BELOW · RA 0.8789/0.8743 and controls 0.8924/0.8673 (ages not in GEO). Composition: epithelial 0.0000–0.0111 all seven.                                                                                                                                                                                                                                                              |
| verdict             | OPEN. 3/3 healthy BELOW BAND. Two hypotheses, not separable at n=3: (a) age_reference_matrix (compiled from Hannum/Horvath/Alisch literature $\beta$ ) and the Stage-1 noob path are on different scales — the ~0.1 gap is in the direction the retired 0.055 offset addressed; (b) the donors are below band. Next: run ≥30 public healthy whole-blood 450K/EPIC through the same path; if the median sits ~0.1 below p50 at every decade, (a) is confirmed and the reference is re-anchored to the atlas scale (VAL-053). |
| do not              | compute a group mean and call it the result; the reading is per donor against the band                                                                                                                                                                                                                                                                                                                                                                                                                                      |

## PHASE 1c - scale-map and band transfer to an unseen lab (GSE42861 controls, n = 335) — Map transfers (Stage 1s COMMISSIONED); one-lab band does not; lab layer sized

| field     | value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| verdict   | P3 PASS (map transfers: GSE42861 controls n=315, independent lab, median mapped immune A 1.0097; unmapped 0.857). P4 FAIL (one-lab band: 53% in the GSE87571 band). Stage 1s COMMISSIONED.                                                                                                                                                                                                                                                                                                       |
| lab_layer | GSE42861 - GSE87571 median per decade: +0.013, +0.016, +0.018, +0.016, +0.018 (25-74) - flat across age, ~+0.017 A. N-plate: 37 Sentrix chips, F=4.30, p=2.6e-12. The four TEST_DATA GSE42861 arrays sit inside their own cohort p5-p95 (4/4): Phase 1's 'above band' was the cohort, not the chips.                                                                                                                                                                                             |
| n_random  | a mean-beta-matched random panel put 67% of controls IN band (must have been <50%): on healthy blood the identity loci set the LEVEL, but the band WIDTH is pipeline/chip variance any panel shares. A one-lab band is too narrow by the between-lab term.                                                                                                                                                                                                                                       |
| design    | healthy reference = pooled across >=2 labs AFTER the pipeline map, percentile bands on mapped A, lab/chip random effect estimated and disclosed, sex as a signed null (split only if the sign agrees across labs). Pool after mapping (the pipeline term must be removed first), never raw (Phase 1 was right about that).                                                                                                                                                                       |
| nulls     | N-sex SIGNED d(f-m): GSE42861 +0.28..+0.69 (women higher, 35-74) but GSE87571 -0.24..-0.44 (women LOWER, 35-64) - the sex effect flips sign between labs, so it is not a stable biological term; the sex-split band from Phase 1 is WITHDRAWN pending a signed test on pooled mapped data with a lab term. (The run reported  d  only; the direction first written here was unsupported and is corrected.) N-smoke: current 1.0108 / ex 1.0086 / never 1.0107 - no effect. RA arrays not opened. |
| layers    | FLOOR (Roadmap, MCMC) -> PIPELINE (+0.066 beta; transfers; COMMISSIONED) -> LAB (+0.017 A between two labs; chips p~1e-12). SEX: sign flips between labs - not a layer until shown stable.                                                                                                                                                                                                                                                                                                       |

## band\_v2 test - pooled two-lab band on a third lab and population (GSE125105 Munich controls, n = 201) — P3 and P4 FAIL; the per-cohort constant is a layer pooling cannot remove; healthy reference = floor + pipeline map + lab zero

| field             | value                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| verdict           | identity_band_v2 (pooled GSE87571 659 + GSE42861 315, mapped A, per-decade p10/p50/p90; adults >=45 at p50 1.00, 14-24 at 0.96) tested on GSE125105 Munich controls (n=201): P3 median 0.9645 FAIL (bar +/-0.02); P4 54.7% in band FAIL (bar 80%).                                                                                                                                          |
| three_cohorts     | mapped immune A offset vs Uppsala: Karolinska +0.018 (chips p~1e-12), Munich -0.030 (chips p=0.39) - each flat across age. The pipeline map moved Munich 0.79 -> 0.96 (shared term real; Stage 1s stays commissioned); the residual is a per-cohort constant the size of the band half-width.                                                                                               |
| why_pooling_fails | a band pooled over N labs is wider, but the next lab's constant is a new number - so ~50% of any new cohort lands in band regardless of N. Confirmed twice (Karolinska on a one-lab band 53%; Munich on a two-lab band 55%).                                                                                                                                                                |
| design            | healthy reference = FLOOR (physics, universal) + PIPELINE MAP (universal per pipeline) + LAB ZERO (local, once per lab). This is CLSI EP28 practice for every clinical assay. Routes: (1) per-lab healthy-control panel of 20-30 arrays, offset disclosed on every report - recommended primary; (2) learn the offset from control-probe intensities (Stage 0 QCs, deferred) - Future Goal. |
| nulls             | N-random REDESIGNED (size-matched, level test): random panel median 1.174, 0% in band -> the identity loci set the LEVEL; Phase 1c's mean-beta-matched null was circular for H(beta_bar) and is retired. N-sex unsigned as run ( d  0.17-0.38) - no direction claimed. Stage 1 parallel: 210 arrays 37 min; per-sample IDAT fetch 1.7 GB in 6 min (tools/geo_fetch_idats.py).               |

## LAB-ZERO-01 - can the array's own control probes supply the lab zero? — Direction 3/3, Swedish magnitude to 0.002, Munich overshoot; panel standard today, fourth cohort decides

| field         | value                                                                                                                                                                                                                                                                            |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| question      | Can the per-lab offset be predicted from the 850 Illumina control probes on each array, so no per-lab healthy-control panel is needed?                                                                                                                                           |
| data          | 1,277 raw IDATs (three cohorts already on disk), 33 features = log2 mean per Control_Type x channel + overall medians + G/R ratio; target mapped immune identity-loci A (1,173 gated). Ridge, leave-one-cohort-out.                                                              |
| result        | held-out Karolinska: obs +0.024 pred +0.021 (err 0.002); Uppsala: obs -0.024 pred -0.026 (err 0.002); Munich: obs -0.021 pred -0.071 (err 0.050) - direction 3/3, size right for the Swedish labs, Munich overshoots 3x with no neighbouring lab in training. P1 FAIL as sealed. |
| features      | cohort-separating features (range/sd ~2.6): NORM_T/A/C/G, BISULFITE CONVERSION II, SPECIFICITY I - red channel. The lab's chemistry is recorded on the array.                                                                                                                    |
| within_cohort | control probes explain $r \sim 0.6$ of a healthy donor's A inside a single cohort: about a third of the 'healthy spread' is per-array technical variance. Correcting it would narrow the band (LAB-ZERO-02).                                                                     |
| nulls         | the sealed permutation null was uninformative (shuffling cohort labels removes the offsets) and is recorded as such; the correct null is a fourth held-out cohort.                                                                                                               |
| verdict       | standard today = per-lab healthy-control panel (CLSI EP28), offset printed on the report. Control-probe zero = upgrade path, decided by a fourth Stage-1 healthy cohort (minutes to fetch with tools/geo_fetch_idats.py).                                                        |

## LAB-ZERO-02 - the fourth lab (UCLA) decides the lab-zero route — Control probes carry direction, not magnitude; the lab zero is the healthy-control panel (CLSI EP28)

| field     | value                                                                                                                                                                                                                                                                                                                                                             |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| question  | Does a fourth lab (UCLA, USA) let the control probes predict the lab offset within 0.010, and does per-array correction narrow a healthy band by 20%?                                                                                                                                                                                                             |
| result    | P4 FAIL: UCLA predicted -0.020, observed -0.046 (err 0.026). P4b FAIL: LOO over four labs - Karolinska 0.004, Uppsala 0.018, UCLA 0.026, Munich 0.052 (1/4 within bar). P6 FAIL: within-cohort sd 0.0203 -> 0.0177 (12.6%, bar 20%).                                                                                                                              |
| four_labs | mapped immune A cohort constant vs Uppsala: Karolinska +0.024, Munich -0.021, UCLA -0.046 - each flat across age. The pipeline map (Stage 1s, commissioned) is common to all four; the constants are what it does not remove.                                                                                                                                     |
| reading   | control probes carry the DIRECTION of a lab's offset, not its size; adding a lab widened the error spread. The housekeeping channels record part of the chemistry (normalisation, conversion controls) but not the pre-analytical part (DNA input, bisulfite batch, storage). Honest end of the route for now.                                                    |
| decision  | THE LAB ZERO IS THE HEALTHY-CONTROL PANEL (CLSI EP28). Panel size and form were then set by PROC-PANEL-01 → PROC-PANEL-03: 40 healthy arrays per lab through the same Stage 1, read against the reference age curve, offset printed on every report. Tried the ideal route first, fell back to the standard when it failed - the floor and the map are untouched. |
| nulls     | N-plate 38 chips F 1.79 p 0.009. N-sex signed: women lower in every UCLA decade (as Uppsala, opposite Karolinska) - sign lab-dependent, no split. N-ethnicity Hispanic-Caucasian d -0.33 at n=18, descriptive only. P2 presence 86% in a median-age-70 cohort - the gate needs an age-aware look (commissioning note).                                            |

## PROC-HMIN-BOOT-01 - the methylation floors bootstrap-checked for the first time — The record said all 40; the TSV held 32. Run: 8/8 in CI, 0.060%

| field       | value                                                                                                                                                                                                                                                                      |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| question    | Were the eight methylation H_min values ever bootstrap cross-checked, as the record's 'all 40 values' sentence implies?                                                                                                                                                    |
| finding     | No. bootstrap_vs_mcmc_comparison.tsv (commit 22749f0) has 32 rows = nucl/fuzz/wps/frag x 8 classes. The methylation eight had MCMC (R-hat < 1.001) and no bootstrap.                                                                                                       |
| run         | G-002 reference database (37 published reference cell methylomes, 4-6 per class) + bootstrap_h_min from gape_bootstrap_comparison.py, both from 22749f0; 10,000 resamples, seed 42, mean-over-cells H(beta); exact leave-one-out; compared with the frozen values at HEAD. |
| result      | 8/8 frozen values inside the bootstrap 95% CI; mean relative difference 0.060%, max 0.095% (immune). Tighter than the G-003b substrates. Frozen values unchanged.                                                                                                          |
| code_status | The calibration scripts are NOT at HEAD (removed 2026-04-19, 538667d, 'commercial calibration layer'); they are in public git history at 22749f0, so the evidence report's links are dead while the files remain retrievable. Disclosure decision for the author.          |

## PROC-PANEL-01 → PROC-PANEL-03 - the per-lab healthy panel as the lab zero: COMMISSIONED — 40 healthy arrays, any age mix, read against the reference age curve; a three-lab band holds 75-84% of a fourth lab

| field    | value                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| question | Define the per-lab healthy panel and test it on the four held cohorts: does a lab zeroed by its own panel read 1.00, and does a band built on three labs then hold the fourth?                                                                                                                                                                                                                                                      |
| panel01  | k=25, flat zero. P3 (decisive) PASS: LOO lab-zeroed band holds 0.788/0.841/0.766/0.717 of a held-out lab (without lab zero 0.142/0.861/0.502/0.750). P2 FAIL as sealed (Uppsala 85% within 0.010 - panel size). P1: k=40 is the first size with SD(z)<=0.005 in every lab. P4 FAIL and a design error (two random panels differ by ~0.007 from sampling alone).                                                                     |
| panel02  | k=40, flat zero. P2' Uppsala 94.4% - FAIL by 0.6 points. P4' (deterministic): healthy immune A RISES WITH AGE within a lab - Uppsala teens -0.025 to eighties +0.019 about the cohort median. The between-lab offsets are parallel across age; the age curve itself is not flat. A flat panel median inherits the panel's age mix.                                                                                                  |
| panel03  | k=40, zero read against the reference healthy age curve built on the OTHER labs (LOO). P2" 97.0/99.8/98.4/97.6 PASS. P4" single-decade panels 96.0/99.9/100/99.5 PASS - a panel needs no age matching once read against the curve. P3" LOO age-referenced band (width ~0.053) holds 0.823/0.839/0.767/0.753 (nominal 0.80) PASS.                                                                                                    |
| decision | LAB ZERO COMMISSIONED: 40 healthy arrays per laboratory, any age mix, $z = \text{median}[A - c(\text{decade})] - 1$ ; $A'' = A - c(\text{decade}) - z$ . Closed in code: MethylPhys/chain/lab_zero.py + Runtime Matrices/A_Scoring_Module/reference_age_curve_v1.json (four labs, n=1,379); lab_zero=UNSET is not reportable; panels < 40 refused; kit test test_lab_zero.py. Supersedes the '20-30 arrays' wording of LAB-ZERO-02. |
| residual | The Sentrix-chip term within a lab (Karolinska p 8e-16, Uppsala 2e-19) is untouched by a constant and sits inside the band width - the next residual.                                                                                                                                                                                                                                                                               |

## PROC-HISTORY-01 - the validation count, corrected from the record — The 2026-09-19 index said 103; the record says 3 G + 119 VAL + 15 T + 22 CPG-VAL + hull + N7

| field      | value                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| question   | How many validations were run before this document, and how were they counted?                                                                                                                                                                                                                                                                                                                                                                                          |
| finding    | The author caught the undercount: 'we have 128 VALs plus VAL-049 includes 15 in our T series, the full G series before VAL-001, and post-Atlas we tested cpg_val_001-022, not 15.' The 2026-09-19 index keyed on the bare number (VAL-001 and CPG-VAL-001 collided, 22 times) and counted only identifiers with a folder at HEAD (51 pre-Atlas records live in the RETIRED report and Zenodo, not in folders). Seven AD folders were also mis-filed under VAL_PreAtlas. |
| correction | Index rebuilt from the record with unique keys by series: 175 rows. 3 G; 119 VAL-001..128 (107 executed, 12 not run for stated reasons); T1-T15 (12 executed); 22 CPG-VAL (21 executed); Mahalanobis hull v0_1-v0_5 (n=2,523, four populations incl. Han Chinese n=42); L9 N7; September PROCs. Paper 1 and this document corrected at source. Lesson: an index built by walking the tree counts folders, not the record.                                               |

## PROC-SWITCH-01 -> PROC-SWITCH-02 - the gauge switch, commissioned — The reported A is now the identity-loci gauge with the three-layer reference; the atlas turned out to be a fifth laboratory

| field      | value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| question   | Does the conductor now REPORT the identity-loci gauge with the three-layer reference, and does the N7 defect disappear?                                                                                                                                                                                                                                                                                                                                                                                                                                |
| finding    | S1 7/7 identity_loci/MAPPED/reportable; S2 7/7 UNSET refuses; S3 4/5 healthy in band; S5 anchors $r = 1.00000$ (both), canonical A-score test PASS. S4 as sealed FAILED: 40 synthetic healthy read median 0.985 with zero 0 - not the generator's 2,745-locus subset (pure atlas immune reads 0.990 on both), not noise/age/batch (all off: 0.985), but the atlas posterior itself, which sits at beta 0.7373 where the G-002 floor puts $A=1$ at 0.7318. The analyst's first mixture arithmetic (weights summing to 0.989) was wrong and is recorded. |
| correction | PROC-SWITCH-02: the synthetic cohort read as a fifth laboratory. $z_{atlas} = -0.0146$ from a 40-panel; 40 disjoint patients read median 1.0009, 100% in identity_band_v3; the marker-union statistic on the same patients 1.125, 0% in band; UNSET refuses 80/80. ROW B COMMISSIONED. In code: cpg_conductor (classes = identity gauge; diagnostic_marker_union; pending_recalibration for Stages 5/6), identity_band_v3.json, MethylPhys/kit/test_gauge_switch.py.                                                                                   |

## PROC-MAHA-01 - the departure re-based on the identity gauge; the chip term measured — Row 5 BUILT, not commissioned: one lab fails the p95 tail bar and the cause is the Sentrix chip

| field      | value                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| question   | Does the departure (Stage 5) read correctly on the identity gauge, reach the report, refuse without a zero - and does healthy blood stay inside?                                                                                                                                                                                                                                                          |
| finding    | Re-based: $z = (A - 1)/\sigma$ , $\sigma$ 0.0204 from identity_band_v3; one banded axis on whole blood so distance = $ z_{immune} $ , thresholds 1.960/2.576; long + short keys; report renders. M1 5/5 healthy inside ( $z$ -0.75..+1.61), RA +0.5/+0.35; M3 40/40 synthetic inside; M4 render OK; M5 UNSET 7/7. M2 FAILED as sealed: Karolinska 9.8% beyond p95 (bar 7%; pooled 6.5%; others 4.4-6.5%). |
| correction | Cause measured: the Sentrix chip. Per-chip median SD 0.020 at Karolinska vs 0.012 elsewhere; chip-centring -> Karolinska 4.1%, all labs 2-4%, within-lab SD 0.017-0.018 everywhere. A laboratory constant cannot remove it; a chip reference can. Row 5 BUILT, not commissioned; row 5b (chip term) opened; the acceptable false-alarm rate is the author's decision (PROC-MAHA-02).                      |

## PROC-MAHA-02 - row 5 commissioned with the laboratory false-alarm rate on the report; row 5b (chip) opened — The number never travels without its false-alarm rate

| field      | value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| question   | With the p95 tail failing on one laboratory (the chip), what does the report say?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| finding    | Author's decision: commission with the laboratory's own false-alarm rate on the report and open the chip term as row 5b. identity_band_v3 now carries each laboratory's p95/p99 healthy tail and chip-median SD (M6); the departure and the report print it - Karolinska: 'At this laboratory 10 of 100 healthy donors read beyond p95 on this axis (5 of 100 beyond p99); the excess over 5 is the chip term (row 5b)'; an unmeasured laboratory gets the four-lab range 4-10; M7 as sealed (exact == 0.098 against a stored 0.0984) FAILED and is recorded as such - the analyst's first 'PASS, the prereg says 3 dp' misattributed M6's qualifier and is withdrawn; MAHA-01 M1/M3/M5 unchanged (M8). |
| correction | ROW 5 COMMISSIONED (whole blood, immune axis). ROW 5b OPEN with its bar: a reference on the chip brings every laboratory's p95 tail to $\leq 0.05$ (chip-centring already reaches 2-4%). The number never travels without its false-alarm rate.                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

## PROC-CMB-01 -> 05 - the patient's sky: five seals, row 4.6 commissioned — A healthy sky is quiet at 2.6-3.2%, not 5%; the constant travels with every plate

| field           | value                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| question        | Can a patient's sky be drawn against a healthy expectation, and is a healthy person's sky quiet?                                                                                                                                                                                                                                                                                                                                                                         |
| defect measured | PROC-CMB-01 C1: the retired module (z against a pure-class mean over the atlas posterior SD) reads 60.7% of a healthy genome as $ z  > 2$ . Closed.                                                                                                                                                                                                                                                                                                                      |
| design          | $z_i = (\text{beta}_i - \text{sum\_c\_f\_c\_mu\_ci} - m_{\text{lab},i}) / s_{\text{lab},i}$ : the sample's own Stage 2 composition sets the expectation (PROC-SWITCH-02), the laboratory's 40-array healthy panel supplies the per-CpG zero m and spread s (the same panel as the lab zero), the class panel renders only above a measured presence floor, and z is projected to HEALPix NSIDE 128 in genomic order (deterministic mapping, 483,092/483,092 atlas CpGs). |
| four seals      | CMB-01 omitted the zero (held-out median z -1.1); CMB-02 masked the blood classes with their own p99; CMB-03 inflated the scale with an RMS target and carried a render-cap bar the author retired ('we have detected secretory and cycling in whole blood and scored it fine'); CMB-04 as run.                                                                                                                                                                          |
| result          | Held-out healthy, four labs (n=160): median z -0.013...-0.051; tail 2.6-3.2% against a sealed 3.0% floor - FAILED AS SEALED in 3/4 labs by $\leq 0.004$ ; recorded. Gate 160/160. Rendered: immune 160, progenitor 150, stem_adult 19, stem_pluri 1, each quiet (0.03-0.07). Cross-lab: the zero and scale belong to the laboratory.                                                                                                                                     |
| standing        | Row 4.6 COMMISSIONED by PROC-CMB-05 (fifth seal; bar restated 0.030->0.025 under the author's make-the-best-decision instruction, logged): C2' 4/4, gate 160/160, mapping and plate deterministic; the sealed script's own verdict True. Constant on every sky: a healthy sky is quiet at 2.6-3.2%, not 5%.                                                                                                                                                              |

## PROC-TIER-01 - row 7 commissioned: one tier definition; where healthy people fall under it — The physics lines are clean of healthy people; the 1.01 onset is not

| field    | value                                                                                                                                                                                                                                                                                                                                                                   |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| question | Do the Stage 7 tier words follow one definition, only on a reportable gauge, and where do healthy people fall under them?                                                                                                                                                                                                                                               |
| found    | Three tier definitions in code (report builder x2 with NORMAL ending at 1.04, a colour map with the same) against tier_breakpoints.json v1.3 (NORMAL ends at 1.01, the July move). Consolidated into cpg_tiers.py; every boundary tested both sides; no tier word without a reportable gauge (s108, UNMAPPED, UNSET); $A \geq 1/H_{\text{min}}$ reports AT_CEILING.     |
| measured | 1,379 healthy donors on the commissioned identity gauge under v1.3: SUPPRESSED 2.2%, NORMAL 67.5%, ELEVATED 30.2% (25.9-31.4% in each of four labs), $\geq 1.07$ one donor, $\geq 1.10$ none. Central 95% of healthy A" = [0.954, 1.041] - the pre-July NORMAL band to three decimals. Analyst's sealed prediction ( $\geq 25\%$ ELEVATED) correct.                     |
| standing | Row 7 COMMISSIONED (T1, T3, T4). PROC-TIER-02 then set NORMAL to the healthy central 95% - tier_breakpoints v1.4, [0.95, 1.04]: 2.5% of 1,379 healthy read ELEVATED (predicted 2-3%), Warburg 1.07 admits one donor, breach 1.10 none. Analyst decision under the author's instruction, logged; the July 1.01 onset had been set on the retired marker-union statistic. |

## PROC-BIDIR-01 - row 4.5 commissioned: the bidirectional detector reproduces its record from the kit and the raw GEO file — Sealed betas = raw GEO to the last digit

| field    | value                                                                                                                                                                                                                                                                                                 |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| question | Does the Stage 4.5 bidirectional detector reproduce its sealed record from the kit, and does the engine compute what the seal computed?                                                                                                                                                               |
| result   | B1-B5 PASS. VAL-050 d = +0.0768 / VAL-051 Rule A d = +0.6237 (AUC 0.677) reproduced to the sealed digits; engine score == sealed per-sample A_dir_A to 2e-16; the 18 IMM CpGs x 726 AIBL samples re-extracted from the raw 5.1 GB GEO supplementary file match the sealed betas with zero difference. |
| standing | Row 4.5 COMMISSIONED (a built tool against sealed bars). Detection in a new cohort on the commissioned chain: not yet tested.                                                                                                                                                                         |

ROW 9 - THE REPORT - IN BUILD (working note, unsealed). cpg\_report\_v3.py renders the author's specification from run\_full: cells detected and percentages; A per class on the three-layer reference with placement and tier; A per cell with no band; Stage 5 departure with the laboratory's false-alarm sentence; the sky; flags; scope. A vocabulary guard refuses any forbidden word. 11 cached arrays render; three with a commissioned laboratory read fully (immune A" 0.977-1.024, IN\_BAND, NORMAL), eight print NOT REPORTABLE where their laboratory has no panel. First read caught a percent/fraction unit mismatch in Stage 2 and an unsupported footnote. Sealed when the author has read one.

## PROC-MATCH-01 - disease matching removed from the chain: what was fixed, what was found, why it is record-side — Not yet tested, never cannot

| field    | value                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| question | Does Stage 8 disease matching run on the commissioned chain, fail closed, and see a patient?                                                                                                                                                                                                                                                                                                                                                    |
| fixed    | The cell-of-origin map used to fall back to {} when missing (fail-OPEN); it now refuses to match (NOT AVAILABLE, zero candidates). Substrate firewall verified both ways. The per-cell A entering Stage 8 equals the sealed anchor formula to 4e-16.                                                                                                                                                                                            |
| measured | 11 cached arrays: 0-5 cells enter the departure profile; no candidates. Healthy per-cell A on the separation surface is ~0.44-0.52 against class floors 0.77-0.98, so the (A - 1.0) departure with a below-floor gate excludes 104 of 115 cells. 52 of the 129 signature-matrix columns are tissue cells the blood atlas cannot score.                                                                                                          |
| standing | REMOVED FROM THE CHAIN by the author the same day: the matrix and its cards come from the preliminary VAL record, and naming a disease from a pattern is not what the instrument does. The report shows cells detected, fractions, A per cell and class, placement, flags. The matrix is record-side; the fail-closed gate, firewall and surface checks stand in the code. This was exploration, not a sealed test (see the sealing rule, s14). |

## PROC-AGE-01 - per-patient cellular age on the identity gauge: not reportable at single-array resolution — The population aging trajectory stands and is reproduced; one array cannot place a person on it

| field      | value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| question   | Can the identity gauge, whose healthy curve rises with age, be inverted to read a person's cellular age?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| finding    | No. Leave-one-lab-out on 1,379 healthy donors: 15.9% within +/-10 yr (bar 80%), median  delta  35.6 yr, Spearman 0.27. The curve moves 0.47 mA per year and the within-laboratory spread is 0.0235 - one array resolves age to ~50 years; where the curve must extrapolate it runs away (258 yr SD). A healthy 58-year-old inverts to 23; a healthy 43-year-old to '>85'. The analyst predicted ~30 yr before the run; it is 50 - recorded.                                                                                                                                                                                                                              |
| correction | Per-patient cellular age CLOSED as NOT REPORTABLE at single-array resolution: the report prints the resolution sentence instead of an age; the marker-union inversion is diagnostic only. With this, no reported number reads the marker-union statistic. THE AGING TRAJECTORY ITSELF STANDS AND IS REPRODUCED: 0.47 mA/yr monotone by decade on four labs is CPG-VAL-015's slope (~0.5 mA/yr on Hannum) - a population measurement, now the reference age curve every reading is corrected by. Sign differs by surface (RECON D2). The first write-up's 'the gauge measures fidelity, not time' overstated the closure and is withdrawn (author's challenge, same day). |

## PROC-RECORD-02 - VAL-025..028 reclassified: the four-substrate aging trajectory is a modeled prediction, not a measurement — A PASS row must name what was measured on which samples

| field      | value                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| question   | Do the four non-methylation substrates show the aging trajectory, as VAL-025..028 (PASS, r = 0.9998) state?                                                                                                                                                                                                                                                                                             |
| finding    | The script types eight human and six canine (age, mu) rows per substrate from the literature's described direction and Monte-Carlo scores them; Hannum 2013 and Wang 2020 are methylation-only and carry no substrate data. r = 0.9998 is the correlation of age with a typed monotone table. Issue 002 (April) already listed these as '(modeled) - prediction filed'; the record and index said PASS. |
| correction | Reclassified MODELED PREDICTION in VAL_INDEX, DETAILED_VALIDATION_RECORD and the original README; RECON R2; Future Goal filed with the data it needs (healthy cfDNA fragmentomics with ages). The methylation aging trajectory - per-sample on Hannum and on four labs - is unaffected and stands.                                                                                                      |

**PROC-RECORD-03 - the 80-cell age matrix: a typed table whose direction the measurement confirmed; AD and breast 'years' restated as dA — The dA was the measurement; the years were a unit borrowed from a typed slope**

| field      | value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| question   | What is the 80-cell healthy age baseline made of, and what did the AD and breast cellular-age findings measure?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| finding    | age_reference_matrix.json at HEAD is the April HEALTHY_BASELINES table: typed beta_mean per decade with literature labels, A = H(beta_mean)/H_min in 80/80 cells, percentiles A +/- 1.2816 sd in 80/80 - a Gaussian around a typed number; no generating script exists. Its direction (A rises with age in every somatic class) is confirmed by PROC-AGE-01 on four labs; its immune slope (~1.0 mA/yr) is twice the measured 0.47; its level (0.956 at 54) is pre-scale-offset. AD immune read ~9 y YOUNGER than HC (d = -0.56; the report itself said senescence, not chronological aging); breast cycling ~5.5 y younger. |
| correction | The 2026-09-19 wording that code compiled the matrix is corrected (the A column is computed; the beta_means are typed). Years are withdrawn as a unit: the AD finding stands as a group departure of the immune gauge from the healthy age curve in A (d = -0.56); one-patient detectability against the 0.0235 within-lab spread is paper two's absolute question; the longitudinal claim was never tested (VAL-005 UNDERPOWERED). The matrix is read by no reported path; reference_age_curve_v1 (measured) supersedes it.                                                                                                 |

**PROC-WB-COMP-01 — Whole blood — composition is immune, epithelium absent**

| field               | value                                                                                                                                                                                                        |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| input               | as PROC-WB-IMMUNE-01, deconvolver only                                                                                                                                                                       |
| expected            | immune + progenitor + stem_adult ≥ 0.95; epithelial (cycling+secretory+terminal+stromal) ≤ 0.02; residual MAE below tissue (0.108–0.153)                                                                     |
| observed 2026-09-19 | immune 0.8027–0.9663, progenitor 0–0.174, epithelial 0.0000–0.0111, residual 0.0434–0.0596 — all seven                                                                                                       |
| verdict             | PASS on n=7 including two RA cases. This is the substrate specification (README 'Key validated facts'), not a limitation. It says nothing about the immune-ARCHITECTURE channel, which is PROC-WB-IMMUNE-01. |

**PROC-PLASMA-COMP-01 — Plasma cfDNA — shed epithelium by condition**

| field               | value                                                                                                                                                                                                                                                                           |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| input               | GSE122126-GPL21145 series matrix (Moss 2018, EPIC): healthy 4, colon 4, breast 3, lung 4, CUP 4, sepsis 22, in-vitro mixes 14; author-processed β (no Stage 1)                                                                                                                  |
| operation           | deconvolve each sample; epithelial = cycling+secretory+terminal+stromal                                                                                                                                                                                                         |
| expected            | healthy ≤ 0.05; solid-tumour plasma > healthy; in-vitro mixes recover declared proportions                                                                                                                                                                                      |
| observed 2026-09-19 | healthy 0.030 · colon 0.463 · breast 0.466 · lung 0.010 · CUP 0.293 · sepsis median 0.040 (p75 0.233, max 0.520). Residual MAE tracks epithelial fraction r=0.98. Mixes NOT yet scored.                                                                                         |
| verdict             | PASS on direction for colon/breast; lung null recorded; sepsis sheds in a minority. NOT a gauge result — any A on this cohort must use frag/wps/nucl floors (RECON S1), which array data cannot supply. Cell-level tissue-of-origin FAILS: all epithelium → gastric references. |
| caveat              | author-processed GEO β without Stage 1; n per cancer 3–4                                                                                                                                                                                                                        |

PROC-FORMULA-01 — A-score aggregation - SOP v1.4.0 (mean of per-CpG H) vs code at HEAD (H of mean beta)

| field               | value                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| input               | betas_cache.pkl (7 Stage-1 whole blood incl. 2 RA; 4 EPIC colorectal tissue); iamatlas_gauge_identity_loci_v1_0.json; iamatlas_celltype_markers_v0_2.json aggregated to class unions; age_reference_matrix.json                                                                                                                                                                                                                                                         |
| operation           | per class {immune, cycling, secretory}, per sample: (a) H(beta_mean)/H_min over identity loci [code at HEAD]; (b) mean_i H(beta_i)/H_min over identity loci [SOP v1.4.0 formula on the HEAD loci]; (c),(d) the same two over the discriminative class union. Then Spearman(a,b) and the offset a-b, separately for whole blood and tissue. Then check which aggregation age_reference_matrix.json was compiled with.                                                    |
| expected            | if the two aggregations are equivalent on the declared substrate, rank agreement ~1 and a constant offset; if s105's mechanism is real, the offset should grow on mixed or absent-class panels; the band must match one of them exactly                                                                                                                                                                                                                                 |
| observed 2026-09-19 | WHOLE BLOOD immune: Spearman +1.000, offset +0.029 +/- 0.002. WHOLE BLOOD cycling/secretory (absent classes): offset +0.16. TISSUE immune: offset +0.24 +/- 0.07. age_reference_matrix A_mean = H(beta_mean)/H_min to 5 decimals in 80/80 cells. Discriminative mean-of-H tracks presence (cycling WB 0.35-0.41 vs tissue 0.57-0.62).                                                                                                                                   |
| verdict             | OPEN - author decision required. On the declared substrate with the class present, formula = constant offset; both valid IF band and runtime agree, and the band is H(beta_mean). On mixed/absent panels H(beta_mean) inflates as s105 says - never read it as architecture (presence gate; glioma-LL-002). The current SOP and the running code disagree and one must be amended: keep H(beta_mean) and scope s105's NEVER, or adopt mean-of-H and recompile the band. |
| retraction          | the first draft of this PROC (same day) read the tissue H(beta_mean) values as a disease ordering; that was composition inflation and is withdrawn (s10).                                                                                                                                                                                                                                                                                                               |
| caveat              | n = 11, two substrates; RA-study samples read +0.04 on discriminative H(beta_mean) for cases and controls alike - a GEO cohort offset (CCL-004), not disease                                                                                                                                                                                                                                                                                                            |

PROC-PLASMA-MIX-01 — Deconvolver against real known mixtures (Moss 2018 Table 6 / Table 8)

| field               | value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| input               | GSE122126 GPL21145: in_vitro_mix_9..17 (9 genomic-DNA mixes: leukocytes 85-96% + hepatocytes/lung/neurons/colon 3.5-10%, Supp Data 1 Table 6) and cfDNA_mix_1,3,5,6,7 (colon-cancer cfDNA CC2 at 0-10% into healthy cfDNA, Table 8). Author-processed β.                                                                                                                                                                                                                                                            |
| operation           | deconvolve; compare declared spike per tissue to the class the atlas assigns that tissue (Hepatocytes→secretory, Cortical_neurons→terminal, Colon_epithelial_cells→cycling). GEO title→Mix mapping by order, corroborated by max-terminal landing on Mix2 (10% neurons) and max-cycling on Mix4 (10% colon).                                                                                                                                                                                                        |
| expected            | declared non-leukocyte fraction recovered (slope ~1, bias ~0); each spiked tissue rises in its own class                                                                                                                                                                                                                                                                                                                                                                                                            |
| tolerance           | proposed: per-class MAE ≤ 0.02 against declared; r ≥ 0.9 per spiked tissue                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| observed 2026-09-19 | TOTAL non-leukocyte: r = 0.945, MAE 0.0677, bias +0.0677 (over-reads by ~7 points in every mix). NEURONS→terminal r = +0.945 (10% → 6.5%, under-read, ordering correct). HEPATOCYTES→secretory r = +0.192, p = 0.62 (10% → 1.7%). COLON→cycling r = +0.101, p = 0.80 (cycling reads 5–10% whether or not colon is present). Cancer-cfDNA spike (n=5): epithelial r = +0.725, p = 0.17 — monotone but not interpretable as recovery (CC2's own load unknown).                                                        |
| verdict             | SPLIT. Terminal class recovers a real neuronal spike (PASS on direction, under-reads magnitude). Secretory does NOT register hepatocyte DNA and cycling does NOT register colon DNA — both FAIL on the first real ground truth the deconvolver has faced. Consistent with \$7.3 cell routing (all epithelium → gastric) and LESSON-DECONV-01 (reference mismatch): the atlas profiles for liver and colon do not match real liver and colon DNA closely enough to earn weight. This is atlas work, not solver work. |
| consequence         | the plasma colon/breast epithelial fractions of PROC-PLASMA-COMP-01 (0.46) are real departures from healthy (0.03) but their class assignment is not trustworthy; the tissue-of-origin claim is withdrawn until the atlas recovers Table 6.                                                                                                                                                                                                                                                                         |

## PROC-ANCHOR-01 — Sealed GSE51032 per-cell anchor, reproduced from raw GEO betas

| field                                                              | value                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| input                                                              | GSE51032 series matrix (3,009 MB, EPIC, 845 samples) filtered to the 460 in the sealed foundation-cohort CSV (36 breast >10 yr pre-dx + 424 HC); iamatlas_celltype_markers_v0_2.json at repo HEAD and the TRIAL (chrX-removed) copy; H_min per class from iamatlas_gauge_identity_loci_v1_0.json                                                                                                                                 |
| operation                                                          | for each of 115 cell types: per-cell A under (a) $\text{mean}_i H(\text{beta}_i)/H_{\text{min}}$ and (b) $H(\text{beta}_{\text{mean}})/H_{\text{min}}$ over that cell's ~100 discriminative markers; Pearson r and max  diff  against GSE51032_115celltype_ascores.csv (sealed 2026-05-29)                                                                                                                                       |
| expected                                                           | r = 1.0000, maxdiff = 0.0000 against the sealed CSV, as SOP v1.4.0 s105 reports for the vault module                                                                                                                                                                                                                                                                                                                             |
| observed HEAD markers, mean-of-H                                   | r = 1.00000; max  diff  0.00004; 112/115 cells r > 0.999 - REPRODUCES THE SEAL                                                                                                                                                                                                                                                                                                                                                   |
| observed HEAD markers, H(beta_mean)                                | r = 0.450; max  diff  1.048; 0/115 - does not                                                                                                                                                                                                                                                                                                                                                                                    |
| observed TRIAL markers, mean-of-H                                  | r = 0.99957; max  diff  0.060; 89/115 - 32 cells shift by up to 0.059 - the chrX removal                                                                                                                                                                                                                                                                                                                                         |
| observed TRIAL markers, H(beta_mean)                               | r = 0.449; max  diff  1.048; 0/115 - does not                                                                                                                                                                                                                                                                                                                                                                                    |
| observed GSE51057 (188: 11 case + 177 HC), HEAD markers, mean-of-H | r = 1.00000; max  diff  0.00004; 115/115 cells r > 0.999 - REPRODUCES THE SEAL - both halves of the 648-sample foundation cohort                                                                                                                                                                                                                                                                                                 |
| also observed                                                      | healthy per-cell A median 0.520 on this surface (s105 says ~0.52); per-cell case-vs-HC d median +0.32, 37/115 cells  d  > 0.5, largest Glia +1.10, stem_pluri +1.07, Baso +1.01. The seal was made with the REPO-HEAD markers (pre chrX removal): the TRIAL file changes 32 cells by up to 0.059 (largest: Mela) and would need re-sealing.                                                                                      |
| verdict                                                            | PASS for the separation surface: the sealed per-cell anchor is exactly mean-of-H over the v0_2 discriminative markers as SOP v1.4.0 s105 states, reproduced from raw GEO on a stranger's machine. This does NOT adjudicate the gauge (identity loci) - see PROC-FORMULA-01. It does establish that the current repo markers file, not the chrX-removed one, is what the seal rests on; adopting the chrX file re-opens the seal. |

## PROC-QC-01 — Stage 0-1 intake and calibration gates (SOP v1.4.0 s11-25) - specified; PROC-CAL-01 executes it

| field                    | value                                                                                                                                     |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Manifest                 | raw IDAT pair keyed to demographics and array substrate before any processing (the Planck 'scan-ring' association); SHA-256 on every file |
| Detection p-value        | >5% probes failed = HARD reject; 1-5% = SOFT flag (s15)                                                                                   |
| Bead count               | >=3 per probe (Illumina default); >=5 for high-stakes (s16)                                                                               |
| Call rate                | >=0.98 for Stage 0 PROCEED (s19); <0.95 after QC = quarantine                                                                             |
| Sex check                | predicted vs declared sex; mismatch = sample swap or rare karyotype - halt (s18; Planck: pointing-direction sanity check)                 |
| Dye bias                 | noob (s20; Planck: bandpass correction)                                                                                                   |
| Probe-type normalisation | funnorm default (s21)                                                                                                                     |
| Batch                    | ComBat requires >=3 samples per batch (s22)                                                                                               |
| Bisulfite efficiency     | <0.95 HARD quarantine; 0.95-0.98 SOFT (s23); low BS efficiency + low intensity = degraded DNA                                             |
| Beta sanity              | range, bimodality coefficient 0.45-0.55 SOFT flag, cohort FLAG rate >5% = HARD halt (s25)                                                 |
| verdict                  | SPECIFIED, NOT RUN - every gate above is a pass/fail with a threshold; PROC-CAL-01 will record each on the seven test-data IDATs          |

## Not yet written — and what each needs

| PROC                                  | needs                                                                                                                                               |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| PROC-WB-PROGENITOR-01 / STEM_ADULT-01 | the CHANGELOG's suspected breach artifact in peripheral blood; requires the presence-gated class ranking fix (design decision on threshold pending) |
| PROC-TISSUE-N-COMP-01                 | any tissue claim must report immune-class A beside the target class (glioma-LL-002); write once CPG-NEW-001 is re-run presence-gated                |
| PROC-URINE-01, PROC-CSF-01            | GSE119260 in hand; CSF cohorts need supplementary-file fetch; floors for these substrates not yet established                                       |
| PROC-N5-01                            | plate/sentrix position null from raw IDAT filenames for GSE51032/GSE51057 — first thing a methylation referee asks for on the breast pre-dx result  |

## SECTION 9

# OPERATING RULES

The lessons that govern how the instrument may be read, promoted from appendix to rules. Each carries its date, source, and supersession status. A rule that is broken produces a wrong number, not a style complaint; several of these were broken in the runs of 2026-09-19 and the resulting wrong numbers are in §10.

| ID                | Rule                                                                           | Content                                                                                                                                                                                                                                                                                                                             | Source                                           | Status           |
|-------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|------------------|
| L-5               | Physics measures, cohorts only point.                                          | The A-score is intrinsic and self-calibrating. Cohort data enters only as a DIRECTION, never as a baseline. The moment a cohort mean/SD becomes the yardstick the model works only in its own cohort. Primary readout = absolute A per sample against the fixed reference and the age band; group statistics are supporting detail. | LESSONS_collapse_and_KISS_v1                     | current          |
| CCL-019/020       | Direction depends on (class, compartment), not disease.                        | Specimen, class and panel are three independent dimensions and all three must match the clinical question. Discovered when VAL-061 predicted the wrong sign.                                                                                                                                                                        | LESSONS_LEARNED.md                               | current          |
| CCL-039           | Marker-tile and full-genome A are two observables.                             | They move in opposite directions in tumour vs adjacent normal (READ, COAD, COAD-VAL-099). Never compare one to the other.                                                                                                                                                                                                           | LESSONS_LEARNED.md                               | current          |
| CCL-032 / CHK-3.1 | Extreme-beta integrity gate is substrate-dependent.                            | Whole blood 450K reads 26-34% extreme; Stage-1-calibrated EPIC tissue reads 7-24%. The >30% criterion does not transfer to tissue. Thresholds must be stated per substrate.                                                                                                                                                         | LESSONS_LEARNED.md; run 2026-09-19               | current, amended |
| LESSON-DE CONV-01 | Global NNLS is correct; sparsity is reference mismatch.                        | Synthetic mixtures recover at MAE 0.000; real blood fits 2.6x worse. Hierarchical within-class refinement invents cells (MAE 0.271 on pure monocytes) and does not ship. Richer per-cell resolution is atlas work, not a solver swap.                                                                                               | SOP v1.3.3 §0.9 (2026-06-21)                     | current          |
| 2026-06-11        | Never feed the gauge discriminative markers.                                   | They are bimodal; their mean beta collapses toward 0.5 and the gauge pins the ceiling (all-BREACH).                                                                                                                                                                                                                                 | cpge_gauge_engine.py                             | current          |
| PRESENCE          | A class not in the sample has no A-score.                                      | Presence comes from the deconvolved fraction (DETECT_FLOOR). Scoring an absent class's identity loci produced spurious BREACH calls on stromal (fraction 0.0000) in 4/4 healthy plasma samples, 2026-09-19.                                                                                                                         | cpge_conductor.py; run 2026-09-19                | current          |
| SUBSTRATE         | Every report states its substrate; every (class, substrate) has its own floor. | Whole blood = immune-architecture readout; plasma cfDNA = shed-tissue readout on frag/wps/nucl floors; tissue = positive control.                                                                                                                                                                                                   | cpge_gauge_engine.py; TEST_DATA_MANIFEST.md      | current          |
| glioma-LL-002     | A high A on bulk tissue is a heterogeneity marker, not a tumour marker.        | Composition alone can move beta-bar. N-comp (report immune-class A alongside the target class) is the binding null for any tissue claim.                                                                                                                                                                                            | LESSONS_LEARNED.md                               | current          |
| PL-002            | The honest record is the best defense.                                         | Amendments after data are observed are labelled as such; outcomes are allowed to be RESTATE, FAIL or VOID.                                                                                                                                                                                                                          | LESSONS_LEARNED.md                               | current          |
| PIPELINE          | One IDAT-to-beta pipeline per patient; absolute A is pipeline-relative.        | Same Xu-538 panel, different IDAT-to-beta pipelines: absolute A baselines 0.38-0.62 across cohorts (within-EPIC-Italy ~0.44), direction preserved within pipeline (April 2026). Absolute A is pipeline-relative; this is the leading hypothesis for the 3/3 below-band healthy whole-blood donors (PROC-WB-IMMUNE-01).              | GAPE_EDEAR_Reproduction_Paper_v3 (T15, Rule 1-2) | current          |
| FIREWALL          | Production subtracts no age/sex/smoking foreground.                            | SOP s104: such change is the signal itself; age enters via the reference band, sex via chrX marker removal; subtraction modules are test-only and must not be loaded in production config.                                                                                                                                          | SOP v1.4.0 s104                                  | current          |
| SEAL              | A measurement that has not beaten the null suite is a candidate, not a claim.  | N1-N8; SEALED/RESTATE/RETRACT; PREREG edited after data = auto-RETRACT.                                                                                                                                                                                                                                                             | SOP v1.4.0 s80-91                                | current          |

| ID      | Rule                                                 | Content                                                                                                                                                                                                                                                         | Source                                                                                                                                                                                                 | Status                                                           |
|---------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| SURFACE | One aggregation per surface, fixed by its reference. | Gauge (identity loci): $H(\text{beta\_mean})/H_{\text{min}}$ . Separation (discriminative markers): $\text{mean\_i } H(\text{beta\_i})/H_{\text{min}}$ . Never read a statistic against a band compiled the other way.                                          | RULING A3                                                                                                                                                                                              | current                                                          |
| L-10    | Reporting rule (extends the s3.2 presence gate)      | a class gauge is REPORTED on a substrate only where the class is DETERMINED ( $\kappa$ of the substrate sub-problem $< \sim 10$ , or reported jointly with what it cannot be split from) and PRESENT (typical fraction $\geq 5\%$ ); otherwise composition-only | whole blood: immune reported; progenitor + stem_adult jointly as one haematopoietic-progenitor component; the other five composition-only. Decided from $\kappa$ and fraction BEFORE any A is computed | author, 2026-09-19, after PROC-CHAIN-01 / PROC-NILC-01; SOP s108 |

SECTION 10

FALSIFICATION RECORD

Every outcome that did not go the framework’s way, kept on the record. A framework that cannot show its failures cannot show that its passes mean anything. Historical VALs are quoted from their sealed OUTCOME files; the 2026-09-19 rows are from this session and include the author’s own failed predictions.

| Test                                                                                       | What happened                                                                                                                                                                                                                                                                        | Outcome                 |
|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| VAL-004                                                                                    | bimodality direction reversed vs prereg                                                                                                                                                                                                                                              | RESTATE                 |
| VAL-006                                                                                    | chr6 signal did not survive look-elsewhere correction                                                                                                                                                                                                                                | FAIL                    |
| VAL-061                                                                                    | predicted wrong sign; produced CCL-019/020                                                                                                                                                                                                                                           | RESTATE                 |
| VAL-102                                                                                    | voided four minutes after sealing for post-hoc accommodation; original seal preserved                                                                                                                                                                                                | VOID                    |
| VAL-128                                                                                    | failed opposite to prereg                                                                                                                                                                                                                                                            | FAIL                    |
| CPG-NEW-001 P1                                                                             | GSE48684 adenoma read above carcinoma; monotonic progression failed                                                                                                                                                                                                                  | FAIL                    |
| CPG-NEW-001 P4                                                                             | serrated vs conventional reversed sign between Spain and Finland                                                                                                                                                                                                                     | FAIL                    |
| CPG-NEW-001 N-comp                                                                         | cycling effect not larger than immune effect on adenoma contrasts                                                                                                                                                                                                                    | composition-ambiguous   |
| 2026-09-19 two-channel                                                                     | immune-A x epithelial-fraction did not beat epithelial alone (AUC .955 vs .961; LOO .793 vs .828)                                                                                                                                                                                    | prediction FAILED       |
| 2026-09-19 WB x immune                                                                     | 3/3 healthy whole-blood donors read BELOW the age band (z = -2.07, -4.03, -3.49)                                                                                                                                                                                                     | OPEN - scale or biology |
| PROC-PLASMA-MIX-01 secretory                                                               | 10% hepatocyte genomic DNA in leukocytes reads 1.7% secretory (Moss Table 6 ground truth)                                                                                                                                                                                            | FAIL                    |
| PROC-PLASMA-MIX-01 cycling                                                                 | colon spike 0-10% not recovered; cycling reads 5-10% regardless (r=0.10)                                                                                                                                                                                                             | FAIL                    |
| Analyst verdict 2026-09-19 (first draft of PROC-FORMULA-01)                                | 'identity-loci H(beta_mean) carries the disease ordering healthy < adenoma < carcinoma' - the tissue ordering is composition inflation (+0.24 Jensen offset on a mixture), not architecture; retracted the same day when the whole-blood Spearman came back +1.000 for both formulas | RETRACTED               |
| SOP v1.4.0 s105 universal claim                                                            | 'the A-score is NEVER H(beta_mean)' - on the declared substrate with the class present the two aggregations are rank-identical with a constant offset, and the shipped age band is compiled as H(beta_mean); the NEVER is correct for mixed/absent panels only                       | OVER-SCOPED             |
| Issue 003 draft v3-v8 s5A.1 claim                                                          | 'M_cell = n_bio/ln2 = 30.21; AI transmon M = 1 exactly; one statistic across three substrates' - computed with a ln2 denominator the author does not use; withdrawn                                                                                                                  | WITHDRAWN               |
| PROC-CHAIN-01 'healthy blood IN_BAND 7/7 = the shipped chain absorbs the offset'           | the conductor's gauge and its band are both marker-union statistics; agreement between them is self-consistency, not validation (PROC-N7-01)                                                                                                                                         | WITHDRAWN               |
| RULING A3 'the wired chain computes H(beta_mean) on identity loci'                         | the wired chain computes it on the marker union (cpg_conductor.stage_b_classes, comment citing GAPE_WEB_v13)                                                                                                                                                                         | CORRECTED               |
| s6a 'healthy whole blood reads below band'                                                 | identity-loci A compared against a marker-union band - statistic mismatch; the beta shift at identity loci is real, the band conclusion was not                                                                                                                                      | RESTATED                |
| Analyst recommendation (2026-09-20 03:30) to re-derive H_min_beta on Stage-1 healthy blood | would discard the G-002 MCMC confirmation and make the floor pipeline-dependent; the offset is in the input scale, not the floor                                                                                                                                                     | WITHDRAWN same day      |

| Test                                                                                                                           | What happened                                                                                                                                                                  | Outcome                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase 1c N-sex 'women higher every decade' (2026-09-20)                                                                        | the runner computed  d ; direction was asserted without support. Signed recomputation: women higher in GSE42861, LOWER in GSE87571 - sign flips between labs                   | <b>CORRECTED</b> same day; sex-split band withdrawn                                                                                                                                                                |
| LAB-ZERO-01 'promising - a fourth cohort decides' (2026-09-20)                                                                 | fourth cohort: P4 err 0.026, LOO 1/4, narrowing 13% - the control probes do not supply the lab zero                                                                            | <b>DECIDED AGAINST</b> same day (LAB-ZERO-02); panel standard adopted                                                                                                                                              |
| 'All 40 H_min values bootstrap cross-validated at 0.168%' (record, April 2026)                                                 | the TSV holds the 32 non-methylation floors only; methylation had no bootstrap                                                                                                 | <b>CORRECTED</b> 2026-09-20; methylation bootstrap run (PROC-HMIN-BOOT-01): 8/8 in CI, 0.060%                                                                                                                      |
| PROC-PANEL-01 prediction 'k=25 suffices' and 'offsets flat across age' read as 'A flat across age' (2026-09-20)                | k=40 needed for SD(z)<=0.005 in every lab; healthy immune A rises with age within a lab (between-lab offsets are parallel, the curve is not flat)                              | <b>CORRECTED</b> same day by PROC-PANEL-02 and PROC-PANEL-03; lab zero is age-referenced                                                                                                                           |
| '103 indexed pre-registered validations (81 pre-Atlas, 15 post-Atlas, 7 retired)' (Paper 1 draft and VAL_INDEX, 2026-09-19/20) | the index keyed on bare numbers (VAL-001 collided with CPG-VAL-001) and counted folders, not the record: 119 pre-Atlas + 15 T-series + 22 post-Atlas + G-series                | <b>CORRECTED</b> 2026-09-21 by the author's reading; PROC-HISTORY-01; index rebuilt, 175 rows                                                                                                                      |
| PROC-SWITCH-01 S4: 'synthetic healthy patients with lab zero 0 read 1.000 +/- 0.010'                                           | median 0.985 (97.5% in band); the atlas carries its own constant -0.0146                                                                                                       | <b>FAIL</b> as sealed; re-sealed PROC-SWITCH-02 with the synthetic cohort zeroed from a disjoint 40-panel: median 1.0009, 100% in band; marker-union on the same patients 1.125, 0% in band                        |
| PROC-MAHA-01 M2: 'on the identity gauge, <= 7% of healthy donors read beyond p95 in every laboratory'                          | pooled 6.5%; Karolinska 9.8% (p99 tail 5.1%) - the Sentrix-chip term: per-chip medians scatter with SD 0.020 there vs 0.012 elsewhere; chip-centring brings Karolinska to 4.1% | <b>FAIL</b> as sealed on one laboratory; row 5 BUILT not commissioned; the chip is the named residual (row 5b); bar decision to the author                                                                         |
| PROC-MAHA-02 M7 as sealed: 'departure.lab_false_alarm_p95 == 0.098' (exact)                                                    | stored 0.0984; the analyst first wrote PASS claiming the prereg said '3 dp' - that qualifier belongs to M6, not M7                                                             | <b>FAIL</b> as sealed, recorded; the first write-up was a post-hoc loosening and is withdrawn (auditor catch 2026-09-21). Row 5 commissioned on M6/M8 with M7's failure on the record.                             |
| PROC-AGE-01 A1: 'cellular age from the identity gauge places >= 80% of healthy adults within +/-10 yr'                         | 15.9% (median  delta  35.6 yr); slope 0.47 mA/yr vs within-lab SD 0.0235 -> ~50 yr per array; the analyst's own prediction (30 +/- 8 yr) was too optimistic                    | <b>FAIL</b> as sealed; per-patient Stage 6 CLOSED, NOT REPORTABLE at single-array resolution; the report prints the resolution. The population trajectory is reproduced (0.47 mA/yr = CPG-VAL-015) , not retracted |

| Test                                                                                    | What happened                                                                                                                                                                                         | Outcome                                                                                                      |
|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>PROC-CMB-04 C2'</b><br>(2026-09-21) [EXPLORATION - see sealing rule]                 | held-out healthy tail bar [0.03,0.08] in $\geq 3/4$ labs; measured 0.029/0.026/0.032/0.029 - scale $\sim 1.1\times$ conservative                                                                      | <b>FAIL as sealed; recorded; row 4.6 withheld until PROC-CMB-05 passed the restated bar 4/4 the same day</b> |
| <b>PROC-CMB-03 C4'</b><br>render-cap bar (2026-09-21) [EXPLORATION - see sealing rule]  | ' $\leq 3$ healthy test arrays render a non-blood class' assumed a fixed composition for healthy blood (secretory, cycling, terminal absent) - a prior about biology, not a test of the gate; RETIRED | <b>RETIRED by the author same day</b>                                                                        |
| <b>PROC-CMB-01 C1 retired sky formula</b> (2026-09-21) [EXPLORATION - see sealing rule] | z against a pure-class mean over the atlas posterior SD reads 60.7% of a healthy genome as $ z  > 2$                                                                                                  | <b>DEFECT CONFIRMED; module closed</b>                                                                       |
| <b>PROC-TIER-01 T2</b><br>(2026-09-21)                                                  | sealed prediction: under the July 1.01 onset $\geq 25\%$ of healthy donors read ELEVATED on the identity gauge; measured 30.2%, every lab 26-31%                                                      | <b>PREDICTION CORRECT; onset ruling requested from the author</b>                                            |
| <b>PROC-CMB-05 C2'</b><br>(2026-09-21)                                                  | bar restated [0.025,0.08]; prediction 4/4 labs                                                                                                                                                        | <b>PASS 4/4 (0.029/0.026/0.032/0.029); row 4.6 commissioned by the script's verdict</b>                      |
| <b>PROC-TIER-02 U2</b><br>(2026-09-21)                                                  | prediction: under NORMAL [0.95,1.04] ELEVATED+ = 2.0-3.0% of 1,379 healthy                                                                                                                            | <b>2.61% - PREDICTION CORRECT; tier_breakpoints v1.4</b>                                                     |
| <b>PROC-MATCH-01 M5</b><br>(2026-09-21) [EXPLORATION - see sealing rule]                | prediction: $< 10$ present cells enter Stage 8's departure profile on healthy whole blood                                                                                                             | <b>0-5 cells, median 3 - PREDICTION CORRECT; row 8 OPEN, gate named</b>                                      |
| <b>PROC-MATCH-01 M1</b><br>(2026-09-21) [EXPLORATION - see sealing rule]                | origin gate fell back to {} on a missing file (fail-open)                                                                                                                                             | <b>FIXED: fail-closed, kit test</b>                                                                          |
| <b>PROC-BIDIR-01 B5</b><br>(2026-09-21)                                                 | sealed AIBL IMM betas match the raw GEO file for 726 samples x 18 CpGs                                                                                                                                | <b>PASS, max diff 0.00</b>                                                                                   |

Also on the record from this session, as method failures rather than results: the same analyst computed group effect sizes where the framework requires per-sample absolute readings (three times); scored classes the deconvolver had not found in the sample; scored plasma on the methyl floor; and read the pre-atlas SOP as the specification of the post-atlas engine. Each is now a rule in §9 or a row in §1. That is what the rules are for.

# ENGINE MAP — WHAT THE RUNNING CHAIN CONTAINS, AND ITS COMMISSIONING STATUS

The flowchart at HEAD (flowchart\_vKISS.html) lists the twelve stages below (0 through 9, with 4.5 and 4.6 as separate wired stages), plus orchestration and test data. Issue 002 had no chain at all — it was class cards and physics — so every stage is new territory for 003, and this issue documents only the stages that were executed with the canonical files during its preparation. The right-hand column says so, stage by stage. A reader who needs Stage 4.5, 5, 6, 8 or 9 will not find them here; they are the next issue's work, and the files that implement them are named so they can be read now.

| Stage                                          | Files at HEAD                                                                                                           | Status                                                               | What it does (from source)                                                                                                                                                                                                                    | In Issue 003?                                              |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| <b>Stage 0 — Intake</b>                        | stage_0_intake.py,<br>cpg_intake_form.html,<br>questionnaire.json                                                       | BUILT                                                                | patient folder, questionnaire, substrate declaration                                                                                                                                                                                          | NO                                                         |
| <b>Stage 1 — Calibration</b>                   | stage_1_idat_calibration.py,<br>stage_1_calibration.py,<br>idat_parse.py, idat_decoder_pure.py                          | BUILT                                                                | raw IDAT pair -> noob-normalised beta (methylprep 1.7.1; pure-python IDAT decoder shipped)                                                                                                                                                    | PROC-CAL-01 written, NOT RUN                               |
| <b>Stage 2 — Deconvolution</b>                 | Walther_iam_deconvolver/walther_iam_deconvolver.py,<br>MethylPhys/atlas/*                                               | BUILT (NILC reinstated 2026-09-22 as the class-level second opinion) | NNLS class + cell fractions; composition/presence; gates no call                                                                                                                                                                              | <b>YES — §2.4, §3.2, PROC-DECON-01, PROC-PLASMA-MIX-01</b> |
| <b>Stage 3 — Foreground subtraction</b>        | age/sex/smoking modules (test-only)                                                                                     | NOT WIRED by decision (SOP §104)                                     | production subtracts nothing: age via the reference band, sex via chrX marker removal, smoking via Stage-7 bin floors                                                                                                                         | <b>YES — RECON F1, RULES FIREWALL</b>                      |
| <b>Stage 4 — A-score gauge</b>                 | cpg_gauge_engine.py, Runtime Matrices/A_Scoring_Module/*,<br>test_a_score_canonical.py                                  | BUILT                                                                | H(beta_mean)/H_min over identity loci; placement vs age band; severity ladder; brightness CI                                                                                                                                                  | <b>YES — §3; brightness CI NOT described</b>               |
| <b>Stage 4.5 — Bidirectional decomposition</b> | Runtime Matrices/Directional Panel/bidirectional_decomposition.py, directional_panels_v1_0.json                         | BUILT (wired 2026-06-27)                                             | composition-independent directional composite; AD detector (sealed VAL-051 Rule A, 7-CpG immune panel); gate composite > 0.40; AIBL-trained, does not transfer to GIFT                                                                        | NO                                                         |
| <b>Stage 4.6 — Patient CMB</b>                 | cpg_patient_cmb.py, Runtime Matrices/cpg_healpix_mapping/*                                                              | BUILT (2026-06-29)                                                   | per-class departure $z=(\beta-\mu)/\text{sd\_class}$ on a HEALPix sky; absent-tissue panels self-masked                                                                                                                                       | NO                                                         |
| <b>Stage 5 — Second chain</b>                  | stage_5_second_chain.py, Mahalanobis_healthy_reference/*,<br>Literature_anchors_Report_building/literature_anchors.json | BUILT                                                                | fires only on a flag; Mahalanobis adjudicator Option A (age-matched class gauge, presence gate $\geq 3\%$ AND outside [0.95,1.04]); RUN-everything residual matched-filter sweep (breast + immune-alarm maps; AD removed); literature anchors | NO — only the 3% floor is mentioned (RECON D2)             |
| <b>Stage 6 — Cellular age</b>                  | iam_cellular_age_scoring.py (in CPG_TRIAL_CODE; NOT in repo engine)                                                     | BUILT 2026-06-30, calibration pending                                | cellular age from class A vs age_reference_matrix                                                                                                                                                                                             | NO — and the file is not in the repository (RECON F2)      |
| <b>Stage 7 — Tier</b>                          | Runtime Matrices/Tier_breakpoints/tier_breakpoints.json                                                                 | BUILT (edge 1.04->1.01, 2026-07-03)                                  | continuous A -> customer tiers                                                                                                                                                                                                                | <b>YES — §3.1, RECON T1-T3</b>                             |

| Stage                             | Files at HEAD                                                                                                                                                                 | Status       | What it does (from source)                                                                                                                                                                                                                                                                                                                         | In Issue 003?                                                                                                                                 |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Stage 8 — Disease matching</b> | Disease Matrix/disease_cell_signature_matrix_v1_13.csv (81 rows, 49 VAL-anchored), disease_origin_cells.json, iamatlas_115_to_matrix_v0_2_mapping.json, Collinearity_Groups/* | BUILT        | Route B directional concordance (weighted matcher over SIGNAL cells  d >=0.20; STRONG needs dc>=0.70, coverage>=0.40, >=3 cells); Mode 2 cell-of-origin presence; Mode 3 systemic-stress wellness read (never a disease call); specificity gate (NLR axis = NON_SPECIFIC_GENERIC). Patient-match loop verified on 381 breast + 142 colon (VAL-093) | NO                                                                                                                                            |
| <b>Stage 9 — Report</b>           | cpg_report_builder.py, cpg_report_builder_v2.py, build_dashboard_v1.py, cpg_gauge.py (Appendix A1 gauge), Crown Jewel and Patient Strawman/*                                  | BUILT        | clinician report; patient straw man on the eight-class grid; crown-jewel reference wall; dashboard                                                                                                                                                                                                                                                 | NO                                                                                                                                            |
| <b>Orchestration</b>              | cpg_conductor.py (2026-07, replaces walther_clinical.py), walther_clinical.py (still the wired chain), run_batch.py, preflight.py, bootstrap.sh                               | BOTH present | conductor = Stage A pure functions; walther_clinical = the full wired chain the report builders still call                                                                                                                                                                                                                                         | PARTIAL — conductor's presence rule only. walther_clinical.py origin-gate fail-open (bare except disables specificity rule) NOT in this issue |
| <b>Test data</b>                  | TEST_DATA/TEST_DATA_MANIFEST.md, harness/*, N7                                                                                                                                | present      | 11 public IDATs with expected outputs; synthetic harness; N7 chain-integrity                                                                                                                                                                                                                                                                       | <b>YES — §2.4, PROC-DECON-01</b>                                                                                                              |

## Findings from the 2026-09 review that are recorded here only as reconciliation rows, not yet as sections

The origin-gate fail-open in walther\_clinical.py (F3); two stale strings describing the retired gauge, one on the patient report (F4); the gauge's missing sign (F5); two inequivalent A-score definitions (F6); the immune H\_min revision history (F7); the seminoma value carried as both 0.67 and 0.755 in different files; the breast pre-diagnostic result (n = 47 cases / 601 controls across GSE51032 + GSE51057; matched-filter  $p = +0.058$ , CI [+0.001, +0.114]) and its unrun plate-position null; the colorectal cohort test CPG-NEW-001 in full (its P1/P4 failures are in §10, its method and N-random/N-comp results are not here); the two-observables test on GSE48684/GSE139404; and the disease-signature matrix v1.13 with its 81 rows. Each is in the project record (CPG\_first\_read.md, CPG\_LEDGER.md, CPG\_breast\_CRC\_learned.md, OUTCOME\_CPG-NEW-001.md) and none has been re-verified for this issue.

## SECTION 12

# FOR THE CLINICIAN - THE INSTRUMENT IN PLAIN TERMS

Two of the source documents (The Cellular Margin; What Is Astro-Genetics) were written for readers who are not physicists. This section carries their content, corrected where the errata in s10 apply, so a physician reading only this section understands what a departure is and is not.

### MARGIN

A cell spends one packet of ATP energy (about 54,000 J/mol) every time it does a unit of ordering work. Thermal noise at 37 C is worth  $RT = 8.314 \times 310.15$  J/mol. The ratio - how many ATP packets a cell has per packet of noise - is about 21. That is the cellular margin: healthy human cells run roughly twenty-one times above the bare thermal floor. It is a constant of the chemistry, the same for every patient; what varies between patients is how much of that margin the cell is actually using, which is what the A-score reads.

### FLOOR

The floor is not a statistical cutoff. It is the minimum disorder a cell of a given type must carry to stay that type - the Landauer cost,  $k_B T \ln 2$  per irreversible write, summed over the sites the cell is committed to. Eight cell architectures, eight floors, calculated once and frozen. A patient is read against the floor, not against other patients.

### READING

$A = 1.0$  is the commitment line. Healthy cells sit on an age-dependent band near it (immune  $\sim 0.91$  at age 4 rising to  $\sim 1.00$  at 95). Frank cancer cells cluster at  $A \sim 1.28$ - $1.32$ ; established colon cancer reads  $\sim 1.15$ . Below the band is also a reading - a cell that has locked down (senescence, or the stem-cell reversion seen in seminoma).

### YOUR SAMPLE

If you send one blood draw to a laboratory and receive one array, the chain will tell you what your blood is made of and where it sits on the identity loci - but it will not report an absolute number against the floor until it knows that laboratory's zero. The zero is a property of the laboratory's instrument, measured once on 40 healthy samples, exactly as a clinical laboratory establishes a reference interval for cholesterol. If the laboratory's healthy samples are already public, the zero is already known; if not, it is one plate of work, once. Two draws a year apart from the same laboratory give you your own trajectory with no zero needed at all, because the constant cancels. The report will say which of these applies; it will not print a number that means nothing.

### SUBSTRATE

The tube decides what can be read. Whole blood carries the immune architecture and nothing else - a healthy person has no epithelial DNA in it, and the instrument correctly says so. Plasma carries what tissues shed. Tissue reads everything present, mixed. Nothing in this document is a diagnosis; a departure is a flag for the clinician to place in context.

### LIFE

One way to say what the instrument measures: life is the continuous purchase of organised pattern against the second law, paid for in metabolic work. The A-score is the receipt.

### ERRATA

Two earlier clinician-facing drafts stated the floor as  $kT$  (it is  $kT \ln 2$ ) and presented the Mahaffey number as a per-patient reading (it is a constant, 20.94; the per-patient quantity is  $A$ ). Both are corrected here.

The stellar analogy the author uses: white dwarfs and neutron stars sit below a mass limit (Chandrasekhar  $1.4 M_{\text{sun}}$ , TOV  $\sim 2.3 M_{\text{sun}}$ ) set by physics, not by a survey of stars; the Sun's eventual core at  $0.54 M_{\text{sun}}$  reads 0.38 of the limit, Procyon B 0.42, PSR J0740+6620 0.90. The cellular floor is the same kind of number: a limit from the physics, against which each object is read individually.

# APPENDIX V - VALIDATION INDEX: EVERY VAL IN THE REPOSITORY, WITH ITS PATH

## THE COMPLETE VALIDATION HISTORY, APRIL-SEPTEMBER 2026 (PROC-HISTORY-01)

| series                           | when                       | count                                                                                                                                     | executed                                             | what it was                                                                                                                                                                                   |
|----------------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| G-series                         | April 2026, before VAL-001 | G-002 (8 methylation floors, 17 chains), G-003b (32 substrate floors), bootstrap of the 32; G-008, E_A,bio, n_bio                         | 3 sealed                                             | the H_min calibration; code restored to Hmin_Calibration/; the methylation bootstrap was first run 2026-09-20 (8/8 in CI)                                                                     |
| VAL-001 -> VAL-128 (pre-Atlas)   | April-May 2026             | 119 identifiers in six families                                                                                                           | 107 executed; 12 not run / gated / excluded / voided | methylation 001-013; five substrates 014-033; drift cascade 037-046 (35/39 predictions); EDEAR disease cards 047-128 across 12 cards, SHA-sealed from VAL-050                                 |
| T1 -> T15 (VAL-049)              | April 2026                 | 15 cohorts, 6 populations                                                                                                                 | 12 executed; T4/T6/T7 dbGaP-gated                    | frozen panel + frozen H_min transferred across US/AU/UY/UK/PL/CN-SG - the first cross-population test                                                                                         |
| CPG-VAL-001 -> 022 (post-Atlas)  | 29 May - 7 June 2026       | 22 slots                                                                                                                                  | 21 executed; 021 deferred                            | breast 001-007 (Mahalanobis d +1.88/+2.10; 2 RESTATED), AD 008-014 (AD up / PSP-CBD down / FTD between), immune-aging 015-020 & 022; L9 nulls N1-N8 per VAL; PREREGs retrospective, so marked |
| Mahalanobis HC hull v0_1 -> v0_5 | 6 June 2026                | 5 versions, 8 cohorts                                                                                                                     | n_HC 601 -> 2,523                                    | four populations incl. Han Chinese (GSE141682, n=42, first Asian); fixed d>=2.0 shown invalid in 112-D, replaced by percentile-of-HC; anchor d fell honestly as the hull broadened            |
| L9 N7 chain integrity            | 5 June 2026                | 3 synthetic cohorts x 250                                                                                                                 | R1 MAE 0.0076-0.0093                                 | synthetic truth through Walther -> A-scoring -> Mahalanobis; its September rerun (PROC-N7-01) found the gauge on the wrong loci                                                               |
| September 2026 procedures        | 18-21 September            | PROC-CAL/DECON/ANCHOR/FORMULA/N7/NILC/SEP/CHAIN/STAGE0/WB-IMMUNE/HMIN-BOOT/PANEL-01..03; PHASE 1/1c; band_v2; LAB-ZERO-01/02; CPG-NEW-001 | all sealed before data                               | the rebuild from scratch: raw IDATs, absolute reading, the scale map, the age curve, the lab zero - this document                                                                             |

What none of the April-June validations could see, and why this document exists: every one was a within-pipeline comparison or a reading against a control centroid. The pipeline-scale offset (predicted in the April caveats tab, measured in September) and the healthy age curve cancel in any difference. They were found by reading single arrays absolutely against the floor. The history is evidence of six months of consistent, sealed testing; it is not a set of results this document re-asserts.

Rebuilt 2026-09-21 from the record, not the tree (PROC-HISTORY-01): 175 rows with unique keys by series - 3 G-series calibrations, 119 pre-Atlas VAL identifiers (107 executed), the 15-cohort T-series of VAL-049, 22 post-Atlas CPG-VALs (21 executed), the five-version Mahalanobis hull lineage, L9 N7, and the September procedures. The 2026-09-19 index had counted 103: it keyed on the bare number (VAL-001 and CPG-VAL-001 collided) and counted only identifiers with a folder at HEAD. Verdicts are recorded from each OUTCOME, not re-verified.

175 rows. Path prefix BP/ = Biological\_Physics/; an empty path means the record lives in the RETIRED evidence report / inventory and the Zenodo deposit, not in a repository folder. Verdicts are recorded from each OUTCOME, not re-verified.

| series | id    | title                                          | date    | result / status                                                                                         | path (repo)                     |
|--------|-------|------------------------------------------------|---------|---------------------------------------------------------------------------------------------------------|---------------------------------|
| G      | G-002 | H_min for 8 architecture classes (methylation) | 2026-04 | 17 chains · R-hat < 1.001 · 800,000 samples · 8 class floors converged   CONVERGED — values proprietary | BP/MethylPhys/hmin_calibration/ |

| series | id                                   | title                                                        | date    | result / status                                                                                                            | path (repo)                         |
|--------|--------------------------------------|--------------------------------------------------------------|---------|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| G      | G-003b                               | H_min for 4 non-methylation substrates × 8 classes           | 2026-04 | 5 chains × 32 walkers × 5,500 steps<br>· R-hat max 1.0009 · 800,000 sampl<br>  CONVERGED — 32 H_min values<br>(8 classes × | BP/MethylPhys/hmin_calibrat<br>ion/ |
| G      | G-003b bootst<br>rap cross-chec<br>k | Non-parametric validation of all 32 H_min values             | 2026-04 | 10,000 bootstrap resamples × 32 =<br>320,000 total · MCMC vs bootstrap<br>ag   CONFIRMED — independent<br>method agreement | BP/MethylPhys/hmin_calibrat<br>ion/ |
| VAL    | VAL-001                              | Cancer signal, 6 types                                       | 2026-04 | 6/6 confirmed   PASS                                                                                                       |                                     |
| VAL    | VAL-002                              | Bulk blood null test (Health ABC, n=20)                      | 2026-04 | Null as predicted; class-stratified<br>best d=0.303 (secretory) p=0.68  <br>PASS (predicted null) — confirms<br>bulk-bl    |                                     |
| VAL    | VAL-003                              | Cancer field effect, 28 types, 4,092 matched pairs           | 2026-04 | 28/28; p=1.32e-15; +20.2%<br>adjacent-normal elevation   PASS                                                              |                                     |
| VAL    | VAL-004                              | OSK rejuvenation (Yamanaka factors), 7/7 predictions         | 2026-04 | 63.8% aging ΔA reversed (RGC);<br>84.8% (SH-SY5Y)   PASS                                                                   |                                     |
| VAL    | VAL-005                              | Longitudinal entropy trajectory (n=17 pilot)                 | 2026-04 | directional signal below threshold<br>(best d=-0.303 p=0.68)  <br>UNDERPOWERED — awaits larger<br>cohort                   |                                     |
| VAL    | VAL-006                              | Aging trajectory (n=656); normal aging does NOT reach A=1.05 | 2026-04 | r=0.9999 p=6.1e-12; A=1.05<br>extrapolates to ~1,075 yr   PASS                                                             |                                     |
| VAL    | VAL-007                              | Tissue-specific cfDNA signal, 9/9 P1                         | 2026-04 | mean ΔA=+0.177; 104,297×<br>bulk-blood improvement   PASS                                                                  |                                     |
| VAL    | VAL-008                              | Specimen matrix, 19 cancer types                             | 2026-04 | 19/19 FLOOR BREACH; mean \  <br>=0.167 (SARC 0.132 → LGG 0.301)                                                            |                                     |
| VAL    | VAL-009                              | Pre-cancer window, cervical (WID-CIN, n=2,254)               | 2026-04 | 3/5 strict; A=1.015 CIN2, A=1.100<br>invasive   PARTIAL PASS (3/5)                                                         |                                     |
| VAL    | VAL-010                              | HCC combined score S = fraction × ΔA (novel)                 | 2026-04 | Cirrhosis S=0.072 vs early-HCC<br>S=0.583; 8.03× separation   PASS<br>(AFP cannot discriminate these)                      |                                     |
| VAL    | VAL-011                              | Pre-cancer window, endometrial (n=306)                       | 2026-04 | Monotonic progression; 1/4 strict  <br>PARTIAL — tissue-dependent<br>threshold                                             |                                     |
| VAL    | VAL-012                              | D+Q senolytic (global-mean proxy)                            | 2026-04 | GAPE ΔA=-0.00079 (decrease) vs<br>Hannum/Horvath/PhenoAge all<br>increase   PASS (direction) — only<br>GAPE moves corre    |                                     |
| VAL    | VAL-013                              | Cross-species: canine, 3/3                                   | 2026-04 | H_min diff=0.004 across 70M yr;<br>r(dog_age,A)=0.9273;<br>osteosarcoma ΔA=+   PASS                                        |                                     |
| VAL    | VAL-014                              | MESA theory — why combining substrates works                 | 2026-04 | inter-substrate r=0.54;<br>d_combined/d_single=1.15× (vs<br>2.0× if independ   PASS                                        |                                     |
| VAL    | VAL-015                              | Four Mahaffey values derivation                              | 2026-04 | all four G-003b MCMC confirmed;<br>R-hat<1.001; 42s runtime   PASS<br>(MCMC)                                               |                                     |

| series | id      | title                                           | date    | result / status                                                                                                         | path (repo) |
|--------|---------|-------------------------------------------------|---------|-------------------------------------------------------------------------------------------------------------------------|-------------|
| VAL    | VAL-016 | Nucleosome occupancy — breast cancer (n=139)    | 2026-04 | $\Delta A_{\text{nucl}}$ FLOOR BREACH (report +0.265 / README +0.55)   PASS [report/README $\Delta A$ differ]           |             |
| VAL    | VAL-017 | Fuzziness — prostate cancer grading (n=26 PDX)  | 2026-04 | $\Delta A_{\text{fuzz}}$ +0.311 (README +0.32); monotonic ARPC→NEPC   PASS                                              |             |
| VAL    | VAL-018 | WPS — 15 tissue types                           | 2026-04 | 15/15; $\Delta A_{\text{WPS}}$ +0.501 (README +0.53); 8 yr before MESA   PASS                                           |             |
| VAL    | VAL-019 | Fragment size — 7 cancer types (n=208)          | 2026-04 | 7/7; $\Delta A_{\text{frag}}$ +0.373 (README +0.37); AUC=0.940   PASS                                                   |             |
| VAL    | VAL-020 | Five-substrate convergence                      | 2026-04 | 5/5 direction; $r_{\text{inter}}$ =0.54; ceiling AUC=1.000   PASS                                                       |             |
| VAL    | VAL-021 | Nucleosome occupancy field effect (22 types)    | 2026-04 | 22/22; p=3.6e-14; mean $\Delta A$ field +0.102 (README +0.218); TGCT inversion   PASS [report/README $\Delta A$ differ] |             |
| VAL    | VAL-022 | Fuzziness field effect (22 types)               | 2026-04 | 22/22; p=6.9e-12; mean $\Delta A$ +0.081 (README +0.084); TGCT inversion   PASS                                         |             |
| VAL    | VAL-023 | WPS field effect (22 types)                     | 2026-04 | 22/22; p=9.1e-12; mean $\Delta A$ +0.160 (README +0.174); TGCT inversion   PASS                                         |             |
| VAL    | VAL-024 | Fragment size field effect (22 types)           | 2026-04 | 22/22; p=9.8e-11; mean $\Delta A$ +0.099 (README +0.102); TGCT inversion   PASS                                         |             |
| VAL    | VAL-025 | Nucleosome occupancy aging (human + 104 canine) | 2026-04 | r=0.9998 human, r=0.986 canine; slope 24.9× (README 53.6×) methylation   MODELED PREDICTION (literature-typed age       |             |
| VAL    | VAL-026 | Fuzziness aging (human + canine)                | 2026-04 | r=0.9995 human, r=0.982 canine; slope 20.2× (README 21.0×)   MODELED PREDICTION (literature-typed age                   |             |
| VAL    | VAL-027 | WPS aging (human + canine)                      | 2026-04 | r=0.9990 human, r=0.983 canine; slope 37.7× (README 40.9×)   MODELED PREDICTION (literature-typed age                   |             |
| VAL    | VAL-028 | Fragment size aging (human + canine)            | 2026-04 | r=0.9962 human, r=0.993 canine; slope 24.0× (README 24.5×)   MODELED PREDICTION (literature-typed age                   |             |
| VAL    | VAL-029 | Nucleosome occupancy — tissue-specific cfDNA    | 2026-04 | FLOOR BREACH; AUC=0.89 (Griffin ER); bulk plasma buried   PASS                                                          |             |
| VAL    | VAL-030 | Fuzziness pre-cancer window                     | 2026-04 | Monotonic dysplasia gradient; A=1.01-1.05 zone   PASS                                                                   |             |
| VAL    | VAL-031 | WPS pre-cancer + field effect                   | 2026-04 | Adjacent-normal WPS depletion (8 yr pre-MESA)   PASS                                                                    |             |
| VAL    | VAL-032 | Fragment size early detection                   | 2026-04 | Pre-dx signal 2 yr before dx; Stage I→IV monotonic   PASS                                                               |             |

| series | id      | title                                                                    | date       | result / status                                                                                        | path (repo)                                                |
|--------|---------|--------------------------------------------------------------------------|------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| VAL    | VAL-033 | Complete 5×6 evidence matrix                                             | 2026-04    | all 5 substrates MCMC-confirmed; 30/30 cells confirmed   PASS                                          |                                                            |
| VAL    | VAL-037 | Cross-class field effect (24 TCGA types, n=1,109 STN)                    | 2026-04    | 3/4; mean $\Delta A_{\text{field}}$ +0.036; 22.9% of tumor signal; 24/24 directionally   PASS (3/4)    | BP/Record/VAL_PreAtlas/VAL_037_results.json                |
| VAL    | VAL-038 | Plasma cfDNA pan-cancer correlation (Zeng 2026, n=1,294, 14 types)       | 2026-04    | 1/3; HONEST NEGATIVE; Spearman $\rho$ =-0.02; confirms VAL-002   HONEST NEGATIVE (confirms prior null) | BP/Record/VAL_PreAtlas/VAL_038_results.json                |
| VAL    | VAL-039 | Spatial field-effect gradient (6 distance-annotated cancers)             | 2026-04    | 4/4; 6/6 monotonic T→N→F→H; far ≥5-10cm still +0.025   PASS                                            | BP/Record/VAL_PreAtlas/VAL_039_spatial_field_gradient.py   |
| VAL    | VAL-040 | Alzheimer's multi-class peripheral drift (7 class combos)                | 2026-04    | 4/4; 4 classes elevated (terminal/immune/secretory/stromal); 7/7 sever   PASS                          | BP/Record/VAL_PreAtlas/VAL_040_results.json                |
| VAL    | VAL-041 | Tissue-of-origin deconvolution localization (10 types)                   | 2026-04    | 4/4; 10/10 top-1 correct; mean max $\Delta A$ +0.174   PASS — Stage 2 anchor                           | BP/Record/VAL_PreAtlas/VAL_041_results.json                |
| VAL    | VAL-042 | Monotonic pre-cancer progression (5 systems)                             | 2026-04    | 4/4; 5/5 monotonic; 4/5 FLOOR BREACH; MARGINAL in 5/5   PASS                                           | BP/Record/VAL_PreAtlas/VAL_042_pre_cancer_progression.py   |
| VAL    | VAL-043 | Cross-species replication (5 canine cancers, n=104 Labradors)            | 2026-04    | 4/4; mean cross-species diff 0.010; canine aging $r$ =0.9995   PASS                                    | BP/Record/VAL_PreAtlas/VAL_043_cross_species.py            |
| VAL    | VAL-044 | Post-treatment reserve depletion (5 trials)                              | 2026-04    | 4/4; 5/5 responder vs non-responder separable; CR → A=1.00   PASS                                      | BP/Record/VAL_PreAtlas/VAL_044_treatment_trajectory.py     |
| VAL    | VAL-045 | Inversion specificity (seminoma vs 5 TGCT histologies)                   | 2026-04    | 2/4; seminoma inversion confirmed (A=0.755); 2.1× divergence distingui   PARTIAL (2/4)                 | BP/Record/VAL_PreAtlas/VAL_045_inversion_specificity.py    |
| VAL    | VAL-046 | Systemic multi-class pre-dx signature (7 cohort-cancer combos) — capston | 2026-04    | 4/4; 9/9 endpoints $\Delta A$ ≥0.008; 3 classes; detectable 2-5 yr pre-dx; mean   PASS — capstone      | BP/Record/VAL_PreAtlas/VAL_046_pre_diagnostic_signature.py |
| VAL    | VAL-047 | Blood pre-dx per-patient (Phase 9 + Phase 12 + Tightening v2) [breast-ep | 2026-04-21 | Breast secretory $d$ =+1.85 (Ph9), CRC inverted; pooled +0.45→+0.71, +1.3   O1 PASS (paradigm)         | BP/Record/VAL_PreAtlas/VAL_047_option3.py                  |
| VAL    | VAL-048 | Framework-derived 650-CpG cycling panel on CRC pre-dx [crc-epic]         | 2026-04-20 | Phase 1 too-strict (clean fail, logged); Phase 2 empirical threshold →   Null (logged)                 |                                                            |
| VAL    | VAL-049 | Cross-population series T1-T15 (see T-series tab) [cross-population]     | 2026-04-21 | 5/5 informative pre-dx cohorts directionally positive (binomial $p$ =0.03   Directional PASS           |                                                            |
| VAL    | VAL-050 | AIBL AD pooled-entropy A-score [ad-immune]                               | 2026-04-23 | $d$ =+0.077 $p$ =0.32 AUC=0.51; 7/18 CpGs FDR<0.05 bidirectional   O3 NULL                             |                                                            |
| VAL    | VAL-051 | AD directional 7-CpG panel (Rule A), sealed holdout recovery [ad-immune] | 2026-04-23 | $d$ =+0.624 $p$ =0.0013 AUC=0.677; Male +0.51, Female +0.71   O1 FULL RECOVERY                         |                                                            |

| series | id      | title                                                                    | date        | result / status                                                                                        | path (repo)                                            |
|--------|---------|--------------------------------------------------------------------------|-------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| VAL    | VAL-052 | AddNeuroMed cross-platform AD replication [ad-immune]                    | 2026-04-23  | raw d=+0.33; age-adj d=+0.12; age R <sup>2</sup> =26%   MIXED (O1 + O3-borderline)                     |                                                        |
| VAL    | VAL-053 | Sex-specific vs unified AD panel [ad-immune]                             | 2026-04-23  | Panel-F worse; Panel-M under-selects; Jaccard 0.10   O4 unified wins                                   |                                                        |
| VAL    | VAL-054 | Age-confounding bound on AIBL holdout [ad-immune]                        | 2026-04-23  | 54b: p=0.003 signal exceeds within-HC variance; 54a flagged non-test   STRONG (54b)                    |                                                        |
| VAL    | VAL-055 | AD extension (ADNI / direct-access AIBL age metadata) [ad-immune]        | (extension) | strengthens with ADNI replication or direct AIBL age metadata   Planned/extension                      |                                                        |
| VAL    | VAL-056 | Lung multi-anchor (Kadota field + Moss plasma + TCGA-LUAD/LUSC) [lung-ep | 2026-04-24  | field effect + top-1/top-2 ratio 60.87×; FLOOR_BREACH   PASS (multi-anchor)                            | BP/Record/VAL_PreAtlas/VAL_056_lung_epic_validation.py |
| VAL    | VAL-057 | AD-directional specificity vs FTD and PSP/CBD (5 analyses) [ad-immune]   | 2026-04-24  | pooled null d=+0.013; male AD d=+0.415; PSP/CBD 5/7 directions preserv   External null + sex recovery  | BP/Record/VAL_PreAtlas/VAL_057_PREREG.md               |
| VAL    | VAL-058 | First own tumor-vs-adjacent-normal tissue arm (secretory) [prostate-epic | 2026-04-24  | paired d=+0.497 [+0.314,+0.681] p=1e-07   O1 PASS                                                      | BP/Record/VAL_PreAtlas/VAL_058_PREREG_AMENDMENT.md     |
| VAL    | VAL-059 | Cross-cohort HCC: ccfDNA vs whole-blood [hcc-epic]                       | 2026-04-24  | ccfDNA d=+0.634 p=0.002 monotonic; whole-blood d=-0.156 NULL   PASS (ccfDNA only)                      | BP/Record/VAL_PreAtlas/VAL_059_PREREG_AMENDMENT.md     |
| VAL    | VAL-060 | Tissue arm TCGA-BRCA (secretory) [breast-epic]                           | 2026-04-24  | paired d=+0.675 [+0.448,+0.902] p=4.4e-09   O1 PASS                                                    | BP/Record/VAL_PreAtlas/VAL_060_SEAL.txt                |
| VAL    | VAL-061 | CRC tumor TIL immune-compartment (Xu-538 in tumor) [crc-epic]            | 2026-04-24  | paired d=+1.066 [+0.585,+1.547] p<1e-05   O2 PASS (strong)                                             | BP/Record/VAL_PreAtlas/VAL_061_outcome.md              |
| VAL    | VAL-062 | CRC tumor cycling-class rescore [crc-epic]                               | 2026-04-24  | paired d=+0.724 [+0.292,+1.156] p=2.2e-04   O1 PASS                                                    | BP/Record/VAL_PreAtlas/VAL_062_results.json            |
| VAL    | VAL-063 | Lung tissue arm (cycling) [lung-epic]                                    | 2026-04-24  | paired d=+1.020 [+0.571,+1.469] p=3.9e-08; ever-smoker +1.283; non-smo   O1 PASS (strong)              | BP/Record/VAL_PreAtlas/VAL_063_prereg.md               |
| VAL    | VAL-064 | HCC tissue arm, etiology-stratified (secretory) [hcc-epic]               | 2026-04-24  | pooled d=+0.498 p=7.4e-04; non-viral +0.664; viral +0.023 NULL   PASS (non-viral)                      | BP/Record/VAL_PreAtlas/VAL_064_stratified.json         |
| VAL    | VAL-065 | Urine arm specimen comparison [prostate-epic]                            | 2026-04-25  | urine vs benign paired d=-2.39 (wrong direction)   O5 UNEXPECTED                                       | BP/Record/VAL_PreAtlas/VAL_065_prereg.md               |
| VAL    | VAL-066 | PDAC tissue arm cohort #1 [pancreatic-epic]                              | 2026-04-25  | d=+1.182 p=8.2e-03; 46.9% positive-direction (bidirectional)   O5 UNEXPECTED                           | BP/Record/VAL_PreAtlas/VAL_066_PREREG_SEAL.txt         |
| VAL    | VAL-067 | PDAC large unpaired tissue case-control [pancreatic-epic]                | 2026-04-25  | d=+0.249 Welch p=0.22; 50.4% positive — cleanest bidirectional evidenc   O3 TISSUE NULL (large cohort) | BP/Record/VAL_PreAtlas/VAL_067_PREREG_SEAL.txt         |
| VAL    | VAL-068 | PDAC multi-substrate [pancreatic-epic]                                   | 2026-04-25  | paired d=-0.310; juice d=-0.720; 52.9% positive; PH64 outlier   O3 TUMOR NULL                          | BP/Record/VAL_PreAtlas/VAL_068_prereg.md               |

| series | id      | title                                                                    | date        | result / status                                                                              | path (repo)                                        |
|--------|---------|--------------------------------------------------------------------------|-------------|----------------------------------------------------------------------------------------------|----------------------------------------------------|
| VAL    | VAL-069 | PDAC directional Xu-538 fallback (324-CpG, z-scored) [pancreatic-epic]   | 2026-04-25  | TCGA d=+1.511 p=6.4e-05 (all 7 positive); GSE74071 d=+0.222 partial (P   O2 PARTIAL RECOVERY | BP/Record/VAL_PreAtlas/VAL-069_PREREG_SEAL.txt     |
| VAL    | VAL-070 | Not run — landscape error caught at runtime [cervical-epic]              | 2026-04-25  | —   NOT RUN (note)                                                                           | BP/Record/VAL_PreAtlas/VAL-070_NOT_RUN_NOTE.md     |
| VAL    | VAL-071 | Cervical landscape survey [cervical-epic]                                | 2026-04-25  | breadth mapped per CCL-029   Landscape                                                       | BP/Record/VAL_PreAtlas/VAL-071_landscape_survey.md |
| VAL    | VAL-072 | Cervical tissue exploratory TCGA-CESC [cervical-epic]                    | 2026-04-25  | paired d=+1.26 CI straddles zero; bidirectional signal   O3 TISSUE NULL (exploratory n=3)    | BP/Record/VAL_PreAtlas/VAL-072_outcome.md          |
| VAL    | VAL-073 | Cervical tissue progression anchor [cervical-epic]                       | 2026-04-25  | Normal vs CIN3 d=+0.7253 p=0.004; monotonic Normal<CIN3<SCC   O1 TISSUE ANCHOR               | BP/Record/VAL_PreAtlas/VAL-073_prereg.md           |
| VAL    | VAL-074 | Cervical tissue replication (HPV-neg normals) [cervical-epic]            | 2026-04-25  | Normal vs CIN3 d=-0.61; not monotonic   O5 NEGATIVE DIRECTION                                | BP/Record/VAL_PreAtlas/VAL-074_results.json        |
| VAL    | VAL-075 | GSE38266 attempted [cervical-epic]                                       | 2026-04-25  | cohort is HNSCC (head/neck), NOT cervical   EXCLUDED (runtime catch)                         |                                                    |
| VAL    | VAL-076 | LBC primary pathway #1 [cervical-epic]                                   | 2026-04-25  | Healthy vs all-lesion d=-0.114 flat   O6 UNEXPECTED (panel transferability)                  | BP/Record/VAL_PreAtlas/VAL-076_PREREG_SEAL.txt     |
| VAL    | VAL-077 | LBC primary pathway #2 (largest LBC) [cervical-epic]                     | 2026-04-25  | d=-0.029 flat   O6 UNEXPECTED → DATA INTEGRITY                                               | BP/Record/VAL_PreAtlas/VAL-077_prereg.md           |
| VAL    | VAL-081 | Cancer-only SCC at large n [cervical-epic]                               | 2026-04-25  | tumors d=-0.43 vs VAL-073 normals; only 6.7% above p95   O5 NEGATIVE DIRECTION               | BP/Record/VAL_PreAtlas/VAL-081_PREREG_SEAL.txt     |
| VAL    | VAL-082 | AML myeloid arm at blood level [heme-epic]                               | 2026-04-25  | $\Delta A = +0.1039$ ; d=+3.71 [+3.23,+4.20] p=0; 98.5% above HC p95   O1 PASS               | BP/Record/VAL_PreAtlas/VAL-082_outcome.md          |
| VAL    | VAL-083 | EnviroGenomarkers pre-dx CLL (n=347; 28 future-CLL 2-15.7 yr pre-dx) [he | (v0.2 queue | signal exists in published analysis; data biobank-gated   Priority v0.2 (gated)              |                                                    |
| VAL    | VAL-084 | MARLIN myeloid cross-cohort replication [heme-epic]                      | (v0.2 queue | reference identified   Queued v0.2                                                           |                                                    |
| VAL    | VAL-085 | CHIP→AML serial trajectory (G-2026-P010) [heme-epic]                     | (v0.2 queue | —   Queued v0.2                                                                              |                                                    |
| VAL    | VAL-086 | ICI immunotherapy response (G-2026-P011) [heme-epic]                     | (v0.2 queue | —   Queued v0.2                                                                              |                                                    |
| VAL    | VAL-087 | Heme v0.2 queue placeholder [heme-epic]                                  | (v0.2 queue | —   Queued v0.2                                                                              |                                                    |
| VAL    | VAL-088 | Glioma Stage 1 immune A-score in blood [glioma-epic]                     | 2026-04-25  | d=+0.91 [+0.61,+1.22]; pre-surg naïve +0.94; LGG>GBM   O1 PASS (revised from O5 inverted)    | BP/Record/VAL_PreAtlas/VAL-088_distributions.png   |
| VAL    | VAL-089 | Glioma tumor tissue arm [glioma-epic]                                    | 2026-04-25  | GBM primary d=+0.24 (wide CI); recurrent +1.17; spheres -1.81   O2 PARTIAL (variance high)   | BP/Record/VAL_PreAtlas/VAL-089_distributions.png   |

| series | id      | title                                                                     | date       | result / status                                                                                       | path (repo)                                               |
|--------|---------|---------------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| VAL    | VAL-090 | Cortical-neuron cfDNA in glioma plasma (Loyfer array atlas) [glioma-epic] | 2026-04-25 | glioma 1.092% vs HC 0.276%; d=+1.96 [+1.62,+2.31]; pre-surg +1.97   O1 PASS                           | BP/Record/VAL_PreAtlas/VAL-090_results.json               |
| VAL    | VAL-091 | AD Stage 2 neuro deconvolution (layered atlas) [ad-immune]                | 2026-04-26 | AIBL d=-0.026; AddNeuroMed d=-0.083; GIFT +0.96 outlier; PSP/CBD -0.51   O4 AD NEURO NULL             | BP/Record/VAL_PreAtlas/VAL-091_PREREG_SEAL.txt            |
| VAL    | VAL-092 | Stage 2 A_terminal on cortical-neuron CpGs (run-everything demo) [framew  | 2026-04-26 | AIBL AD d=-0.228; PSP d=-0.433 (replicable BELOW_NORMAL); glioma blood   O1 DRIFT DISCRIMINATOR       | BP/Record/VAL_PreAtlas/VAL-092                            |
| VAL    | VAL-093 | Stage 2 25-tile per-class A-score (Loyfer array) [breast-epic]            | 2026-04-26 | ≥3/4 secretory tiles \   d\                                                                           | BP/Record/VAL_PreAtlas/VAL-093                            |
| VAL    | VAL-094 | EpiSCORE BreastRef sub-cell-type resolution [breast-epic]                 | 2026-04-26 | 7 sub-types behave as one signal (within 0.10-0.16)   O2 DISTRIBUTED (resolution-collapse)            | BP/Record/VAL_PreAtlas/VAL-094_095_096_seal_manifest.json |
| VAL    | VAL-095 | UniLIFE 19-cell vs Salas 6-cell Stage 3 head-to-head [breast-epic]        | 2026-04-26 | aTreg >10yr d=+1.26/+0.79; aBnv 0-2yr +0.44/+0.49 (CIs exclude 0)   O1 RESOLUTION GAIN                | BP/Record/VAL_PreAtlas/VAL-095                            |
| VAL    | VAL-096 | TTD-window stratification on Loyfer 25-tile [breast-epic]                 | 2026-04-26 | breast tile +0.43/+0.49 at 0-2yr only; immune inversion near-dx   O1 (partial) + O4 amplifies near-dx | BP/Record/VAL_PreAtlas/VAL-096                            |
| VAL    | VAL-097 | Never-smoker LUAD 25-tile cross-cohort [lung-epic]                        | 2026-04-28 | Lung_cells d=-0.27 (2nd-weakest); 11/25 tiles >3 SD; 22/25 same direct   O5 BASELINE DOMINATED        | BP/Record/VAL_PreAtlas/VAL-097                            |
| VAL    | VAL-098 | Early-onset rectal subsite tissue arm [crc-epic]                          | 2026-04-28 | cycling d=+0.612 [+0.227,+1.882]; Colon tile -2.501; Rectum-NOS +0.750   O1 CYCLING RECTAL CONFIRMED  | BP/Record/VAL_PreAtlas/VAL-098                            |
| VAL    | VAL-099 | TCGA-COAD age-stratified re-analysis [crc-epic]                           | 2026-04-28 | d=+0.7241 reproduces VAL-062 byte-for-byte; under-50 ΔA=+0.0357 (descr   O1 AGE-STRATIFIED CONFIRMED  | BP/Record/VAL_PreAtlas/VAL-099                            |
| VAL    | VAL-100 | Under-50 buffy-coat polyp Stage 1 (EPIC v2.0) [crc-epic]                  | 2026-04-28 | extreme 3.9%/middle 6.8% FAILS raw-β; +15.13 anchor-SD off; d=+0.236 d   O5 DATA INTEGRITY FLAG       | BP/Record/VAL_PreAtlas/VAL-100                            |
| VAL    | VAL-101 | HCC 25-tile etiology stratification (Marcus-analog stratum) [hcc-epic]    | 2026-04-28 | extreme 26.6%/middle 9.1% trips CHK-3.1; Hepatocytes tile d=-1.521 (de   O5 DATA INTEGRITY FLAG       | BP/Record/VAL_PreAtlas/VAL-101                            |
| VAL    | VAL-102 | Re-seal attempt under platform-tuned threshold [hcc-epic]                 | 2026-04-28 | —   VOIDED before execution                                                                           | BP/Record/VAL_PreAtlas/VAL-102                            |
| VAL    | VAL-106 | TCGA HM450 sesame CHK-3.1A baseline [calibration]                         | 2026-04-28 | full-genome f_extreme ~55.87%; KIRC vs PRAD p=1.55e-05   O3 CALIBRATION DEGENERATE → anchor           | BP/Record/VAL_PreAtlas/VAL-106                            |
| VAL    | VAL-107 | CHK-3.1B threshold on 8,100-CpG marker subset [cardio-epic (calib)]       | 2026-04-28 | subset f_extreme ~63%; threshold extreme≥55%, middle≤8.5%   O2 PLATFORM DIVERGENCE                    | BP/Record/VAL_PreAtlas/VAL-107                            |
| VAL    | VAL-108 | Ischemic stroke 3-subtype (TOAST) [cardio-epic]                           | 2026-04-28 | every d<0.5; max \   d\                                                                               | BP/Record/VAL_PreAtlas/VAL-108                            |

| series | id      | title                                                                    | date       | result / status                                                                                             | path (repo)                                               |
|--------|---------|--------------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| VAL    | VAL-109 | PAH cultured pulmonary endothelial cells [cardio-epic]                   | 2026-04-28 | Vascular tile control vs hPAH d=+0.79; Stage 1 +0.65   O2 VASCULAR TILE DIFFERENTIATING                     | BP/Record/VAL_PreAtlas/VAL-109                            |
| VAL    | VAL-110 | Ascending aorta dissection / BAV+dilation [cardio-epic]                  | 2026-04-28 | Stage 1 immune normal vs BAV d=+1.08; Vascular tile NOT differentiatin   d\                                 | BP/Record/VAL_PreAtlas/VAL-110                            |
| VAL    | VAL-111 | EpiSCORE HeartRef Stage 2 atlas integration [cardio-epic]                | 2026-04-29 | all 5 cardiac tiles ~0.46-0.51 regardless of substrate; max discrimina   O3 TISSUE FLOOR DOMINATED          | BP/Record/VAL_PreAtlas/VAL-111                            |
| VAL    | VAL-112 | Run-everything multi-atlas calibration + rescoring [cardio-epic]         | 2026-04-29 | PAH convergent across 3 atlases (Caggiano heart +1.42); stroke converg   d\                                 | BP/Record/VAL_PreAtlas/VAL-112_run_everything             |
| VAL    | VAL-113 | Caggiano CelFiE TIM array-bridged calibration [cardio-epic]              | 2026-04-29 | 254-CpG / 19-tile atlas calibrated and scored   Calibration sealed                                          |                                                           |
| VAL    | VAL-114 | Wave-1 calibration on Hannum GSE40279 (n=656) [cardio-epic]              | (cardio Wa | cohort-substrate precheck baked in   Wave-1 calibration                                                     |                                                           |
| VAL    | VAL-115 | Wave-1 promotion-path calibration [cardio-epic]                          | (cardio Wa | promotion-path reference   Wave-1 calibration                                                               |                                                           |
| VAL    | VAL-116 | Wave-1 calibration close-out [cardio-epic]                               | (cardio Wa | —   Wave-1 calibration                                                                                      |                                                           |
| VAL    | VAL-117 | EpiSCORE ProstateRef Phase B calibration [prostate-epic]                 | 2026-04-30 | all CHK gates clear; LE tile mean 0.4254 sd 0.0041 q5 0.4190   O1 (calibration sealed)                      | BP/Record/VAL_PreAtlas/VAL-117_prostateref_calibrate      |
| VAL    | VAL-118 | Phase C run-everything multi-atlas rescoring [prostate-epic]             | 2026-04-30 | LE tile d=-0.767; other 5 tiles +0.48 to +1.31; Salas Mono +0.771; Xu-   O1 + O2 (LE_NEGATIVE) + O4         | BP/Record/VAL_PreAtlas/VAL-118_prostateref_phaseC         |
| VAL    | VAL-119 | EpiSCORE BladderRef Phase B calibration [bladder-epic]                   | 2026-05-01 | max within-cohort tile range 0.0694 (Epi tightest 0.0410); 4 cell type   O1 CALIBRATION SEALED              | BP/Record/VAL_PreAtlas/VAL-119_bladderref_calibrate       |
| VAL    | VAL-120 | Stage 1 Xu-538 on bladder [bladder-epic]                                 | 2026-05-01 | panel coverage 51.1% pass; diagnostic paired d=+1.8977 p=3.14e-08 (not   O4 DATA INTEGRITY (panel coverage) | BP/Record/VAL_PreAtlas/VAL-120_bladder_stage1_xu538       |
| VAL    | VAL-121 | Stage 2 multi-atlas direction check [bladder-epic]                       | 2026-05-01 | Loyfer Bladder +1.91 vs BladderRef Epi -1.46 (same pairs); all 14 Loyf   O2 DIRECTION AMBIGUOUS             | BP/Record/VAL_PreAtlas/VAL-121_bladder_stage2_multi atlas |
| VAL    | VAL-122 | Stage 3 immune fine-tune [bladder-epic]                                  | 2026-05-01 | 6/6 Salas IDOL tiles +0.49 to +1.24 (broad infiltration)   O1 IMMUNE DIFFERENTIATING                        | BP/Record/VAL_PreAtlas/VAL-122_bladder_stage3_immune      |
| VAL    | VAL-123 | BoccellatoStomachRef HM450 Phase B calibration [gastric-esophageal-epic] | 2026-05-02 | calibration sealed; 738,115 CpGs × 6 tiles   O1 BOCCELLATO CALIBRATION SEALED                               | BP/Record/VAL_PreAtlas/VAL-123_boccellato_calibrate       |
| VAL    | VAL-124 | EpiSCORE EsoRef bridged Phase B calibration [gastric-esophageal-epic]    | 2026-05-02 | cross-tile separation 0.0990 (largest of any EpiSCORE bridge); 2,464 C   O1 CALIBRATION SEALED              | BP/Record/VAL_PreAtlas/VAL-124_esoref_calibrate           |
| VAL    | VAL-125 | EpiSCORE OERef bridged Phase B calibration (cross-card arm) [gastric-eso | 2026-05-02 | 4/9 strict floors cleared; cross-tile separation 0.0407; 5,396 CpGs ×   O2 PARTIAL FLOORS                   | BP/Record/VAL_PreAtlas/VAL-125_oeref_calibrate            |

| series       | id          | title                                                                    | date       | result / status                                                                                                   | path (repo)                                                    |
|--------------|-------------|--------------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| VAL          | VAL-126     | module_1 STAD Phase C run-everything (8 atlases) [gastric-esophageal-epi | 2026-05-02 | f_extreme 0.4399 = 5.02 anchor-SD shift (tier-3); within-cohort MSI d=   O5 SUBSTRATE BASELINE T3 + O1 WITHIN-COH | BP/Record/VAL_PreAtlas/VAL-126_stad_phase_c                    |
| VAL          | VAL-127     | module_2 ESCA Phase C run-everything (8 atlases) [gastric-esophageal-epi | 2026-05-02 | ESCC +2.64 / EAC +3.70; d_ESCC-EAC=-1.06 p=1.5e-11; Barrett+ d=+4.50 v   O1 SUBTYPE DISCRIMINATION + O5 T3 + O1 B | BP/Record/VAL_PreAtlas/VAL-127_esca_phase_c                    |
| VAL          | VAL-128     | module_3 Crohn's-pathway blood (8 atlases) [gastric-esophageal-epic]     | 2026-05-02 | max \   d_CD-HC\                                                                                                  | BP/Record/VAL_PreAtlas/VAL-128_crohns_blood                    |
| T (VAL-0 49) | T1          | GSE40279 Hannum US healthy (n=656, 450K)                                 | 2026-04-21 | healthy distribution; mean A=0.493 flat across decades   PASS (calibration baseline)                              |                                                                |
| T (VAL-0 49) | T2          | GSE104942 Australian HBOC (n=191, 450K)                                  | 2026-04-21 | d=+0.291 p=0.056 [+0.03,+0.60]; WPB-only subset d=+0.664 p=0.048   PASS (weak positive)                           |                                                                |
| T (VAL-0 49) | T3          | GSE148663 Uruguayan sporadic (n=32, 450K)                                | 2026-04-21 | d=-0.373 p=0.33; post-dx design, outlier-driven   OUTLIER (design-incomparable)                                   |                                                                |
| T (VAL-0 49) | T5          | GSE89093 TwinsUK MZ (paired)                                             | 2026-04-21 | pooled d=+0.16; 0-2yr window positive   PASS                                                                      |                                                                |
| T (VAL-0 49) | T8          | NHANES 1999-2002                                                         | 2026-04-21 | HR per SD = 1.58 (Cox on published clocks)   PASS                                                                 |                                                                |
| T (VAL-0 49) | T9          | GSE283951 Polish                                                         | 2026-04-21 | d=+0.285   PASS                                                                                                   |                                                                |
| T (VAL-0 49) | T10         | GSE37965 Heyn UK (paired EpiTwin)                                        | 2026-04-21 | d=+0.177   PASS                                                                                                   |                                                                |
| T (VAL-0 49) | T11         | GSE243529 Singapore                                                      | 2026-04-21 | d=+0.120   PASS                                                                                                   |                                                                |
| T (VAL-0 49) | T12         | GSE314261 St Jude                                                        | 2026-04-21 | attempted, not interpretable   NOT INTERPRETABLE                                                                  |                                                                |
| T (VAL-0 49) | T13         | GSE51057 (secretory)                                                     | 2026-04-21 | d=+0.83 at >10yr   PASS                                                                                           |                                                                |
| T (VAL-0 49) | T14         | GSE51032 (secretory)                                                     | 2026-04-21 | d=+0.28 pre-dx; CRC divergence   PASS                                                                             |                                                                |
| T (VAL-0 49) | T15         | NHANES 1999-2002                                                         | 2026-04-21 | DunedinPoAm HR=2.14 (blinded prospective)   PASS                                                                  |                                                                |
| T (VAL-0 49) | T4          | NOWAC Norwegian pre-dx                                                   | 2026-04-21 | not deposited / dbGaP-gated   NOT RUN (gated)                                                                     |                                                                |
| T (VAL-0 49) | T6          | Sister Study extension                                                   | 2026-04-21 | dbGaP-gated   NOT RUN (gated)                                                                                     |                                                                |
| T (VAL-0 49) | T7          | Framingham                                                               | 2026-04-21 | dbGaP-gated   NOT RUN (gated)                                                                                     |                                                                |
| CPG-VAL L    | CPG-VAL-001 | Breast per celltype fanout                                               | 2026-06-03 | Basophil d = +1.58 / +1.01 (top hit, hidden in 51-cell immune class).   O1_PRIMARY_VALIDATED; SUBSTANTIVELY SEAL  | BP/Record/VAL_PostAtlas/CPG_VAL_001_Breast_per_celltype_fanout |

| series  | id          | title                  | date       | result / status                                                                                                             | path (repo)                                                |
|---------|-------------|------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| CPG-VAL | CPG-VAL-002 | Breast mahalanobis     | 2026-06-03 | $d = +1.876 / +2.097$ (very large effect, beats Xu-538 by +0.75 on GSE51   O1_PRIMARY_VALIDATED; SUBSTANTIVELY SEAL         | BP/Record/VAL_PostAtlas/CPG_VAL_002_Breast_mahalanobis     |
| CPG-VAL | CPG-VAL-003 | Breast residual map    | 2026-06-03 | 1,392 concordant CpGs across cohorts; 1,389 NEW (not in Xu-538); 5.4:1   O1_PRIMARY_VALIDATED; SUBSTANTIVELY SEAL           | BP/Record/VAL_PostAtlas/CPG_VAL_003_Breast_residual_map    |
| CPG-VAL | CPG-VAL-004 | Breast bimodality      | 2026-06-03 | Direction reversed: 1,096 GAIN bimodality vs 396 lose. 35 double-confi   O3_RESTATE_DIRECTION_REVERSED_THEN_VALID           | BP/Record/VAL_PostAtlas/CPG_VAL_004_Breast_bimodality      |
| CPG-VAL | CPG-VAL-005 | Breast PC axes         | 2026-06-03 | PC2 captures T-cell suppression axis; case-vs-HC $d = -0.67 / -0.58$ acr   O1_PRIMARY_VALIDATED; SUBSTANTIVELY SEAL         | BP/Record/VAL_PostAtlas/CPG_VAL_005_Breast_PC_axes         |
| CPG-VAL | CPG-VAL-006 | Breast chr6 mhc        | 2026-06-03 | chr6 MHC most-deviant uncorrected ( $z = +2.81$ , $p = 0.009$ ), but does NOT   O4_RESTATE_INSUFFICIENT_POWER_OR_CORRECT    | BP/Record/VAL_PostAtlas/CPG_VAL_006_Breast_chr6_mhc        |
| CPG-VAL | CPG-VAL-007 | Breast age subtraction | 2026-06-03 | Mahalanobis signal retained at $d = +0.255$ post-age-subtraction. Termin   O1_PRIMARY_VALIDATED; SUBSTANTIVELY SEAL         | BP/Record/VAL_PostAtlas/CPG_VAL_007_Breast_age_subtraction |
| CPG-VAL | CPG-VAL-008 | AD AIBL per celltype   |            | 20 Bonferroni-sig negative effects across immune class; top Eosino $d =$   PRELIMINARY — substantively complete             | BP/Record/VAL_PostAtlas/CPG_VAL_008_AD_AIBL_per_celltype   |
| CPG-VAL | CPG-VAL-009 | AD AIBL mahalanobis    |            | $d = +0.20$ (modest — confirms AD's signal is targeted not universal)   PRELIMINARY — substantively complete                | BP/Record/VAL_PostAtlas/CPG_VAL_009_AD_AIBL_mahalanobis    |
| CPG-VAL | CPG-VAL-010 | AD AddNeuroMed         | 2026-06-02 | Eosino $d = -0.46$ (replicates AIBL $d = -0.43$ ). Universal Mahalanobis at   PRELIMINARY (not yet sealed under v4 pro      | BP/Record/VAL_PostAtlas/CPG_VAL_010_AD_AddNeuroMed         |
| CPG-VAL | CPG-VAL-011 | AD age subtraction     | 2026-06-02 | Raw $d = -0.004$ (correctly null at baseline). Post-subtraction $d = -0.1$   PRELIMINARY                                    | BP/Record/VAL_PostAtlas/CPG_VAL_011_AD_age_subtraction     |
| CPG-VAL | CPG-VAL-012 | AD PC axes             | 2026-06-02 | PC1 (not PC2) is the T-cell axis in AIBL — same biology as breast PC2,   PRELIMINARY                                        | BP/Record/VAL_PostAtlas/CPG_VAL_012_AD_PC_axes             |
| CPG-VAL | CPG-VAL-013 | AD residual map        | 2026-06-02 | Top CpG cg19459094 $d = -0.493$ ; cross-cohort Spearman $\rho = 0.231$ ( $p = 1$   PRELIMINARY — candidate panel staged, no | BP/Record/VAL_PostAtlas/CPG_VAL_013_AD_residual_map        |
| CPG-VAL | CPG-VAL-014 | AD GIFT specificity    | 2026-06-02 | AD $d = +0.68$ ( $p = 0.001$ ); PSP/CBD $d = -0.38$ ( $p = 2 \times 10^{-11}$ ), BELOW-norma   PRELIMINARY                  | BP/Record/VAL_PostAtlas/CPG_VAL_014_AD_GIFT_specificity    |

| series           | id                                 | title                                                                    | date       | result / status                                                                                                   | path (repo)                                                          |
|------------------|------------------------------------|--------------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| CPG-VAL          | CPG-VAL-015                        | Immune Aging Trajectory                                                  | 2026-06-07 | PASS                                                                                                              | BP/Record/VAL_PostAtlas/CPG_VAL_015_Immune_Aging_Trajectory          |
| CPG-VAL          | CPG-VAL-016                        | Cross Disease Universal Alarm                                            | 2026-06-07 | DIRECTIONAL                                                                                                       | BP/Record/VAL_PostAtlas/CPG_VAL_016_Cross_Disease_Universal_Alarm    |
| CPG-VAL          | CPG-VAL-017                        | Inflammaging Pooled HC                                                   | 2026-06-07 | NULL                                                                                                              | BP/Record/VAL_PostAtlas/CPG_VAL_017_Inflammaging_Pooled_HC           |
| CPG-VAL          | CPG-VAL-018                        | Menarche Immune                                                          | 2026-06-07 | NULL                                                                                                              | BP/Record/VAL_PostAtlas/CPG_VAL_018_Menarche_Immune                  |
| CPG-VAL          | CPG-VAL-019                        | Immune Bidirectional                                                     | 2026-06-07 | PASS                                                                                                              | BP/Record/VAL_PostAtlas/CPG_VAL_019_Immune_Bidirectional             |
| CPG-VAL          | CPG-VAL-020                        | Immune Hannum full chain                                                 | 2026-06-06 | O_CHAIN_INTEGRITY_PASS + O_REFERENCE_CAL                                                                          | BP/Record/VAL_PostAtlas/CPG_VAL_020_Immune_Hannum_full_chain         |
| CPG-VAL          | CPG-VAL-021                        | Weight Loss DEFERRED                                                     | 2026-06-07 | DEFERRED                                                                                                          | BP/Record/VAL_PostAtlas/CPG_VAL_021_Weight_Loss_DEFERRED             |
| CPG-VAL          | CPG-VAL-022                        | Smoking Cessation Reversibility                                          | 2026-06-07 | NULL                                                                                                              | BP/Record/VAL_PostAtlas/CPG_VAL_022_Smoking_Cessation_Reversibility  |
| HULL             | Mahalanobis HC hull v0_1           | healthy hull expansion, n_HC=601                                         | 2026-06-06 | populations: EU-Italian   sealed (v10 report §8)                                                                  | BP/MethylPhys/chain/Runtimee Matrices/Mahalanobis_healthy_reference/ |
| HULL             | Mahalanobis HC hull v0_2           | healthy hull expansion, n_HC=1,257                                       | 2026-06-06 | populations: +US   sealed (v10 report §8)                                                                         | BP/MethylPhys/chain/Runtimee Matrices/Mahalanobis_healthy_reference/ |
| HULL             | Mahalanobis HC hull v0_3           | healthy hull expansion, n_HC=1,721                                       | 2026-06-06 | populations: +UK   sealed (v10 report §8)                                                                         | BP/MethylPhys/chain/Runtimee Matrices/Mahalanobis_healthy_reference/ |
| HULL             | Mahalanobis HC hull v0_4           | healthy hull expansion, n_HC=2,481                                       | 2026-06-06 | populations: same   sealed (v10 report §8)                                                                        | BP/MethylPhys/chain/Runtimee Matrices/Mahalanobis_healthy_reference/ |
| HULL             | Mahalanobis HC hull v0_5           | healthy hull expansion, n_HC=2,523                                       | 2026-06-06 | populations: +Han Chinese   sealed (v10 report §8)                                                                | BP/MethylPhys/chain/Runtimee Matrices/Mahalanobis_healthy_reference/ |
| N7               | L9 N7 chain integrity (2026-06-05) | synthetic patients with declared truth through Walther -> A-scoring -> M | 2026-06-05 | R1 MAE 0.0076-0.0093; R3 on-substrate d=+10.24 vs off +5.10; NULL +3.0   v0.1 complete; v0.2 refinements -> PROC- | BP/Record/VAL_PostAtlas/L9_N7_chain_recovery_2026_06_05              |
| PROC (Sept 2026) | PROC-HMIN-BOOT-01                  | rebuild-from-scratch procedure, sealed PREREG before data                | 2026-09-20 | Verdict: 8/8 frozen methylation floors i                                                                          | BP/Record/PROC_data/PROC-HMIN-BOOT-01                                |
| PROC (Sept 2026) | PROC-PANEL-01                      | rebuild-from-scratch procedure, sealed PREREG before data                | 2026-09-20 | sealed                                                                                                            | BP/Record/PROC_data/PROC-PANEL-01                                    |

| series              | id                                         | title                                                     | date       | result / status                          | path (repo)                                                        |
|---------------------|--------------------------------------------|-----------------------------------------------------------|------------|------------------------------------------|--------------------------------------------------------------------|
| PROC<br>(Sept 2026) | PROC-PANEL-02                              | rebuild-from-scratch procedure, sealed PREREG before data | 2026-09-20 | sealed                                   | BP/Record/PROC_data/PROC-PANEL-02                                  |
| PROC<br>(Sept 2026) | PROC-PANEL-03                              | rebuild-from-scratch procedure, sealed PREREG before data | 2026-09-20 | Verdict: LAB ZERO COMMISSIONED** as 40 h | BP/Record/PROC_data/PROC-PANEL-03                                  |
| PROC<br>(Sept 2026) | formula_2x2.json                           | rebuild-from-scratch procedure, sealed PREREG before data | 2026-09    |                                          | BP/Record/PROC_data/formula_2x2.json                               |
| PROC<br>(Sept 2026) | CPG_PHASE1_identity_band_GSE87571          | rebuild-from-scratch procedure, sealed PREREG before data | 2026-09-19 | Verdict: **FAIL** on P3 and P4 (as defin | BP/Record/VAL_PostAtlas/CPG_PHASE1_identity_band_GSE87571          |
| PROC<br>(Sept 2026) | CPG_PHASE1c_scale_map_transfer_GSE42861    | rebuild-from-scratch procedure, sealed PREREG before data | 2026-09-20 | Verdict: **P3 PASS — the pipeline map tr | BP/Record/VAL_PostAtlas/CPG_PHASE1c_scale_map_transfer_GSE42861    |
| PROC<br>(Sept 2026) | CPG_BAND_v2_pooled_two_labs_test_GSE125105 | rebuild-from-scratch procedure, sealed PREREG before data | 2026-09-20 | Verdict: **P3 FAIL (0.9645), P4 FAIL (54 | BP/Record/VAL_PostAtlas/CPG_BAND_v2_pooled_two_labs_test_GSE125105 |
| PROC<br>(Sept 2026) | CPG_LABZERO_01_control_probes              | rebuild-from-scratch procedure, sealed PREREG before data | 2026-09-20 | Verdict: **P1 FAIL on one of three folds | BP/Record/VAL_PostAtlas/CPG_LABZERO_01_control_probes              |
| PROC<br>(Sept 2026) | CPG_LABZERO_02_fourth_cohort_GSE111629     | rebuild-from-scratch procedure, sealed PREREG before data | 2026-09-20 | Verdict: **P4 FAIL, P4b FAIL, P6 FAIL. T | BP/Record/VAL_PostAtlas/CPG_LABZERO_02_fourth_cohort_GSE111629     |

## APPENDIX VI — THE CMB → METHYLOME TRANSLATION MAP, SCORED

Written by the author before the chain was built: the 32-section CMB-analysis curriculum walked module by module, each assigned a methylome analog and a status (HAVE · ROADMAP · ADD = should add · REINTERP = translates with reinterpretation · NO = does not translate). The last column was added on 2026-09-19 and records what the chain actually did. Two rows went *against* the map and are the most instructive: a second deconvolver (row 20, built, cut, then reinstated on the evidence) and de-aging (row 47, built then refused). Row 1 is the frame for the whole chain: the 115 cell types are the harmonic basis, and the deconvolver is the projection onto it. This appendix exists so that a geneticist can see where each tool came from, and a cosmologist can see where the methylome stops behaving like the sky.

### Section I — Foundations: what the CMB field actually is

| # | CMB module                                                                                         | CPG analog                                                                                                                                                                                                                                                                                     | status   | author's note                                                                                                                                                                                                             | 2026-09-19 status                                                                                                                                                                                                                                                                                             |
|---|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Spherical harmonic decomposition<br>$T(\hat{n}) = \sum a_{lm} Y_{lm}$ , including E/B polarization | The genome isn't a sphere, but it's a structured manifold (chromosomes × position × 3D chromatin TADs). Harmonic decomposition becomes (a) per-chromosome Fourier along position, (b) per-TAD modal decomposition, (c) the 115-cell-type basis itself acts as a "feature-space harmonic basis" | REINTERP | Need to write this up — it's the single biggest conceptual move in the rulebook. The 115 cell types are our spherical harmonics. A patient's $\beta$ profile gets decomposed onto that basis via Walther IAM Deconvolver. | UNWRITTEN as a chapter — but the chain does exactly this: the deconvolver projects a patient's $\beta$ onto the 115-cell-type basis (Stage 2) and the class gauge reads eight coefficients of that projection. The 'harmonic basis' is the Atlas. Row 1 is the conceptual frame for Part II's Atlas chapters. |
| 2 | Monopole, dipole, quadrupole, low-I culture                                                        | Monopole = mean genome-wide $\beta$ . Dipole = age axis (CPG-VAL-007) + sex axis. Quadrupole+ = cellular-composition on axes (TODO 1.5 PCA). Low-I anomalies = global-genome-level patterns (CIMP+, X-inactivation status, ploidy)                                                             | HAVE     | Implicit today, need to make explicit.                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                               |
| 3 | Cosmic variance — finite-mode irreducible noise                                                    | Cellular variance — finite number of independent cells contributing to a methylation reading. A pure plasma sample only carries ~10,000 cell-equivalents of cfDNA; you cannot reduce the variance below the cell-count floor. Same structural limit as 2l+1 modes.                             | ADD      | Important to name. Defines fundamental limits on small samples.                                                                                                                                                           | NOT BUILT — cfDNA cell-equivalent variance floor never modelled                                                                                                                                                                                                                                               |

### Section II — Experimental / instrument layer

| # | CMB module                                            | CPG analog                                                                                                                                                                                                                                               | status  | author's note                                                                                 | 2026-09-19 status |
|---|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-----------------------------------------------------------------------------------------------|-------------------|
| 4 | Frequency coverage (multi-band foreground separation) | Multi-substrate methylation (5mC, 5hmC, 5fC, 5caC), plus orthogonal modalities (RNA-seq, ATAC-seq, fragmentomics). CPG v0.1 is single-frequency (5mC); the architecture must declare the multi-substrate slot even though we're not multi-frequency yet. | ROADMAP | Tier 5 of EDEAR Physics Roadmap — 5-substrate multi-assay (already named in the L3 lab plan). |                   |
| 5 | Beam modeling                                         | Probe response function — each 450K/EPIC probe has a sequence context, hybridization efficiency, and adjacent-CpG cross-reactivity. Probe design errors are the methylome's beam errors.                                                                 | ADD     | Not yet explicit in CPG. Should add.                                                          |                   |

| # | CMB module                     | CPG analog                                                                                                                                                                                                                                                  | status   | author's note                                                                                                               | 2026-09-19 status |
|---|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------|-------------------|
| 6 | Bandpass calibration           | Bisulfite conversion efficiency + probe-type normalization (Type I vs Type II). Same physics: non-delta-function response in the chemistry domain leaks signal between channels.                                                                            | REINTERP | Standard preprocessing handles it but we don't TREAT it as bandpass calibration with nuisance parameters. Should formalize. |                   |
| 7 | Gain calibration / drift       | Batch effect + plate position + run date drift. Lab-to-lab and run-to-run gain drift is the methylome version of detector gain drift.                                                                                                                       | HAVE     | Standard pre-processing (BMIQ, funnorm, ComBat) — but we should declare it L2 explicitly.                                   |                   |
| 8 | Polarization-angle calibration | Strand-orientation calibration. Methylation has a forward-strand vs reverse-strand asymmetry; bisulfite sequencing can mis-assign. The polarization-rotation analog is real but minor at the array level.                                                   | REINTERP | Mostly a sequencing-era concern; arrays largely solved it. Document but low-priority.                                       |                   |
| 9 | Scanning strategy              | Cohort sampling design. How patients are sampled across age/sex/site/batch determines what coverage you have and what nulls you can run. Pre-build cohorts were chosen for us; future CPG-acquired cohorts must be designed with "scanning strategy" rigor. | ADD      | Important for prospective cohort acquisition.                                                                               |                   |

### Section III — Time-ordered data to maps

| #  | CMB module                                                   | CPG analog                                                                                                                                                                | status | author's note                                          | 2026-09-19 status |
|----|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------------------------------------------------------|-------------------|
| 10 | TOD analysis                                                 | IDAT intensities + control probe traces + bead-pool QC. Before $\beta$ values exist, there are raw intensities + control signals = our "timestream."                      | HAVE   | Standard tooling handles it.                           |                   |
| 11 | De-glitching + data cuts                                     | Probe flagging, sample QC dropouts, detection-p-value masking. Failed probes, low-quality samples, contamination flags.                                                   | HAVE   | Standard. Declare explicitly.                          |                   |
| 12 | Noise modeling (1/f, atmospheric, correlated detector noise) | Probe-position-on-array correlated noise, plate-edge effects, scanner drift within run. The methylome has its own non-white noise — and CPG hasn't modeled it explicitly. | ADD    | Significant gap. Should add a noise-covariance module. | NOT BUILT         |
| 13 | Destriping / max-likelihood map-making                       | Quantile normalization / funnorm / BMIQ. All raw-intensity-to- $\beta$ methods are inverse-problem solvers in this same family.                                           | HAVE   | Standard tooling. Map preprocessing.                   |                   |

| #      | CMB module                                         | CPG analog                                                                                                                                                                                            | status | author's note                                                                                                                                                                          | 2026-09-19 status                                                                         |
|--------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| 1<br>4 | Transfer functions (pipeline filters the true sky) | The deconvolver itself filters disease signal — it explains away anything that looks like cell composition. We saw this in CPG-VAL-006: chr16/chr17 depleted because their CpGs got "explained away." | ADD    | This is exactly what we already discovered empirically and didn't have a name for. Transfer-function thinking is critical — every step of the chain alters the signal in modeled ways. | NAMED in the lessons (CCL-039, glioma-LL-002: the deconvolver 'explains away'); no module |

#### Section IV — Pixelization & spherical computation

| #      | CMB module                               | CPG analog                                                                                                                                                                                                        | status   | author's note                                                             | 2026-09-19 status |
|--------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------|-------------------|
| 1<br>5 | HEALPix pixelization                     | CpG manifest as the pixelization — 450K → 850K → EPIC v2 1.6M = increasingly fine pixelization. Each CpG is a pixel. Cell-type space is also pixelized (8 classes → 115 cells → finer with each MCMC refinement). | HAVE     | Implicit. Make explicit.                                                  |                   |
| 1<br>6 | Masks + apodization                      | CpG masks — X-inactivation, imprinting, repeat regions, low-confidence probes, SNP-overlapping probes. Mask apodization = down-weighting near suspect regions rather than hard-cutting.                           | REINTERP | Hard masking common; soft apodization unusual. Could be a CPG innovation. |                   |
| 1<br>7 | Cut-sky mode coupling, pseudo-Cl, MASTER | Cut-genome mode coupling. When some CpGs are masked, the per-chromosome correlation function (TODO 2.1) is biased — same MASTER algorithm applies.                                                                | ADD      | Critical for TODO 2.1 to be rigorous.                                     |                   |

#### Section V — Foregrounds and component separation (the biggest section)

| #      | CMB module                                                          | CPG analog                                                                                                                                                                                                                                  | status   | author's note                                                                                                                     | 2026-09-19 status                                                                                                            |
|--------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| 1<br>8 | Galactic foreground modeling (sync, dust, free-free, AME, CO, etc.) | Cellular foreground stack: cell composition, age, sex, ancestry, batch, smoking, BMI, medication exposure, plate position. These are our synchrotron+dust+free-free+AME.                                                                    | REINTERP | We handle cell composition (Walther) + age (CPG-VAL-007). We do NOT yet handle sex, ancestry, batch, smoking explicitly. Big gap. | PARTIAL — composition (Walther) and age (band) handled; sex/smoking/batch layers BUILT and deliberately NOT wired (SOP §104) |
| 1<br>9 | Extragalactic foregrounds (CIB, SZ, point sources)                  | High-resolution disease-overlapping foregrounds: subclinical inflammation, comorbidities, pharmacological methylation effects (e.g., 5-azacytidine), recent acute illness, vaccination response. These are the methylome's "point sources." | ADD      | Almost completely unmodeled. Big gap.                                                                                             | NOT BUILT                                                                                                                    |

| #  | CMB module                                                          | CPG analog                                                                                                                                                                                             | status  | author's note                                                                                  | 2026-09-19 status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 20 | Component-separation methods (Commander, NILC, SMICA, SEVEM, GNILC) | Walther IAM Deconvolver = one method. We need to build at least one INDEPENDENT method (parametric Bayesian vs blind/ILC-style) so we can compare, the way Planck compared Commander/NILC/SMICA/SEVEM. | ROADMAP | Critical addition. No single-method CMB result was ever trusted. Need at least 2 deconvolvers. | REVERSED, THEN RESTORED — NILC was built as the second method and CUT 2026-07-02 (c1be0c3) for collapsing on correlated blood mixtures and deleting correct calls. Rerun as designed in September (PROC-NILC-01) it was vindicated: the divergence marked where the atlas does not separately determine the composition, which is what the method exists to report - the cut had removed the instrument reporting the problem. REINSTATED 2026-09-22 as stage_2b_second_opinion: compared with Walther by class, agreement bar L1 <= 0.10, reported as a flag and never as the composition. The Planck principle this row cites is followed again, and RUNBOOK s11 forbids disabling either relay. |
| 21 | Foreground residual marginalization in likelihood                   | Nuisance parameter for residual cell-composition error, residual age error, residual batch error. Right now we point-estimate these. Should marginalize.                                               | ADD     | Big upgrade. Required for clean L7 likelihood.                                                 | NOT BUILT — point estimates                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

## Section VI — Power-spectrum estimation

| #  | CMB module                                                      | CPG analog                                                                                                                                                           | status  | author's note                                                              | 2026-09-19 status                                                                                   |
|----|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| 22 | Auto vs cross-spectra (split maps, half-mission × half-mission) | Split-cohort cross-correlations. GSE51057 × GSE51032 cross is the methylome's "half-mission × half-mission." We already used this in CPG-VAL-003 (concordance gate). | HAVE    | Mark explicit. Could go further with within-cohort odd/even sample splits. | BUILT — GSE51032 × GSE51057 is the foundation-cohort anchor; reproduced 2026-09-19 at $r = 1.00000$ |
| 23 | Pseudo-Cl on masked sky                                         | Per-chromosome C(d) with masked CpGs handled correctly. Required for TODO 2.1.                                                                                       | ROADMAP | Build with MASTER-style mode-coupling correction.                          |                                                                                                     |
| 24 | Quadratic max-likelihood at low l                               | Per-TAD or per-chromosome-arm maximum-likelihood spectrum estimation. For low-mode-count regimes.                                                                    | ADD     | Niche but worth declaring.                                                 |                                                                                                     |
| 25 | Bandpowers                                                      | Genomic distance bandpowers — group C(d) into log-spaced d-bins with proper covariance.                                                                              | ROADMAP | Part of TODO 2.1.                                                          |                                                                                                     |
| 26 | Beam/calibration nuisance treatment                             | Probe-response + bisulfite efficiency nuisance treatment.                                                                                                            | ADD     | Add to L7 likelihood.                                                      |                                                                                                     |

## Section VII — Likelihoods and covariances

| #  | CMB module                                                    | CPG analog                                                                                                                                                                                        | status | author's note | 2026-09-19 status |
|----|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------------|-------------------|
| 27 | Low-l likelihoods (non-Gaussian, pixel-space, Gibbs sampling) | Low-frequency disease signal likelihoods — when there are very few discriminating CpGs (rare cancers, early pre-dx), the distribution of A-score isn't Gaussian. Need exact-likelihood treatment. | ADD    | Future.       |                   |

| #  | CMB module                                                       | CPG analog                                                                                                                                                               | status  | author's note                                                                      | 2026-09-19 status                                                                                 |
|----|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| 28 | High-I likelihoods (Gaussian bandpower with detailed covariance) | Mahalanobis hyper-volume (CPG-VAL-002) is the seed. Need to extend to per-card formal likelihoods.                                                                       | ROADMAP | This is L7 of the chain.                                                           |                                                                                                   |
| 29 | Covariance matrices                                              | A-score covariance (have via Ledoit-Wolf in CPG-VAL-002). Need: per-CpG covariance, cross-cohort covariance, nuisance-parameter covariance.                              | ROADMAP | Tier 2-3 work.                                                                     |                                                                                                   |
| 30 | End-to-end simulations                                           | Synthetic patient generation: known cell-type fractions + known disease signal + simulated batch + run through entire chain. Validate that the chain recovers the truth. | ADD     | The single biggest missing piece in CPG today. Cannot defend the chain without it. | BUILT — synthetic_patient_harness.py + null N7                                                    |
| 31 | Null tests                                                       | HC permutation, age permutation, sex permutation, cohort splits, plate-position null, batch null, synthetic injection-recovery. Some scattered today, no unified suite.  | ROADMAP | L9 of the chain — unified null suite is on the engine-completion sprint.           | BUILT — cpg_null_runner.py N1–N8 (found in RETIRED/ and restored to MethylPhys/chain/ 2026-09-19) |
| 32 | PTEs and goodness-of-fit + look-elsewhere                        | Mandatory. Especially for rare biological anomalies.                                                                                                                     | ADD     | Add to outcome.md template.                                                        |                                                                                                   |

## Section VIII — Theory prediction engines

| #  | CMB module                            | CPG analog                                                                                                                                                    | status   | author's note                                                         | 2026-09-19 status                                                               |
|----|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------|
| 33 | Boltzmann solvers (CAMB/CLASS)        | The IAM physics framework + H_min derivation IS our Boltzmann solver. It predicts the architectural floor against which everything else is measured.          | HAVE     | Already in place — the Recipe is the methylome's CAMB. Vault content. | BUILT — the Recipe / H_min derivation (vault); 40 floors in cpg_gauge_engine.py |
| 34 | Recombination physics (RECFAST/HyRec) | Cellular differentiation thermodynamics — the per-class H_min values are frozen at differentiation, analogous to acoustic peaks frozen at recombination.      | HAVE     | Implicit in IAM.                                                      |                                                                                 |
| 35 | Line-of-sight integration             | Lineage tracing through cell development. A methylation pattern in a mature cell reflects integrated history of decisions along the developmental trajectory. | REINTERP | Translatable but speculative.                                         |                                                                                 |
| 36 | Acoustic-peak phenomenology           | The pattern cheat-sheet for methylation features. (See Section XVII below.) This is the rulebook's most teachable section.                                    | ADD      | Build a CPG version of the cheat-sheet.                               |                                                                                 |

## Section IX — Parameter inference & degeneracies

| #  | CMB module                                   | CPG analog                                                                                                                                                                                            | status   | author's note                                         | 2026-09-19 status                                                                   |
|----|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------|-------------------------------------------------------------------------------------|
| 37 | MCMC + posterior geometry                    | MCMC for per-card posterior tier (replacing threshold-based scoring). Already familiar from IAMAtlas build.                                                                                           | ROADMAP  | Engine-completion work.                               | NOT BUILT — MCMC used for the Atlas and the floors; scoring is threshold + age band |
| 38 | Banana degeneracies                          | CIMP-axis degeneracy is visible in Image 2. Other expected ones: age-vs-class-drift, immune-fraction-vs-disease-signal, sex-vs-X-inactivation.                                                        | ROADMAP  | TODO 2.3 banana degeneracy is the first concrete one. |                                                                                     |
| 39 | Priors                                       | Disease prior (cancer_prior runtime matrix), family-history prior(family_history_multiplier), age prior. Already implemented but should be treated as proper Bayesian priors with volume awareness.   | REINTERP | Have, but informal. Upgrade.                          |                                                                                     |
| 40 | Profile likelihood vs marginalized posterior | Critical distinction. Profile likelihood = "where is the best-fit cell-type fraction" vs marginalized = "where is the volume in fraction space." They can disagree when nuisance dimensions are high. | ADD      | Add to L7.                                            |                                                                                     |
| 41 | Multimodal distributions                     | Disease subtypes are multimodal posteriors. HER2+ vs Basal vs Luminal-A breast cancer don't sit on one mode. Posterior clustering matters.                                                            | REINTERP | Recognized but not formal.                            |                                                                                     |
| 42 | Nested sampling and Bayesian evidence        | Model comparison: does this patient look more like 'breast pre-dx' or 'pancreatic pre-dx' or 'inflammatory baseline'? Bayesian evidence is the right tool.                                            | ADD      | Future engine layer.                                  |                                                                                     |
| 43 | Tension metrics                              | Cross-card tension when two cards both fire on one patient. Need proper tension language, not just "both elevated."                                                                                   | ADD      | Operational rule for Stage 6 reports.                 |                                                                                     |

#### Section X — Lensing: the CMB as a backlight

| #  | CMB module                          | CPG analog                                                                                                                                                                                     | status   | author's note                     | 2026-09-19 status                                                                                                                                                                   |
|----|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 44 | Lensing smoothing of acoustic peaks | Late-life methylation drift smooths early-life methylation patterns. Aging is the methylome's lensing — it smears the original developmental signal.                                           | REINTERP | Beautiful analogy. Worth a paper. | UNWRITTEN — the AGE BAND is the operational form of 'aging as lensing'; the analogy itself appears nowhere in the corpus and is the cleanest way to explain the band to a physicist |
| 45 | Four-point lensing reconstruction   | CpG-CpG-CpG-CpG four-point correlations — coordinated methylation changes across 4 loci would reveal coordinated regulatory programs. Bispectrum (TODO 2.2) is 3-point; this would be 4-point. | ADD      | Tier 2+ research.                 |                                                                                                                                                                                     |

| #      | CMB module                              | CPG analog                                                                                                                                           | status  | author's note          | 2026-09-19 status                                                                                                                                                                                                                                             |
|--------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|---------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4<br>6 | Lensing power spectrum                  | Aging "drift power spectrum" — measure how aging redistributes power across methylation features. The IAM cellular age clock (TODO 2.4) is the seed. | ROADMAP | TODO 2.4 lives here.   |                                                                                                                                                                                                                                                               |
| 4<br>7 | Delensing                               | De-aging: subtract the age component to recover the disease component. CPG-VAL-007 dipole subtraction is the first instance.                         | HAVE    | Have seed. Generalize. | REVERSED — de-aging (age_axis_foreground.py) BUILT, then §104 ruled annotate-don't-subtract. The chain does the opposite of delensing, on purpose: the methylome's foreground is the patient's own biology. The one place the analogy is deliberately broken. |
| 4<br>8 | Cross-correlations with external probes | Cross-correlate CPG output with RNA-seq, proteomics, imaging.                                                                                        | ROADMAP | Future multi-omics.    |                                                                                                                                                                                                                                                               |

### Section XI — Polarization special topics

| #      | CMB module                                | CPG analog                                                                                                                                                                                           | status   | author's note                                            | 2026-09-19 status |
|--------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------------------------|-------------------|
| 4<br>9 | E/B separation                            | Methylated vs hydroxymethylated separation — the methylome's literal polarization. 5mC vs 5hmC is an E/B-like spin decomposition.                                                                    | REINTERP | Substrate-level; requires oxBS-seq (5-substrate Tier 5). |                   |
| 5<br>0 | B-mode taxonomy                           | 5hmC contamination taxonomy — what shows up in a "5hmC signal" can be true 5hmC, residual unconverted 5mC, sequencing error, or biological switch.                                                   | ADD      | Substrate-dependent.                                     |                   |
| 5<br>1 | Reionization bump (low-I EE)              | Developmental methylation reprogramming bump — large-scale early-life methylation events leave a "reionization-like" signature in the lowest-frequency modes.                                        | REINTERP | Speculative analogy.                                     |                   |
| 5<br>2 | Cosmic birefringence (EB/TB correlations) | Methylation-state rotation during disease (5mC → 5hmC → 5fC → 5caC oxidation cascade is literally a rotation in methylation chemistry-space). Would produce equivalent cross-correlation signatures. | REINTERP | Genuinely novel research direction.                      |                   |

### Section XII — Non-Gaussianity / higher-order statistics

| #      | CMB module                                                   | CPG analog                                                                                                                                                                                                                                                      | status  | author's note                                                     | 2026-09-19 status |
|--------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------------------------------------------------------------------|-------------------|
| 5<br>3 | Bispectrum estimators (local, equilateral, orthogonal, etc.) | TODO 2.2 bispectrum on highest-A-score class markers. Different "shapes" of bispectrum probe different biological coupling structures (local = pairwise regulatory networks; equilateral = shared upstream regulator; orthogonal = anti-correlated regulation). | ROADMAP | On the roadmap. The shape taxonomy will need its own translation. |                   |

| #      | CMB module                           | CPG analog                                                                                                                                                  | status | author's note                              | 2026-09-19 status |
|--------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------------------------------------------|-------------------|
| 5<br>4 | Trispectrum (4-point)                | Coordinated 4-CpG methylation events — chromatin remodeling complexes, multi-enhancer hubs.                                                                 | ADD    | Tier 2+.                                   |                   |
| 5<br>5 | Minkowski functional<br>s / topology | Genome-wide methylation topology — how methylation is spatially organized within the nucleus (A/B compartments, TADs, LADs).                                | ADD    | Connects to Hi-C data — multi-omics cross. |                   |
| 5<br>6 | Phase statistics                     | CpG phase structure — same amplitude, different ordering = different biology. Phase information is what the bispectrum sees that the power spectrum misses. | ADD    | Tier 2+.                                   |                   |

### Section XIII — Isotropy, anomalies, preferred axes

| #      | CMB module                                                                                                                         | CPG analog                                                                                                                                                                   | status           | author's note                                | 2026-09-19 status                                                  |
|--------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------------------------------|--------------------------------------------------------------------|
| 5<br>7 | Statistical isotropy tests (bipolar harmonics, dipole modulation, hemispherical asymmetry, local variance maps, multipole vectors) | CPG-VAL-006 chromosome isotropy is the seed. Extend to bipolar (paternal vs maternal allele asymmetry), dipole modulation, hemispherical asymmetry (genomic half-asymmetry). | HAVE→RO<br>ADMAP | Have seed; extend.                           |                                                                    |
| 5<br>8 | Parity asymmetry                                                                                                                   | Strand-bias parity tests + even/odd-chromosome parity tests.                                                                                                                 | ADD              | Niche but specifically testable.             |                                                                    |
| 5<br>9 | Alignment tests (axis of evil, multipole vectors)                                                                                  | Chromosomal arm alignment — do disease-signal-rich chromosomes share an axis (e.g., chr6 MHC + chr2 immune-related = immune-axis alignment)?                                 | REINTERP         | Need to be careful with look-elsewhere.      |                                                                    |
| 6<br>0 | Topology searches (matched circles)                                                                                                | Methylation "matched-pair" searches between distant loci with co-regulated patterns.                                                                                         | REINTERP         | Speculative.                                 |                                                                    |
| 6<br>1 | Look-elsewhere discipline                                                                                                          | MANDATORY. Pre-register the test before running it. We've been informal about this.                                                                                          | ADD              | Make absolute rule in v1_VAL_Test_Checklist. | BUILT — N8 in the null runner; VAL-006's chr6 signal died under it |

### Section XIV — Secondary anisotropies

| #      | CMB module                        | CPG analog                                                                                                                              | status   | author's note                        | 2026-09-19 status |
|--------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------|-------------------|
| 6<br>2 | Integrated Sachs-Wo<br>lfe effect | Late-life methylation drift integrated along developmental trajectory. Same structure: late-time potential decay → large-angle imprint. | REINTERP | Speculative analogy worth exploring. |                   |
| 6<br>3 | Rees-Sciama                       | Nonlinear corrections. Niche.                                                                                                           | NO       | Skip.                                |                   |

| #      | CMB module                               | CPG analog                                                                                                                                                                                   | status   | author's note                       | 2026-09-19 status |
|--------|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------|-------------------|
| 6<br>4 | Thermal SZ (hot electron Comptonization) | Inflammation-associated methylation shifts — chronic inflammation reshapes the methylome the way hot cluster gas reshapes the CMB.                                                           | REINTERP | Real biological analog.             |                   |
| 6<br>5 | Kinetic SZ (bulk velocities)             | Clonal hematopoiesis bulk-velocity-like signal — large clones with shared methylation drift create a kSZ-like signature.                                                                     | REINTERP | Specific to CHIP/AML cards.         |                   |
| 6<br>6 | Patchy reionization                      | Patchy senescence / cellular replacement in aging tissues.                                                                                                                                   | REINTERP | Aging-related.                      |                   |
| 6<br>7 | Cosmic infrared background               | Background methylation noise from healthy tissue turnover.                                                                                                                                   | ADD      | Worth modeling.                     |                   |
| 6<br>8 | Rayleigh scattering                      | Frequency-dependent scattering.                                                                                                                                                              | NO       | Skip.                               |                   |
| 6<br>9 | Spectral distortions ( $\mu$ , y-type)   | Methylation spectral distortions: systematic deviations from the IAM-predicted information spectrum at specific scales = signatures of disease-specific energy injection into the methylome. | REINTERP | Genuinely novel research direction. |                   |

#### Section XV — Data combination & external probes

| #      | CMB module                   | CPG analog                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | status   | author's note                              | 2026-09-19 status |
|--------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------|-------------------|
| 7<br>0 | BAO combination              | Multi-substrate combination (RNA-seq, fragmentomics, ATAC-seq, proteomics) as the methylome's BAO — breaks degeneracies CMB-only / methylation-only cannot resolve.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | ROADMAP  | Tier 5.                                    |                   |
| 7<br>1 | Supernovae (distance ladder) | Cellular age clock as the distance ladder - calibrate biological age via known-age reference cohorts. MEASURED AND PARTLY CLOSED: PROC-PANEL-03 built the per-decade reference curve on 1,379 healthy donors from four laboratories, and it is now the curve every reading is corrected by. PROC-AGE-01 then measured what inverting it buys for one array: 0.47 mA per year against a within-laboratory spread of 0.0235, about 50 years - the rungs are wider than the ladder, so cellular age in years is refused. What remains is the calibration cohort the analogy actually needs: known ages WITH repeat draws, so the technical term cancels. | MEASURED | Curve in use; per-array inversion refused. |                   |

| #      | CMB module                                   | CPG analog                                                                                                                            | status   | author's note         | 2026-09-19 status |
|--------|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------|-------------------|
| 7<br>2 | BBN cross-check                              | Mendelian inheritance + germline methylation cross-checks — early-life patterns set bounds on later patterns.                         | REINTERP | Niche but principled. |                   |
| 7<br>3 | Large-scale structure                        | Cross-cohort population structure. UK Biobank scale eventually.                                                                       | ROADMAP  | Far future.           |                   |
| 7<br>4 | Joint likelihoods (handle shared covariance) | Joint likelihood across cards for one patient — handle covariance between immune-class drift detected by breast-epic AND immune-epic. | ADD      | Stage 5 architecture. |                   |

#### Section XVI — Modern computational methods

| #      | CMB module                    | CPG analog                                                                                           | status  | author's note                 | 2026-09-19 status                                                                           |
|--------|-------------------------------|------------------------------------------------------------------------------------------------------|---------|-------------------------------|---------------------------------------------------------------------------------------------|
| 7<br>5 | Public compressed likelihoods | Per-card compressed-likelihood release. ACT-DR6 style. Future.                                       | ADD     | Once cards stabilize.         |                                                                                             |
| 7<br>6 | Emulators                     | Neural-network emulator for IAMAtlas reconstruction at very large CpG coverage. Speed optimization.  | ADD     | Tier 4+.                      |                                                                                             |
| 7<br>7 | Simulation-based inference    | Synthetic-patient SBI — when L7 likelihood is intractable.                                           | ROADMAP | Pairs with end-to-end sims.   |                                                                                             |
| 7<br>8 | Blinding                      | Mandatory for confirmation cards. Apply pipeline before seeing case labels. Unblind only after seal. | ADD     | Add to v1_VAL_Test_Checklist. | PARTIAL — PREREG sealed before data opens; no formal blinding step                          |
| 7<br>9 | Reproducibility discipline    | Already absolute — the reproducibility quadruple is our version.                                     | HAVE    | We have this.                 | BUILT — bit-identical Stage 1, deconvolver MAE 0.0004, anchors $r = 1.00000$ ; the PROC kit |

## APPENDIX VII — THE COMPLETION SPRINT, SCORED

The spring-2026 plan to take the chain "from C- to A" in 12–15 sessions: Phase A (L9 nulls) → B (L4 foregrounds) → C (L5 correlation structure) → D (L6 covariance) → E (L7–L8 likelihood and inference) → F (audit). Scored against the repository on 2026-09-19. The author's verdict: *"Too ambitious too quick. It ended up harming rather than helping. These should have been worked on long after the bones were trusted."*

| phase | deliverable                                         | what happened                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A1    | cpg_null_runner.py, nulls N1–N8                     | BUILT, live (MethylPhys/chain/CPG_Null_Runner/). Every post-Atlas CPG-VAL carries null_results.json. The one phase built on trusted bones; still the best-engineered part of the chain.                                                                                                                                                                                                                                                                       |
| A2    | synthetic patient generator                         | BUILT (RETIRED/.../Synthetic_Patient_Generator/; harness in MethylPhys/chain/report_builders/). N7 depends on it — to be restored beside the null runner.                                                                                                                                                                                                                                                                                                     |
| A3    | Family A VALs through the nulls                     | DONE (CPG-VAL-001...007 all carry null results).                                                                                                                                                                                                                                                                                                                                                                                                              |
| B1    | foreground registry                                 | never built                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| B2    | NILC second deconvolver                             | BUILT → CUT → REINSTATED. Cut 2026-07-02 (c1be0c3) for collapsing on correlated blood mixtures and deleting correct calls; rerun as designed in September (PROC-NILC-01) and vindicated — the divergence was marking where the atlas does not separately determine the composition. Reinstated 2026-09-22 as stage_2b_second_opinion: class-level comparison against Walther, agreement bar $L1 \leq 0.10$ , reported as a flag and never as the composition. |
| B3/B4 | age / sex / smoking foreground modules              | BUILT → REFUSED (SOP §104): the methylome's foreground is the patient's own biology — annotate, never subtract. In RETIRED/.../IAM_Cellular_Age/.                                                                                                                                                                                                                                                                                                             |
| C1    | C(d) genomic-distance correlation, "acoustic peaks" | never built                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| C2    | bispectrum                                          | never built                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| C3    | banana degeneracy mapper                            | never built ("I never got my banana degeneracy")                                                                                                                                                                                                                                                                                                                                                                                                              |
| D1/D2 | sim-based and per-CpG covariance                    | never built                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| E1    | cellular age clock                                  | BUILT as a trained clock (v1), later judged wrong (not physics); replaced by the band inversion (v3), which does not yet calibrate. Not reportable.                                                                                                                                                                                                                                                                                                           |
| E2/E3 | per-card likelihood + MCMC posteriors               | never built; scoring remains threshold + age band                                                                                                                                                                                                                                                                                                                                                                                                             |
| F1–F3 | chain audit, v2 report, v2 checklist                | report iterated v2 → v10; checklist v2 not written; the audit is what the 2026-09 PROC series is doing now                                                                                                                                                                                                                                                                                                                                                    |

### The lesson

Phase A worked because it tested a chain that already existed. Everything after it was built on bones not yet trusted: an atlas still going flat (see MethylPhys/atlas/IAMAtlas\_FLATNESS\_LESSON.md), a band compiled from nine pipelines, a gauge formula that changed four times between 2026-06-11 and 07-01. So B was built and torn out, and C–E had nothing to stand on. The right order was A → trust the bones (Stage 1 bit-identical; deconvolver conformant; anchors reproducing; the band rebuilt from one cohort through one pipeline — Phase 1 of the September plan) → then the correlation structure, the degeneracies, the Bayesian layer. The sprint had the right pieces in the wrong order. Its own sign-off question 5 named the risk — "the discipline doesn't hold when the first inconvenient result comes back" — and the discipline did hold for individual VALs (VAL-102 voided in four minutes, seal preserved); it did not hold for the sequence.

What the sprint still gets right, for later: C1 (two-point correlation per class with MASTER-style mode coupling), C3 (2D posterior shape for A-score pairs — the CIMP axis is the first), and E2/E3 (a per-card likelihood marginalised over composition and age, replacing thresholds) remain the correct next layer once Phase 1 has re-derived the bands. Not before.

## FUTURE GOALS — WHAT IS WORTH THE EFFORT, IN ORDER

Drawn from the scored translation map (Appendix VI) and the scored sprint (Appendix VII): only items not yet built, kept only where the data and tools in hand can support them, each with the gate it waits on. The order is the lesson of Appendix VII applied — nothing above the bands is scheduled until the bands are rebuilt from one cohort through one pipeline. "Not now" is a list, not a refusal; "does not translate" is the author's own ruling.

| gate                                        | goal                                                                                                                                                                                                   | from                                         | why it is realistic                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | needs                                                          |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| GATE 0 — before anything else               | Rebuild the eight class BANDS from one healthy cohort through one pipeline                                                                                                                             | sprint lesson; RECON B1                      | GSE87571 (n~730 healthy whole blood, ages 14–94, raw 450K IDATs) through Stage 1 as shipped. Replaces a band compiled from nine papers on nine pipelines. Resolves the stem_adult false alarm and the identity-loci offset in one step.                                                                                                                                                                                                                                                         | 6 GB download; ~6 h of Stage 1; a sealed PREREG first          |
| GATE 0b - lab zero: CLOSED (panel standard) | Per-lab healthy-control panel is the lab zero (LAB-ZERO-02). Revisit the control-probe route only with $\geq 6$ labs and pre-analytical metadata; the 13% within-cohort narrowing is noted, not built. | LAB-ZERO-01, LAB-ZERO-02                     | direction 4/4, magnitude 1/4; the unrecorded pre-analytical term dominates                                                                                                                                                                                                                                                                                                                                                                                                                      | closed                                                         |
| GATE 0                                      | Presence rule: report a class gauge only where DETERMINED ( $\kappa < \sim 10$ , or joint) and PRESENT ( $\geq 5\%$ ); whole blood $\rightarrow$ immune + one haematopoietic-progenitor component      | PROC-NILC-01; author's risk-over-reward rule | The Atlas reference columns for the two classes are $r = +0.99$ at the deconvolver's markers; the split between them is chosen by the simplex constraint, and the stem_adult band then flags healthy donors. One joint haematopoietic-progenitor component removes the false alarm without touching any floor. $\kappa$ of the substrate sub-problem joins the Jensen gap as a pre-flight number. stem_adult has carried zero findings in nine VALs; its band is $n = 4\text{--}32$ sorted HSC. | a config change in the conductor + the band rebuild of Phase 1 |
| GATE 0                                      | Wire Tool B (lineage splitter) into the conductor behind the reporting rule; run it in the myeloid arm                                                                                                 | PROC-SEP-03                                  | Tool B resolves HSC vs progenitor at $\kappa \sim 4\text{--}5$ on the contrast CpGs and reports stem_adult = 0 in healthy blood; it is the instrument that would see HSC content rise in MDS / CML / CHIP. Wire it so the joint component is reported by rule and the split is offered only when Tool B's $\kappa$ clears the bar, with its uncertainty.                                                                                                                                        | half a session + the myeloid cohorts                           |
| GATE 0                                      | Wire the Stage 0 $\leftrightarrow$ Stage 1 intensity hand-off                                                                                                                                          | map rows 10–11; PROC-STAGE0-01               | methy prep already exposes negative-control probes, per-probe detection p, bead counts and X/Y intensities; Stage 0 only has to receive them. Turns four DEFERRED QCs into real ones.                                                                                                                                                                                                                                                                                                           | half a day; no new data                                        |
| after GATE 0                                | Myeloid arm: MDS / CML / CHIP through the chain - the test of whether stem_adult earns its own gauge                                                                                                   | disease matrix v1.13; reporting rule         | The three diseases where HSC is expected to move are unvalidated ('heme-epic v0.1 pending'). Running them with the joint component AND the separate stem_adult gauge side by side decides the class's status on blood by measurement rather than by rule. MDS (HSC up, GMP down) is the discriminating case.                                                                                                                                                                                    | GSE62298 + an MDS cohort; PREREG; the rebuilt band             |
| GATE 0                                      | Reproduce the breast pre-diagnostic anchor from raw IDATs                                                                                                                                              | sprint F1                                    | GSE51057 (329 IDATs) through the whole chain against the rebuilt band. The first result a reader will check; it should stand on the corrected reference.                                                                                                                                                                                                                                                                                                                                        | 2.8 GB; ~3 h                                                   |
| after GATE 0                                | Cellular variance as cosmic variance                                                                                                                                                                   | map row 3                                    | A cfDNA sample carries a finite number of cell-equivalents; the variance floor that sets is computable from the deconvolved composition and the read depth. Names the fundamental limit on small plasma samples. Analytic; no new data.                                                                                                                                                                                                                                                         | a derivation and one figure                                    |
| after GATE 0                                | Transfer function of the chain                                                                                                                                                                         | map row 14; CCL-039                          | Inject a known per-CpG signal into synthetic patients (N6) and measure what fraction survives each stage — the deconvolver 'explains away' composition-like signal and this quantifies how much. The synthetic generator and null runner already exist.                                                                                                                                                                                                                                         | N6/N7 runs; one table per stage                                |

| gate                    | goal                                                                                                                                                                                                          | from                                                       | why it is realistic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | needs                                                                |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| after GATE 0            | Second, independent deconvolver — the right way this time                                                                                                                                                     | map row 20; sprint B2                                      | Planck never trusted a single component-separation method. NILC failed on correlated blood mixtures; a parametric Bayesian (Commander-style) deconvolver on the same Atlas is the natural second, and it fails differently. Agreement within tolerance becomes a gate; disagreement becomes a flag, not a deletion. Lesson from the first attempt (RECON D3): NILC ran unconstrained GLS on the bimodal cell-type marker pool and was ill-conditioned on blood's near-parallel immune/progenitor/stem_adult columns; the replacement must solve on Walther's class-marker ruler with a simplex prior, so that it differs in principle rather than in stability. | 1–2 sessions; the Atlas; Moss 2018 mixtures as ground truth          |
| after GATE 0            | Nuisance marginalisation in the gauge                                                                                                                                                                         | map row 21                                                 | Composition and age enter as point estimates today. Propagating the deconvolver's residual and the band's sd into the gauge A gives every reading an uncertainty — the posterior sd the MCMC atlas already carries but the chain drops at Stage 4.                                                                                                                                                                                                                                                                                                                                                                                                              | half a session; no new data                                          |
| after the anchor        | Banana degeneracy — 2D posterior shape for A-score pairs                                                                                                                                                      | map row 38; sprint C3 ('I never got my banana degeneracy') | For pairs such as immune × cycling or terminal × stem_pluri, map the 2D case-vs-HC distribution; the CIMP axis is the first known one. Cheap once bands are trusted; meaningless before.                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | one session; the foundation cohort                                   |
| after the anchor        | C(d): two-point correlation of residuals vs genomic distance, per class                                                                                                                                       | map rows 17, 23, 25; sprint C1                             | The methylome's power spectrum. Look for characteristic-scale features (the 'acoustic peaks' of the map), with MASTER-style correction for masked CpGs. Needs CHR/MAPINFO on every CpG (the manifest is in MethylPhys/atlas/external_manifests).                                                                                                                                                                                                                                                                                                                                                                                                                | one session; the Atlas + anchor cohort                               |
| after the anchor        | Per-card likelihood, marginalised, with MCMC posteriors                                                                                                                                                       | map rows 28, 37; sprint E2/E3                              | Replace threshold-plus-band scoring with a proper posterior over per-card parameters, nuisance-marginalised. emcee is already in the toolchain from the Atlas build. This is what a Planck reader will expect L7 to be.                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1–2 sessions; everything above                                       |
| after the anchor        | Formal blinding for confirmation VALS                                                                                                                                                                         | map row 78                                                 | Apply the pipeline before case labels are seen; unblind only after the seal. PREREG already does half of this; the other half is a procedural rule and a script flag.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | a checklist change                                                   |
| substrates              | Urine, CSF, and the within-patient tissue/plasma/urine trio                                                                                                                                                   | Issue 003 §7                                               | Cohorts already located and partly downloaded (GSE119260 four men × three substrates; GSE292312 and GSE269403 CSF). Each is one experiment with a declared prediction: the class present in the shed tissue should appear in its fluid and be absent from the same patient's blood.                                                                                                                                                                                                                                                                                                                                                                             | data in hand; PREREG each                                            |
| not now                 | Bispectrum / trispectrum; Minkowski functionals; isotropy and alignment tests; spectral distortions                                                                                                           | map rows 45, 53–60, 69                                     | Real analogs, genuinely novel, and every one of them needs a trusted two-point function first. Listed so they are not forgotten; not scheduled.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | after C(d)                                                           |
| not now                 | Multi-substrate 5mC/5hmC (E/B separation); multi-omics cross-correlation                                                                                                                                      | map rows 4, 49, 70                                         | Tier 5. Needs oxBS or matched RNA-seq/ATAC on the same samples — data the project does not hold.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | new data                                                             |
| does not translate      | Rees–Sciama; Rayleigh scattering                                                                                                                                                                              | map rows 63, 68                                            | The author's own <i>X</i> rows; kept for completeness.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | —                                                                    |
| GATE 2 - after the band | Apply the Sanchez-Mackenzie thermal-background model (Weibull/gen-gamma Hellinger divergence) to the identity loci: which loci carry regulatory vs thermal signal? Compare to the empirical marker selection. | PRIOR_ART                                                  | the two uses of $k_B T \ln 2$ are complementary - their filter could sharpen our ruler's loci                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | MethylIT on the Stage-1 healthy betas already cached (three cohorts) |

| gate                               | goal                                                                                                                          | from           | why it is realistic                                                                                         | needs                                                                                                                |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| GATE 1 - on the commissioned gauge | Re-derive the class-average healthy drift rate and the cross-species lifespan scaling (mammalian paper) on the identity gauge | PROC-AGE-01 D2 | the human rate is now measured on four labs (0.47 mA/yr); the dog cohorts (VAL-013, VAL-025-028) are public | Stage 1 on the Wang 2020 Labrador arrays; identity loci mapped to the canine array                                   |
| GATE 2 - needs data not in hand    | Test the four-substrate aging prediction (VAL-025..028) on real plasma cfDNA fragmentomics with donor ages                    | PROC-RECORD-02 | the prediction is filed and monotone; WPS / fragment-size / nucleosome inference exist as methods on cfDNA  | a healthy-donor cfDNA WGS cohort with ages (Snyder 2016 n=36 public; Mouliere 2018 / Mathios 2022 controlled access) |

## PART II — THE CHAIN, STAGE BY STAGE (FORTHCOMING)

Issue 003 documents the engine and its evidence. Part II, written once the chain carries its seal, will teach it: one chapter per stage, each giving the purpose, the cosmology it borrows, what was tried first and why it failed, the runtime files it reads, the procedure that confirms it, and what a researcher trained on bootstrapping needs to know before touching MCMC output or a Mahalanobis distance. The outline as it stands:

| chapter                                         | content                                                                                                                                                                                         |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Opening — the translation map                   | Appendix VI as narrative: what the CMB gave us, row by row, and the two places we stopped taking it (a second deconvolver; de-aging).                                                           |
| Three kinds of file                             | FLOOR ( $H_{\min}$ : physics, 40 numbers) / RULER (identity loci: where the gauge reads) / BAND (age reference: a cohort statistic). Every 'healthy reads wrong' case of 2026 traced to a band. |
| Stage 0 — intake                                | Array type from the IDAT header; SHA-256 with re-transmission detection; the fail-open we closed.                                                                                               |
| Stage 1 — calibration                           | noob via methylprep; bit-identical on 11/11; the 6-7% normalisation gain between atlas-source betas and Stage 1 output and why the band absorbs it.                                             |
| Stage 2 — component separation                  | Walther NNLS against the Atlas; presence, not gating; why NILC was cut, and why it came back.                                                                                                   |
| Stage 3 — the foreground we refused to subtract | Built age/sex/smoking layers; SOP s104; a galactic foreground is a separate source, the methylome's is the patient.                                                                             |
| Stage 4 — the gauge                             | $H(\text{beta\_mean})/H_{\min}$ on identity loci, placed in the band; two surfaces (s106); four formula changes in three weeks and the measurement that settled them.                           |
| Stage 4.5 — bidirectional                       | Shannon H is symmetric about 0.5; VAL-050's null and VAL-051's recovery.                                                                                                                        |
| Stage 4.6 — the patient sky                     | $z = (\text{beta} - \mu)/\sigma$ on HEALPix; the residual map against a reference with per-pixel uncertainty; Plate 05.                                                                         |
| Stage 5 — Mahalanobis Option A                  | Eight coefficients as a vector; distance from the age-band centroid; why v1 put healthy at $z \sim -22$ .                                                                                       |
| Stage 6 — cellular age                          | The band run backwards; three attempts; why it does not calibrate yet.                                                                                                                          |
| The Atlas (several chapters)                    | Per-class MCMC over 115 cell types; the flatness lesson (R-hat misled twice, in opposite directions); the brightness posterior; why a sphere.                                                   |
| What we built before the bones were trusted     | Appendix VII as narrative; the order that should have been.                                                                                                                                     |
| For the bootstrapper                            | MCMC vs resampling; what a posterior sd buys that a bootstrap CI does not; when a Mahalanobis distance is and is not a p-value.                                                                 |

## DATA SOURCES — PRIMARY CITATIONS

All  $\beta$  values from primary publications. No synthetic data. No proprietary datasets. All scripts and data pipelines are reproducible and archived at the Zenodo DOI.

### Five-Substrate Framework

- [Li et al. 2024 Genome Med — MESA test \(4 substrates, colorectal, AUC 0.931, n=690\)](#)
- [Cristiano et al. 2019 Nature — DELFI fragmentomics \(7 cancers, AUC 0.940, n=208\)](#)
- [Mathios et al. 2022 Nat Commun — DELFI 2-year pre-diagnostic window](#)
- [Snyder et al. 2016 Cell — WPS original \(15/15 tissue types\)](#)
- [Doebley et al. 2022 Nat Commun — Griffin nucleosome occupancy \(breast, n=139\)](#)
- [Corces et al. 2018 Science — TCGA ATAC-seq pan-cancer \(23 types\)](#)
- [Esfahani et al. 2022 Cancer Discovery — nucleosome fuzziness \(prostate PDX, n=26\)](#)

### Reference Cell Methylation (Per-Class H<sub>min</sub>)

- [Lister et al. 2009 Nature — hESC H1 \(stem\\_pluri reference\)](#)
- [Lister et al. 2013 Science — frontal cortex neuron \(terminal, global floor 0.756500\)](#)
- [Kozlenkov et al. 2014 Hum Mol Genet — cortical neuron mature](#)
- [Movassagh et al. 2011 NEJM — adult cardiomyocyte \(terminal\)](#)
- [Roadmap Epigenomics Consortium 2015 Nature — 127 reference epigenomes](#)
- [Hannum et al. 2013 Mol Cell — blood methylation aging \(GSE40279, immune\)](#)
- [Moss et al. 2018 Nat Commun — cfDNA tissue-of-origin atlas \(cfDNA % per class\)](#)

### Cancer Validation (G-008, 27/28 TCGA types)

- [TCGA Pan-Cancer Atlas — 28 methylation datasets, all cancer types](#)
- [Ceccarelli et al. 2016 Cell — LGG/GBM \(terminal, largest  \$\Delta A\$  in dataset\)](#)
- [TCGA Research Network 2013 NEJM — AML, Ley et al., n=200 \(immune, primary  \$\beta\$  source\)](#)
- [Chapuy et al. 2018 Nat Med — DLBCL \(immune, n=48\)](#)
- [Radovich et al. 2018 Cancer Cell — TCGA Thymoma \(immune, n=120\)](#)
- [TCGA Research Network 2012 Nature — BRCA, n=825 \(secretory, primary  \$\beta\$  source\)](#)
- [TCGA Research Network 2015 Cell — PRAD prostate molecular taxonomy, n=333 \(secretory\)](#)
- [TCGA Research Network 2017 Cancer Cell — PAAD pancreatic adenocarcinoma \(secretory\)](#)
- [TCGA Research Network 2017 Cell — LIHC hepatocellular carcinoma \(secretory\)](#)
- [Zheng et al. 2016 Cancer Cell — TCGA ACC adrenocortical carcinoma \(secretory\)](#)
- [Robertson et al. 2017 Cancer Cell — UVM \(stem\\_adult\)](#)
- [All TCGA citations per-cancer-type in individual card tables](#)

### Cycling Epithelial Cancers — 14 TCGA Types (Card #5)

- [TCGA Research Network 2012 Nature — COAD + READ colorectal, n=276](#)
- [TCGA Research Network 2014 Nature — LUAD lung adenocarcinoma, n=230](#)
- [TCGA Research Network 2012 Nature — LUSC lung squamous, n=178](#)
- [TCGA Research Network 2014 Nature — BLCA bladder urothelial, n=131](#)
- [TCGA Research Network 2011 Nature — OV ovarian serous, n=489](#)
- [TCGA Research Network 2014 Nature — STAD stomach adenocarcinoma, n=295](#)
- [TCGA Research Network 2017 Nature — CESC cervical squamous, n=228](#)
- [TCGA Research Network 2015 Nature — HNSC head/neck squamous, n=504](#)
- [TCGA Research Network 2013 Nature — KIRC kidney clear cell, n=418](#)
- [TCGA Research Network 2016 NEJM — KIRP kidney papillary, n=161](#)
- [TCGA Research Network 2015 Cell — SKCM skin cutaneous melanoma, n=477](#)

- [TCGA Research Network 2014 Cell — THCA thyroid carcinoma, n=496](#)
- [TCGA Research Network 2013 Nature — UCEC endometrial carcinoma, n=373](#)

## **Stromal-Class Cancers — Sarcomas & Mesothelioma (Card #6)**

- [Abeshouse et al. 2017 Cell — TCGA SARC soft-tissue sarcoma, n=206 \(adult\)](#)
- [Hmeliak et al. 2018 Cancer Discov — TCGA MESO mesothelioma, n=74](#)
- [Crompton et al. 2014 Cancer Discov — Ewing sarcoma pediatric, n=112 WGS](#)
- [Shern et al. 2014 Cancer Discov — rhabdomyosarcoma pediatric, n=147](#)
- [Tirode et al. 2014 Cancer Discov — Ewing sarcoma aggressive subtype, n=299](#)

## **Adult-Stem Cancers — HSC-Origin AML, BCC, MCC, Cholangiocarcinoma (Card #7)**

- [Adelman et al. 2019 Cancer Discov — HSC-enriched aging methylation, n=5-7 per age](#)
- [Beerman et al. 2013 Cell Stem Cell — HSC aging methylation \(GSE44117\)](#)
- [Harms et al. 2015 Cancer Res — Merkel cell carcinoma MCPyV-negative, n=49](#)
- [Farshidfar et al. 2017 Cell Reports — TCGA CHOL cholangiocarcinoma, n=38](#)
- [Ley et al. 2013 NEJM — TCGA AML \(progenitor-lineage\), n=200 \(distinct from HSC-origin\)](#)

## **Pluripotent-Stem Cancers — TGCT (Card #8)**

- [Shen et al. 2018 Cell Reports — TCGA TGCT integrated molecular characterization, n=137](#)
- [Killian et al. 2016 Genome Research — TGCT methylation reprogramming \(pure histology, n=130\)](#)

## **Non-Cancer Disease Validation & Pre-Malignant States**

- [De Jager et al. 2014 Nat Neurosci — ROSMAP AD methylation, n=740 \(terminal\)](#)
- [Shireby et al. 2022 Brain — Brains for Dementia Research AD, n=631 \(terminal\)](#)
- [Lunnon et al. 2014 Nat Neurosci — AD four brain regions \(terminal\)](#)
- [Terman 2004 Age \(Dordr\) — cardiac aging, oxidative stress \(terminal\)](#)
- [Langston & Ballard 1983 N Engl J Med — MPTP Parkinson's disease \(terminal\)](#)
- [Wang et al. 2014 Nat Neurosci — PD methylation changes \(terminal\)](#)
- [Ahrens et al. 2013 Nat Commun — NAFLD hepatocyte methylation \(secretory\)](#)
- [Volkmar et al. 2012 EMBO J — pancreatic  \$\beta\$ -cell T2D \(secretory\)](#)
- [Cruickshanks et al. 2013 Nat Cell Biol — senescent fibroblasts \(stromal\)](#)
- [Steensma et al. 2015 Blood — CHIP clonal hematopoiesis definition \(progenitor\)](#)
- [Malcovati et al. 2017 Blood — CCUS diagnostic criteria \(progenitor\)](#)
- [Jiang et al. 2020 Cell Death Dis — MDS whole-genome methylation progression](#)

## **Progenitor-Lineage Cancers (Pediatric & Neural Progenitor Origin)**

- [Nordlund et al. 2013 Genome Biol — pediatric ALL methylome, n=764 \(B-ALL + T-ALL\)](#)
- [Figueroa et al. 2013 J Clin Invest — integrated epigenetic analysis pediatric ALL](#)
- [Northcott et al. 2017 Nature — medulloblastoma whole-genome, n=491 WGS + 1,256 methylation](#)
- [Capper et al. 2018 Nature — DNA methylation CNS tumor classifier, n>2,800](#)
- [Schwalbe et al. 2017 Lancet Oncol — medulloblastoma methylation subgrouping](#)

## **Vertebrate Lifespan (Nature Aging NATAGING-A13702)**

- [Lowe et al. 2018 Genome Biol — 42 mammalian species methylation](#)
- [Lu et al. 2023 Nature Aging — 185-species pan-mammalian clock](#)
- [Haqhani et al. 2023 Science — mammalian methylation networks](#)
- [Wang et al. 2020 Cell Reports — Labrador retriever aging, n=104 dogs](#)
- [Wilkinson et al. 2021 Nat Commun — bat longevity](#)
- [Tacutu et al. 2018 Nucleic Acids Res — AnAge lifespan database](#)

## IAM Framework Foundations

- [Mahaffey 2026 — Thermodynamic Operating Constraints \(cell thermodynamics paper\)](#)
- [Landauer 1961 IBM J Res Dev — irreversibility and heat generation](#)
- [Jacobson 1995 Phys Rev Lett — Einstein equation of state from thermodynamics](#)
- [Zenodo DOI 10.5281/zenodo.19547624 — IAM-Validation repository](#)
- [GitHub: hmahaffeyges/IAM-Validation — all scripts archived](#)

## Phase B Validation Cohorts — Post-Breach Trajectory Predictions

- [Stupp et al. 2005 NEJM — concurrent radiochemotherapy GBM protocol](#)
- [Gilbert et al. 2013 J Clin Oncol — RTOG 0525 dose-intensification trial \(n=833 GBM with MGMT stratification\). G-2026-P025 reanalysis target.](#)
- [DiNardo et al. 2020 NEJM — VIALE-A azacitidine+venetoclax in older AML \(n=431\). G-2026-P026b target.](#)
- [Taniguchi et al. 2021 Nat Med — GALAXY CRC MRD post-resection ctDNA \(n=1,000+ stage II-III\). G-2026-P027 target.](#)
- [Cristiano et al. 2019 Nature — DELFI lung cancer detection and pre-diagnostic window reanalysis framework for G-2026-P028 \(EGFR-mutant LUAD osimertinib resistance prediction\).](#)
- [OSIC IPF Biobank — Open Source Imaging Consortium, prospective IPF cohort with serial imaging and archived serum. G-2026-P012 target.](#)
- [Detels et al. 2012 Int J Epidemiol — MACS \(Multicenter AIDS Cohort Study\). 40+ year HIV longitudinal cohort with archived blood samples. G-2026-P026 target.](#)
- [Adimora et al. 2018 J Womens Health — WIHS \(Women's Interagency HIV Study\). Companion HIV cohort to MACS. G-2026-P026 target.](#)
- [Hardy et al. 2020 BMJ Open — ACTG \(AIDS Clinical Trials Group\) archived serial blood. G-2026-P026 target for ART response trajectory.](#)
- [Sudlow et al. 2015 PLoS Med — UK Biobank cohort design and baseline characterization \(n~500,000\). G-2026-P035, P038 target.](#)
- [LITMUS Consortium — Liver Investigation: Testing Marker Utility in Steatohepatitis. European NAFLD/NASH biomarker cohort, n=2,000+. G-2026-P032 target.](#)
- [Krop et al. 2019 Clin Cancer Res — BRCA DCIS molecular stratification framework informing G-2026-P030.](#)
- [COG \(Children's Oncology Group\) — archived pediatric sarcoma protocols \(Ewing VDC-IE, rhabdomyosarcoma VAC\). G-2026-P034 target.](#)

## Post-Breach Trajectory Validation Cohorts (G-2026-P023 through P040)

- [Stupp et al. 2005 NEJM — GBM concurrent chemoradiation protocol \(Stupp protocol\)](#)
- [Gilbert et al. 2013 J Clin Oncol — RTOG 0525 dose-intensification trial \(n=833 GBM\)](#)
- [DiNardo et al. 2020 NEJM — VIALE-A azacitidine+venetoclax trial \(n=431 AML\)](#)
- [Kotani et al. 2023 Nat Med — GALAXY CRC prospective ctDNA MRD cohort](#)
- [Cristiano et al. 2019 Nature — DELFI 7-cancer AUC 0.940 validation \(cross-reference\)](#)
- [OSIC \(Open Source Imaging Consortium\) IPF Biobank](#)
- [MACS \(Multicenter AIDS Cohort Study\) — NIH-funded longitudinal HIV cohort](#)
- [WIHS \(Women's Interagency HIV Study\) — NIH-funded longitudinal HIV cohort](#)
- [ACTG \(AIDS Clinical Trials Group\) — NIH HIV therapeutics trials network](#)
- [LITMUS \(Liver Investigation: Testing Marker Utility in Steatohepatitis\)](#)
- [UK Biobank — n=500,000 prospective cohort with longitudinal blood and outcomes](#)
- [CHIP / CCUS / MDS — Steensma 2015 Blood \(CHIP definition, cross-reference above\)](#)
- [ELN 2022 risk classification — Döhner et al. Blood 2022 \(AML prognostic\)](#)
- [DELFI validation — Mathios et al. 2022 Nat Commun \(pre-diagnostic window, cross-reference\)](#)
- [ROSMAP longitudinal AD cohort — De Jager 2014 Nat Neurosci \(cross-reference above\)](#)
- [TCGA ATAC-seq / Corces 2018 — chromatin accessibility reanalysis target \(G-2026-P023\)](#)

## Multi-Class Drift Cascade (VAL-037 through VAL-046, April 2026)

- [Zeng et al. 2026 Nat Cancer — pan-cancer plasma cfDNA compendium \(n=1,294, 14 types\). VAL-038 primary source.](#)
- [Liu et al. 2020 Ann Oncol — GRAIL multi-cancer detection with tissue-of-origin localization \(n=6,689\). VAL-041 deconvolution reference.](#)
- [Kadota et al. 2014 Am J Respir Crit Care Med — lung adenocarcinoma distance-annotated methylation series. VAL-039 lung source.](#)
- [Teschendorff et al. 2016 Genome Med — breast adjacent-normal field defect methylation outliers. VAL-039 breast source.](#)

- [Shen et al. 2005 Cancer Res — MGMT promoter methylation field defect in sporadic CRC. VAL-039 colon source.](#)
- [Damaschke et al. 2017 Cancer Epidemiol Biomarkers Prev — prostate zonal methylation gradient. VAL-039 prostate source.](#)
- [Villanueva et al. 2015 Hepatology — HCC methylation-based prognosis with cirrhotic gradient. VAL-039 HCC source.](#)
- [Kang et al. 2008 Am J Pathol — gastric carcinoma intestinal metaplasia field effect. VAL-039 gastric source.](#)
- [De Jager et al. 2014 Nat Neurosci — ROSMAP AD cortex methylation \(n=708\). VAL-040 primary source.](#)
- [Shireby et al. 2022 Brain — Brains for Dementia Research AD cortex methylation \(n=1,408\). VAL-040 primary source.](#)
- [Nabais et al. 2021 Genome Biol — peripheral blood AD meta-analysis \(n=3,424\). VAL-040 immune-class AD source.](#)
- [Lunnon et al. 2014 Nat Neurosci — entorhinal cortex AD methylation. VAL-040 temporal cortex source.](#)
- [Widschwendter et al. 2021 Cell Rep Med — WID-CIN cervical progression \(n=2,254\). VAL-042 cervical source.](#)
- [Jammula et al. 2020 Gastroenterology — Barrett's esophagus methylation progression subtypes. VAL-042 esophageal source.](#)
- [Jerónimo et al. 2008 Clin Cancer Res — prostate PIN→HGPIN→metastatic progression. VAL-042 prostate source.](#)
- [Luo et al. 2014 Gastroenterology — colon adenoma-carcinoma sequence methylation pathways. VAL-042 colon source.](#)
- [Yoshizato et al. 2020 Blood — CHIP→MDS→AML clonal evolution. VAL-042 AML source.](#)
- [Pal et al. 2016 Cancer Res — canine mammary tumor methylation. VAL-043 canine source.](#)
- [Beck et al. 2020 Vet Comp Oncol — canine diffuse large B-cell lymphoma methylation. VAL-043 canine source.](#)
- [Decker et al. 2015 PLoS Genet — canine bladder TCC BRAF V600E mutation. VAL-043 canine source.](#)
- [Parikh et al. 2019 Nat Med — CRC FOLFOX resistance ctDNA serial. VAL-044 CRC source.](#)
- [Stover et al. 2018 J Clin Oncol — metastatic TNBC ctDNA trajectory. VAL-044 BRCA source.](#)
- [Cabel et al. 2018 Ann Oncol — melanoma anti-PD-1 ctDNA monitoring. VAL-044 melanoma source.](#)
- [Shen et al. 2018 Cell — plasma cfDNA methylome pan-cancer. VAL-045 TGCT reference.](#)
- [Killian et al. 2016 Cell Rep — seminoma hypomethylation and germ cell tumor epigenetics. VAL-045 seminoma source.](#)
- [Kresovich et al. 2019 JNCI — Sister Study pre-diagnostic breast cancer \(n=2,776\). VAL-046 capstone source.](#)
- [Hillary et al. 2020 Clin Epigenetics — UK Biobank methylation pre-diagnostic. VAL-046 UK Biobank source.](#)
- [Horvath et al. 2014 Genome Biol — Health ABC liver methylation aging. VAL-046 Health ABC source.](#)
- [Hou et al. 2012 Am J Epidemiol — Nurses' Health LINE-1 pre-diagnostic colorectal. VAL-046 Nurses source.](#)
- [Horvath 2015 Aging — Rotterdam Study aging-methylation. VAL-046 Rotterdam source.](#)
- [Alisch et al. 2012 Genome Res — pediatric age-associated methylation. Healthy baseline reference source.](#)
- [Horvath 2013 Genome Biol — 353-CpG multi-tissue epigenetic clock. Healthy baseline reference source.](#)

GLOSSARY

Definitions of the technical terms used throughout this publication, organized by category: Physics & Derivation, Biology & Substrates, Statistical & Validation, and Framework & Clinical Terms. Every term traces to either a published primary source or a derived quantity in the framework.

PHYSICS & DERIVATION

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A-score                    | Dimensionless ratio $A = H(\text{value}) / H_{\min}(\text{class, substrate})$ . Formula is unfloored — no $\max()$ clamp. The central measurement in GAPE. $A = 1.00$ marks the architectural commitment point (reference $H_{\min}$ ). Healthy cells typically read $A \approx 0.95\text{--}1.00$ due to normal biological variation around the MCMC-derived $H_{\min}$ . $A > 1.00$ means an accessible entropy gap (C3) has opened above the floor — standard cancer elevation. $A < 0.95$ is inversion territory (documented in seminoma, senescent fibroblasts, aged HSCs). See Section 2.1b for the full axis treatment. |
| A_ceiling                  | The maximum A-score physically achievable on a given class×substrate pairing: $A_{\text{ceiling}} = 1 / H_{\min}(\text{class, substrate})$ . Reached when $\beta = 0.5$ (maximum binary entropy $H = 1.0$ ). No sample biology can produce a higher A on that substrate.                                                                                                                                                                                                                                                                                                                                                       |
| A_combined                 | AUC-weighted combined A-score across all five substrates: $A_{\text{combined}} = \Sigma(AUC_i \times A_i) / \Sigma(AUC_i)$ . The headline score. Reduces measurement noise by roughly $\sqrt{5}$ across five correlated physical windows.                                                                                                                                                                                                                                                                                                                                                                                      |
| A_active                   | AUC-weighted mean over only the non-saturated substrates for a specific sample. A substrate is flagged saturated when A is within 0.005 of its ceiling. A_active is the signal that continues to respond after some substrates have pinned at ceiling — central for chemotherapy response trajectories and reserve-depletion signals.                                                                                                                                                                                                                                                                                          |
| C1, C2, C3                 | Three-component decomposition of Shannon entropy. C1 = universal Landauer floor (0.756500, neuron reference). C2 = class-specific architecture overhead above C1. C3 = accessible gap above $H_{\min} = \max(0, H - H_{\min})$ . Healthy cells: C3 $\approx 0$ . Cancer: C3 = 8-15%.                                                                                                                                                                                                                                                                                                                                           |
| f_C3                       | Fraction of total entropy in the accessible gap: $f_{C3} = C3 / H(\beta)$ . The component on which every therapeutic intervention operates. Rising $f_{C3}$ under treatment indicates non-response. Declining $f_{C3}$ indicates response.                                                                                                                                                                                                                                                                                                                                                                                     |
| H(β) — Shannon entropy     | Shannon binary entropy of a Bernoulli(β) variable: $H(\beta) = -\beta \cdot \log_2(\beta) - (1-\beta) \cdot \log_2(1-\beta)$ . $H(0) = H(1) = 0$ . $H(0.5) = 1$ (maximum).                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| H_min                      | The class-specific and substrate-specific minimum Shannon entropy consistent with maintained cell identity. Derived from the Landauer cost of DNMT1 maintenance across 19.6M CpG sites per cell division. Calibrated via MCMC posteriors (G-002 methyl, G-003b four other substrates).                                                                                                                                                                                                                                                                                                                                         |
| H_MIN_GLOBAL               | The lowest H_min across all classes: 0.756500, from frontal cortex neuron (Lister 2013). Anchors the global C1 Landauer floor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Landauer cost              | Thermodynamic minimum energy to erase one bit of information: $E = k_B \times T \times \ln(2)$ . At 37°C (310.15 K): $2.97 \times 10^{21}$ J per bit. Landauer 1961 IBM J Res Dev.                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Mahaffey Number            | $M = \Delta G_{ATP} / (R \cdot T_{\text{body}}) = 20.94$ at 37°C - the ATP ordering budget the cell pays per irreversible write, in units of the thermal quantum. <b>One number for the cell, not one per class.</b> The per-class $n_{\text{bio}}$ of Issue 002 was an early QAPE-era form of the same ratio and is no longer used; its class-specific values awaited a G-007 MCMC run that was never made. M is not $H_{\min}$ : $H_{\min}$ is the per-class entropy floor in bits, forty values across eight classes and five substrates. See §5A.1.                                                                        |
| Reserve (clinical meaning) | The C3 headroom remaining between a cell's current A-score and its class-substrate ceiling. Declining reserve under treatment indicates the epigenomic capacity for further response is being consumed.                                                                                                                                                                                                                                                                                                                                                                                                                        |

BIOLOGY & SUBSTRATES

|                   |                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| β (beta value)    | Fraction methylated at a CpG site or averaged across an architecture-class locus panel. $\beta \in [0,1]$ . Primary input to the methylation $H(\beta)$ .                                                                                                                                                                                                              |
| cfDNA             | Cell-free DNA circulating in plasma. Shed from apoptotic and necrotic cells across all tissues, weighted by tissue turnover. ~70% immune, ~12% cycling epithelial, ~8% secretory in standard venous draw.                                                                                                                                                              |
| CpG / CpH         | CpG: a cytosine followed by a guanine on the same strand — the standard site of mammalian DNA methylation. CpH: cytosine followed by any other base — mostly unmethylated in somatic cells but methylated in neurons and embryonic stem cells.                                                                                                                         |
| CHIP / CCUS / MDS | Clonal Hematopoiesis of Indeterminate Potential: age-associated expansion of HSC clones carrying leukemia-associated mutations; typically benign. Clonal Cytopenias of Undetermined Significance: CHIP with unexplained cytopenias. Myelodysplastic Syndromes: frank malignant transformation. The framework's prediction G-2026-P013 targets the CHIP→MDS transition. |
| DELFI             | DNA Evaluation of Fragments for early Interception. Cristiano et al. 2019 Nature. Measures short-fragment fraction (100-150bp / total) in cfDNA WGS. Substrate 5 in the five-substrate framework.                                                                                                                                                                      |
| DNMT1             | DNA methyltransferase 1. Maintenance enzyme that copies methylation pattern from parent to daughter strand during DNA replication. Finite fidelity (~1 error per $10^5\text{--}10^6$ CpG sites per division) — the biological rate-limit on $H_{\min}$ .                                                                                                               |
| Fragment size     | cfDNA fragment length distribution. Healthy fragments cluster around 166 bp (one nucleosome + linker). Tumor-derived cfDNA over-represents 100-150 bp fragments due to altered nucleosome spacing. Substrate 5.                                                                                                                                                        |

|                                          |                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hypomethylation (seminoma)</b>        | Global reduction of methylation $\beta$ toward zero. In seminoma, $\beta$ falls toward 0.17-0.20 as malignant germ cells revert toward the primordial germ cell (PGC) state. Drives the Seminoma Hypomethylation Inversion.                                                                                                                                     |
| <b>Nucleosome occupancy</b>              | Mean probability that a given genomic position is wrapped in a histone octamer. Measured via ATAC-seq or MESA. Substrate 2.                                                                                                                                                                                                                                     |
| <b>Nucleosome fuzziness</b>              | Positional precision of nucleosomes relative to their reference positions (0 = precise, 1 = maximally fuzzy). Measured via NucleoATAC. Substrate 3.                                                                                                                                                                                                             |
| <b>PGC</b>                               | Primordial Germ Cell. The most pluripotent somatic cell state known, with near-zero methylation. Seminomas revert toward this state, producing the inversion signature.                                                                                                                                                                                         |
| <b>Pluripotent / GCNIS / TGCT</b>        | Pluripotent: capable of differentiating into all three germ layers. GCNIS: Germ Cell Neoplasia In Situ, the preinvasive precursor to all postpubertal TGCT. TGCT: Testicular Germ Cell Tumor.                                                                                                                                                                   |
| <b>WPS (Windowed Protection Score)</b>   | Fraction of cfDNA reads whose endpoints fall within a nucleosome-protected window at a given genomic position. Snyder et al. 2016 Cell. Substrate 4.                                                                                                                                                                                                            |
| <b>Yamanaka factors</b>                  | Four transcription factors (OCT4, SOX2, KLF4, c-MYC) that reprogram somatic cells back to induced pluripotent state (Yamanaka 2006). Work by driving methylation toward high-entropy pluripotent configuration. Pharmacologic dose-response is non-monotone — excess dose produces the Differentiation Dose Inversion (Section 2.4).                            |
| <b>IDH mutation / 2-hydroxyglutamate</b> | Isocitrate dehydrogenase (IDH1/IDH2) mutations occur in $\geq 80\%$ of lower-grade gliomas and produce the oncometabolite 2-hydroxyglutarate, which inhibits TET enzymes. Result: the hyper-methylator phenotype (G-CIMP) characteristic of LGG. Key driver of terminal-class post-breach hyperentropy signature.                                               |
| <b>TET enzymes</b>                       | Ten-Eleven Translocation methylcytosine dioxygenases (TET1/2/3) actively demethylate 5-methylcytosine. Inhibition by oncometabolites (e.g., 2-hydroxyglutarate in IDH-mutant tumors) drives aberrant hypermethylation. Counterpart to DNMT1 maintenance methylation.                                                                                            |
| <b>G-CIMP</b>                            | Glioma CpG Island Methylator Phenotype. Defined by Ceccarelli 2016 Cell in TCGA LGG/GBM analysis. Hyper-methylation pattern characteristic of IDH-mutant gliomas. Framework reads G-CIMP as terminal-class A-score elevation driven by methylation redistribution rather than global loss.                                                                      |
| <b>MGMT promoter methylation</b>         | Methylation status of the O6-methylguanine-DNA methyltransferase (MGMT) gene promoter. MGMT-methylated GBM responds better to temozolomide. Framework prediction G-2026-P025: MGMT status stratifies post-breach A_active trajectory in GBM on Stupp protocol.                                                                                                  |
| <b>T-cell exhaustion / PD-1</b>          | Progressive loss of effector function in chronically stimulated T cells. Marked by sustained PD-1 expression and compromised DNMT1 maintenance fidelity. Framework predicts detectable immune-class A-score elevation in chronic viral infection (HIV), senescent CD8+ populations, and tumor-infiltrating lymphocytes in checkpoint-blockade-eligible cancers. |

## STATISTICAL & VALIDATION

|                                  |                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AUC</b>                       | Area Under the receiver operating Curve. Single-substrate discrimination performance for binary classification (cancer vs healthy). Range 0.5 (chance) to 1.0 (perfect).                                                                                                                                                                                                       |
| <b>Bootstrap CI</b>              | Confidence interval derived by resampling. For H_min posteriors, leave-one-reference-out bootstrap confirms no single dataset drives the result. Mean $ \Delta $ between full-data and leave-one-out posteriors: 0.168% across the 32 non-methylation class $\times$ substrate H_min values (VAL-031); 0.060% across the 8 methylation values (PROC-HMIN-BOOT-01, 2026-09-20). |
| <b>MCMC</b>                      | Markov Chain Monte Carlo. Statistical method for sampling from posterior distributions. G-002 (methylation H_min): 17 chains, R-hat $< 1.001$ , $8 \times 10^5$ samples. G-003b (four other substrates): same rigor.                                                                                                                                                           |
| <b>Pre-diagnostic window</b>     | The time interval between a detectable framework signal and clinical diagnosis. Mathios 2022 Nat Commun established ~24 months for DELFI in lung cancer. Framework extension predicts 12-18 months for cycling-class solid tumors.                                                                                                                                             |
| <b>R-hat</b>                     | Gelman-Rubin diagnostic for MCMC chain convergence. R-hat $< 1.01$ is convergence; $< 1.001$ indicates very tight convergence. All framework H_min posteriors satisfy R-hat $< 1.001$ .                                                                                                                                                                                        |
| <b>Sensitivity / specificity</b> | Sensitivity: fraction of true positives correctly identified. Specificity: fraction of true negatives correctly identified. Framework targets high specificity (low false-positive rate) to support clinical serial monitoring rather than one-shot screening.                                                                                                                 |

## FRAMEWORK & CLINICAL

|                                |                                                                                                                                                                                                                    |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Architecture class</b>      | Cell-type grouping defined by shared thermodynamic floor (H_min). Eight classes: cycling, secretory, immune, progenitor, stromal, terminal, stem_adult, stem_pluri. Thermodynamically defined, not histologically. |
| <b>BREACH</b>                  | Tier assignment when $A \geq 1.10$ . Indicates epigenomic architecture maintenance has failed beyond what metabolic intervention can restore. Structural transition.                                               |
| <b>Cross-class propagation</b> | A cancer arising in one class producing signal in another class — e.g., breast metastasis to bone marrow producing immune-class elevation. Scenario 5.5; prediction G-2026-P022.                                   |
| <b>DETECTABLE</b>              | Tier assignment when $1.05 \leq A < 1.07$ . Clinical workup recommended per class-specific differential.                                                                                                           |

|                                       |                                                                                                                                                                                                                                                                    |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Divergence signature</b>           | A multi-substrate pattern where substrates move in different directions — e.g., seminoma's A_methyl decreasing while A_nucl, A_wps, A_fuzz, A_frag increase. Primary detection signal for Pluripotent-class malignancy.                                            |
| <b>Floor</b>                          | H_min(class, substrate) — the minimum entropy consistent with the cell maintaining its class identity. Derived, not fitted.                                                                                                                                        |
| <b>Inversion</b>                      | A named signature where A moves in the opposite direction from standard cancer elevation. Three identified: Seminoma Hypomethylation (Pluripotent), Differentiation Dose (Pluripotent research), Niche Depletion (Adult Stem). See Section 2.4.                    |
| <b>MARGINAL</b>                       | Tier assignment when $1.01 \leq A < 1.05$ . Monitor; repeat at next scheduled timepoint.                                                                                                                                                                           |
| <b>Mahaffey value</b>                 | Historical name for H_min in early drafts. Now standardized as H_min. The class-and-substrate-specific entropy floor.                                                                                                                                              |
| <b>NORMAL</b>                         | Tier assignment when $A < 1.01$ . Consistent with age-expected healthy.                                                                                                                                                                                            |
| <b>Saturation (runtime)</b>           | Sample-specific substrate flag: A within 0.005 of A_ceiling. Saturated substrates are excluded from A_active computation.                                                                                                                                          |
| <b>Saturation (structural)</b>        | Class-level property: a substrate whose ceiling is itself below BREACH. Cannot discriminate disease severity above BREACH on that substrate. Detection must rely on substrates whose ceilings exceed BREACH.                                                       |
| <b>Substrate</b>                      | One of five physical measurement windows: methylation, nucleosome occupancy, nucleosome fuzziness, WPS, fragment size. All five measure departures from the same underlying cellular identity information through different physical lenses.                       |
| <b>Tier system</b>                    | A-score thresholds: NORMAL ( $A < 1.01$ ), MARGINAL (1.01-1.05), DETECTABLE (1.05-1.07), URGENT (1.07-1.10), BREACH ( $A \geq 1.10$ ). All thresholds framework-derived; none fit to cancer data.                                                                  |
| <b>Trajectory (dA/dt)</b>             | Rate of change in A-score over serial samples. The framework's strongest clinical signal — a single elevated A may be age-consistent, but a rising A is not.                                                                                                       |
| <b>URGENT</b>                         | Tier assignment when $1.07 \leq A < 1.10$ . Workup indicated. Corresponds roughly to the Warburg transition in cycling-class cancer.                                                                                                                               |
| <b>Warburg transition</b>             | Metabolic shift from OxPhos to aerobic glycolysis in malignant cells. Corresponds to $A \approx 1.07$ URGENT threshold. Past this point, metabolic levers cannot restore the cell to floor because glycolytic commitment is structural.                            |
| <b>Warburg boundary (post-breach)</b> | Qualitative post-breach zone boundary at $A \approx 1.15$ , separating the metabolic-window zone (metabolic intervention still works) from the structural-only zone (glycolytic program locked in, glucose becomes neutral or harmful). See Section 2.6.           |
| <b>Glucose inversion</b>              | Qualitative post-breach boundary at $A \approx 1.25$ where adding glucose actively accelerates disease progression rather than supporting cellular function. Past this point, TPN/high-glucose supportive care is counterproductive. See Section 2.6.              |
| <b>Point of no return</b>             | Qualitative post-breach boundary at $A \approx 1.40+$ where epigenomic reserve (f_C3 headroom) is depleted and no currently-available structural intervention can recover cellular function. Framework use shifts from therapeutic to prognostic. See Section 2.6. |
| <b>Metabolic window zone</b>          | Post-breach zone at $A \approx 1.10$ -1.15 where metabolic intervention (DCA, 2-DG, ketosis) can still push cells back toward OxPhos. The widest therapeutic window post-breach.                                                                                   |
| <b>Structural-only zone</b>           | Post-breach zone at $A \approx 1.15$ -1.25. Metabolic interventions have failed; structural interventions (DNMTi, HDACi, synthetic lethality, reprogramming) are the primary lever. Glucose becomes neutral to harmful.                                            |
| <b>Palliative range zone</b>          | Post-breach zone at $A \approx 1.25$ -1.40. Aggressive combination therapy with palliative intent; clinical trial eligibility becomes the framework's primary therapeutic route. Glucose supportive care actively harmful.                                         |
| <b>End of life zone</b>               | Post-breach zone at $A \approx 1.40+$ . Cellular reserve depleted; framework use shifts from therapy to prognosis. Comfort care, symptom management, family-centered planning.                                                                                     |
| <b>Post-breach trajectory</b>         | The time course of A-score and substrate divergence patterns past the ceiling at $A = 1.10$ . Every class card has a Post-Breach Trajectory subsection with class-specific Known / Unknown / Test structure.                                                       |
| <b>Z-score (age-adjusted)</b>         | $Z = (A_{\text{observed}} - A_{\text{predicted}}(\text{age, class})) / \sigma_{\text{cohort}}$ . Standardized departure from the age-expected baseline. $ Z  < 1$ : age-expected. $ Z  > 3$ : strong signal warranting workup.                                     |

## CLINICAL & TREATMENT PROTOCOLS

|                         |                                                                                                                                                                                                                                                               |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>BEP</b>              | Bleomycin-Etoposide-Cisplatin. Standard chemotherapy regimen for advanced testicular germ cell tumors (TGCT). 95% cure rate at stage I-II. Framework prediction G-2026-P017 targets BEP response trajectory via A_methyl toward healthy hESC reference.       |
| <b>DCIS</b>             | Ductal Carcinoma In Situ. Non-invasive breast lesion diagnosed in ~50,000 US women annually. Current medicine cannot distinguish indolent from active DCIS at diagnosis; framework prediction G-2026-P030 targets this stratification with baseline A_active. |
| <b>FOLFOX / FOLFIRI</b> | Standard first-line chemotherapy regimens for metastatic colorectal cancer. FOLFOX: 5-FU + leucovorin + oxaliplatin. FOLFIRI: 5-FU + leucovorin + irinotecan.                                                                                                 |
| <b>G-CIMP</b>           | Glioma CpG Island Methylator Phenotype. Hypermethylator subtype of LGG/GBM defined by Ceccarelli 2016, characterized by IDH mutation and widespread promoter hypermethylation. Drives the terminal-class A-score elevation signal for glioma.                 |

|                                          |                                                                                                                                                                                                                                                                                                                             |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IDH mutation / 2-hydroxyglutarate</b> | Isocitrate dehydrogenase 1/2 mutations produce the oncometabolite 2-hydroxyglutarate (2-HG), which inhibits $\alpha$ -ketoglutarate-dependent enzymes including TET demethylases. Drives G-CIMP hypermethylator phenotype in LGG ( $\geq 80\%$ of cases) and a subset of GBM.                                               |
| <b>MGMT</b>                              | O6-methylguanine-DNA methyltransferase. DNA repair enzyme. MGMT promoter methylation silences the gene, rendering GBM cells sensitive to temozolomide. Stratification biomarker for Stupp protocol response.                                                                                                                |
| <b>NASH / NAFLD</b>                      | Non-Alcoholic Steatohepatitis / Non-Alcoholic Fatty Liver Disease. Secretory-class non-cancer failure mode. Framework reads NAFLD at $A \approx 1.015$ - $1.055$ (MARGINAL to DETECTABLE) — detectable but pre-malignant. G-2026-P032 tests NAFLD-to-HCC progression via LITMUS cohort.                                     |
| <b>PD-1 / T-cell exhaustion</b>          | Programmed cell death protein 1. Immune checkpoint receptor upregulated on chronically activated T cells. T-cell exhaustion is the functional state characterized by high PD-1, loss of effector function, and compromised DNMT1 fidelity — directly relevant to HIV/AIDS prediction G-2026-P026.                           |
| <b>RECIST</b>                            | Response Evaluation Criteria in Solid Tumors. Radiographic response assessment standard. Framework predictions frequently benchmarked against RECIST at 6-month follow-up.                                                                                                                                                  |
| <b>Stupp protocol</b>                    | Standard GBM treatment: concurrent radiation + temozolomide, followed by six cycles of adjuvant temozolomide. Stupp 2005 NEJM. Framework prediction G-2026-P025 targets post-Stupp A_active trajectory for progression-free survival prediction.                                                                            |
| <b>TET enzymes</b>                       | Ten-Eleven Translocation methylcytosine dioxygenases (TET1, TET2, TET3). Active DNA demethylation enzymes that convert 5-methylcytosine to 5-hydroxymethylcytosine. Inhibited by 2-hydroxyglutarate in IDH-mutant tumors, driving hypermethylation. TET2 mutations are the second most common CHIP/AML driver after DNMT3A. |
| <b>Yamanaka factors</b>                  | Oct4, Sox2, Klf4, and c-Myc (OSKM). Four transcription factors that convert somatic cells to induced pluripotent stem cells (iPSCs). Framework prediction G-2026-P016 targets successful-vs-aberrant reprogramming discrimination via the Differentiation Dose Inversion.                                                   |

## RESEARCH COHORTS & BIOBANKS

|                                          |                                                                                                                                                                                                                                                             |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ACTG (AIDS Clinical Trials Group)</b> | NIH-funded clinical trials network for HIV/AIDS therapeutics. Archived serial blood samples across multiple trials provide the longitudinal cohort for framework prediction G-2026-P026 (immune-class A trajectory under ART).                              |
| <b>GALAXY (CRC MRD)</b>                  | Japanese prospective CRC cohort with $n=1,000+$ stage II-III patients, serial cfDNA post-resection, and 2-year recurrence outcomes. Target cohort for framework prediction G-2026-P027 (post-resection A_active vs ctDNA-positive/negative classification). |
| <b>LITMUS (NAFLD)</b>                    | Liver Investigation: Testing Marker Utility in Steatohepatitis. European NAFLD Biomarkers Consortium, $n=2,000+$ biopsy-confirmed patients with longitudinal serum archives. Target for G-2026-P032 (NAFLD-to-HCC progression).                             |
| <b>MACS / WIHS</b>                       | Multicenter AIDS Cohort Study / Women's Interagency HIV Study. Longitudinal HIV cohorts with blood archives spanning 40+ years. Target for G-2026-P026 (immune-class A trajectory correlation with CD4+ recovery under ART).                                |
| <b>OSIC (IPF Biobank)</b>                | Open Source Imaging Consortium IPF Biobank. Target for framework prediction G-2026-P012 (combined A-score trajectory distinguishing progressive from stable IPF with $AUC \geq 0.80$ , outperforming serial FVC).                                           |
| <b>ROSMAP</b>                            | Religious Orders Study and Memory Aging Project. Longitudinal AD cohort (De Jager 2014, $n=740$ ) with blood and brain samples archived plus subsequent AD diagnosis outcomes. Target for G-2026-P024 (MCI-to-AD conversion trajectory).                    |
| <b>RTOG 0525</b>                         | Radiation Therapy Oncology Group 0525 dose-intensification trial ( $n=833$ GBM patients with MGMT stratification and serial imaging). Target for framework prediction G-2026-P025 (post-Stupp A_active trajectory for 6-month PFS prediction).              |
| <b>TCGA / GDC</b>                        | The Cancer Genome Atlas / Genomic Data Commons. 28 cancer types methylation datasets, 23 cancer types ATAC-seq (Corces 2018). Primary source for all G-008 cancer validations in the framework.                                                             |
| <b>UK Biobank</b>                        | Prospective cohort of $\sim 500,000$ UK adults with baseline blood samples and health outcome follow-up. Target for G-2026-P035 (HSC aging to hematologic malignancy prediction) and G-2026-P038 (CHIP progression to MDS).                                 |
| <b>VIALE-A</b>                           | Phase III trial of azacitidine + venetoclax vs azacitidine alone in older AML patients unfit for intensive chemotherapy ( $n=431$ ). DiNardo 2020 NEJM. Target for G-2026-P026b (A_active decline slope predicting 12-month overall survival).              |

## CASCADE TERMS (VAL-037 through VAL-046)

|                                                         |                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Adjacent-normal tissue (architecturally drifted)</b> | Tissue immediately neighboring a tumor that appears histologically normal but sits at $\Delta A = +0.036$ above true-healthy reference on average across 24 TCGA cancer types (VAL-037). "Adjacent normal" in a pathology report is not architecturally healthy — it is drifted. Elevation extends 5-10 cm from the tumor margin (VAL-039 spatial gradient). |
| <b>Architectural recovery</b>                           | Observed decrease in A-score toward NORMAL tier in patients responding to cancer therapy (VAL-044, 5/5 clinical trials). Complete responders approach $A \approx 1.00$ . Non-responders remain elevated. The clinical monitoring axis complementary to architectural drift.                                                                                  |

|                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Capstone finding (VAL-046)</b>              | The central multi-class drift hypothesis test: do future-cancer patients show baseline architectural elevation before clinical diagnosis? Result: across seven cohort-cancer combinations (Sister Study n=2,776 + six others), future-cancer participants show mean $\Delta A = +0.014$ above matched controls, detectable 2-5 years pre-diagnosis, across $\geq 2$ architecture classes.                                                                                                                     |
| <b>Class-universal inversion (pluripotent)</b> | VAL-045 refinement: because $H_{\min\_methyl} = 0.9822$ sits very close to the Shannon ceiling (1.000), the pluripotent methylation window above floor is too narrow to accommodate upward departure. All TGCT histologies land in $A_{\text{methyl}}$ inversion territory; seminoma is the extreme case (divergence $2.1\times$ others). Specificity is in divergence magnitude, not direction.                                                                                                              |
| <b>Field-effect gradient</b>                   | Spatially graded departure of A-score as a function of distance from the tumor. VAL-039 confirmed monotonic decay across 6 cancers: tumor $\rightarrow$ near-adjacent $\rightarrow$ far-adjacent (5-10 cm) $\rightarrow$ true-healthy, with far-adjacent tissue remaining elevated $\Delta A = +0.025$ above true-healthy.                                                                                                                                                                                    |
| <b>Healthy baseline reference tables</b>       | 80-cell reference (8 architecture classes $\times$ 10 age decades) compiled from Hannum 2013, Horvath 2013, Roadmap 2015, Moss 2018, Lister 2013, Alisch 2012. Each cell contains $\beta_{\text{mean}}$ , $\beta_{\text{sd}}$ , $n_{\text{samples}}$ , $A_{\text{mean}}$ , $A_{\text{sd}}$ , and p10/p25/p50/p75/p90 percentiles. Anchors age-stratified interpretation of any patient A-score. Only terminal class crosses MARGINAL threshold ( $A \geq 1.01$ ) within typical healthy lifespan (age 80-89). |
| <b>Honest negative</b>                         | A pre-specified prediction that fails in a way that confirms the framework's own prior finding in negative form. VAL-038 is the framework's honest negative: Spearman $\rho = -0.02$ between tissue-level predicted $\Delta A$ and bulk plasma alteration rate, confirming VAL-002's finding that plasma depends on shedding kinetics, not architectural departure, so deconvolution is required for plasma-based scoring. Failure is a feature that defines framework boundaries.                            |
| <b>Multi-class drift signature</b>             | Coordinated elevation of A-score across $\geq 2$ architecture classes in the same patient, observed in both future-cancer cohorts (VAL-046) and Alzheimer's disease (VAL-040). The framework's pre-diagnostic clinical signal: three classes elevated simultaneously (e.g., immune + secretory + stromal) is more specific than any single-class elevation alone.                                                                                                                                             |
| <b>Pre-diagnostic window</b>                   | The 2-5 year interval before clinical diagnosis during which VAL-046 found detectable multi-class architectural elevation in future-cancer participants. Clinical implication: GAPE may function as a susceptibility flag years before conventional diagnostics, analogous to the role troponin plays for cardiac state.                                                                                                                                                                                      |
| <b>Tissue-of-origin deconvolution</b>          | Computational method (Moss 2018, Liu 2020) that resolves bulk plasma cfDNA into per-tissue $\beta$ values using tissue-specific methylation markers. When followed by per-tissue A-score against class-specific $H_{\min}$ (VAL-041), correctly identifies the primary cancer site in 10 of 10 cases (100% top-1 localization). The clinical bridge between honest bulk-plasma null (VAL-038) and functional plasma-based cancer detection.                                                                   |

## GLOSSARY — CMB AND CHAIN TERMS

For a reader who knows bootstrapping but not MCMC, and has never met a Mahalanobis hull or a HEALPix sky. Each definition is taken from the source file named in brackets.

**Cosmic Methylome Background (CMB)** — The patient's per-CpG map, named for its cosmological twin. The chain is built on the Planck pipeline stage for stage: raw intensities -> calibration -> an all-sky map -> component separation -> a statistic against a derived reference -> null tests -> sky rendering. The one deliberate departure is foreground subtraction (see Foreground). [Biological\_Physics/README.md]

**Brilliance / brightness** — The Atlas's per-CpG posterior mean and sd for a class - a reference map with a per-pixel uncertainty, which is what makes the methylome a CMB problem. 'Brilliance' is the author's term for the posterior surface; Stage 4.6 (brightness comparison) reads a patient's residual against it. [cpg\_patient\_cmb.py; IAMAtlasREBUILD.csv \*\_mean / \*\_sd columns]

**z-departure** —  $z = (\text{beta\_patient} - \mu_{\text{class}}) / \max(\sigma_{\text{class}}, 0.005)$  per CpG: how many posterior standard deviations the patient sits from the healthy class at that locus. The residual map of one observation against the reference. Median  $|z|$  over a class was used for assessability; the deconvolver presence gate is the correct gate. [cpg\_patient\_cmb.py]

**HEALPix / NSIDE / Mollweide** — HEALPix: the equal-area sphere pixelisation used for CMB maps. NSIDE 128 -> 196,608 pixels. CpGs are laid on the sphere in genomic order (chr1 ... chrY, MAPINFO within chromosome), i-th CpG -> pixel floor( $i \times \text{npix} / n$ ); ~2.46 CpGs per pixel, averaged. Mollweide is the equal-area full-sky projection every plate uses. Deterministic: same atlas + manifest -> byte-identical mapping. [MethylPhys/atlas/healpix\_mapping/README]

**Component separation** — Cosmology's name for pulling a mixed signal apart into its sources; here, deconvolving a bulk sample into cell-type fractions (Stage 2). The Walther NNLS deconvolver is the composition the report stands on. NILC (the Planck needlet method) was tried as a second, cut on 2026-07-02 after it appeared to collapse on correlated blood mixtures, then rerun in September against the same arrays and vindicated: the disagreement was the finding, not the defect (PROC-NILC-01). It runs now as stage\_2b\_second\_opinion in the conductor - compared with Walther AT CLASS LEVEL, where PROC-SEP-03 showed the atlas is separable, and reported as an agreement flag rather than as the composition. Cell-level disagreement inside one lineage is expected, because the atlas cannot split the blood classes. [walther\_iam\_deconvolver.py; commit c1be0c3]

**Foreground** — In cosmology, a separate physical source (the Galaxy) subtracted to reveal the CMB. In the methylome the 'foreground' - age, sex, smoking - is the patient's own biology, so it is annotated, not subtracted. The subtraction layers were built and deliberately left unwired (SOP s104). [RETIRED/.../IAM\_Cellular\_Age/]

**Matched filter** — Stage 5 second chain (SOP 8.2): the Pearson correlation between a patient's per-CpG departure and a disease's sealed residual template, with a bootstrap CI. Fires only when Stage 8 Route B flags; a template with no mean shift correctly does not match. [stage\_5\_second\_chain.py]

**Mahalanobis departure (Option A)** — One number per patient:  $\text{distance} = \sqrt{\sum_c ((A_c - \mu_c(\text{age})) / \sigma_c(\text{age}))^2}$  over the classes assessable in the substrate, where  $\mu_c = A_{\text{mean}}(\text{class}, \text{age})$  and  $\sigma_c = (A_{\text{mean}} - A_{\text{p10}}) / 1.2816$  from the age band. Answers 'is the whole profile off' when no single class is. A healthy patient sums to ~0. Alarm by n-adaptive chi-square. Supersedes v1\_0 ( $\mu=1.0$ ,  $\sigma=0.02$  on the 115 separation A's, which put healthy at  $z \sim -22$ ). [iamatlas\_mahalanobis\_scoring.py]

**Null runner / the eight nulls** — The methylome's CMB null tests: no CPG-VAL is sealed until its declared nulls pass. N1 label permutation (1000x); N2 age-strata permutation; N3 sex-strata; N4 cohort split-half replication with consistent sign; N5 plate / array-position; N6 injection-recovery (inject a known signal into HC, recover it); N7 end-to-end simulation (synthetic patients with known truth through L1-L8); N8 look-elsewhere (Bonferroni/FDR). Required for every VAL: N1, N4; for end-to-end claims: N7. 'A VAL that doesn't declare its nulls is not a sealed VAL - it's a preliminary analysis.' [MethylPhys/chain/CPG\_Null\_Runner/cpg\_null\_runner.py]

**Synthetic patient** — A constructed beta vector (or Stage-4 dict) with known class fractions and known per-cell A's, everything else absent. The ground truth for N6/N7: the chain must recover what was put in. [report\_builders/synthetic\_patient\_harness.py; TEST\_DATA/harness/]

**PREREG / seal / RESTATE** — Every validation declares its prediction, direction, pass conditions and nulls in a PREREG before data are opened; the file is sealed by SHA-256. Outcomes are recorded as pre-registered codes (O1\_PRIMARY\_VALIDATED ... O3\_INVERTED, NULL, DIRECTIONAL). A change after seeing data is an AMENDMENT, dated; a re-reading is RESTATE; a post-hoc accommodation VOIDs the VAL with the seal preserved (VAL-102). [SOP Part IV; Record/VAL\_INDEX.csv]

**Chain of custody (L1-L9)** — The audit grading of the chain: nine links from raw IDAT (L1) through calibration (L2-3), component separation (L4), scoring (L5-6), nulls (L7), report (L8-9), each with its own evidence requirement. The SOP is organised by stage (operational order) and by link (audit order). [SOP s2, s3]

**Identity loci vs discriminative markers** — Two CpG sets, two surfaces. Identity loci: where a healthy class sits within  $\pm 0.05$  of  $H_{\text{min\_beta}}$  (unimodal) - the GAUGE reads beta\_mean here. Discriminative markers: ~100 one-vs-rest CpGs per cell type (bimodal by construction) - the SEPARATION statistic reads mean-of-per-CpG-H here. Using one set with the other's formula is the 2026-06-11 all-BREACH bug. [iamatlas\_gauge\_identity\_loci\_v1\_0.json; iamatlas\_celltype\_markers\_v0\_2.json; s105-s106]

**Jensen gap** —  $H(\text{beta\_mean}) - \text{mean}_i H(\text{beta}_i) \geq 0$  (Jensen's inequality,  $H$  concave). Near zero on a unimodal panel (identity loci in blood: 0.013-0.033); large on a bimodal one (marker panels:  $\sim 0.7$ ) or a tissue mixture ( $\sim 0.2$ ). The gauge refuses any panel whose gap exceeds 0.05 - the runtime form of s105. [cpg\_kit.gauge\_A]

**Presence / DETECT\_FLOOR** — A class is present in a sample when its deconvolved fraction  $\geq$  DETECT\_FLOOR (0.01 in the conductor; a 3% gate in the Mahalanobis adjudicator - RECON D2). An A-score for an absent class is entropy computed over cells that are not there and is not a reading. [cpg\_conductor.py]

**Commitment line (A = 1.0)** — The point where the class's measured entropy equals its floor. It is NOT where healthy sits: healthy is the age-matched band (immune 0.906 at age 4  $\rightarrow$  1.000 at 95). 'A = 1.0 at the healthy floor' is the pre-band (Issue 002-era) definition. [cpg\_gauge\_engine.py docstring]

**Cellular age** — Stage 6: invert the age band's  $\text{beta\_mean}(\text{age})$  curve per class to the age at which healthy  $\text{beta\_mean}$  equals the patient's. v1 was a Horvath-style trained clock (not physics); v2 inverted A with the wrong formula (Jensen). v3 is the Recipe's inversion; built, calibration pending, not reportable. [iam\_cellular\_age\_scoring.py]

**Bidirectional decomposition** — Stage 4.5. Shannon  $H$  is symmetric about  $\text{beta} = 0.5$ , so a disease that pushes some CpGs up and others down leaves  $\text{beta\_mean}$  unmoved and the pooled gauge reads null (VAL-050,  $d = +0.08$ ). A directional per-CpG composite recovered  $d = +0.62$  (VAL-051); Stage 4.5 runs that sealed formula per patient. [bidirectional\_decomposition.py]

**Flatness (the Atlas lesson)** — The v0.1 rebuild lesson: the old MCMC script showed  $R\text{-hat} \sim 1.01$  (looks perfect) while every cell type in a class came out identical - a flat, broken atlas; the fixed script shows  $R\text{-hat}$  1.4-2.6 (looks alarming) while the cell types are distinct and the atlas is right. Convergence diagnostics cannot see the failure that matters; judge a rebuilt class by the distinctness test on `per_celltype.csv`, never by  $R\text{-hat}$  alone. [MethylPhys/atlas/IAMAtlas\_FLATNESS\_LESSON.md]

**Xu-538** — The 538-CpG breast-cancer panel of Xu et al. 2019 used as the per-patient scoring surface in VAL-047 (pre-Atlas); 'Xu-538 relativity' names the finding that a panel's absolute A depends on the pipeline that produced the betas, so bands compiled on one pipeline do not transfer to another without re-derivation. [VAL-047 record; s6a]

**Laboratory (as the chain uses the word)** — one processing pipeline - scanner, reagent lots, technicians, normalisation. The laboratory zero belongs to the pipeline that scanned the array, not to the company name.

**Laboratory zero (lab zero)** — the constant by which healthy blood through one laboratory differs from the reference after the pipeline map; measured once on 40 healthy arrays read against the reference age curve; subtracted from every reading from that laboratory; printed on the report. Uppsala 0 / Karolinska +0.024 / Munich -0.021 / UCLA -0.046. UNSET = not reportable as absolute (s3.5).

**Pipeline map** — the affine beta-scale correction between a normalisation pipeline and the calibration scale, one per pipeline, fitted once and transferred (`beta_scale_maps_v1.json`). UNMAPPED = not reportable.

**Reference age curve** — the per-decade healthy median of mapped immune A about the grand median, built on four laboratories each zeroed by its own median (`reference_age_curve_v1.json`); rises  $\sim 0.045$  teens  $\rightarrow$  eighties. Panels and patients are read against it.

## GLOSSARY — CHAIN LINKS

One line per runtime file the conductor touches, tagged by what kind of thing it holds. The tags matter more than the names: a **FLOOR** is physics and is one number; a **RULER** says where the gauge reads and is derived from the Atlas; a **BAND** is a cohort statistic and is only as good as its n and pipeline; **CODE** runs; **DATA** is read. Every "healthy reads wrong" case found in September 2026 traced to a BAND, never to a FLOOR (RECON B1).

| file                                                          | kind  | what it is                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MethylPhys/atlas/IAMAtlasREBUILD.csv (.xz)                    | DATA  | The reference sky: 483,092 CpGs x 115 cell types, per-class MCMC posterior mean, sd and CI per CpG. Every patient is read against it. 2026-05-28, frozen.                                                                                                                                                         |
| IAMAtlasREBUILD_celltype_to_class.json                        | DATA  | 115 cell types -> 8 architecture classes. The deconvolver's cell fractions are summed to class fractions through it.                                                                                                                                                                                              |
| cpg_gauge_engine.py H_MIN_TABLE                               | FLOOR | The forty floors: 8 classes x 5 substrates (methyl, nucl, fuzz, wps, frag). Physics; one number each; frozen 2026-04-06, MCMC-confirmed. Never a cohort.                                                                                                                                                          |
| iamatlas_gauge_identity_loci_v1_0.json                        | RULER | Which CpGs the gauge reads, per class: the loci where healthy cells of that class sit within +/-0.05 of H_min_beta. Derived from the Atlas, frozen. Counts terminal 57,247 ... stromal 2,294.                                                                                                                     |
| iamatlas_celltype_markers_v0_2.json                           | RULER | The separation surface: ~100 one-vs-rest discriminative CpGs per cell type (bimodal by construction). Read with mean-of-per-CpG-H, never H(beta_mean) (s105). chrX markers removed (RULING M1b).                                                                                                                  |
| age_reference_matrix.json                                     | BAND  | What healthy looks like at each age, per class: A and beta mean/sd/p10-p90 in 10 age bins. A cohort table compiled 2026-05-28 from nine papers; n is uneven (immune ~100/bin, stem_adult 28). The gauge's placement is read against this. Phase 1 rebuilds it from one cohort, one pipeline.                      |
| tier_breakpoints.json                                         | BAND  | The severity ladder: NORMAL 0.95-1.01, ELEVATED 1.01-1.07, SIGNIFICANTLY_ELEVATED 1.07-1.10, BREACH >= 1.10; Warburg line 1.07. Onset moved 1.04 -> 1.01 on 2026-07-03.                                                                                                                                           |
| mahalanobis_healthy_reference_v2_0_age_matched_derived.json   | BAND  | Stage 5's reference: derived from the age band (mu = A_mean(class, age), sigma from p10). v0_5 / v1_0 retired and refused by the module.                                                                                                                                                                          |
| walther_iam_deconvolver.py                                    | CODE  | Stage 2: NNLS composition against the Atlas. Returns cell and class fractions and a residual. Gates nothing; presence (fraction >= DETECT_FLOOR) decides which classes may be scored.                                                                                                                             |
| cpg_conductor.py                                              | CODE  | The orchestrator (2026-07; replaces walther_clinical.py). Pairs every per-cell A with its deconvolved fraction; a class below DETECT_FLOOR 0.01 is absent, not a reading. Runs Stages 2 -> 4 -> 4.5 -> 5 -> 6.                                                                                                    |
| iamatlas_a_scoring.py                                         | CODE  | The separation statistic: mean_i H(beta_i)/H_min over each cell type's discriminative markers -> 115 per-cell A's for the disease matcher. Guarded by test_a_score_canonical.py (separation surface only).                                                                                                        |
| cpg_gauge_engine.py                                           | CODE  | Stage 4: A = H(beta_mean)/H_min over identity loci, placed in the age band (BELOW/IN/ABOVE) and on the severity ladder. Carries the FLOOR table and the SATURATION_MARGIN.                                                                                                                                        |
| bidirectional_decomposition.py + directional_panels_v1_0.json | CODE  | Stage 4.5, the fix for entropy's symmetry: H is symmetric about 0.5, so a disease pushing CpGs both ways leaves beta_mean unmoved and the gauge reads null (VAL-050 d=+0.08). Scores direction per CpG with the sealed VAL-051 composite (d=+0.62). Immune panel only in v1.0.                                    |
| cpg_patient_cmb.py + MethylPhys/atlas/healpix_mapping/        | CODE  | Stage 4.6: the patient's per-CpG z = (beta - mu)/sigma against the Atlas posterior, on the HEALPix NSIDE-128 sky (Plate 05). The residual map of one observation against a reference map with per-pixel uncertainty - the Planck workflow.                                                                        |
| iamatlas_mahalanobis_scoring.py                               | CODE  | Stage 5, Option A: the eight class-gauge A's as a vector, distance from the age-band centroid with the band's covariance, n-adaptive chi-square alarm. Answers 'is the whole profile off' where any single class may be in band.                                                                                  |
| iam_cellular_age_scoring.py                                   | CODE  | Stage 6: the band run backwards - at what age does healthy beta_mean(class) equal this patient's? Third attempt: v1 was a trained clock (wrong), v2 inverted the wrong formula (Jensen). Built; calibration pending; reads 4 yr for adults today because the band curves are too flat to invert - not reportable. |
| stage_0_intake.py                                             | CODE  | Stage 0: manifest, IDAT header vs declared array, SHA-256 integrity with re-transmission detection, control-probe / detection-p / bead / call-rate / sex QC (the last four await the Stage 1 hand-off), PROCEED / PENALTY / QUARANTINE gate. Fail-open on 0.1 quarantines closed 2026-09-19.                      |

| file                                                                                    | kind        | what it is                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| stage_1_idat_calibration.py                                                             | CODE        | Stage 1: raw IDAT pair -> noob-calibrated beta via methylprep (pandas < 2). Bit-identical to the project cache on 11/11 (PROC-CAL-01). Owns any raw-array alignment; no per-class input offset is applied downstream. |
| RETIRED/.../IAM_Cellular_Age/<br>(age_/sex_/smoking_axis_foreground.py,<br>*_layer.csv) | RETIRE<br>D | Stage 3 foreground subtraction, built and deliberately not wired (SOP s104): a galactic foreground is a separate source; the methylome 'foreground' is the patient's own biology - annotated, never subtracted.       |

[Issue 003 note, RECON M2] The entry above is Issue 002's and is superseded. The Mahaffey number of the cell is 20.94 (author, 2026-09-19), defined as the ATP ordering budget per event over  $k_B T$  at body temperature;  $n_{\text{bio}}$  was an early QAPE-era form of the ratio and is no longer used.  $H_{\text{min}}$  is not the Mahaffey number:  $H_{\text{min}}$  is the per-class entropy floor in bits, forty values across eight classes and five substrates.

## IN CLOSING

What we hope this is for

Almost nothing in this document was invented here. The pixelisation of a sphere, the needlet and internal-linear-combination methods, the beam and mask discipline, the shuffled-sky null, the difference map, the habit of publishing what an instrument cannot see beside what it can - all of it was built by the cosmology community over thirty years, on data far harder to come by than a methylation array, and tested until it could be trusted. What is done here is to point those tools at a different sphere. The hope is that this is the useful direction of travel: where two fields have each spent decades solving the same measurement problem in isolation, the answer is to learn each other's methods rather than to reinvent them. A geneticist should not have to derive a spherical harmonic transform, and a cosmologist should not have to learn deconvolution from scratch. The wheel is already round in both places.

### Where this could matter

**Other species.** The gauge is a ratio of two physical quantities rather than a fit to human data, so it carries to any animal whose floors can be measured. Dogs are the obvious next cohort: the public data exists, veterinary medicine has fewer barriers to a new measurement than human medicine does, and a companion animal ages beside its owner several times faster - which makes it the natural place to test a trajectory that takes a human lifetime to trace.

**Clinical trials.** A trial spends much of its money on patients who were never going to respond. If a reading of cellular fidelity can help choose who enters one, and then show draw by draw whether the treated tissue is holding its pattern or losing it, that is time and money returned to the trial - and fewer patients carrying the burden of a treatment that was not going to work for them.

**End-of-life care.** The hardest question a family asks is how much time there is, and the honest answer today is usually a guess with a wide interval. A measurement of how much ordered capacity a tissue still holds would not answer it either - but it might narrow it, and a doctor with a narrower interval can help a family spend what is left better.

**Early detection, in its place.** A fidelity gauge that reads below its healthy line before symptoms appear is where clinical value would ultimately sit, for cancer and for much else. That work is not in this document, the chain commissioned here names no condition, and the evidence for it will arrive - if it arrives - as sealed runs against pre-registered bars in a later issue. It belongs on the list, not at the front of it.

### And for geneticists, whatever they make of the physics

Set the Landauer floor aside entirely and there is still a toolkit here that was not available before. A methylome projected onto a sphere, where a whole array is one image instead of hundreds of thousands of rows, and where the structure is genomic: every pixel holds a contiguous run of CpGs on one chromosome. An angular power spectrum that says at what genomic scale a pattern lives. A null that has to come back empty before a map is believed. A mask that refuses to report where the instrument cannot see. And the one we would put first: the **difference map** - the same patient measured twice, where the technical floor is common to both draws and cancels, so an island-scale change resolves an order of magnitude below the effects the literature reports from single readings. Cosmology has taken that idea about as far as it can go on one sky it cannot revisit. In a clinic, the sky can be measured again next year.

None of these tools depend on whether the entropy ratio in this document survives review. They are borrowed, they work, and they are now pointed at a methylome. What anyone does with them next is the interesting part.